

Nullexit: Unified Project Document

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1 Project Directory Tree

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|-- .gitignore
|-- .host_ips
|-- LICENSE
|-- README.md
|-- Recover-Gateway.applescript
|-- Toggle-Gateway.applescript
|-- agent.md
|-- benchmarks
|   |-- benchmark.sh
|   `-- test_load.py
|-- black_list.txt
|-- cloud-init.yaml
|-- devref.md
|-- diagrams.md
|-- docker
|   `-- tor
|       |-- Dockerfile
|       `-- entrypoint.sh
|-- docker-compose.yml
|-- launchd
|   `-- com.nullexit.wake-recovery.plist
|-- post-rules.txt
|-- recover.sh
|-- scripts
|   |-- Dockerfile.routing-fix
|   |-- Dockerfile.rule-compiler
|   |-- Toggle-Gateway.bat
|   |-- check_circuits.py
|   |-- common.sh
|   |-- crypto.sh
|   |-- diagnose-host-leak.sh
|   |-- dns-proxy.py
|   |-- fix-docker-bridge-collision.sh
|   |-- fix-ssh-delay.sh
|   |-- generate_tex.py
|   |-- logger
|   |   |-- go.mod
|   |   `-- main.go
|   |-- pf.conf
|   |-- recover-linux.sh
|   |-- reinstall-host-tailscale.sh
|   |-- routing-fix.sh
|   |-- rule-compiler
|   |   |-- go.mod
|   |   `-- main.go
|   |-- setup-common.sh
|   |-- setup-linux.sh
|   |-- toggle-linux.sh
|   |-- unlock-files.sh
|   `-- watcher.sh
```

```
|-- setup.sh  
|-- toggle.sh  
|-- tor-bridges.txt  
`-- white_list.txt
```

2 README.md

```
1 # nullexit: Tailscale + Cloudflare WARP Docker Gateway
2
3 > **Last updated:** July 11, 2026 [Architecture & Flow Diagrams ](./diagrams.md) [Full Dev Reference ](./
  devref.md)
4
5 **nullexit** is a chained network gateway that routes all Tailscale exit-node traffic through a Cloudflare WARP
  VPN tunnel double-encrypting every packet, hiding your ISP metadata, and providing network-wide DNS ad-
  blocking (AdGuard Home) and kernel-level IP threat blocking (`ipset`/`iptables`) for every device on your
  mesh.
6
7 ---
8
9 ## 1. Prerequisites
10 - Docker and Docker Compose installed.
11 - A Tailscale account.
12 - **macOS:** Install the standalone Tailscale CLI via Homebrew (`brew install tailscale`) and register `
  tailscaled` as a system service (`brew services start tailscale`). No `.app` GUI install required.
13 - **OS & Python:** Developed and tested against **macOS 26.5.2**. The project has zero external Python
  dependencies (no `requirements.txt`) and explicitly targets the default Apple-provided **Python 3.9.6** (`/
  usr/bin/python3`). While the scripts could function on newer versions, `nullexit` explicitly avoids
  experimenting with or modifying the built-in system Python environment to prevent unexpected OS-level side
  effects.
14
15 ## 2. Installation & Setup
16
17 Run the interactive setup script for your OS. It downloads `wgcf`, generates fresh Cloudflare WARP WireGuard
  keys, prompts for a Tailscale Auth Key, and writes your `.env`.
18
19 ```bash
20 ./setup.sh # macOS
21 ./scripts/setup-linux.sh # Linux
22 ```
23
24 ### Reference `.env`
25 ```env
26 WIREGUARD_PRIVATE_KEY=<Generated>
27 WIREGUARD_PUBLIC_KEY=<Generated>
28 WIREGUARD_ADDRESSES=<Generated>
29 TS_AUTHKEY=tskey-auth-...
30 GATEWAY_RULE_PROFILE=medium # light | medium | heavy
31 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
  have a healthy path.
32 GATEWAY_MSS=1120
33 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
  internet is NOT hijacked).
34 GATEWAY_HIJACK_HOST=true
35 GATEWAY_USE_EXIT_NODE=true
36 WARP_FAIL_THRESHOLD=6 # consecutive warp=off polls before auto-shutdown (default 6 = 30s)
37 KILL_SWITCH=false # enforce strict PF lock that breaks SSH if VPN fails
38 # On campus/enterprise networks (WPA2-Enterprise / 802.1x), AP Client Isolation blocks direct
39 # LAN traffic between devices. Tailscale attempts a local P2P upgrade anyway, which causes
40 # macOS gvproxy to SNAT-mangle the reply's source port poisoning the phone's WireGuard
41 # endpoint and blackholing 100% of its traffic. Setting this to false drops those local
42 # hole-punch packets so the phone safely falls back to Tailscale DERP relays.
43 # watcher.sh auto-detects 802.1x and AP isolation on every Wi-Fi change and overrides
44 # this setting automatically you only need to set this manually if auto-detection fails.
45 TAILSCALE_ALLOW_LAN_P2P=false # auto-managed; true only on trusted hotspots/home networks
46 ADGUARD_USER=admin # Username for AdGuard Home (defaults to admin if not set)
47 ADGUARD_PASSWORD=nullexit # Shared password for AdGuard Home and Tor ControlPort (defaults to nullexit
  if not set)
48 BLOCKED_COUNTRIES="kp il" # dynamically block country IP ranges via ipdeny.com
49 TOR_EXCLUDE_EXIT_NODES="" # comma-separated country codes to exclude as Tor exit nodes (e.g. {us},{gb
  })
50 ```
51
52 ## 3. Deploy the Gateway
53
54 **macOS (Recommended):**
55 ```bash
56 ./toggle.sh
57 ```
58 Running it again stops the gateway and restores your network. Handles the full lifecycle: Colima VM, containers
  , DNS hijacking, Tailscale exit-node routing, rule compilation, and sleep prevention.
59
60 **Linux / Manual:**
61 ```bash
62 docker compose up -d
63 ```
```

```

64
65 ## 4. Post-Deployment
66
67 ### Approve the Exit Node
68 1. Go to [Tailscale Admin Machines](https://login.tailscale.com/admin/machines).
69 2. Find the `tailscale` device ``...` **Edit route settings** enable as Exit Node.
70
71 ### Verify Traffic Flow
72 On any Tailscale client device, select this gateway as the exit node, then visit [api.ipify.org](https://api.
73 ipify.org) or run:
74 ```bash
75 curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(\warp|ip|colo)='
76 # expect: warp-on
77 ```
78
79 ### AdGuard DNS Setup
80 To ensure client devices correctly route DNS through the AdGuard container (and do not bypass it via `
81 tailscaled`'s internal resolver), you must configure your Tailscale Admin Console:
82 1. Go to the DNS tab in the [Tailscale Admin Console](https://login.tailscale.com/admin/dns).
83 2. Enable MagicDNS.
84 3. Under Nameservers, add a Custom nameserver and enter the gateway's Tailscale IPv4 address (run `
85 tailscale ip -4` on the gateway machine to find it, e.g., `100.x.x.x`).
86 4. Click the three dots (...) next to the nameserver you just added and enable "Use with exit node".
87 5. Delete any other Global Nameservers (like Google or Cloudflare).
88 6. Toggle ON Override local DNS.
89
90 Traffic will now be correctly forced into the AdGuard sinkhole, and mobile devices will be blocked from
91 bypassing it via encrypted DNS (DoH/DoT).
92
93 > [!WARNING]
94 > Disable "Private DNS" on mobile devices. Android's Private DNS (DoT/DoH) bypasses port 53 interception
95 entirely. Go to `Settings Connections More connection settings Private DNS Off`.
96
97 > [!WARNING]
98 > Bandwidth doubling: Streaming 1GB of video on a remote device consumes 1GB download + 1GB upload on the
99 host's network. Be mindful on metered connections.
100
101 ### AdGuard Dashboard
102 Navigate to `http://<gateway-tailscale-ip>:3000` credentials: admin / nullexit.
103
104 ### Access Any Mesh Device From Anywhere (SFTP/SSH)
105
106 Once the gateway exit node is online, any device on your Tailscale mesh can reach any other mesh device,
107 anywhere in the world as long as both devices are on the mesh and the target device has an SSH server
108 running. No port forwarding, no public IPs, no firewall holes. You can SSH into any device using its
109 MagicDNS hostname (e.g. `ssh username@my-macbook`) or its assigned Tailscale IP (`100.x.x.x`).
110
111 > Security Tip (Zero Local Attack Surface): It is highly recommended to disable macOS's native "Remote
112 Login" and "Screen Sharing" in System Settings to prevent exposing those ports (22, 5900) to physical Wi-Fi
113 networks (which scanners can fingerprint). The gateway enforces `tailscale up --ssh=true` and the `pf`
114 Kill-Switch automatically blocks local Wi-Fi access to these ports. Since both MagicDNS and Tailscale IPs
115 route through the exact same encrypted tunnel, they are equally secure but MagicDNS is recommended simply
116 for ease of use.
117
118 ### Extreme OPSEC: The Secret Tor-over-WARP Proxy
119 `nullexit` includes a completely isolated, headless Tor infrastructure proxy designed for terminals and hacking
120 tools (e.g., `curl`, `sqlmap`, `nmap`).
121 To protect against local malware fingerprinting:
122 1. Procedural Ports: The proxy ports are randomized into the ephemeral range (10000-65000) on every single
123 boot using a strict kernel socket collision-avoidance check (`netstat`).
124 2. Honey-Port Tripwire: The gateway spawns a fake trap port. If any malware attempts a sequential port-scan
125 on `localhost` to find the proxy, it trips the Honey-Port, instantly triggering a macOS desktop
126 notification and a critical log alert.
127
128 > [!NOTE]
129 > Threat Model limitation: Port randomization and the Honey-Port only defeat blind, generic port scanners.
130 They do not protect against targeted local malware that runs `lsof -i` or reads the `/tmp/nullexit-ports.
131 env` file directly. To actually prevent malicious local processes from reaching the proxy, you must run a
132 host-level outbound firewall (like LuLu 4.3.0+ or Little Snitch) with localhost/loopback filtering
133 explicitly enabled to gate access by process identity.
134
135 To use the Tor proxy, run `cat /tmp/nullexit-ports.env` to find your TOR SOCKS_PORT for the current session,
136 and point your hardened browser (LibreWolf/Tor Browser) or CLI tool to 127.0.0.1:TOR SOCKS_PORT.
137
138 > [!WARNING]
139 > Preventing DNS Leaks (SOCKS5 vs. SOCKS5-Hostname):
140 > When routing command-line tools through Tor, you must force the tool to delegate DNS resolution to the
141 Tor proxy. Using plain SOCKS5 (e.g., `socks5://` or curl's --socks5 flag) will resolve hostnames locally
142 * using the host's DNS settings (AdGuard/WARP) before establishing the connection. While this DNS traffic
143 is encrypted inside the WARP tunnel, the destination domain name is still leaked to AdGuard and Cloudflare,

```

```

    undermining your anonymity.
119 >
120 > Use these correct protocols and configurations:
121 > - curl: Use --socks5-hostname or the socks5h:// scheme (e.g., curl -x socks5h://127.0.0.1:
    $TOR SOCKS_PORT https://check.torproject.org). Do not use --socks5 or socks5://.
122 > - sqlmap: Pass the proxy URL using the socks5h:// scheme to ensure remote resolution: --proxy="
    socks5h://127.0.0.1:$TOR SOCKS_PORT".
123 > - nmap: Nmap's built-in --proxies option resolves hostnames locally. To safely scan targets without
    DNS leaks, tunnel it using proxychains-ng configured with proxy_dns enabled (add socks5 127.0.0.1 <
    PORT> to /etc/proxychains.conf or a local config, then run proxychains4 nmap <args>).
124
125
126 ##### Transparent .onion Browsing & Dark Web Access
127 In addition to the randomized SOCKS5 proxy port, nullexit provides seamless, transparent .onion domain
    browsing for all devices on your Tailscale mesh:
128 - Zero-Config DNS: AdGuard Home automatically intercepts queries for .onion domains and resolves them to
    Tor's virtual mapping subnet.
129 - Tailscale Auto-Routing: The virtual subnet is mapped to the 198.18.0.0/15 benchmark range (RFC 2544).
    Since this range is not a private IP subnet, Tailscale's Exit Node mode automatically routes it to the
    gateway (bypassing local LAN exclusions).
130 - Transparent NAT Proxying: Kernel firewall rules in the gateway redirect all TCP traffic destined for
    198.18.0.0/15 directly into Tor's transparent port (9040).
131 - Exit Node Exclusion: Supports the ability to exclude specific countries from being used as Tor exit nodes
    via the TOR_EXCLUDE_EXIT_NODES environment variable in .env.
132
133 ##### Enable in Web Browsers
134 To prevent accidental DNS leaks, web browsers block .onion lookups by default. To browse dark web sites
    natively:
135 - Firefox: Go to about:config, search for network.dns.blockDotOnion, and set it to false. Type any
    onion address (e.g. duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion) directly into the URL
    bar and it will load.
136 - Mobile Browsers (iOS/Android): Connect to your Tailscale exit node, open Firefox (with blockDotOnion
    disabled), and browse onion sites natively on the go.
137
138 ##### Hardening: Using obfs4 Tor Bridges
139 If you want to disguise your Tor traffic from the WARP exit node provider, you can optionally enable Tor
    bridges:
140 1. Obtain private obfs4 bridge lines from [bridges.torproject.org](https://bridges.torproject.org) or the
    official Telegram bot.
141 2. Create a file named tor-bridges.txt in the root of the nullexit folder and paste your bridge lines
    inside it (one per line).
142 3. Set TOR_USE_BRIDGES=true in your .env file and restart the gateway. The proxy will refuse to start if
    the bridges file is empty or missing.
143
144 ##### Example: Browse your Android phone's files from your Mac
145
146
147 1. On your Android phone, install [Termux](https://termux.dev/) and [Termux:Boot](https://f-droid.org/
    packages/com.termux.boot/) from F-Droid.
148 2. In Termux, install and start the SSH server:
149     ```bash
150     pkg install openssh
151     sshd
152     ```
153     Termux defaults to port 8022 (ports below 1024 require root on Android).
154 3. Find your phone's Tailscale IP:
155     ```bash
156     tailscale ip -4 # e.g. 100.87.42.87
157     ```
158 4. Stop Android from killing Termux do ALL of these (Samsung One UI is especially aggressive):
159     - Settings Apps Termux Battery Unrestricted
160     - Device Care Battery Background usage limits Sleeping apps / Deep sleeping apps remove Termux if
    listed
161     - In Termux, run termux-wake-lock to hold a persistent wakelock
162 5. On your Mac/PC, open [Cyberduck](https://cyberduck.io/) (or any SFTP client: WinSCP, FileZilla, FE File
    Explorer on iOS) and connect to:
163     - Server: 100.87.42.87 (your phone's Tailscale IP)
164     - Port: 8022
165     - Protocol: SFTP (SSH File Transfer Protocol)
166     - Username: (your Termux username, usually u0_a123 run whoami in Termux to check)
167     - Password: (your Termux password set with passwd in Termux if you haven't already)
168
169 That's it you can now browse, download, and upload files to your phone from any device on the mesh, no matter
    where either device is physically located. This works identically for any mesh device: Mac Mac, Windows
    Linux, phone NAS, etc.
170
171 > Troubleshooting: If the connection hangs for ~30 seconds before failing, your phone's SSH server isn't
    running. Open Termux and run sshd again. If Android keeps killing it despite Unrestricted battery, the
    termux-wake-lock command is your fix.
172
173 ---

```

```

174
175 ## 5. Multi-Layer Firewall & Kill-Switch
176
177 ### Layer 1 DNS Sinkhole (AdGuard Home)
178 Blocks ads and tracking domains before they are downloaded. In automated Lighthouse tests on ad-heavy sites:
179 - **Time to Interactive:** ~22% faster (51.0s 41.8s)
180 - **Speed Index:** ~20% faster (15.3s 12.8s)
181
182 Customize blocking via `black_list.txt` and `white_list.txt`. Rule profiles (`light` / `medium` / `heavy`) are
    set in `.env`.
183
184 ### Layer 2 Kernel IP Firewall (`ipset`/`iptables`)
185 On every startup, the `rule-compiler` fetches threat-intelligence feeds (Spamhaus DROP, Feodo Tracker, Emerging
    Threats, CINS) and compiles **~16,700 unique malicious IPs/CIDRs** into the kernel `FORWARD` chain
    blocking both outbound C2 connections and inbound attack traffic. Zero configuration required.
186
187 ### Layer 3 Geo-IP Blocking
188 Dynamically blocks all traffic to and from specific countries using live IP ranges from `ipdeny.com`.
189 To add countries to your blocklist, open your `.env` file and set the `BLOCKED_COUNTRIES` variable with 2-
    letter ISO country codes (e.g. `BLOCKED_COUNTRIES="kp il cn ru"`), then restart the gateway.
190
191 ### Layer 4 macOS PF Kill-Switch
192 A native macOS Packet Filter (`pf`) kill-switch that enforces a strict default-deny policy on your physical Wi-
    Fi interface (`en0`). When `KILL_SWITCH=true` is set in your `.env`, it drops all outgoing traffic except
    the Cloudflare WARP endpoints and Tailscale DERP relays.
193
194 - **Failsafe Design:** If the VPN tunnel crashes or Docker dies, your Mac completely loses internet access
    rather than leaking your real IP.
195 - **Remote-Access Safe:** Carefully designed to whitelist Tailscale's control plane (`192.200.0.0/24`), `utun*`
    tunnel traffic, and local LAN subnets. This guarantees that your remote SSH sessions and local AirDrop
    will survive even when the kill switch engages.
196 - **Setup Requirement:** The toggle scripts need permission to manipulate the network and firewall in the
    background without prompting you for a password. You must configure your passwordless sudo config by
    running exactly:
197 ```bash
198 echo "$USER ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /usr/bin/dsccacheutil, /usr/bin/killall,
    /usr/bin/pkill, /bin/kill, /sbin/route, /sbin/ifconfig, /usr/bin/true, /opt/homebrew/bin/brew, /usr/local/
    bin/brew, /usr/bin/python3 -I $PWD/scripts/dns-proxy.py, /opt/homebrew/bin/python3 -I $PWD/scripts/dns-
    proxy.py, /usr/local/bin/python3 -I $PWD/scripts/dns-proxy.py" | sudo tee /etc/sudoers.d/nullexit
199 ```
200
201 ---
202
203 ## 6. Toggle Scripts
204
205 ### macOS `toggle.sh`
206 Fully automated lifecycle script. Also supports a native `.app` launcher:
207 ```bash
208 osascript -o "Toggle Gateway.app" Toggle-Gateway.applescript
209 ```
210
211 Optional `.env` settings:
212 - `GATEWAY_BYPASS_PING=true` proceed even if pre-flight connectivity checks fail.
213 - `GATEWAY_USE_EXIT_NODE=false` DNS-only mode, skip exit-node routing.
214 - `GATEWAY_HIJACK_HOST=false` skip DNS hijacking on the host (provides VPN/adblocking only to external
    Tailscale peers).
215 - `GATEWAY_MSS=1360` override TCP Maximum Segment Size for the tunnel (default is calculated automatically).
216 - `HOST_LEAK_PROBE=false` disable background host-egress leak prober (default: true).
217 - `HOST_MTU=1200` override the host Tailscale interface MTU. Defaults to 1200 to fit inside the 1280-byte WARP
    tunnel.
218 - `KILL_SWITCH=true` enforce strict network lock that breaks SSH if the VPN fails.
219 - `STOP_COLIMA_ON_EXIT=true` fully shut down the Colima VM on toggle-off (saves battery on dedicated hosts).
220 - `TAILSCALE_ALLOW_LAN_P2P=true` allow direct Tailscale LAN P2P connections between devices on the same
    network.
221 - **Default: `false`** On campus or enterprise Wi-Fi (WPA2-Enterprise / 802.1x), AP Client Isolation blocks
    local device-to-device traffic. When Tailscale still attempts a local P2P upgrade, macOS's `gvproxy` layer
    SNAT-mangles the reply's source port. WireGuard's endpoint roaming treats this as a legitimate address
    change and updates the phone's outbound path to the newly-mangled port which has no listener. This
    blackholes 100% of the phone's traffic while the DERP relay path is abandoned. Setting this to `false`
    preemptively drops all RFC1918-destined Tailscale UDP from the gateway so the phone receives a clean
    failure and safely falls back to DERP.
222 - **Auto-managed:** `watcher.sh` automatically detects WPA2-Enterprise via `system_profiler SPAirPortDataType`
    (the `airport` binary was removed in macOS 14.4) and probes for AP isolation by pinging the default
    gateway (`route -n get 1.1.1.1`) on every network change. It writes the detected value to `lan_p2p_detected`
    in the repo root, which `routing-fix.sh` re-reads every 30 seconds **no restart required**
    Only set this manually if auto-detection fails. See `devref.md 36` for the full SNAT endpoint poisoning
    analysis and `devref.md 40` for the `airport` removal background. *(Note: Direct P2P between the gateway
    container and the host Mac itself is always permitted via the Docker bridge to bypass DERP relays,
    regardless of this setting).*
223 - Set to `true` on trusted home networks or Windows/mobile hotspots (no AP isolation) for a faster, lower-
    latency direct path.

```



```

224 - `WARP_FAIL_THRESHOLD=3` number of consecutive failed checks before forcing a gateway teardown (default: 6
    checks = 30s).
225 - `WARP_ENDPOINT_1` / `WARP_ENDPOINT_2` override the Cloudflare WARP WireGuard endpoints.
226
227 ### Linux `scripts/toggle-linux.sh`
228 ```bash
229 ./scripts/toggle-linux.sh # toggle ON/OFF
230 ./scripts/recover-linux.sh # recovery tool
231 ```
232
233 ### Sleep Prevention (macOS)
234 When the gateway is ON, `caffeinate -i` runs in the background to prevent idle system sleep, keeping Docker and
    all services alive even on battery. The display can still sleep normally.
235
236 ### Auto-Recovery (macOS post-wake / post-roam)
237 Install the launchd LaunchAgent so the gateway self-heals after sleep/wake and Wi-Fi roams:
238 ```bash
239 cp ./launchd/com.nullexit.wake-recovery.plist ~/Library/LaunchAgents/
240 sed -i '' "s|__NULLEXIT_HOME_|$(pwd)|" ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
241 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
242 ```
243 See `devref.md 11.16` for the full deep dive.
244
245 ---
246
247 ## 7. Networking Notes
248
249 - Port conflicts: None. Tailscale uses dynamic UDP ports; WARP uses outbound UDP:2408. The gateway binds
    `0.0.0.0:41642/udp` on the host for Tailscale's WireGuard listener this is intentional and required to
    receive inbound hole-punch packets from peers on the internet. All other internal ports (AdGuard :5335,
    SOCKS5 :1080) are bound to `127.0.0.1` only and are not reachable from the LAN.
250 - Tailscale P2P vs DERP: On the same Wi-Fi, Tailscale establishes a fast P2P connection. On cellular (CGNAT
    ), it always falls back to a DERP relay (~80-200ms). See `devref.md 5.1`.
251 - AirDrop / AirPlay: Unaffected the gateway only touches standard Wi-Fi/Ethernet interfaces, not `awdl0`.
252 - VPN coexistence: Do not run a local VPN client (WARP, Mullvad, NordVPN) simultaneously with the exit
    node. Both fight for the default route and will blackhole your connection.
253 - Banking & financial sites: May intermittently block logins through WARP. Banks use anti-fraud databases
    that flag datacenter IP ranges (like Cloudflare's `104.28.x.x`) as VPN/proxy traffic. Success fluctuates
    depending on which WARP IP you land on. If a site blocks you, either disable the exit node temporarily for
    that session or use the bank's mobile app (which often uses different fraud detection).
254 - MSS clamping (Bufferbloat): Double-tunnelling adds ~120-160 bytes of overhead per packet. This is now
    automatically handled natively via the macOS `pf` firewall, which universally applies TCP MSS Clamping to
    all outbound packets to strictly prevent MTU fragmentation and bufferbloat. No manual `.env` configuration
    is required. Additionally, the host's Tailscale interface MTU is explicitly lowered to 1200 (configurable
    via `HOST_MTU` in `.env`) to guarantee packets fit inside the container's WARP tunnel without fragmenting.
255
256 ---
257
258 ## 8. Privacy & Security Hardening
259
260 The stack shifts trust across four layers: WARP hides traffic from your ISP, AdGuard blocks tracking at DNS,
    Tailscale encrypts device-to-device, and `ipset` blocks known malicious IPs at the kernel.
261
262 For higher security, the gateway supports these drop-in upgrades:
263
264 | Layer | Default | Hardened |
265 |---|---|---|
266 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, no-logs, anonymous payment) |
267 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH |
268 | Coordination | Tailscale (US) | Headscale (self-hosted) |
269 | Auth | Persistent OAuth keys | Ephemeral auth keys |
270 | Content | WireGuard asymmetric | WireGuard + Pre-Shared Keys (PSK) |
271
272 #### Python Runtime Airgap (Isolated Mode)
273 Never install third-party `pip` packages (like `requests` or `pyyaml`) into the Python environment that
    `nullexit` uses. All python scripts in this project (e.g., the `dns-proxy.py` daemon) are strictly
    engineered to run on the zero-dependency Python Standard Library. To strictly isolate the environment from
    supply-chain attacks, `nullexit` executes these scripts using Python's Isolated Mode (`python3 -I`).
    This deliberately blinds the execution environment to any malicious user-installed `pip` packages on the
    host, permanently air-gapping the gateway from PyPI.
274
275 See `devref.md 8` for the full threat model, Snowden programme analysis, and trust assumptions.
276
277 ---
278
279 ## 9. Debugging
280
281 - output.log All stderr from `toggle.sh`, `recover.sh`, `setup.sh`, and the rule-compiler is written
    here. The WARP Watcher (`WARP_FAIL_THRESHOLD`) and the Host Leak Probe (`HOST_LEAK_PROBE`) also write all
    their events here `LEAK`, `ROTATE`, `HOST-PROBE failed`, and WARP shutdown notices. Check this first.

```

```

282 - ``bash scripts/diagnose-host-leak.sh`` One-shot host-routing diagnostic. Runs 8 checks and classifies your
      state into one of four known scenarios (System Extension conflict, SOCKS5 fallback, IPv6 leak, route-table
      freeze) or OK, and prints the exact fix command. Check 5 uses route -n get 1.1.1.1 to ask the kernel
      which interface real traffic actually resolves through (more accurate than netstat -rn | grep default on
      macOS). Pass --fix to apply the remediation automatically. Pass --watch (or --watch 30) to run a full
      baseline diagnostic then continuously monitor warp/IPv6/default-route for leaks, alerting on any state
      change.
283 - ``bash scripts/host-leak-probe.sh`` Sub-second host-egress prober (enabled automatically via 
      HOST_LEAK_PROBE=true in .env). Polls cdn-cgi/trace every 300ms directly from the host NIC not via 
      docker exec so it catches flash-leaks invisible to the in-container WARP Watcher. All state changes, curl
      errors, and probe timeouts land in output.log. Grep with grep LEAK output.log.
284 - ``bash scripts/fix-docker-bridge-collision.sh`` One-shot fix for Docker bridge IP conflicts. If your local
      Wi-Fi router assigns you an IP in the 172.17.0.0/16 range, it will violently collide with Docker's
      default internal bridge, permanently killing your internet. This script detects the collision and auto-
      patches your Colima VM to use a safe subnet (e.g. 10.200.0.1/24).
285 - ``devref.md`` Complete architecture reference, routing deep-dives, and full resolved-issues log. Read this
      before modifying any routing or firewall logic.
286
287 > [!CAUTION]
288 > Two infinite routing loops are possible. (1) If a hotspot device providing internet to the gateway host
      is *also* using the gateway as its exit node, packets bounce forever between the two. (2) When the host's
      default route is redirected to utun*, WARP endpoint packets can loop back into the tunnel. Both are
      documented in devref.md 11.10 and 11.18 and are automatically handled by toggle.sh.
289
290 ---
291
292 ## 10. Unified Project Document (Quine)
293
294 The project includes a generate_tex.py script that compiles the entire source code, configurations, and
      documentation into a single, unified LaTeX document (nullexit_unified.tex / .pdf).
295
296 Funnily enough, this document acts as a "project-level quine." It encapsulates the entirety of its own source
      code, its documentation, and the exact Python code required to generate itself. It serves as a self-
      contained, long-term archiving strategy providing everything needed to reconstruct and understand the 
      nullexit system from a single file.
297
298 ---
299
300 ## 11. Acknowledgements
301 - ``[SyameimaruKoa](https://github.com/SyameimaruKoa):`` Dual-stack TCP MSS clamping, SIGHUP state-tracking
      in the routing sidecar, and TS_AUTH_ONCE integration.
302
303 ## 12. License
304 GNU Affero General Public License v3. See [LICENSE](./LICENSE).
305
306 ## 13. Changelog
307
308 - ``July 10, 2026`` ``Bug fixes``
309
310 1. Resolved a race condition where Wi-Fi roaming caused the host IP to leak as the raw ISP IP (132.x) after
      every network switch. The WARP Watcher was firing nuclear shutdown while recover.sh --post-wake was
      still intentionally recreating the warp container. Fixed by adding a /tmp/nullexit-warp-inhibit.marker
      that pauses the watcher's failure counter during post-wake recovery. See devref.md 25.
311
312 2. Resolved a startup race condition where the pre-flight connectivity check [1/3] Gateway reachable via
      Tailscale would fail because the host's tailscaled had not yet propagated the netmap immediately after 
      tailscale up --reset. Upgraded the check to a 5-attempt retry loop with a 1-second delay. See devref.md
      26.
313
314 3. Resolved a concurrency collision where the network changes from a nuclear recover.sh teardown triggered
      watcher.sh to run recover.sh --post-wake in parallel, corrupting container states. Implemented a shared
      lifecycle lock (/tmp/nullexit-toggle.lock) across both scripts to safely skip post-wake logic during
      teardowns. See devref.md 27.
315
316 4. Resolved an infinite auto-recovery loop after a tunnel failure shutdown. When the WARP Watcher triggered
      nuclear recover.sh to teardown the gateway, the active-state marker file /tmp/nullexit-gateway-active.
      marker was leaked on disk. The resulting network configuration changes from the teardown then triggered 
      watcher.sh, which saw the marker and immediately restarted the gateway, looping indefinitely. Fixed by
      explicitly deleting the marker file at the start of nuclear recover.sh. See devref.md 28.
317
318 5. Resolved a pre-flight check false negative during cold boots where the [1/3] Gateway reachable via
      Tailscale check would fail. Because gluetun blocks UDP STUN traffic to the public internet, Tailscale is
      forced to use a slower DERP-assisted handshake which takes ~30-35 seconds on cold boots. Increased the
      retry loop limit to 45 attempts (45s) to give the handshake sufficient time to complete. See devref.md
      29.
319
320 6. Resolved a startup race condition during restarts where docker compose build would fail with DNS
      resolution errors because physical Wi-Fi was power-cycled during teardown and Colima booted before the host
      obtained a DHCP lease. Implemented a unified wait_for_dhcp_settle helper (configured with a 30s timeout
      to tolerate macOS DHCP lease latency) in scripts/common.sh and added it to the start of toggle.sh. See
      devref.md 30.
321
322 7. Resolved a bug where enabling the PF Kill-Switch during SOCKS5 fallback mode blocked all host traffic (
      including tailscaled daemon connections to the control plane) because SOCKS5 does not route non-TCP/
      system traffic. Removed kill-switch activation from the SOCKS5 fallback path in toggle.sh. See devref.md
      31.
323
324 8. Resolved a bug where host-side tailscale up calls passed the --reset flag, which aggressively cleared
      macOS Tailscale credentials and logged the user out. Removed the --reset flag from all host-side

```

```

317     invocations to preserve authenticated sessions during exit-node state toggling. See `devref.md 32`.
318 9. Resolved a bug where container connectivity check failures in `toggle.sh` and `recover.sh` were discarded
    to `/dev/null`. Redirected stderr of Docker exec checks to `output.log` to prevent diagnostic blind spots
    during tunnel outages. See `devref.md 33`.
319 - **July 6, 2026** Edge-case security and stability hardening: DNS Watcher interface independence, robust lock
    -file PID verification, fail-closed crypto integrity check, strict `iptables` pinning for tailscale0tun0,
    Docker port binding to `127.0.0.1` to prevent LAN exposure, routing verification via `netstat -nr`. See `
    devref.md 12`.
319 - **July 4, 2026** Colima bridged networking, SOCKS5 proxy migrated natively to Tailscale, headless prompt
    crashes fixed, proxy bypass domains added to fix LAN P2P. Python dependencies completely eliminated: `
    logger` and `rule-compiler` ported to blazing-fast Go multi-stage Docker builds. Added `crypto.sh` to
    enforce cryptographic signature validation on all core bash scripts. Massively refactored `toggle.sh` and `
    recover.sh` to eliminate ~200 lines of duplicated code by migrating network and process logic to `scripts/
    common.sh`. Added a concurrency mutex to `toggle.sh`, resolved unbound `TS_AUTHKEY` check crashes on setup,
    and integrated Go validation to the CI pipeline.
320 - **July 3, 2026** **Critical Security Updates:** Addressed the overnight silent IP leak and improved geo-
    blocking. Introduced `scripts/host-leak-probe.sh` (a sub-second host-egress leak detector) and a
    statistical threshold auto-shutdown watcher. Added `unlock-files.sh` to safely reset file permissions.
    Resolved numerous startup regressions and system bugs.
321 - **July 2, 2026** Two infinite routing loops resolved (`devref.md 11.18`): host-VM tunnel loop (WARP bypass
    routes now via gateway IP + manual `utun` redirection) and Docker Compose subnet takeover (`10.200.1.0/24`
    lock). `--accept-routes=true` now explicit on all `tailscale up --reset` calls.
322 - **July 2, 2026** Auto-recovery daemon documented in 6 above (`scripts/watcher.sh` + launchd plist). See `
    devref.md 11.16`.
323 - **July 2, 2026** Linux scripts (`scripts/toggle-linux.sh`, `recover-linux.sh`, `setup-linux.sh`) documented
    in 6.
324 - **July 1, 2026** `recover.sh --post-wake` ships; wired to watcher daemon. Triggered on every sleep/wake or
    network-state change.
325 - **June 28, 2026** 8 internal scripts moved to `scripts/`. User-facing files (`toggle.sh`, `recover.sh`, `
    setup.sh`, `docker-compose.yml`) stay at repo root.
326 - **June 28, 2026** All hardcoded install paths removed. `SCRIPT_DIR`-relative resolution and `
    _NULLEXIT_HOME` placeholder used throughout.
327
328 ### Cryptographic Integrity Verification
329
330 nullexit uses a built-in cryptographic integrity checker designed to provide tamper-*evidence* against
    accidental edits, logic bugs, or non-privileged malware. During setup, a 256-bit `NULLEXIT_SEED` is
    injected into your `.env` file. The core entrypoint scripts (`toggle.sh`, `recover.sh`, etc.) are hashed
    using HMAC-SHA256, and their signatures are stored in `.signatures`.
331
332 *(Note: This is not a strict boundary against an attacker who already has write access to the repo, as they
    could simply read the seed and re-sign the scripts. It is designed as a strict footgun-catcher).*
333
334 If any script is modified without authorization, the gateway will loudly fail to boot. If you make intentional
    edits to the bash scripts, you must re-sign them by running:
335 ```bash
336 ./scripts/crypto.sh --sign
337 ```

```

3 agent.md

```

1 # nullexit Detailed Agent Analysis
2
3 > Complete per-file breakdown of every function, constant, duplication, and bug found.
4 > **Note (July 2026):** `sync-rules.py` and `logger.py` have been fully ported to Go (`scripts/rule-compiler/
    main.go` and `scripts/logger/main.go`) to dramatically improve performance and reduce container footprint
    via multi-stage Alpine builds. The analysis below reflects their previous Python states.
5
6 ## Important Agent Instruction
7 - **NEVER use `cat << EOF` or terminal redirections to write or modify files.** You must always use your native
    file-editing tools (`replace_file_content` or `multi_replace_file_content`). Using `cat EOF` leads to
    duplicated text, broken markdown headers, and structural corruption (which previously destroyed the section
    numbering in `devref.md`).
8 - **Always run `bash scripts/crypto.sh --sign` after making any modifications to the core bash scripts.** This
    is required to update the cryptographic HMAC-SHA256 signatures; otherwise, the startup integrity checks
    will fail and block execution.
9 - **Always run `git diff` and print the changes to the user *before* requesting or executing any git commit
    command.** The user must be shown the diff so they can review and understand the changes. Based on this
    diff, formulate a highly precise, detailed, and descriptive commit message (explaining exactly what was
    changed and why) rather than a generic summary, and present it to the user alongside the diff.
10 - **Always run `git status` before `git diff` to identify all modified and untracked files, ensuring no files (
    such as new scripts or configuration assets) are missed before formulating commit diffs and staging.**
11 - **Always use the `--restart` flag when asked to restart the toggle (e.g., run `./toggle.sh --restart`).**
12 - **NEVER execute any `sudo` commands (especially `sudo route`, `sudo dscacheutil`, or any network-modifying
    commands) without explicitly explaining to the user what the command does and why it is needed FIRST. Do
    not assume implicit permission. Wait for the user's approval before running it.**

```

```

13 - **When fixing an error and using logs to debug, always show the user the reasoning and the specific logs that
    support this theory.**
14 - **When the user tells you to "push", this means read git diffs and prepare commit messages that correspond to
    these changes (do not ignore any changes). If the changes can be atomized into multiple commits for easier
    understanding, do so.**
15 - **When the user says "latex this", it means you should run `python3 scripts/generate_tex.py` to regenerate
    the unified LaTeX document.**
16 - **NEVER apply changes to running gateway containers directly via `docker compose up` or `docker compose
    restart`.** Recreating or restarting containers (especially the `warp` container which hosts the network
    namespace) breaks the shared network stack for all other services (`adguardhome`, `tailscale`, `routing-fix`
    `), severing host connectivity and freezing macOS DNS. If you need to apply changes or rebuild containers,
    cleanly stop the gateway first using `./toggle.sh`, or trigger `recover.sh --post-wake` to rebuild and re-
    verify the network stack safely in the correct dependency order. Do NOT run `./toggle.sh --restart` or any
    network-rebuilding commands yourself if the execution could sever host network access; instead, explain the
    changes and ask the USER to run `./toggle.sh --restart` (or `./toggle.sh` stop and start) themselves to
    safely apply the modifications.

17
18 ## How to Verify Gateway is Working
19 Whenever modifications are made to the gateway scripts, routing, or containers, execute these verification
    checks in order:
20 1. **Container Health:** Run `docker compose ps` to ensure all containers (`warp`, `adguardhome`, `routing-fix`
    ``, `tailscale`) are `running` and `healthy`.
21 2. **DNS Hijacking Status:** Run `networksetup -getdnsservers "Wi-Fi"` (or appropriate network service) and
    ensure it is set exclusively to the gateway's Tailscale static IP (typically `100.100.21.8`).
22 3. **DNS Query Interception:** Run `dig @100.100.21.8 google.com` (using the gateway static IP from `.`
    gateway_ip`) to ensure AdGuard Home is active, intercepting, and resolving queries.
23 4. **Double-Tunnel Egress:** Run `curl -s https://www.cloudflare.com/cdn-cgi/trace | grep warp` to verify the
    egress is routed through Cloudflare WARP (`warp=on`).
24 5. **Host Leak Monitoring:** Run `tail -n 30 output.log` and verify the background watchers and host leak
    probe are reporting healthy status without `LEAK` warnings.

25
26 ---
27
28 ## Part 1: macOS Main Scripts
29
30 ### toggle.sh (1281 lines, 56KB)
31
32 **Functions (27 total):**
33
34 | # | Function | Lines | Purpose |
35 |---|---|---|---|
36 | 1 | `run_with_timeout()` | L3064 | Run a command with a timeout watchdog; kills after N seconds |
37 | 2 | `run_gui_cmd()` | L6880 | Run a GUI command as the logged-in console user |
38 | 3 | `write_gateway_active_marker()` | L8890 | Write timestamp to `/tmp/nullexit-gateway-active.marker` |
39 | 4 | `clear_gateway_active_marker()` | L9295 | Remove gateway active marker + TUNNEL_FAILED_CLOSED.marker |
40 | 5 | `start_sleep_prevention()` | L108154 | Start `caffeinate` in background with shutdown trap to revert DNS
    |
41 | 6 | `stop_sleep_prevention()` | L157167 | Stop caffeinate process via PID file |
42 | 7 | `start_dns_watcher()` | L173191 | Background loop to re-hijack DNS every 30s for Wi-Fi roaming |
43 | 8 | `stop_dns_watcher()` | L193203 | Stop DNS watcher via PID file |
44 | 9 | `start_warp_watcher()` | L220279 | Background WARP liveness monitor; triggers recover.sh after N
    consecutive failures |
45 | 10 | `stop_warp_watcher()` | L281291 | Stop WARP watcher via PID file |
46 | 11 | `start_host_leak_probe()` | L297310 | Start host-egress leak probe (300ms polling) |
47 | 12 | `stop_host_leak_probe()` | L312322 | Stop host leak probe via PID file |
48 | 13 | `cleanup_handler()` | L325382 | Error/signal handler: resets DNS, stops all watchers, tears down
    containers |
49 | 14 | `restart_tailscaled_daemon()` | L449463 | Restart tailscaled via brew services (user then system
    fallback) |
50 | 15 | `disconnect_tailscale_host()` | L465477 | Disconnect host Tailscale with retry on failure |
51 | 16 | `get_active_service()` | L480508 | Detect active macOS network service name (e.g., "Wi-Fi") |
52 | 17 | `add_warp_bypass_routes()` | L524546 | Add static routes for WARP endpoints (162.159.192.1,
    162.159.193.1) |
53 | 18 | `remove_warp_bypass_routes()` | L548552 | Remove WARP bypass routes |
54 | 19 | `setup_exit_node_routing()` | L558577 | Override default route to point through Tailscale utun interface
    |
55 | 20 | `reset_dns()` | L588595 | Reset DNS to 1.1.1.1 on ACTIVE_SERVICE + ENO_SERVICE |
56 | 21 | `cleanup_network_state()` | L600645 | Nuclear cleanup: disable proxies, flush DNS cache, flush routes,
    power-cycle Wi-Fi, kill sharing services |
57 | 22 | `enable_socks_proxy()` | L663682 | Enable system-wide SOCKS5 proxy on macOS |
58 | 23 | `disable_socks_proxy()` | L684689 | Disable SOCKS5 proxy |
59 | 24 | `stop_local_dns_proxy()` | L702710 | Kill local Python DNS proxy |
60 | 25 | `start_local_dns_proxy()` | L713768 | Start Python DNS proxy (UDP:53 TCP:5354) |
61 | 26 | `force_dns_to_gateway()` | L780820 | Force DNS to a specific IP with 3-retry verification loop |
62 | 27 | `is_gateway_active()` | L833854 | Check if gateway is running (containers, DNS, SOCKS proxy) |

63
64 **Constants:**
65
66 | Constant | Value | Line |
67 |---|---|---|
68 | `LOG_FILE` | `$PWD/output.log` | L8 |

```

```

69 | `PID_FILE` | `/tmp/nullexit-cafeinate.pid` | L105 |
70 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L169 |
71 | `WARP_WATCHER_PID_FILE` | `/tmp/nullexit-warp-watcher.pid` | L170 |
72 | `HOST_LEAK_PROBE_PID_FILE` | `/tmp/nullexit-host-leak-probe.pid` | L171 |
73 | `SOCKS_PROXY_PORT` | `1080` | L661 |
74 | `PATH` export | `/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH` | L396 |
75 | Gateway active marker | `/tmp/nullexit-gateway-active.marker` | L89 |
76 | TUNNEL_FAILED_CLOSED marker | `${SCRIPT_DIR}/TUNNEL_FAILED_CLOSED.marker` | L94 |
77 | WARP endpoints | `162.159.192.1`, `162.159.193.1` | L535-544 |
78 | DNS fallback | `1.1.1.1` | L592 |
79 | DNS search domain | `ts.net` | L794 |
80 | `LOCK_FILE` | `/tmp/nullexit-toggle.lock` | L62 |
81
82 ---
83
84 ### recover.sh (566 lines, 23KB)
85
86 **Functions (7 total):**
87
88 | # | Function | Lines | Purpose |
89 | ---|-----|-----|-----|
90 | 1 | `step()` | L40 | Print bold step header |
91 | 2 | `ok()` | L41 | Print green success message |
92 | 3 | `warn()` | L42 | Print yellow warning message |
93 | 4 | `fail()` | L43 | Print red failure message |
94 | 5 | `die()` | L44 | Print red error and exit |
95 | 6 | `run_with_timeout()` | L5674 | Run command with timeout (bash-native, no GNU coreutils) |
96 | 7 | `restart_tailscaled_daemon()` | L118129 | Restart tailscaled daemon via brew services |
97
98 **Constants:**
99
100 | Constant | Value | Line |
101 | ---|-----|-----|
102 | Color codes | `RED`, `GREEN`, `YELLOW`, `BOLD`, `NC` | L37-38 |
103 | `SCRIPT_DIR` | `${dirname "${BASH_SOURCE[0]}"}` | L50 |
104 | `PATH` export | `/opt/homebrew/bin:/usr/local/bin:$PATH` | L77 |
105 | `PID_FILE` | `/tmp/nullexit-cafeinate.pid` | L387 |
106 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | L407 |
107 | `HOST_LEAK_PROBE_PID_FILE` | `/tmp/nullexit-host-leak-probe.pid` | L423 |
108 | WARP endpoints | `162.159.192.1`, `162.159.193.1` | L329-336 |
109 | TUNNEL_FAILED_CLOSED marker | `${SCRIPT_DIR}/TUNNEL_FAILED_CLOSED.marker` | L51 |
110
111 **Duplicated logic from toggle.sh:**
112 *(Note: As of July 2026, **ALL** of the below duplicated logic has been successfully extracted to `scripts/`
113   common.sh`!)*
114 - `run_with_timeout()` simpler subshell version vs toggle's PID-tracking version
115 - `restart_tailscaled_daemon()` near-identical (uses warn()/ok() vs echo)
116 - Active network service detection inline at L217-233 (toggle has `get_active_service()` function)
117 - ENO service resolution inline at L230-233 (same as toggle L515-516)
118 - WARP bypass routes inline at L329-337 (toggle has `add_warp_bypass_routes()`)
119 - Exit node routing setup inline at L340-345 (toggle has `setup_exit_node_routing()`)
120 - DNS cache flushing L296-302 (same as toggle L611-616)
121 - Wi-Fi power-cycling L442-461 (similar to toggle L626-637)
122 - Proxy disabling L282-288 (same as toggle L604-608)
123 - Sharing services reset L586 (same as toggle L642)
124 - PID file stop pattern L387-399, L407-419, L423-435 (same as toggle L157-167, L193-203, L312-322)
125 - KILL_SWITCH check L194 (same as toggle L141, L469)
126 - .gateway_ip reading L138-142, L209-213 (same pattern as toggle L1080)
127
128 ---
129
130 ### setup.sh (410 lines, 18KB)
131
132 **Functions (4 total):**
133
134 | # | Function | Lines | Purpose |
135 | ---|-----|-----|-----|
136 | 1 | `step()` | L10 | Print bold blue step header |
137 | 2 | `ok()` | L11 | Print green success message |
138 | 3 | `warn()` | L12 | Print yellow warning message |
139 | 4 | `die()` | L13 | Print red error and exit |
140
141 **Constants:**
142
143 | Constant | Value | Line |
144 | ---|-----|-----|
145 | Color codes | `RED`, `GREEN`, `YELLOW`, `BLUE`, `BOLD`, `NC` | L7-8 |
146 | `RECOMMENDED_TS_VERSION` | `1.98.5` | L71 |
147 | `ADGUARD_PASSWORD` | `nullexit` | L215 |
148 | Default .env values | `GATEWAY_RULE_PROFILE=medium`, `GATEWAY_MSS=1120`, etc. | L240-248 |

```

```

149 ---
150
151 ## Part 2: Linux Scripts
152
153 ### scripts/toggle-linux.sh (931 lines, 39KB)
154
155 **Functions (17 total):**
156
157 | # | Function | Line | Purpose |
158 |---|-----|-----|-----|
159 | 1 | `run_with_timeout()` | L19 | Runs command with timeout watchdog, kills on expiry |
160 | 2 | `run_gui_cmd()` | L57 | macOS code uses `stat -f '%Su' /dev/console`, DEAD CODE on Linux |
161 | 3 | `start_sleep_prevention()` | L74 | Uses `systemd-inhibit` (Linux) to block sleep |
162 | 4 | `stop_sleep_prevention()` | L116 | Kills sleep prevention process via PID file |
163 | 5 | `start_dns_watcher()` | L130 | Background daemon polling DNS every 30s, re-hijacks if changed |
164 | 6 | `stop_dns_watcher()` | L149 | Kills DNS watcher process via PID file |
165 | 7 | `cleanup_handler()` | L162 | Trap handler for ERR/INT/TERM/HUP restores DNS, kills bg processes |
166 | 8 | `disconnect_tailscale_host()` | L273 | Disconnects host Tailscale from mesh |
167 | 9 | `get_active_service()` | L283 | Gets active network interface via `ip route` (Linux) |
168 | 10 | `reset_dns()` | L304 | Restores DNS via `resolvectl revert` (Linux) |
169 | 11 | `cleanup_network_state()` | L314 | Flushes DNS cache, routes, power-cycles network interface |
170 | 12 | `enable_socks_proxy()` | L361 | Stub prints message that Linux global SOCKS is DE-specific |
171 | 13 | `disable_socks_proxy()` | L367 | Stub prints skip message |
172 | 14 | `stop_local_dns_proxy()` | L383 | Kills Python DNS proxy processes |
173 | 15 | `start_local_dns_proxy()` | L394 | Starts Python DNS proxy (UDP:53 TCP:5354) + hijacks DNS |
174 | 16 | `force_dns_to_gateway()` | L461 | 3-attempt loop to set + verify DNS via `resolvectl` |
175 | 17 | `is_gateway_active()` | L495 | Checks if containers running or DNS was hijacked |
176
177 **Shared constants with macOS toggle.sh:**
178
179 | Constant | Value | Both |
180 |-----|-----|-----|
181 | `SOCKS_PROXY_PORT` | `1080` | toggle-linux L359, toggle.sh L661 |
182 | `PID_FILE` | `/tmp/nullexit-cafeinate.pid` | toggle-linux L71, toggle.sh L105 |
183 | `DNS_WATCHER_PID_FILE` | `/tmp/nullexit-dns-watcher.pid` | toggle-linux L128, toggle.sh L169 |
184 | ARP threshold | `15` devices | toggle-linux L518, toggle.sh L863 |
185 | Colima memory | `0.6` (600MB) | toggle-linux L578, toggle.sh L928 |
186 | Colima swap | `400MB` | toggle-linux L587, toggle.sh L937 |
187 | Tailscale wait loop | `60` iterations | toggle-linux L613, toggle.sh L1010 |
188 | NoState stuck threshold | `40` consecutive | toggle-linux L628, toggle.sh L1025 |
189
190 ---
191
192 ### scripts/recover-linux.sh (257 lines, 10KB)
193
194 **Functions (6 total):**
195
196 | # | Function | Line | Purpose |
197 |---|-----|-----|-----|
198 | 1 | `step()` | L24 | Print formatted step header |
199 | 2 | `ok()` | L25 | Print green success message |
200 | 3 | `warn()` | L26 | Print yellow warning message |
201 | 4 | `fail()` | L27 | Print red failure message |
202 | 5 | `die()` | L28 | Print red error + exit 1 |
203 | 6 | `run_with_timeout()` | L33 | Simplified timeout (immediate SIGKILL, no PID tracking) |
204
205 ---
206
207 ### scripts/setup-linux.sh (416 lines, 18KB)
208
209 **Functions (4 total):**
210
211 | # | Function | Line | Purpose |
212 |---|-----|-----|-----|
213 | 1 | `step()` | L10 | Print formatted step header |
214 | 2 | `ok()` | L11 | Print green success message |
215 | 3 | `warn()` | L12 | Print yellow warning message |
216 | 4 | `die()` | L13 | Print red error + exit 1 |
217
218 **~95% identical to macOS setup.sh.** Only differences:
219 1. Desktop shortcuts (L366-391 Linux `.desktop` files vs macOS `.app` via `osacompile`)
220 2. Final instructions text
221 3. .env template: macOS adds `GATEWAY_USE_EXIT_NODE`, `WARP_FAIL_THRESHOLD`, `HOST_LEAK_PROBE`, `KILL_SWITCH`
222    Linux omits these
223
224 ---
225
226 ## Part 3: Utility Scripts
227
228 ### scripts/diagnose-host-leak.sh (722 lines, 38KB)

```



```

229 **Purpose:** Comprehensive host-routing diagnostic. 8 checks classify host IP leaks. Supports `--fix` and `--
      watch` modes.
230
231 **Duplicated logic:**
232 - `resolve_active_service()` identical pattern as toggle.sh, recover.sh, fix-docker-bridge-collision.sh
233 - `run_with_timeout()` reimplemented timeout
234 - Color codes RED/GREEN/YELLOW/BOLD/NC with tty detection
235 - WARP check via `cloudflare.com/cdn-cgi/trace`
236 - `export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"`
237 - .gateway_ip reading pattern
238
239 ### scripts/fix-docker-bridge-collision.sh (402 lines, 18KB)
240
241 **Purpose:** Fixes Docker bridge subnet collisions with `10.200.1.0/24`.
242
243 **Duplicated logic:**
244 - `resolve_active_iface()` same interface detection as diagnose-host-leak.sh
245 - Color codes, step()/ok()/warn()/fail()/die() formatting helpers
246
247 ### scripts/fix-ssh-delay.sh (63 lines, 2.7KB)
248
249 **Purpose:** One-shot fix for ~20s SSH delay. Standalone, minimal duplication (uses hardcoded / instead of
      color functions).
250
251 ### scripts/host-leak-probe.sh (110 lines, 5.5KB)
252
253 **Purpose:** Sub-second host-egress leak prober at 300ms intervals.
254
255 **Duplicated logic:**
256 - WARP check via cdn-cgi/trace (same curl + awk as toggle.sh WARP Watcher)
257 - Color codes (hardcoded; status updates and warnings are conditioned on TTY detection to prevent log pollution
      )
258 - LOG_FILE = `PROJECT_ROOT/output.log`
259
260 ### scripts/reinstall-host-tailscale.sh (740 lines, 31KB)
261
262 **Purpose:** Complete uninstall + reinstall of host Tailscale.
263
264 **Duplicated logic:**
265 - `with_timeout()` yet another bash timeout implementation (polling-based, different from diagnose and toggle
      versions)
266 - Color codes + logging functions
267 - System Extension detection logic (similar to diagnose-host-leak.sh)
268
269 ### scripts/routing-fix.sh (266 lines, 10.5KB)
270
271 **Purpose:** Runs INSIDE warp container. Maintains routing table 200, FORWARD rules, country blocking, IP
      blocklists, tunnel health checks.
272
273 **Duplicated logic:**
274 - (Fixed) WARP_ENDPOINT defaults are now centralized in common.sh
275 - WARP health check via cdn-cgi/trace (another instance)
276 - iptables dual-backend pattern (for-loop over `iptables iptables-legacy`)
277
278 ### scripts/watcher.sh (269 lines, 13KB)
279
280 **Purpose:** Long-running launchd daemon for post-wake/post-roam recovery.
281
282 **Key function `detect_lan_p2p_mode()`:**
283 Added July 10, 2026. Called on startup and on every network state change (Listener 2 NET: events). Determines
      whether the current Wi-Fi network safely allows direct Tailscale LAN P2P connections:
284 1. **WPA2-Enterprise check:** `system_profiler SPAirPortDataType` reads the `Security:` field for the current
      association. Enterprise = 802.1x = always AP-isolated sets `allow=false`.
285 2. **AP isolation probe:** `route -n get 1.1.1.1` finds default gateway IP, then `ping -c 3 -t 1 -q "$gw" if
      0 responses on a non-empty network, AP isolation is active sets `allow=false`.
286 3. **Output:** Writes `true` or `false` to `lan_p2p_detected` in the repo root.
287 4. **Consumer:** `routing-fix.sh` reads this file every 30 seconds and enforces/relaxes the RFC1918 DROP rule
      accordingly no restart required.
288
289 **Duplicated logic:**
290 - PATH export (similar to toggle.sh/recover.sh)
291 - Marker file pattern (`tmp/nullexit-gateway-active.marker`)
292
293 ### scripts/dns-proxy.py (92 lines, 2.9KB) Clean, standalone
294
295 ### scripts/logger.py (144 lines, 5.9KB) Clean, standalone
296
297 ---
298
299 ## Part 4: Cross-Cutting Issues
300

```

```

301 ### Functions Identical Across macOS Linux
302
303 | Function | macOS toggle | macOS recover | macOS setup | Linux toggle | Linux recover | Linux setup |
304 |-----|:-----|:-----|:-----|:-----|:-----|:-----|
305 | `run_with_timeout()` | | | identical | identical | |
306 | `stop_local_dns_proxy()` | | | identical | | |
307 | `start_local_dns_proxy()` | | | ~95% similar | | |
308 | `is_gateway_active()` | | | ~80% similar | | |
309 | `step/ok/warn/die` | | | identical | identical |
310
311 ### Platform-Adapted Functions (same logic, different OS commands)
312
313 | Function | Linux Command | macOS Command |
314 |-----|-----|-----|
315 | `get_active_service()` | `ip route get 1.1.1.1 \| awk '{print $5}'` | `route get default` + `networksetup -
listnetworkserviceorder` |
316 | `reset_dns()` | `resolvectl revert` | `networksetup -setdnsservers` |
317 | `cleanup_network_state()` | `ip link set dev down/up` | `networksetup -setairportpower off/on` + proxy
disable |
318 | `force_dns_to_gateway()` | `resolvectl dns` + `resolvectl domain -ts.net` | `networksetup -setdnsservers` +
`-setsearchdomains ts.net` |
319 | `start_sleep_prevention()` | `systemd-inhibit --what=sleep` | `caffeinate -i` |
320 | `enable/disable_socks_proxy()` | Stub (no-op on Linux) | `networksetup -setsocksfirewallproxy` |
321 | DNS cache flush | `resolvectl flush-caches` | `dscacheutil -flushcache` + `killall -HUP mDNSResponder` |
322
323 ### Features Missing From Linux Scripts
324
325 These exist in macOS toggle.sh but NOT in toggle-linux.sh:
326
327 1. `write_gateway_active_marker()` / `clear_gateway_active_marker()`
328 2. `restart_tailscaled_daemon()`
329 3. `start_warp_watcher()` / `stop_warp_watcher()` WARP liveness monitor
330 5. `add_warp_bypass_routes()` / `remove_warp_bypass_routes()` / `setup_exit_node_routing()`
331 6. `LOG_FILE` structured logging (timestamps to output.log)
332 7. MSS clamping injection (step 4c)
333 8. Rule count reporting
334 9. `--post-wake` mode in recover
335 10. Container teardown on failure in cleanup_handler
336
337 ---
338
339
340
341 ## Part 6: Constant Duplication Map
342
343 | Value | Files |
344 |-----|-----|
345 | `162.159.192.1` (WARP endpoint) | docker-compose.yml, routing-fix.sh (now dynamically loaded via .env/common.
sh!) |
346 | `5335` (AdGuard DNS port) | docker-compose.yml, AdGuardHome.yaml, post-rules.txt |
347 | `5354` (mapped DNS port) | docker-compose.yml, dns-proxy.py |
348 | `41642` (Tailscale WireGuard) | docker-compose.yml, routing-fix.sh, post-rules.txt |
349 | `1080` (SOCKS5 port) | docker-compose.yml, toggle.sh |
350 | `1120` (MSS clamp) | post-rules.txt, .env |
351 | `admin:nullexit` (AdGuard creds) | AdGuardHome.yaml (bcrypt) |
352 | `10.200.1.0/24` (Docker subnet) | (Fixed) no longer duplicated |
353 | `America/NewYork` (timezone) | docker-compose.yml (4 services) |
354 | `/tmp/nullexit-caffeinate.pid` | (Fixed) centralized in common.sh |
355 | `/tmp/nullexit-dns-watcher.pid` | (Fixed) centralized in common.sh |
356 | `/tmp/nullexit-host-leak-probe.pid` | toggle.sh, recover.sh |
357 | `/opt/homebrew/bin:...` (PATH) | (Fixed) centralized in common.sh |
358
359 ---
360
361 ## Part 7: Refactoring Outcome (Implemented July 2026)
362
363 **Decision: "Scripts-only" architecture**
364 Instead of introducing a new `lib/` directory, all shared code was moved into the existing `scripts/` directory
to keep the project hierarchy flat and simple.
365
366 ### 1. `scripts/common.sh` (288 lines, 10KB)
367 Extracts the fundamental primitives and networking logic used across orchestrator scripts:
368 - **Formatters:** `step()`, `ok()`, `warn()`, `fail()`, `die()`
369 - **Timeout Wrapper:** `run_with_timeout()` (Canonical version with PID tracking, graceful SIGTERM, and SIGKILL
fallback)
370 - **Daemon Lifecycle:** `restart_tailscaled_daemon()`, `stop_pidfile_daemon()` (collapsed caffeinate, DNS
watcher, WARP watcher, and host leak probe teardown logic)
371 - **Network Interface/Routing:** `get_active_service()`, `get_en0_service()`, `bounce_wifi_interfaces()`,
`add_warp_bypass_routes()`, `remove_warp_bypass_routes()`
372 - **Proxy/DNS Cleanup:** `disable_all_proxies()`, `flush_dns_cache()`, `reset_sharing_services()`

```



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373 - **Configuration Helpers:** `read_adguard_ip()`, `is_kill_switch_enabled()`, `get_warp_endpoint_1()`, `
    get_warp_endpoint_2()` (which centralizes the hardcoded 162.159.192.1 / .193.1 into .env logic)
374
375 ### 2. `scripts/setup-common.sh` (~350 lines)
376 Extracts the heavy, duplicated installation logic shared between `setup.sh` (macOS) and `scripts/setup-linux.sh`
    `:
377 - Docker and Colima prerequisite checks
378 - `wgcf` binary download and WARP key generation
379 - `.env` configuration file generation
380 - Interactive Tailscale Auth Key and AdGuard Rule Profile prompts
381
382 ### Sourcing Pattern
383 - macOS root scripts (`toggle.sh`, `recover.sh`, `setup.sh`) source via:
384   `source "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/common.sh"`
385 - Linux sibling scripts (`scripts/toggle-linux.sh`, etc.) source via:
386   `source "$(dirname "${BASH_SOURCE[0]}")/common.sh"`
387
388 ## Agent Verbs / Protocols
389
390 ### `/sweep` (or `sweep the gateway`)
391 When the user invokes this verb, the agent MUST perform a full end-to-end connectivity, health, security, and
    performance audit of the gateway:
392 1. **WARP Validation:** Run `curl -s https://www.cloudflare.com/cdn-cgi/trace | grep warp=` (must equal `on`).
393 2. **Tor Proxy & Control Validation:**
394   - Source the dynamically generated ports from `/tmp/nullexit-ports.env` (or identify them using `docker
    compose ps`).
395   - **SOCKS5 Check:** Run `curl -s --socks5-hostname 127.0.0.1:$TOR SOCKS5_PORT https://check.torproject.org/
    api/ip` to confirm the SOCKS5 proxy successfully connects and routes through the Tor network.
396   - **Control Port Check:** Query the Tor Control Port using netcat: `echo "PROTOCOLINFO" | nc 127.0.0.1:
    $TOR_CONTROL_PORT`. Verify it returns a standard `250-PROTOCOLINFO` response, confirming the Control Port
    is active and communicating.
397   - **Circuit & Node Lookup:** Query the Tor Control Port for the active path (`GETINFO circuit-status`). For
    each active built circuit, parse the relay fingerprints, lookup their IP addresses (`GETINFO ns/id/<
    fingerprint>`), and resolve their country codes using Tor's local GeoIP database (`GETINFO ip-to-country/<
    IP>`). Include the list of active circuits, showing the role (Guard/Middle/Exit), nickname, IP, and country
    of each hop in the final sweep report.
398   - **Transparent Onion Validation:** Run `nslookup duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.
    onion` to verify it resolves to an IP within `198.18.0.0/15` benchmark range. Verify transparent dark web
    routing by running `curl -sI -H "Host: duckduckgogg42xjoc72x3sjasowoarfbgcmvfimaftt6twagswzczad.onion" http
    ://<resolved-ip>/` (should return HTTP 301/200).
399 3. **DNS Leak Validation:**
400   - Audit the upstream resolver currently utilized by the host. Run `curl -s https://edns.ip-api.com/json` to
    fetch details of the DNS resolver processing the client's requests.
401   - Verify that the resolver IP and organization returned belong to Cloudflare (e.g., AS13335) or match the
    WARP tunnel's exit range, and that **no** DNS traffic leaks to the user's raw physical ISP DNS.
402 4. **Performance Benchmark Audit:**
403   - Run the python benchmark script `python3 benchmarks/test_load.py`.
404   - Report the overall load speed and the ratio of successfully fetched vs. blocked/failed subresources for
    the targeted domains to ensure ad-blocking and MTU encapsulation are performing correctly.
405 5. **Tailscale Mesh Validation:**
406   - Run `ping -c 1 100.100.21.8` to ensure the gateway is reachable.
407   - Run `tailscale status` to identify all active/online nodes in the mesh.
408   - For every online peer returned in the status output, execute a ping check (`ping -c 1 <peer-ip>`) to
    verify bidirectional connectivity and confirm healthy routing to all active mesh devices.
409 6. **Log Audit:** Run `tail -n 100 output.log | grep -iE "(error|fail|warn)"` to catch any underlying component
    failures or warnings.
410
411 ### `/latex` (or `generate latex`)
412 When the user invokes this verb, the agent MUST run `python3 -I scripts/generate_tex.py` to regenerate the
    documentation source code (`nullexit_unified.tex`). The agent MUST NOT attempt to compile the `.tex` file
    into a PDF (the user handles PDF compilation locally).

```

4 devref.md

```

1 # nullexit Development Reference & Resolved Issues
2
3 > **Last updated:** July 10, 2026
4 > **Purpose:** Provide any LLM or developer with complete project understanding, debugging history, and
    resolved issues so they can make informed changes without re-reading every file.
5 > **Diagrams:** See [diagrams.md](./diagrams.md) for system architecture, toggle.sh flowcharts, monitoring
    layer, traffic sequence, recover.sh decision tree, and the full failureself-healing map.
6
7 ---
8
9 ## 1. What Is nullexit?
10

```

```

11 nullexit is a **Tunnel-in-Tunnel VPN gateway** that chains **Tailscale** (mesh VPN) through **Cloudflare WARP**
    (exit VPN). It runs on a macOS host inside Docker containers managed by **Colima** (lightweight Linux VM).
    The goal: any device on the user's Tailscale mesh can use this gateway as an exit node, and all traffic
    exits through Cloudflare WARP achieving double encryption, ISP invisibility, and network-wide ad-blocking
    via **AdGuard Home**.
12
13 ### Traffic Flow
14 ---
15 Phone/Laptop  Tailscale mesh (WireGuard)  Mac host  Docker container
16   Tailscale decrypts  Gluetun re-encrypts  WARP tun0  Cloudflare  Internet
17 ---
18
19 ### SOCKS5 Failover Path (if exit node is unavailable)
20 ---
21 DNS:  Browser  host 127.0.0.1:53  Python proxy  TCP:5354  AdGuard  WARP  DNS
22 TCP:  Apps  macOS  SOCKS5 proxy (127.0.0.1:1080)  container  tun0  WARP  Internet
23 ---
24
25 ---
26
27 ## 2. Architecture
28
29 ### Containers (all share `warp`'s network namespace via `network_mode: service:warp`)
30 | Container | Image | Role |
31 |-----|-----|-----|
32 | `warp` | `qmcgaw/gluetun:v3.41.1` | Gluetun WireGuard client  Cloudflare WARP. Owns the network namespace.
    Strict firewall. |
33 | `tailscale` | `tailscale/tailscale:v1.98.4` | Advertises as exit node on the mesh. Also provides the built-in
    SOCKS5 proxy fallback via `TS_SOCKS5_SERVER=:1080`. |
34 | `routing-fix` | `alpine:3.20` | Sidecar that maintains routing tables + iptables rules every 30 seconds. |
35 | `rule-compiler` | `golang:1.22-alpine` / `alpine:3.20` | One-shot startup container that compiles 500k+ DNS
    block rules in <2s and immediately exits. |
36 | `adguardhome` | `adguard/adguardhome:v0.107.77` | DNS sinkhole for ads/trackers. Listens on port 5335.
    Upstream DNS: `127.0.0.1:53` (through WARP). |
37
38 ### Port Mappings (on host via Colima)
39 | Host Port | Container Port | Protocol | Purpose |
40 |-----|-----|-----|-----|
41 | 5354 | 5335 | TCP+UDP | AdGuard DNS |
42 | (Tailscale IP):3000 | 3000 | TCP | AdGuard web UI (Internal to Mesh) |
43 | 41641 | 41641 | UDP | Tailscale WireGuard direct |
44 | 1080 | 1080 | TCP | SOCKS5 proxy |
45
46 ### Host Components
47 - **Colima VM** Linux VM running Docker (600MB RAM + 400MB swap)
48 - **tailscaled** Standalone Tailscale daemon via `brew install tailscale` + `brew services start tailscale` (
    NOT the GUI `.app`)
49 - **macOS network settings** DNS, SOCKS proxy, routing all manipulated by `toggle.sh`
50
51 ---
52
53 ## 3. Key Files
54
55 | File | Lines | Purpose |
56 |-----|-----|-----|
57 | `toggle.sh` | ~1265 | **Main script.** Detects state, toggles gateway ON/OFF. Handles DNS hijacking,
    Tailscale exit node, SOCKS proxy, sleep prevention, timeouts, cleanup. |
58 | `setup.sh` | 406 | **One-time setup.** Installs deps (Docker, Tailscale, wgcf), generates WARP keys, writes
    `.env`, configures AdGuard via API, starts containers. |
59 | `docker-compose.yml` | 194 | Service definitions for all 5 containers. |
60 | `scripts/routing-fix.sh` | 201 | Maintains routing tables (table 200 for SOCKS5, table 52 for CGNAT) and
    FORWARD iptables rules in both nftables and legacy backends. Loads and enforces the IP blocklist via `ipset`
    . Runs in a 30-second loop. |
61 | `post-rules.txt` | 18 | Gluetun iptables rules loaded at container start. FORWARD accept for tailscale0, NAT
    MASQUERADE on tun0, DNS redirect to port 5335, IPv6 drop, TCP MSS clamping. |
62
63 | `scripts/dns-proxy.py` | 91 | Local DNS proxy. UDP:53 TCP:5354 with proper 2-byte length prefix. Fallback
    when exit node is unavailable. |
64 | `scripts/rule-compiler/` | ~800 | Unified threat compiler (ported to Go from Python). **DNS pipeline:**
    fetches remote blocklists, deduplicates against AdGuard native filters, optimizes subdomains (~50%
    reduction), compiles to AdGuard syntax. **IP pipeline:** fetches threat-intel feeds (Spamhaus, Feodo, ET,
    CINS), strips private/Tailscale ranges, outputs atomic `ipset` restore file. Built dynamically in Docker
    multi-stage build. |
65 | **`adguard/work/`** | | **Bind-mounted container data folder. Off-limits from outside Docker  READ-ONLY-
    EXCEPT-VIA-`sync-rules`. See README 6.** |
66 | `recover.sh` | ~557 | Nuclear recovery script (gain: use `--post-wake` for non-destructive refresh after
    sleep/wake or Wi-Fi roam keeps the gateway live while still re-hijacking DNS + refreshing Tailscale exit-
    node + force-recreating warp if unhealthy). Resets DNS on ALL services, disables proxies, flushes routes,
    stops containers, kills sleep prevention, power-cycles Wi-Fi. Cryptographically verified. |
67 | `scripts/diagnose-host-leak.sh` | 580+ | **One-shot host-routing diagnostic.** Runs 7 checks (Tailscale,
    SOCKS5, CDN-cgi/trace, IPv6 leak, default route, output.log hints, gateway IP), classifies into ONE of

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three known leak scenarios (A=SOCKS5 fallback, B=IPv6 leak, C=route freeze) or OK, writes a timestamped
report to `host-leak-diagnostic-<UTC>.txt`, and prints a ready-to-run fix on stdout. Pass `--fix` to apply
the matched remediation and re-verify egress. Pass `--watch` (or `--watch 30`) to run a full baseline then
continuously monitor warp/IPv6/default-route every N seconds, alerting on any state change. See 10.30. |
68 | `scripts/host-leak-probe.sh` | ~110 | **Continuous sub-second host-egress prober.** Launched automatically by
`toggle.sh` when `HOST_LEAK_PROBE=true` in `.env` (default). Polls `cdn-cgi/trace` every 300ms directly
from the host NIC via `curl` not via `docker compose exec` so it detects flash-leaks invisible to the in-
container WARP Watcher. Logs only on state change (`LEAK`, `ROTATE`, `HOST-PROBE failed/timeout`) to `
output.log`. curl errors (`2>>output.log`) also land there. Stopped gracefully (SIGTERM) by `toggle.sh`
STOP path, `cleanup_handler`, and `recover.sh`. See 10.30.1. |
69 | `scripts/fix-docker-bridge-collision.sh` | ~290 | **One-shot fix for 9.13.** Detects Docker's `172.17.0.0/16`
bridge colliding with the host Wi-Fi subnet (the symptom that produces `default 172.17.0.1 UGScg en0` in `
netstat -rn`), picks a non-colliding `docker.bip` from a small candidate set, idempotently edits `~/colima
/default/colima.yaml` with a timestamped backup, restarts Colima, rebinds Wi-Fi DHCP, verifies the new
default route is no longer Docker's bridge, and re-runs `toggle.sh` + `scripts/diagnose-host-leak.sh`.
Supports `--dry` and `--skip-toggle`. |
70 | `scripts/watcher.sh` | 194 | **Long-running post-wake + post-roam daemon.** Launched by launchd LaunchAgent;
subscribes to `com.apple.powermanagement` events via `log stream` and to network-state changes via `scutil
n.watch`. Both fire `recover.sh --post-wake`. Single-instance lock + SCRIPT_DIR-relative resolve of `
RECOVER`. See 10.29. |
71 | `scripts/common.sh` | ~40 | **Shared script primitives.** Centralizes formatted echo statements (`step`, `ok
`, `warn`, `die`) and the safe background `run_with_timeout` execution wrapper. Sourced by all orchestrator
scripts across macOS and Linux. |
72 | `scripts/setup-common.sh` | ~350 | **Shared installation logic.** Centralizes logic for verifying Docker,
installing Tailscale/wgcf, generating `.env`, and prompting for auth keys. Sourced by `setup.sh` and `setup
-linux.sh`. |
73 | `scripts/toggle-linux.sh` | ~900 | **Linux sibling of `toggle.sh`.** Sources `common.sh`. Native Linux
equivalents for every macOS-specific primitive `nmcli radio wifi off/on` instead of `networksetup -
setairportpower`, `ip link set ... down/up` fallback, systemd-resolved wiring. Paired with `.desktop`
shortcuts. |
74 | `scripts/recover-linux.sh` | ~250 | **Linux sibling of `recover.sh`.** Sources `common.sh`. Same nuclear
semantics (flush, restart, verify) but using `nmcli` / `ip link` instead of macOS `networksetup`. |
75 | `scripts/setup-linux.sh` | ~80 | **Linux sibling of `setup.sh`.** Sources `setup-common.sh`. Distro detection
(apk/dnf/pacman), installs Linux-specific dependencies, creates `.desktop` shortcuts. |
76 | `scripts/Toggle-Gateway.bat` | ~300 | **Windows Toggle helper.** Moved from root to `scripts/`. Helps invoke
the framework on Windows if applicable. |
77 | `scripts/reinstall-host-tailscale.sh` | ~740 | **Nuke-and-reinstall tool for host Tailscale.** Completely
purges Tailscale, its settings, macOS Background Items (`.btm` database), and stale System Extensions.
Wipes ALL Login Items via `sftool reset-login-items` (requires sudo). Useful for crisis recovery but
destructive to other login items. |
78 | `scripts/fix-ssh-delay.sh` | ~100 | **Fixes SSH connection delays.** One-shot script to address SSH delays
caused by DNS or routing issues, useful in crisis situations. |
79 | `scripts/logger/` | ~100 | **Internal logger piped from `scripts/routing-fix.sh`** inside the `warp`
container. Ported to Go from Python. Built via Docker multi-stage build. |
80 | `black_list.txt` | 71 | Custom domains to block (ads, trackers, telemetry). Supports `$important` modifier. |
81 | `white_list.txt` | 222 | Domains to force-allow (YouTube, Apple services, etc.). Always wins over blocks. |
82 | `.env` | ~14 | WARP WireGuard keys, Tailscale auth key, rule profile. **Contains secrets & NULLEXIT_SEED.** |
83 | `.gateway_ip` | 1 | Static gateway Tailscale IP (fallback for dynamic resolution). |
84 | `scripts/unlock-files.sh` | ~20 | **One-shot stale-permission fix.** Uses atomic rename (`cp` + `mv`) to
replace locked inodes (mode `000`/`0444` from old `chmod` decisions) with fresh writable ones no `chmod`
called. |
85 | `scripts/crypto.sh` | ~50 | **Cryptographic integrity enforcement.** Uses `NULLEXIT_SEED` from `.env` to sign
core bash scripts (HMAC-SHA256) and strictly verify them on start to prevent manipulation. Pass `--sign`
or `--verify`. |
86 | `scripts/pf.conf` | 35 | **macOS native firewall ruleset.** Enforces the default-deny kill-switch, allows
local LAN / Tailscale traffic, and applies TCP MSS Clamping (`max-mss 1160`) to prevent WireGuard MTU
fragmentation. |
87
88 ---
89
90 ## 4. toggle.sh Flow
91
92 ### State Detection
93 `is_gateway_active()` returns true if EITHER containers are running OR host DNS is not `1.1.1.1`. This dual
check prevents the script from starting when it should stop (e.g., containers crashed but DNS is still
hijacked).
94
95 ### START Path (gateway OFF ON)
96 1. **Reset DNS to 1.1.1.1** Prevents deadlocks during startup
97 2. **Disconnect host Tailscale** `tailscale down` (prevents exit-node routing during container boot)
98 3. **Compile DNS rules** `docker compose run --rm rule-compiler` (Go binary inside multi-stage Docker build)
99 4. **Boot Colima VM** `colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged` if not
running
100 - **Why Bridged Networking?** The `--network-mode bridged` and `--network-address` flags are critical. By
default, Colima creates an isolated network. Bridged mode forces the VM to receive its own IP address
directly from your local router's DHCP server, placing it on the same physical subnet as your host machine.
This bypasses macOS network isolation and is absolutely required for peer-to-peer (P2P) connections, mDNS,
and local LAN service discovery to function properly across the container boundary.
101 - **Enterprise Wi-Fi Fallback & Dynamic Roaming:** `toggle.sh` dynamically queries `system_profiler
SPAirPortDataType` on boot. If it detects a **WPA2-Enterprise** connection (802.1X), it forces Colima to
fall back to `--network-mode shared`. This prevents aggressive network intrusion systems on corporate/

```

```

university Wi-Fi switches from terminating the host's Wi-Fi connection due to "MAC spoofing" (detecting
multiple MAC addresses originating from a single authenticated port). P2P connections are impossible on
these networks anyway due to AP Client Isolation.
102 - **Dynamic Roam Downgrading:** `toggle.sh` writes its selected network mode to `/tmp/nullexit_colima_mode
.txt`. When roaming between networks (e.g., Home to University), `recover.sh` actively checks this state
against the new network's security requirement (`lan_p2p_detected`). If a mismatch is detected (e.g.,
bridged Colima on an Enterprise network), `recover.sh` automatically forces a full `toggle.sh --restart` in
the background to safely transition the Virtual Machine's network mode and protect the host from de-
authentication.
103 5. **Configure VM swap** 400MB swap file inside the VM to prevent OOM
104 6. **Clean corrupted AdGuard config** Remove empty `AdGuardHome.yaml`
105 7. **Start containers** `docker compose up -d`
106 8. **Wait for gateway Tailscale** Poll `tailscale status` for "offers exit node" (up to 60s, abort at 40
consecutive NoState)
107 9. **Resolve gateway IP** From `.gateway_ip` or `docker compose exec tailscale tailscale ip -4`
108 10. **Connect host to mesh** Verify `tailscaled` is running (auto-start if needed), then `tailscale up --reset
--ssh=true --accept-dns=false --accept-routes=true --exit-node=` (`--accept-routes=true` must be explicit
`--reset` reverts it to `false` otherwise, silently preventing the default route from switching to `utun
*`)
109 11. **Pre-flight checks** [1/3] `tailscale ping` gateway, [2/3] `dig +tcp AdGuard DNS`, [3/3] WARP container
internet
110 12. **If all pass** Set exit node + hijack DNS to gateway IP (single server, no fallback)
111 13. **If any fail** Enable SOCKS5 proxy + local DNS proxy as fallback
112 14. **Final DNS re-force** `force_dns_to_gateway` called again after exit-node transition (tailscaled clobbers
DNS briefly)
113 15. **Enable sleep prevention** `caffeinate -i` in background to prevent idle sleep
114
115 ### STOP Path (gateway ON OFF)
116 1. Disconnect host Tailscale (`tailscale down`)
117 2. `docker compose down -t 5`
118 3. Reset DNS to 1.1.1.1
119 4. Stop local DNS proxy
120 5. **Stop sleep prevention** Kill caffeinate process
121 6. Full network state cleanup (proxies off, DNS cache flush, route flush, Wi-Fi power cycle)
122
123 ### Safety Mechanisms
124 - `run_with_timeout()` Pure-bash watchdog for every system command (15s120s)
125 - `cleanup_handler()` Trap on ERR/INT/TERM/HUP restores DNS to 1.1.1.1, kills caffeinate, and tears down
Tailscale
126 - `force_dns_to_gateway()` Sets DNS and VERIFIES via `networksetup -getdnsservers` (3 attempts with backoff)
127 - `SUCCESS_RUN` flag Cleanup only runs if script didn't complete successfully
128 - `start_sleep_prevention()` / `stop_sleep_prevention()` Manages `caffeinate -i` via PID file at `/tmp/
nullexit-caffeinate.pid`
129
130 ---
131
132 ## 5. Routing & Firewall Architecture (Inside Container)
133
134 ### IP Rules (`ip rule show`)
135 | Priority | Match | Table | Purpose |
136 |-----|-----|-----|-----|
137 | 99 | `to 100.64.0.0/10` | 52 | CGNAT range Tailscale routes (injected by routing-fix) |
138 | 100 | `from 172.18.0.2` | 200 | Container's own traffic (SOCKS5 proxy) tun0 |
139 | 101 | all | 51820 | Gluetun's WireGuard fwmark tun0 |
140
141 ### Table 200 (SOCKS5 proxy traffic)
142 - `162.159.192.1 via $DOCKER_GW dev eth0` WARP endpoint exception (prevents tunnel loop)
143 - `default dev tun0` Everything else through WARP
144
145 ### iptables (TWO separate stacks!)
146 - **nftables backend** (`iptables`) Used by Gluetun's `post-rules.txt`. FORWARD policy DROP.
147 - **legacy backend** (`iptables-legacy`) Used by Tailscale's `ts-forward`. FORWARD policy ACCEPT.
148 - Both stacks apply to every packet. FORWARD RELATED,ESTABLISHED must exist in BOTH.
149
150 ### post-rules.txt (loaded by Gluetun)
151 ```
152 FORWARD: ACCEPT for tailscale0 in/out
153 INPUT: ACCEPT for tailscale0
154 NAT POSTROUTING: MASQUERADE on tun0
155 MANGLE: TCP MSS clamp to 1180
156 NAT PREROUTING: Redirect DNS (53) 5335 on tailscale0 and eth0
157 ip6tables: DROP FORWARD on tailscale0
158 ```
159
160 ---
161
162 ### 5.1 Tailscale Local Network Discovery (DERP Bypass)
163 By default, Gluetun's strict NAT swallows all of Tailscale's UDP hole-punching packets, forcing Tailscale to
fall back to incredibly slow DERP relay servers (often adding 500ms+ latency).
164

```

```

165 To fix this, we use `FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8` in `docker-compose.yml`
    (on the `warp` container). This creates `ip rule` bypasses (Priority 99, Table 199) that route all packets
    destined for private IP ranges *outside* the WARP tunnel directly to the host's `eth0`.
166 - **172.16.0.0/12** is especially critical because many modern Wi-Fi routers (and Docker itself) assign IPs in
    this block (e.g., `172.17.x.x`).
167 - This bypass allows Tailscale's UDP packets to reach devices on the same Wi-Fi directly, establishing a fast
    peer-to-peer connection while the actual web traffic inside the tunnel remains fully routed through WARP.
168
169 ---
170
171 ### 5.2 Firewalling & Per-Device Access Control
172 Because every mesh device's traffic passes through the `warp` container's network namespace, this namespace
    serves as the ultimate choke point for network-wide firewall rules.
173
174 **Where to Put Rules**
175 The right place to add firewall rules is `scripts/routing-fix.sh`. It runs a 30-second loop inside the
    namespace and already handles re-injecting rules that Gluetun resets on reconnect. **Do not use `post-rules
    .txt` for dynamic rules**, as Gluetun flushes and resets it upon VPN reconnects.
176
177 **The Dual iptables Backend Constraint (CRITICAL)**
178 The container runs two simultaneous iptables backends: `iptables` (for the nftables backend used by Gluetun)
    and `iptables-legacy` (for Tailscale). Every rule **must** be added to both stacks, or it will silently
    fail for certain traffic.
179
180 **Avoiding Duplication**
181 Because `scripts/routing-fix.sh` runs in a loop, you must always check for a rule's existence with `-C` before
    appending with `-A`. Otherwise, your rules will duplicate every 30 seconds.
182
183 **Per-Device Identification & Filtering**
184 Each mesh device has a stable `100.x.x.x` Tailscale IP, which is visible as the `src` IP in the `FORWARD` chain
    . This is how you identify devices for filtering.
185 * **Domain-level:** AdGuard Home (port 3000, credentials `admin/nullexit`) provides a REST API that supports
    per-client blocking rules based on their Tailscale IP.
186 * **IP-level:** Use `iptables` in `scripts/routing-fix.sh` (e.g., `iptables -I FORWARD -s 100.x.x.x -d <
    blocked_ip> -j DROP`).
187 * **Country-level:** Add `ipset` to the `routing-fix` apk install step in `docker-compose.yml`, and load CIDR
    ranges from `ipdeny.com` into an ipset, then block that set in the `FORWARD` chain.
188
189 ## 6. macOS Quirks to Remember
190
191 1. **Colima SSH tunnel is TCP-only** UDP ports (like 5354/udp) are NOT accessible from host. Use `dig +tcp` or
    the Python DNS proxy (UDPTCP converter).
192 2. **docker compose exec -T` injects `\"r\"` Always `tr -d 'r'` when capturing output for comparisons or `
    networksetup` commands.
193 3. **brew services` can get stuck** Fix with `sudo brew services restart tailscale`. Common state: `brew
    services list` shows `started` but `tailscale status` shows "Tailscale is stopped."
194 4. **No `timeout` command** macOS lacks GNU `timeout`. Use the `run_with_timeout()` bash function.
195 5. **`sudo -v` prompts even with NOPASSWD** Use `sudo -n` (non-interactive) everywhere.
196 6. **DNS is scoped to en0** macOS scutil DNS resolver is scoped to en0 (Wi-Fi). DNS changes on other
    interfaces (like USB Ethernet) are ignored unless en0's service is also updated. The script always sets DNS
    on BOTH `ACTIVE_SERVICE` and `ENO_SERVICE`.
197 7. **tailscaled clobbers DNS during exit-node transition** `force_dns_to_gateway` is called a second time at
    the end of the ENABLE branch to counteract this.
198 8. **tailscale up --reset** resets all unspecified flags to defaults Every `--reset` call MUST also pass `--
    accept-dns=false` explicitly or tailscaled re-enables DNS management.
199 9. **docker compose ps` CWD pitfall** `docker compose` commands require a `docker-compose.yml` in the current
    working directory. Use `docker ps` for raw container listing from any CWD; `docker compose ps` requires
    the project directory.
200
201 ---
202
203 ## 7. Environment Variables
204
205 ### `.env` File
206 | Variable | Purpose |
207 |-----|-----|
208 | `WIREGUARD_PRIVATE_KEY` | WARP WireGuard private key (from `wgcf generate`) |
209 | `WIREGUARD_PUBLIC_KEY` | WARP WireGuard public key |
210 | `WIREGUARD_ADDRESSES` | WARP WireGuard address (e.g., `172.16.0.2/32`) |
211 | `TS_AUTHKEY` | Tailscale auth key for the container |
212 | `NULLEXIT_SEED` | 256-bit seed for HMAC-SHA256 script integrity signing (generated by `setup.sh`) |
213 | `GATEWAY_RULE_PROFILE` | Rule compilation tier: `light`, `medium`, `heavy` |
214 | `GATEWAY_BYPASS_PING` | (Optional) `true` to proceed even if pre-flight checks fail |
215 | `GATEWAY_USE_EXIT_NODE` | (Optional) `false` to skip exit node, DNS-only mode |
216 | `GATEWAY_HIJACK_HOST` | (Optional) `false` to skip DNS hijacking on the host (VPN/adblocking for Tailscale
    peers only) |
217 | `GATEWAY_MSS` | (Optional) TCP MSS clamp value (default 1120); 1180 for speed on healthy paths |
218 | `WARP_FAIL_THRESHOLD` | (Optional) Consecutive `warp=off` polls before auto-shutdown (default 6 = 30s; 3 = 15
    s) |
219 | `WARP_ENDPOINT_1` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 1 (default `162.159.192.1`) |
220 | `WARP_ENDPOINT_2` | (Optional) Override Cloudflare WARP WireGuard endpoint IP 2 (default `162.159.193.1`) |

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221 | `KILL_SWITCH` | (Optional) `true` to enforce strict PF lock drops all host traffic if VPN fails. Breaks SSH
    | if VPN dies. |
222 | `HOST_LEAK_PROBE` | (Optional) `false` to disable the 300ms host-egress leak prober (default: `true`) |
223 | `STOP_COLIMA_ON_EXIT` | (Optional) `true` to fully shut down the Colima VM on toggle-off (saves battery on
    | dedicated hosts) |
224 | `ADGUARD_USER` | (Optional) AdGuard Home web UI username (default: `admin`) |
225 | `ADGUARD_PASSWORD` | (Optional) Shared password for AdGuard Home and Tor ControlPort (default: `nullexit`) |
226 | `BLOCKED_COUNTRIES` | Space-separated 2-letter ISO codes to block via ipdeny.com CIDR ranges (e.g. `kp il cn
    | ru`) |
227
228 ---
229
230 ## 8. Diagnostic Commands
231
232 ### Container Health
233 ```bash
234 docker ps --format '{{.Names}} {{.Status}}'
235 docker compose exec -T tailscale tailscale status
236 docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace
237 ```
238
239 ### SOCKS5 Proxy
240 ```bash
241 curl --socks5-hostname 127.0.0.1:1080 --max-time 5 -s https://www.cloudflare.com/cdn-cgi/trace
242 ```
243
244 ### AdGuard DNS
245 ```bash
246 dig +tcp @127.0.0.1 -p 5354 google.com +short +timeout=5
247 ```
248
249 ### Firewall & Routing
250 ```bash
251 # nftables FORWARD chain
252 docker compose exec -T warp iptables -L FORWARD -v --line-numbers
253 # Legacy FORWARD chain
254 docker compose exec -T warp iptables-legacy -L FORWARD -v --line-numbers
255 # IP rules + routing tables
256 docker compose exec -T warp ip rule show
257 docker compose exec -T warp ip route show table 199
258 docker compose exec -T warp ip route show table 200
259 ```
260
261 ### Host macOS
262 ```bash
263 tailscale status
264 brew services list | grep tailscale
265 networksetup -getdnsservers Wi-Fi
266 networksetup -getsocksfirewallproxy Wi-Fi
267 sysctl net.inet.ip.forwarding
268 ```
269
270 ---
271
272 ## 9. Development Issue Tracker
273
274 This section documents issues encountered during development, their status, and resolutions.
275
276 ### 9.1 Exit Node Return Path (`rx 0`) RESOLVED
277 **Symptom:** Gateway showed `tx 936 rx 0` transmitted but never received return traffic.
278
279 **Root Cause:** `docker-compose.yml` included `FIREWALL_OUTBOUND_SUBNETS=100.64.0.0/10`. Gluetun created a
    | strict routing rule (`ip rule add to 100.64.0.0/10 lookup 199`, priority 99) that forced all Tailscale
    | CGNAT traffic to bypass the VPN via the Docker bridge. Return packets were sent to the macOS host instead
    | of back through `tailscale0`.
280
281 **Fix:** Removed `FIREWALL_OUTBOUND_SUBNETS` entirely. `scripts/routing-fix.sh` now injects `lookup 52`, and
    | return packets correctly flow back into `tailscale0`. The SOCKS5 proxy was repurposed as a bulletproof
    | failover.
282
283 ### 9.2 Gluetun Resets nftables FORWARD Rules
284 **Symptom:** FORWARD RELATED,ESTABLISHED rule disappears after Gluetun health check.
285
286 **Fix:** `scripts/routing-fix.sh` re-adds to both iptables backends every 30 seconds. Legacy backend (policy
    | ACCEPT) acts as safety net.
287
288 ### 9.3 docker compose exec `\r` Injection
289 **Fix:** All `docker compose exec` output piped through `tr -d '\r'`.
290
291 ### 9.4 Colima VM Memory / Swap Thrashing Latency

```



```

292 **Symptom:** After running nullexit flawlessly for over 24 hours straight to test stability, the 0.5GB VM RAM
    configuration began experiencing severe network latency on the Tailscale mesh later in the day.
293
294 **Root Cause:** Services (like AdGuard, Tailscale, Docker logs) slowly accumulate cache and state over extended
    periods. Under the tight 512MB physical RAM limit, this slow creep forced the Linux kernel to aggressively
    swap memory to the 512MB SSD swap file. The constant disk I/O thrashing blocked process execution, causing
    DNS resolutions and packet forwarding to delay significantly (making the internet feel "very slow").
295
296 **Proof (Empirical Observation):** While in this OOM-thrashing state, direct DNS queries through the Tailscale
    mesh (`dig @100.100.21.8 google.com`) would intermittently hang or take several seconds to resolve. After
    restarting with the 600MB configuration, the exact same query immediately dropped to a lightning-fast **41
    ms** response time. (Note: Querying the mapped host port via `127.0.0.1:5354` will always time out due to
    Gluetun's strict leak-prevention firewall blocking non-VPN ingress traffic).
297
298 **Fix:** Increased VM base physical RAM to 600MB (`--memory 0.6`) and reduced the swap file to 400MB. This
    provides enough native RAM headroom for long-running state caches without forcing heavy I/O swapping.
    Subdomain deduplication still reduces blocklists by ~60%. Memory profiles (`light`/`medium`/`heavy`) let
    users trade off coverage for memory.
299
300 ### 9.5 tailscaled Daemon Broken State
301 **Symptom:** `brew services list` shows `started` but `tailscale status` shows "stopped".
302
303 **Fix:** `sudo brew services restart tailscale`. Toggle script now detects and auto-restarts.
304
305 ### 9.6 Stderr Invisibly Swallowed
306 **Symptom:** Failures were silent due to stderr redirected to `output.log`.
307
308 **Fix:** Tailscale wait loop now shows output. Critical commands show errors inline.
309
310 ### 9.7 Missing iptables in routing-fix Container
311 **Root Cause:** `alpine:3.20` does not have iptables installed. Commands failed silently with `2>/dev/null ||
    true`.
312
313 **Fix:** Updated `docker-compose.yml` to `apk add --no-cache iptables iproute2` before `scripts/routing-fix.sh`
    runs.
314
315 ### 9.8 DOCKER_GW Variable Parsing Bug
316 **Root Cause:** `ip route show default` printed two lines; `awk '{print $3}'` picked up both, causing route
    commands to fail.
317
318 **Fix:** Added `head -1` to the parsing chain.
319
320 ### 9.9 Native SSH & SFTP Security (Zero Local Attack Surface)
321 **Context:** macOS "Remote Login" (`sshd`) and "File Sharing" (`smbd`) open ports on the local network (e.g.
    `172.x.x.x`), exposing the host to brute-force attacks on public Wi-Fi.
322
323 **Implementation:** The script strictly enforces `tailscale up --ssh=true` whenever the mesh state is reset.
    This empowers the user to completely disable native macOS Remote Login and File Sharing. This provides an
    incredible zero-trust security advantage: even if an attacker steals your Mac username and password, they
    **cannot** access your files remotely. Disabling the native services completely closes the listening ports
    on the local Wi-Fi interface (`172.x.x.x`), rendering the Mac inaccessible to local network logins.
    Meanwhile, Tailscale intercepts port 22 traffic exclusively over the encrypted mesh and authenticates
    cryptographically based on your Tailscale SSO identity. Stolen Mac passwords are mathematically useless
    without first bypassing your Tailscale Two-Factor Authentication and registering a device onto the mesh.
324
325 **Cross-Platform File Exchange:** To easily browse and exchange files securely, simply use an SFTP client and
    connect to your Mac's MagicDNS name on port 22. Recommended clients: **WinSCP** or **FileZilla** (Windows),
    **Cyberduck** (macOS), and **FE File Explorer** or **Solid Explorer** (iOS/Android).
326
327 ### 9.10 Changing macOS Local Hostname
328 **Context:** Users may wish to change their Mac's computer name/hostname in macOS System Settings.
329 **Effect:** Changing the Mac's hostname will **not** break Nullexit (which relies on internal routing/localhost
    ). However, Tailscale will automatically detect this change and update the machine name on the Tailnet.
330 - The underlying Tailscale `100.x.x.x` IPv4 address will remain exactly the same.
331 - The **MagicDNS name** will change to match the new hostname. Users must remember to update their SFTP/SSH
    clients to use the new MagicDNS address.
332
333 ### 9.11 Logging Architecture
334 For debugging, logs are strictly segmented based on the component's lifecycle:
335 - **`output.log` (Host-side):** Contains all standard error (`stderr`) and verbose output from the host scripts
    (`toggle.sh`, `recover.sh`, `setup.sh`). Since the `rule-compiler` container is ephemeral and deleted
    after running, its logs (and any Python errors from `scripts/sync-rules.py`) are extracted and appended to
    this file before deletion.
336 - **Log Rotation Policy:** On startup, `toggle.sh` checks if `output.log` exceeds **50MB** (`52,428,800`
    bytes). If it does, it performs a metadata-only rename (`mv output.log output.log.old`), preserving history
    while resetting the active log to 0 bytes. This rename-based rotation avoids the heavy disk I/O and CPU
    overhead of rewriting a massive file line-by-line.
337 - **Egress Probe Logging:** The background host-egress leak prober checks if stdout is a TTY (`[ -t 1 ]`) and
    will only print live status updates (using carriage return `\r`) or duplicate console warnings in
    interactive mode. It automatically detects macOS/BSD vs. Linux environments to dynamically construct
    timestamps with date and time (omitting milliseconds on macOS to prevent high CPU utilization from spawning

```

```

sub-second processes).
338 - **docker logs <container>` (Guest-side):** The persistent containers (`warp`, `tailscale`, `routing-fix`, `adguardhome`) use the Docker `json-file` logging driver with a strict `max-size` (1m-10m) to prevent VM disk exhaustion. Use standard `docker logs` to view them.
339
340 ### 9.12 Chrome Remote Desktop Connection Failures
341 **Context:** When attempting to access a remote machine via Chrome Remote Desktop (CRD), the connection fails or hangs if Tailscale is active **on the remote device**. The connection works fine even if Tailscale (and the exit node) is active **on the local client/viewer device**, but only if Tailscale is disabled on the remote machine.
342
343 **Why:** Having Tailscale active on the remote machine creates virtual network interfaces and DNS/routing modifications that conflict with Chrome Remote Desktop's WebRTC bindings and signaling traffic.
344
345 **Workaround:**
346 - **Use Tailscale Native RDP/RustDesk (Recommended):** Connect directly to the remote device's Tailscale IP (`100.X.Y.Z`) via an RDP or RustDesk client. This is faster and avoids third-party relay conflicts.
347 - **Disable Tailscale on the Remote Device:** If you must use CRD, disable Tailscale on the remote machine before connecting.
348
349 ### 9.13 Docker Default Bridge vs Host Wi-Fi Subnet Collision (The 172.17.0.0/16 Subnet Overlap)
350 **Context:** When attempting to ping a local Windows client (`172.17.22.187`) or the default gateway (`172.17.0.1`) directly over Wi-Fi, the Mac host returns `No route to host` (displaying a `REJECT` / `!` flag in the routing table). Tailscale on the client also drops offline shortly after starting when using the exit node, failing to establish direct P2P connections and falling back to slow DERP relays.
351
352 **Root Cause & Subnet Overlap Mechanism:**
353 1. **Docker Default Subnet:** By default, the Docker daemon initializes its default bridge network (`docker0`) on the **172.17.0.0/16** IP range.
354 2. **Subnet Conflict:** The host's physical Wi-Fi network happens to be assigned the exact same **172.17.0.0/16** range by the local network router.
355 3. **Routing Clash:** Colima launches a Linux VM on the Mac host to run the Docker engine. Because Docker inside that VM creates `docker0` and claims the `172.17.0.0/16` subnet, any packet sent to `172.17.x.x` from within the containers (such as Tailscale local discovery) is captured by the VM's internal virtual bridge interface (which is down/empty) and is blackholed. On the Mac host, macOS dynamically marks routes as `REJECT` (!) when local ARP resolution fails, causing local pings to immediately abort with `No route to host`.
356 4. **Tailscale Relay Saturation:** Because local peer-to-peer discovery is blackholed, Tailscale falls back to public DERP relay servers (e.g. `relay "tor"` in Toronto). When exit-node routing is enabled, all client web traffic is routed through the relay. Public relays impose strict rate limits and throttle high-volume traffic, resulting in packet drops. This saturation drops Tailscale's control packets, causing the connection to time out and showing the client as offline.
357
358 **Fix:**
359 The canonical script for this is `scripts/fix-docker-bridge-collision.sh`. **One command applies the entire fix:**
360
361 ```bash
362 bash scripts/fix-docker-bridge-collision.sh          # apply
363 bash scripts/fix-docker-bridge-collision.sh --dry   # preview changes without applying
364 ```
365
366 It auto-detects the host's actual Wi-Fi subnet (no more guessing whether the campus DHCP handed out `172.x`, `10.x`, or `192.168.x`), picks a non-conflicting `docker.bip` from a small candidate set (defaulting to `172.26.0.1/24` if no overlap is matched), backs up `~/colima/default/colima.yaml` with an ISO-8601 timestamp, edits the file **in place** (idempotent replaces an existing `docker.bip` if present, appends under an existing `docker:` block, or creates a fresh block), restarts Colima, rebinds Wi-Fi DHCP, verifies the new default route is no longer Docker's bridge, and re-runs both `./toggle.sh` and `scripts/diagnose-host-leak.sh` to print the post-fix verdict.
367
368 For reference, the underlying manual fix is: edit the Colima configuration file (`~/colima/default/colima.yaml`) on your Mac to define:
369
370 ```yaml
371 docker:
372   bip: 172.26.0.1/24  # any /24 that does NOT overlap your Wi-Fi subnet
373 ```
374
375 Then, restart Colima to apply the new subnet inside the VM:
376
377 ```bash
378 colima restart
379 ```
380
381 This forces Docker to use `172.26.x.x` for its default bridge, freeing the physical `172.17.x.x` range and letting pings and direct Tailscale peer-to-peer connections flow normally.
382
383 ---
384
385 ### 9.14 Cloudflare WARP 24-Second Startup Delay (UDP Session Rate-Limiting)
386 **Context:** When running `toggle.sh` to turn the gateway ON, `docker compose up -d` often hangs for exactly 24-25 seconds waiting for the `warp` container to become `healthy`. Users might mistakenly blame the Go `rule-compiler` processing 400k+ rules for this delay, as Docker prints `rule-compiler Exited 24.3s` at the end of the wait.

```


384 ****Root Cause:**** The ``rule-compiler`` actually finishes its work in <2 seconds. The 24-second blockage is entirely due to Cloudflare WARP's edge server rate-limiting. WireGuard is a stateless protocol with no "disconnect" packet. When the toggle is turned OFF, the old UDP session state remains active on Cloudflare's backend for 20-30 seconds. When the toggle is instantly turned back ON, Docker starts a new WireGuard connection from a new ephemeral UDP source port but uses the ****same static cryptographic key**** (from ``.env``). Cloudflare detects this as a potential replay attack or key-sharing abuse, and intentionally drops all packets (rate-limits) for the new connection until the old session's ghost state fully expires (~20-25 seconds).

385 ****Result:**** Gluetun's healthcheck fails repeatedly every 2 seconds until Cloudflare lifts the penalty box, at which point traffic flows and Docker unblocks the rest of the startup sequence. This is a known, expected behavior of reusing static keys rapidly on Cloudflare's network and cannot be bypassed.

386

387 **### 9.15 Background Sudo Failures**

388 ****Context:**** The ``--post-wake`` script is intentionally designed to skip the ``sudo -v`` interactive prompt because it is launched by ``watcher.sh`` running via ``launchd`` in the background (which has no TTY and cannot accept password input). Thus, it relies entirely on ``sudo -n`` (non-interactive sudo) for all its internal routing changes.

389 ****Problem:**** If the user has not configured the ``/etc/sudoers.d/nullexit`` file properly, any automated background wake event will silently fail: ``sudo -n`` commands drop out, the WARP bypass routes fail to apply, and the ``warp`` container loses internet connectivity.

390

391 **### 9.16 Tailscale Hijacking the Default Route (PHYSICAL_GW Bug)**

392 ****Problem:**** When ``--post-wake`` runs, it clears the routing table using ``sudo route -n flush`` to ensure a clean slate after the Wi-Fi card roams or wakes up. Because it skips bouncing the Wi-Fi interface (to make the reconnect faster), macOS's DHCP client does not automatically rebuild the physical default route. To counteract this, the script must manually restore the physical default route. Originally, the script tried to capture the physical gateway via ``route get default``. However, because the gateway is already running and Tailscale is active, the default route is often already pointing to ``utunX``. When this happens, ``route get default`` does not output a ``gateway:`` field (it only outputs the ``interface:``), causing the captured ``PHYSICAL_GW`` variable to be empty.

393 When ``PHYSICAL_GW`` is empty, the physical default route is never restored, causing the ``en0`` interface to be isolated from the physical network. The WARP bypass routes then fall back to targeting ``-interface en0`` (instead of routing via the gateway), which breaks WireGuard's remote connections completely.

394 ****Fix:**** The script now bypasses the OS routing table entirely and directly queries the hardware DHCP lease (``ipconfig getpacket en0``) to reliably extract the physical router IP, ensuring the physical route is correctly restored even when Tailscale is active.

395

396 **### 9.17 Tailscale MagicDNS Bypassing the AdGuard PREROUTING Intercept**

397 ****Context:**** The ``routing-fix.sh`` script applies an iptables NAT rule (``PREROUTING -i tailscale0 -p udp --dport 53 -j REDIRECT``) to intercept all incoming DNS queries from connected Tailscale peers and force them into the AdGuard container on port 5335.

398 ****Problem:**** When remote clients (like Android or iOS) have "Use Tailscale DNS settings" turned on, they query Tailscale's MagicDNS IP (``100.100.100.100``). This packet enters the exit node, but the ``tailscaled`` daemon running ***inside*** the gateway container intercepts the ``100.100.100.100`` packet internally. ``tailscaled`` then generates a brand new, local DNS request to the upstream DNS server to resolve the query. Because this new request is generated ***locally*** by the daemon (not arriving externally via the ``tailscale0`` interface), it bypasses the ``PREROUTING`` chain completely. The request glides straight out the WARP tunnel to Cloudflare/Google, entirely bypassing the AdGuard sinkhole.

399 ****Fix:**** The only way to fix this blindspot is to force ``tailscaled`` to use AdGuard as its upstream. In the Tailscale Admin Console, the gateway's Tailscale IPv4 address (e.g., ``100.x.x.x``) must be added as the sole Custom Global Nameserver, and "Override local DNS" must be enabled. Additionally, to block Android devices from bypassing this via Private DNS (DNS-over-TLS), outbound Port 853 traffic is explicitly REJECTED in both ``routing-fix.sh`` and ``post-rules.txt``.

400

401 **### 9.18 Network Bufferbloat & MTU Optimizations (Double-VPN UDP Crash)**

402 ****Symptom:**** When running aggressive UDP bandwidth tests (such as macOS ``networkQuality``, which uses QUIC/HTTP3), the ``nullexit`` tunnel completely crashes or exhibits severe lag (6,000ms+ bufferbloat).

403

404 ****Root Cause:**** The bug is a cascading MTU mismatch. Tailscale generates a UDP packet. Cloudflare WARP (also WireGuard) receives the packet, wraps it in another layer of encryption, pushing the packet size beyond the standard 1500 byte limit. Unlike TCP, UDP packets cannot be resized dynamically via MSS negotiation, resulting in massive IP packet fragmentation. The Apple Virtualization framework (``vz``), which Colima uses for bridged networking, silently drops heavily fragmented UDP packets under high load. This blackholes the entire UDP upload stream, instantly dropping the connection.

405

406 ****Fix (Partial):**** To prevent MTU fragmentation for standard web traffic, ****TCP MSS Clamping**** is strictly enforced via the macOS ``pf`` firewall. By adding ``scrub out all max-mss 1160`` to ``scripts/pf.conf``, macOS unilaterally shrinks all outgoing TCP headers to fit comfortably inside the double-encrypted tunnel, bypassing Path MTU Discovery blackholes. ``TS_DEBUG_MTU=1200`` was also added to ``docker-compose.yml`` to minimize internal fragmentation.

407 ****Unresolved:**** The only way to solve the UDP crash bug natively is to artificially cap the maximum tunnel bandwidth using SQM (``fq_codel``). Because a static bandwidth cap would artificially throttle high-speed networks, ``nullexit`` leaves the bandwidth uncapped, optimizing perfectly for TCP while accepting UDP fragmentation under extreme edge-case load tests.

408 ****Why not Dynamic SQM (Aurate)?*** Implementing an EMA/PID-controlled aurate daemon (like ``sqm-aurate``) requires a constant 100ms polling loop to measure the kernel packet queue, which severely degrades laptop battery life by preventing deep C-state CPU idling. Furthermore, macOS's native firewall (``dummynet``) only supports ``fq_codel`` and does not support the newer Linux ``CAKE`` algorithm, which is the only algorithm that offers kernel-native, event-driven ``aurate-ingress`` (zero polling penalty). Therefore, dynamic SQM is computationally hostile to battery-powered macOS hosts.

409

```

410 ## 10. Resolved Issues (from README)
411
412 These bugs and edge cases were discovered and resolved during development.
413
414 ### 10.1 DNS Proxy: TCP Wire Format Mismatch (socat / dnsmasq)
415
416 **Problem:** When the Tailscale data plane is unavailable (DERP relay only), the script falls back to a local
DNS proxy that forwards queries from host UDP:53 to Docker's TCP:5354 (AdGuard). The `socat` and `dnsmasq`
approaches both failed.
417
418 - **socat** forwards raw bytes it does not prepend the 2-byte length prefix that DNS-over-TCP requires.
AdGuard parses the stream incorrectly and returns garbage.
419 - **dnsmasq** tries UDP first to reach the upstream (`127.0.0.1#5354`), but Colima's SSH tunnel only forwards
TCP. With a single upstream server, dnsmasq doesn't fall back to TCP even with `--timeout=1`.
420
421 **Solution:** Replaced both with a **Python DNS proxy** (`scripts/dns-proxy.py`, ~25 lines) that properly
handles the DNS-over-TCP wire format: reads a UDP query from the host, prepends the 2-byte length prefix,
sends it over TCP to AdGuard, strips the prefix from the response, and sends it back over UDP.
422
423 *Architectural Note: Why is the DNS proxy in Python while the SOCKS5 proxy was moved to a compiled Go Docker
container?
424 The SOCKS5 proxy handles heavy data streaming (e.g., video, downloads). Python introduces massive CPU/battery
overhead for raw socket streaming, which is why we rely on the compiled Go implementation built directly
into the `tailscaled` daemon via `TS_SOCKS5_SERVER=:1080`.
425 Conversely, the DNS proxy only handles intermittent UDP queries (a few bytes per minute) generated solely by
the local host machine. The Python overhead for this is effectively 0.001 Watts. We deliberately kept `dns-
proxy.py` in Python because it runs natively on the macOS/Linux hostrewriting it in Go would force users to
install a Go compiler (`brew install go`) on their host machine for zero noticeable battery gain. (Note:
If this architecture is ever adapted to serve as a public DNS resolver for thousands of users or for
massive automated web-scraping clusters, `dns-proxy.py` must be rewritten in Go to survive the concurrent
packet flood).
426
427 ### 10.2 Exit Node Routing Conflict & The SOCKS5 Failover
428
429 **Problem:** For days, Tailscale's exit node refused to return traffic back to the client (`rx 0`), leading to
the assumption that Tailscale's userspace forwarding was bypassing `iptables` and ignoring the WARP `tun0`
interface. In a desperate attempt to fix this, a **SOCKS5 proxy** (`scripts/socks5-proxy.py`) was
introduced as a replacement.
430
431 **The Real Root Cause:** Tailscale's userspace forwarding was not the problem. The issue was an environment
variable in Gluetun (`FIREWALL_OUTBOUND_SUBNETS=100.64.0.0/10`) that instructed the VPN container to hijack
all Tailscale CGNAT traffic and forcibly route it out the unencrypted Docker bridge (`eth0`). This
blackholed all return packets back to the client.
432
433 **Solution:** Removed the offending `FIREWALL_OUTBOUND_SUBNETS` variable, immediately restoring the Tailscale
exit node. Instead of removing the SOCKS5 proxy, we repurposed it into a **bulletproof failover** for when
restrictive Wi-Fi networks block Tailscale UDP ports.
434
435 Architecture after the fix:
436 ```
437 PRIMARY (Exit Node):
438 DNS/TCP/UDP: Apps Tailscale Exit Node gateway container tun0 WARP internet
439
440 FAILOVER (SOCKS5 - if Tailscale is blocked by local Wi-Fi):
441 DNS: Browser 127.0.0.1:53 Python proxy TCP:5354 AdGuard WARP
442 TCP: Apps SOCKS5:1080 gateway container tun0 WARP internet
443 Mesh: SSH/ping 100.x.x.x utun5 WireGuard peers (independent of proxy)
444 ```
445
446 ### 10.3 Pre-Flight Check 1: Wrong `tailscale ping` Flag
447
448 **Problem:** Check 1 of the pre-flight connectivity checks used `--c 1` (double dash) but `tailscale ping`
accepts `-c 1` (single dash). The invalid flag caused the command to exit with code 1, marking the check as
FAIL even though the gateway was reachable.
449
450 **Fix:** Changed `--c 1` to `-c 1`.
451
452 ### 10.4 Pre-Flight Check 1: DERP Relay Exit Code
453
454 **Problem:** Even with the correct `-c 1` flag, `tailscale ping` exits with code 1 when the connection goes
through a DERP relay ('direct connection not established') even though a pong was successfully received
(28ms via DERP(tor)). The check treated relayed connections as failures.
455
456 **Fix:** Changed the check from relying on exit code to piping stdout through `grep -q "pong"`. This accepts
relayed connections as valid (they work fine for the exit node use case), while still failing on true
timeouts or unreachable peers.
457
458 ### 10.5 Container Default Route Bypasses WARP
459
460 **Problem:** Inside the WARP container's network namespace, the main routing table's default route was via `
eth0` (Docker bridge), not `tun0` (WARP tunnel). While gluetun uses policy routing (rule 101 table 51820

```

```

`tun0`) for most traffic, traffic originating from the container's own IP (`172.18.0.2`) hits rule 100 (
from 172.18.0.2 lookup 200), whose default was also via `eth0`. This caused the SOCKS5 proxy's outbound
connections to bypass WARP.
461
462 **Fix:** Updated the `routing-fix` container to:
463 1. Change the main table's default route to `default dev tun0`
464 2. Change table 200's default route to `default dev tun0`
465 3. Add a specific route for the WARP WireGuard endpoint (`162.159.192.1`) through `eth0` via the Docker gateway
   in table 200, preventing a tunnel loop
466 4. Re-assert all routes every 30 seconds (gluetun may reset them on health checks)
467 5. Dynamically detect the Docker subnet from `ip route show dev eth0` instead of hardcoding `172.18.0.0/16`
468
469 ### 10.6 IPv6 Exit-Node Leak
470
471 **Problem:** Tailscale supports IPv6, but WARP/glutun is IPv4-only. IPv6 packets forwarded through the exit
   node would leak through the Docker bridge unencrypted.
472
473 **Fix:** Added `ip6tables -A FORWARD -i tailscale0 -j DROP` to `post-rules.txt`, blocking all IPv6 forwarded
   traffic from the Tailscale interface.
474
475 ### 10.7 Hardcoded Docker Subnet in Routing Fix
476
477 **Problem:** The `routing-fix` container hardcoded `172.18.0.0/16` and `172.18.0.1` for the Docker bridge
   subnet and gateway. If Docker's bridge network changed (after `docker network prune`, Colima restart, or on
   a different machine), these routes would be wrong.
478
479 **Fix:** Detection is now dynamic using `ip route show`:
480 - `DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
481 - `DOCKER_GW=$(ip route show default | awk '{print $3}')
482
483 This extracts the actual subnet and gateway directly from the routing table, working with any subnet size
   (`/16`, `/24`, `/20`, etc.).
484
485 ### 10.8 SOCKS5 Proxy Threading Race Condition
486
487 **Problem:** The `forward` function in `scripts/socks5-proxy.py` called `select.select([src, dst], [], [])`
   which raised `ValueError: file descriptor cannot be a negative integer (-1)` when one thread closed a
   socket while the other thread was selecting on it.
488
489 **Fix:** Added `ValueError` exception handling around `select.select()` and `recv()` calls. When a file
   descriptor becomes negative, the thread breaks out of the forwarding loop cleanly instead of crashing.
490
491 ### 10.9 Docker Healthcheck: Multi-line Python in CMD Array
492
493 **Problem:** The socks-proxy healthcheck used a `CMD` array with a multi-line Python string. YAML collapsed the
   newlines, causing the `#` comment to comment out the rest of the code and the `assert` statement to be
   mangled into `as`.
494
495 **Fix:** Changed from `["CMD", "python3", "-c", "..."]` to `["CMD-SHELL", "python3 -c '...']` with a single-
   line Python expression. Uses a real SOCKS5 handshake (sends `[5, 1, 0]`, expects `[5, 0]`) instead of a
   simple TCP port check.
496
497 ### 10.10 Sudo Credential Caching Removed
498
499 **Problem:** The script used `sudo -v` at startup to cache sudo credentials, plus a background `SUDO_KEEPER_PID`
   loop refreshing them every 60 seconds. This required interactive password entry even with NOPASSWD
   sudoers configured, because `sudo -v` itself prompts.
500
501 **Fix:** Removed the entire `sudo -v` + `SUDO_KEEPER_PID` section. All privileged commands now use `sudo -n` (
   non-interactive), which works silently because the user has NOPASSWD rules in `/etc/sudoers.d/nulllexit` for
   the specific commands needed (`networksetup`, `python3`, `dscacheutil`, `killall`, `pkill`, `route`, `
   ifconfig`).
502
503 ### 10.11 macOS Application Firewall Silently Blocking AirDrop & Continuity
504
505 **Problem:** Users of complex network extensions (Tailscale, LuLu, Docker, etc.) often run into an issue where
   AirDrop, AirPlay, and Universal Clipboard silently fail, even with the gateway inactive and Wi-Fi/Bluetooth
   correctly configured. The internal Apple Wireless Direct Link (`awdl0`) interface appears healthy, but
   connections never establish.
506
507 **Root Cause:** The built-in macOS Application Firewall has a specific list of explicitly blocked applications.
   Apple's own core networking services (e.g., `/usr/libexec/sharingd`, `/usr/libexec/rapportd`, `/usr/
   libexec/avconferenced`) can be silently added to the "Block incoming connections" list either through
   accidental "Deny" clicks on permission prompts, execution of aggressive privacy-hardening scripts, or a
   known macOS bug that retains strict blocks even after toggling the "Block all incoming connections" setting
   off. Since these are background daemons, macOS provides no visible error.
508
509 **Fix:** The firewall rules must be cleared via the terminal (or System Settings > Network > Firewall > Options
   ):
510 ```bash
511 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/sharingd

```

```

512 sudo /usr/libexec/ApplicationFirewall/socketfilterfw --unblockapp /usr/libexec/rapporstd
513 ...
514 Once unblocked, `sharingd` immediately resumes broadcasting mDNS/Bonjour discovery packets, and AirDrop
    restores instantly without requiring a reboot.
515
516 ### 10.12 Hard Reboots Leave Stale Docker Socket in Colima VM
517
518 **Problem:** If the macOS host machine is subjected to a hard reboot, sudden power loss, or a kernel panic, the
    Colima VM running in the background is abruptly killed. Upon next boot, running `toggle.sh` resulted in
    the contradictory output:
519 `Colima is already running.` followed by `Cannot connect to the Docker daemon at unix:///var/run/colima/default/
    docker.sock.`
520
521 **Root Cause:** The hard crash prevents the inner Docker daemon from executing its graceful shutdown routines.
    It leaves behind a corrupted `docker.sock` file inside the VM. On boot, the outer VM spins up (hence "
    already running"), but the inner Docker daemon sees the stale socket, assumes another instance is running,
    and crashes.
522
523 **Fix:** Added an Auto-Healing routine directly into `toggle.sh` (Step 4b). The script now explicitly tests the
    Docker daemon with `docker ps`. If the daemon is unresponsive despite Colima reporting as active, it
    assumes a stale socket and automatically runs `colima restart` in the background to cleanly rebuild the VM
    and socket before proceeding.
524
525 ### 10.13 Graceful Shutdowns Leave DNS Hijacked
526
527 **Problem:** The gateway overrides the macOS host's DNS settings (via `networksetup`) to route queries to the
    AdGuard container. Because macOS persists `networksetup` changes to disk, shutting down the Mac while the
    gateway is running causes the Mac to boot up later with hijacked DNS. Since the Colima VM and Docker
    containers don't auto-start on boot, the user would have no internet access until they manually run the
    toggle script.
528
529 **Root Cause:** The `toggle.sh` script exits after the gateway starts. There was no background daemon listening
    for macOS system shutdown signals to revert the network settings.
530
531 **Fix:** Wrapped the background `caffeinate` process (used to prevent system sleep) in a bash `trap` command.
    The trap listens for `SIGTERM`, `SIGINT`, and `SIGHUP`. When the user initiates a graceful shutdown, macOS
    sends `SIGTERM` to the background trap, which instantly executes `recover.sh` to flush the host DNS back to
    `1.1.1.1` right before the machine powers off.
532 *Note:* We explicitly use `recover.sh` instead of `toggle.sh` because during a shutdown, the OS limits process
    cleanup time and is already independently sending termination signals to Docker and Colima. Trying to run
    `docker compose down` inside a shutdown trap creates a race condition that can hang the script until macOS
    forcefully `SIGKILL`s it. `recover.sh` abandons container management and surgically flushes the macOS
    network settings in milliseconds, guaranteeing completion before the OS pulls the plug.
533
534 ### 10.14 Wi-Fi Edge Packet Loss via QUIC (UDP) Fragmentation
535
536 **Problem:** Applications that heavily utilize HTTP/3 (QUIC) over UDP such as Facebook, Instagram, and Google
    services experience massive stuttering and stalled image loading when the client device is far away from the
    router (weak Wi-Fi signal). The issue did not occur when close to the router.
537
538 **Root Cause:** The `nullexit` gateway uses a double-encrypted tunnel (Tailscale + WARP). To prevent packets
    from fragmenting when entering these tunnels, a firewall rule in `post-rules.txt` uses `--set-mss 1120` to
    safely clamp TCP packets. However, because QUIC runs entirely over UDP, it bypasses the TCP MSS handshake
    completely. QUIC blasts maximum-MTU (1280 byte) UDP packets. The inner `tailscale0` interface (MTU 1200)
    forces these large UDP packets to fragment into two pieces. When the client is on the edge of Wi-Fi range (
    where 5-10% packet loss is common), losing *either* UDP fragment destroys the *entire* image frame. This
    exponential amplification of packet loss stalled QUIC streams.
539
540 **Fix & Trade-offs:** Appended an `iptables` rule to `post-rules.txt` that explicitly blocks `UDP --dport 443`
    (QUIC). When modern web apps detect QUIC is blocked, they instantly fall back to HTTP/2 (TCP 443). Because
    TCP is routed through the MSS clamp, the packets are resized *before* they leave, entirely eliminating IP
    fragmentation and restoring high-speed loading over weak Wi-Fi.
541 *Consequences:* This adds a minor latency penalty (TCP 3-way handshake) compared to QUIC's zero-RTT connection.
    It may also force WebRTC video calls (Google Meet/Discord) to fall back to TCP TURN servers, slightly
    inflating real-time video latency. However, in a constrained double-tunnel mesh, the absolute stability
    gained from eliminating fragmentation vastly outweighs these trade-offs.
542
543 ### 10.15 Threat Model: Local Proxy Discovery
544
545 The Tor SOCKS proxy and Honey-Port Tripwire only bind to `127.0.0.1` (loopback). The real security boundary
    here is "can an unprivileged local process reach the loopback interface," not "does the malware know the
    port number."
546
547 The port-randomization and the Honey-Port trap only catch blind or generic port-scanners that do not already
    know where to look. They **do not protect** against a process that directly reads the `/tmp/nullexit-ports.
    env` file, greps the `toggle.sh` memory space, or runs `lsof -i` to inspect open socket bindings.
548
549 Because modifying file ownership on `/tmp/nullexit-ports.env` via `chown`/`chmod` to restrict read-access
    introduces an unavoidable Time-Of-Check to Time-Of-Use (TOCTOU) race condition (the file is created user-
    owned before privilege escalation, meaning file descriptors can be snatched), it is deliberately left as a
    standard user-owned file.

```

```

550
551 The actual, proper mitigation for preventing a malicious local process from reaching the proxy is not hiding
    the port it is running a host-level outbound firewall with strict localhost/loopback filtering enabled (e.g
    ., LuLu 4.3.0+ or Little Snitch). These firewalls gate access by cryptographic process identity (binary
    signature) at the kernel/network-extension level, rather than by port secrecy.
552
553 ##### Verifying Localhost Filtering (The Rogue Binary Test)
554
555 To empirically validate that the host firewall properly intercepts local proxy hijacking, you can compile and
    execute a completely unknown, un-whitelisted "rogue" binary to attempt a connection to the proxy port:
556
557 ```c
558 // lulu-test.c
559 #include <stdio.h>
560 #include <stdlib.h>
561 #include <arpa/inet.h>
562
563 int main(int argc, char *argv[]) {
564     int port = atoi(argv[1]);
565     int sock = socket(AF_INET, SOCK_STREAM, 0);
566     struct sockaddr_in server = { .sin_family = AF_INET, .sin_port = htons(port) };
567     server.sin_addr.s_addr = inet_addr("127.0.0.1");
568
569     printf("Rogue binary attempting connection to 127.0.0.1:%d...\n", port);
570     if (connect(sock, (struct sockaddr *)&server, sizeof(server)) < 0) {
571         printf("Connection BLOCKED by firewall.\n");
572         return 1;
573     }
574     printf("Connection SUCCESSFUL (Firewall bypassed!).\n");
575     return 0;
576 }
577 ```
578 Compile with `gcc -o lulu-test lulu-test.c`. When executed against the `$TOR SOCKS_PORT`, a properly configured
    host firewall (with "Allow Loopback" disabled in LuLu Preferences) will immediately pause the socket
    execution and generate a desktop alert for the unrecognized binary signature. This physically stops
    targeted local malware from exfiltrating data through the Tor tunnel, proving the threat model mitigation.
579
580 ---
581
582 ## 11. Deep Dive: Exit Node Return Path Analysis (June 26, 2026)
583
584 This section preserves the detailed packet-level analysis that led to identifying and fixing the `rx 0` bug.
585
586 ### The exit node was probably never going to work in this container topology
587
588 Here's the packet path traced by reading `ip rule show` and the routing tables live:
589
590 **Outbound (Phone-1 internet):**
591 1. Phone-1 sends packet to gateway via Tailscale mesh
592 2. Gateway's tailscale0 (kernel TUN, `TS_USERSPACE=false`) decrypts packet enters FORWARD chain
593 3. Packet has src=`100.87.42.87` (Phone-1), dst=some internet IP
594 4. FORWARD chain (nftables): Rule 1 matches (`-i tailscale0 -j ACCEPT`)
595 5. Routing decision for the forwarded packet. Which ip rule matches?
596 - Rule 98: `to 172.18.0.0/16` no (dst is internet)
597 - Rule 99: `to 100.64.0.0/10` no (dst is internet)
598 - Rule 100: `from 172.18.0.2` **NO** (src is `100.87.42.87`, not the container IP!)
599 - Rule 101: `not fwmark 0xca6c lookup 51820` **YES** (no fwmark on forwarded packets)
600 6. Table 51820: `default dev tun0` packet goes out through WARP
601 7. NAT POSTROUTING: `MASQUERADE on tun0` src becomes WARP IP
602 8. Packet exits through WARP to internet (in theory)
603
604 **Return (internet Phone-1):**
605 1. Return packet arrives on tun0, conntrack de-MASQUERADES dst back to `100.87.42.87`
606 2. Routing decision for dst=`100.87.42.87`:
607 - Rule 99: `to 100.64.0.0/10 lookup 199` **YES**
608 3. Table 199: `100.64.0.0/10 via 172.18.0.1 dev eth0` **sends it to the Docker gateway!**
609 4. **This is wrong.** The packet should go to tailscale0 (to be re-encrypted and sent to Phone-1), but table
    199 sends it out eth0 to the Docker bridge host.
610
611 **Table 199 is the smoking gun.** It has one route: `100.64.0.0/10 via 172.18.0.1 dev eth0`. This sends ALL
    Tailscale CGNAT return traffic out the Docker bridge to the host not back through tailscale0. The return
    packet leaves the container's network namespace entirely, hits the host's routing stack, and gets dropped
    because the host has no idea what to do with a raw packet for `100.87.42.87`.
612
613 ### Why table 199 exists and why it's wrong for exit node traffic
614
615 Table 199 + the CGNAT rule (pref 99) was designed so that the **container itself** can reach other Tailscale
    peers (e.g., for mesh management, DERP relay). For that use case, routing CGNAT traffic via the Docker
    bridge to the host (which has its own tailscaled) makes sense.
616

```

```

617 But for **forwarded exit node traffic**, the return packet needs to go back through tailscale0 inside the
    container not out to the host. The CGNAT rule intercepts the return packet before it can be routed back to
    tailscale0 through the main table or Tailscale's own routing (table 52, which is at pref 5270).
618
619 ### The SOCKS5 path works because it dodges all of this
620
621 SOCKS5 proxy creates connections **from the container's own IP** (`172.18.0.2`). So:
622 - ip rule 100 (`from 172.18.0.2 lookup 200`) matches
623 - Table 200 has `default dev tun0`
624 - Return traffic comes back to `172.18.0.2` on tun0, conntrack de-NATs, proxy sends response to client
625
626 No FORWARD chain involved. No CGNAT rule involved. No table 199. It just works.
627
628 ### Summary of findings
629
630 | Finding | Status | Evidence |
631 |-----|-----|-----|
632 | Table 199 CGNAT route hijacks exit node return traffic | **Fixed** | Changed `lookup 199` to `lookup 52` in `
    scripts/routing-fix.sh`. Return packets now route via tailscale0. |
633 | FORWARD RELATED,ESTABLISHED missing | **Fixed** | Installed `iptables` via apk in `docker-compose.yml`. Rule
    now injects successfully. |
634 | DOCKER_GW variable parsing error | **Fixed** | Added `head -1` in `scripts/routing-fix.sh`. |
635 | `FIREWALL_OUTBOUND_SUBNETS` was the root cause | **Fixed** | Removed from `docker-compose.yml`. Gluetun no
    longer hijacks CGNAT routing. |
636 | Host tailscaled error state from aggressive `tailscale down` | **Mitigated** | Toggle script now detects and
    auto-restarts. |
637
638 ### 11.1 Double-Tunneling MSS Clamping (Stalling Bug)
639
640 **Problem:** Traffic double-wrapped through Tailscale (WireGuard) and WARP (WireGuard) suffered 120 bytes of
    MTU overhead (60 + 60). The default MSS clamp of 1180 was still slightly too large, causing large packets
    to fragment or drop, which resulted in mysterious web page stalls on strict-MSS endpoints.
641
642 **Fix:** Lowered the TCP MSS clamp in `post-rules.txt` to `1120` to guarantee double-tunneled payloads fit
    safely within standard internet MTUs.
643
644 ### 11.2 Linux Native Wi-Fi Bouncing
645
646 **Problem:** The generated `scripts/recover-linux.sh` incorrectly retained the macOS `networksetup` commands,
    which would crash and fail to bounce Wi-Fi on Linux hosts.
647
648 **Fix:** Swapped the macOS logic for native Linux commands. The script now attempts `nmcli radio wifi off/on`
    first, and gracefully falls back to `ip link set wlan0 down/up` if NetworkManager is unavailable.
649
650 ### 11.3 TS_AUTH_ONCE Cloud Footgun & Ephemeral Keys
651
652 **Decision:** The compose file uses `TS_AUTH_ONCE=true` to prevent authentication loops. However, on headless
    cloud VPS deployments, if a standard auth key expires (usually 90 days), the container will silently drop
    off the mesh. We documented this trade-off explicitly in `docker-compose.yml` and recommended the use of **
    Tailscale Ephemeral Keys** for cloud deployments, which automatically clean themselves up and bypass this
    issue.
653
654 ### 11.4 Hardcoded AdGuard Credentials
655
656 **Decision:** AdGuard is deployed with the hardcoded credentials `admin / nullexit`. This is perfectly safe
    behind a NAT on a local macOS laptop. However, if deployed on a cloud host with exposed ports, this becomes
    a severe security risk. We added a stark warning comment in `docker-compose.yml` to ensure cloud users
    change this before deployment.
657
658 ### 11.5 Routing Fix: 30-second polling vs Netlink Socket
659
660 **Decision:** Instead of building a complex Netlink socket listener (`ip monitor`) to react instantly to
    iptables and routing flushes from Gluetun, we retained the 5-second `sleep` loop in `scripts/routing-fix.sh`
    .
661 - **Reasoning:** Building a netlink listener in pure bash on Alpine is incredibly brittle and complex (
    especially for iptables events). The 30-second polling loop is bulletproof. The only trade-off is a
    potential 1-4 second stutter if Gluetun reconnects, which was deemed an acceptable edge case for absolute
    reliability.
662
663 ### 11.6 Cache Poisoning and URL Rot in scripts/sync-rules.py
664
665 **Problem:** If a remote blacklist URL went permanently offline (404 Not Found), the Python `urllib` request
    would return the 404 HTML string. The script would aggressively regex-parse this HTML, find no domains, and
    overwrite the healthy local cache with a 0-domain file. This effectively disabled ad-blocking until the
    URL was fixed.
666
667 **Fix:** Implemented a defensive programming sanity check in `scripts/sync-rules.py`. Before overwriting the
    cache, the script now verifies that the compiled domain count is greater than 10 (and 1 for IP feeds). If
    it falls below this threshold (indicating a catastrophic 404 failure across multiple URLs), it aborts the
    cache overwrite, raises a `ValueError`, and keeps the previous day's healthy cache intact.
668

```



```

669 ### 11.7 Cellular Networks & PMTUD Blackholes (The 1280 Byte Limit)
670
671 **Experiment:** Conducted a live packet sweep from the PC-1 host to Phone-1 connected via a cellular network +
    Tailscale DERP relay using `ping -D -s <size>`.
672
673 **Result:**
674 - Packets with a payload of `1252` bytes (+ 28 bytes IP/ICMP headers = **1280 bytes total**) succeeded
    perfectly with ~60ms latency.
675 - Packets with a payload of `1253` bytes (**1281 bytes total**) and above resulted in a silent timeout.
676
677 **Analysis:** The cellular network (or the Tailscale DERP relay) enforces a strict MTU of 1280 bytes.
    Critically, it acts as a "PMTUD Blackhole" it silently drops packets larger than 1280 bytes instead of
    returning an ICMP "Fragmentation Needed" warning. If the exit node tries to send a standard 1500-byte
    internet packet to the phone over this link, it vanishes, causing web pages to stall permanently.
678
679 **Conclusion:** This empirically validates the absolute necessity of the TCP MSS clamping rule in `post-rules.
    txt`. By artificially clamping the MSS to `1120` (or `1180`), we ensure the TCP payload + headers + double-
    WireGuard overhead never exceeds the strict 1280-byte ceiling of the cellular mesh link.
680
681 ### 11.8 Gluetun nf_tables Parser Crash (Bash Interpolation Failure)
682
683 **Problem:** Attempting to make the TCP MSS dynamically configurable by using a standard bash variable inside `
    post-rules.txt` (`iptables ... --set-mss ${GATEWAY_MSS:-1120}`) caused the Gluetun container to
    catastrophically crash during startup. Gluetun's internal parser executes `post-rules.txt` line by line
    without a shell, meaning the variable was never expanded. It literally attempted to inject the string `${
    GATEWAY_MSS:-1120}` into iptables, resulting in a fatal `bad value for option "--set-mss"` error.
684
685 **Fix:** Removed the bash variable from `post-rules.txt` and reverted it to a pure hardcoded integer. To retain
    dynamic `.env` configuration, we moved the interpolation logic to `toggle.sh`. Right before Docker starts,
    the host script parses `GATEWAY_MSS` from the `.env` file and uses a cross-platform `sed` command to
    dynamically overwrite the integer directly inside `post-rules.txt`. This perfectly mimics manual
    configuration and completely bypasses Gluetun's strict parser.
686
687 ### 11.9 Seamless Wi-Fi Roaming & The macOS DNS Wipeout Loophole
688
689 **Problem:** The Tailscale, WireGuard, and Colima NAT stacks are natively designed for connectionless roaming
    and will seamlessly survive when the macOS host switches Wi-Fi networks (e.g., roaming from home Wi-Fi to a
    cellular hotspot). However, the internet completely breaks upon network transition. This occurs because
    macOS intentionally wipes out all custom DNS settings (`networksetup -setdnsservers`) and reverts to the
    new network's default DHCP nameserver whenever the active Wi-Fi BSSID changes. Since the Mac is locked to
    the exit node, its DNS requests are swallowed by WARP and dumped onto the public internet, which cannot
    route private DHCP IPs. This results in a total DNS failure.
690
691 **Solution:** Implementing a silent background "DNS Watcher" daemon (`nullexit-dns-watcher`) in `toggle.sh`.
    When the gateway starts, it spawns a background polling loop that checks the active Wi-Fi DNS every 30
    seconds. If macOS resets the DNS during a network change, the watcher instantly detects the drift and
    forcefully re-injects `100.100.21.8` via `networksetup`. When the gateway stops, the watcher process is
    cleanly killed. This allows true, seamless roaming across physical networks without ever dropping the
    gateway state.
692
693 ### 11.10 The Infinite Recursive Routing Loop (Hotspot Paradox)
694
695 **Problem:** A critical network topology crash occurs if the host Mac connects to a Mobile Hotspot (e.g., a
    Windows PC or Phone) that is *also* connected to the Tailscale mesh and has "Use Exit Node" enabled.
    Because the Mac is hosting the Exit Node, it routes its internet traffic to the Hotspot. The Hotspot
    intercepts the Mac's traffic on its way to the cellular tower and routes it *back* to the Exit Node (the
    Mac) via Tailscale. The Mac receives it, tries to send it out again, and the Hotspot intercepts it again.
    This creates an infinite, recursive encryption loop that instantly destroys network bandwidth and drops all
    connections.
696
697 **Fix:** This is a physical routing paradox that cannot be resolved in software. The upstream physical router
    providing internet to the Mac must be allowed to route its traffic directly to the physical WAN
    interface. The user must manually disable "Use Exit Node" on the upstream device providing the hotspot.
698
699 **Advanced Workaround:** If the upstream hotspot is a Windows machine and the user explicitly wants it to
    remain connected to the Exit Node, a permanent static route can be injected into the Windows routing kernel
    to forcefully bypass the Tailscale WFP interception for WARP packets. By binding the route to the physical
    adapter's Interface Index (IF), it becomes a permanent, roaming-aware bypass.
700
701 On Windows (Admin Command Prompt):
702 ```cmd
703 route print (find Interface List, note the IF number for the active Wi-Fi adapter, e.g. 15)
704 route -p add 162.159.192.1 mask 255.255.255.255 0.0.0.0 IF 15
705 route -p add 162.159.193.1 mask 255.255.255.255 0.0.0.0 IF 15
706 ```
707
708 On Linux (Root Terminal):
709 ```bash
710 ip route add 162.159.192.1 dev wlan0
711 ip route add 162.159.193.1 dev wlan0
712 ```

```



```

713 On macOS (Root Terminal):
714 ```bash
715 route add -host 162.159.192.1 -interface en0
716 route add -host 162.159.193.1 -interface en0
717 ```
718
719
720 *(Note: If the upstream router is an iPhone or Android device, you cannot inject static routes without
    jailbreak/root. You must simply disable "Use Exit Node" in the mobile Tailscale app).*
721
722 This punches a tiny hole straight through the paradox, letting the Mac's WARP packets escape to the physical
    internet while keeping the rest of the upstream router's traffic securely trapped in the Tailscale Exit
    Node.
723
724 ### 11.11 AdGuard Filter Redundancy & Memory Optimization
725
726 **Problem:** AdGuard Home does not perform cross-list deduplication in memory. When it loads its own native
    subscription filters (e.g., `AdGuard DNS filter`) alongside our massive `compiled_rules.txt`, overlapping
    rules are loaded twice, wasting significant RAM in the Colima VM. The UI would show ~500k rules,
    representing the sum of all lists rather than unique domains.
727
728 **Fix:** Updated `scripts/sync-rules.py` to intelligently cross-reference AdGuard's configuration and
    deduplicate our list against it.
729 1. The script now reads `adguard/conf/AdGuardHome.yaml` to dynamically fetch the exact URLs of any enabled
    native AdGuard filters.
730 2. The parsing engine was upgraded to normalize basic AdGuard syntax (`||domain`) back into raw base domains
    during the build process.
731 3. The script subtracts the native AdGuard domains from our custom blocklist *before* compiling, immediately
    purging ~83,000 completely redundant rules.
732 4. The normalized base domains then pass through our subdomain optimizer, which squashes an additional ~50% of
    the remaining rules.
733
734 This reduces the final compiled output from ~325k to ~281k rules on the `heavy` profile, saving ~45k rules from
    being loaded into memory twice and reducing the overall RAM footprint of the `adguardhome` container.
735
736 ### 11.12 Kernel-Level IP Blocklist (Threat Intelligence Firewall)
737
738 **Problem:** DNS sinkholing cannot stop malware or botnets that bypass DNS entirely by hardcoding direct IP
    addresses (a common technique for C2 communication). These connections pass straight through AdGuard unseen
    .
739
740 **Fix:** Added a second compilation pipeline to `scripts/sync-rules.py` that runs on every startup alongside
    the DNS pipeline.
741 1. The compiler concurrently fetches four curated threat-intelligence feeds: **Feodo Tracker** (abuse.ch
    botnet C2 IPs), **Spamhaus DROP** (IPs allocated to criminal organizations), **Emerging Threats** (
    Proofpoint active C2 and scanners), and **CINS** (active brute-force sources).
742 2. All entries are normalized via Python's `ipaddress` module. Any IP or CIDR that overlaps with RFC1918
    private ranges, loopback, link-local, or the Tailscale CGNAT range (`100.64.0.0/10`) is stripped to prevent
    accidentally locking users out of their own LAN or mesh.
743 3. The cleaned list (16,721 unique IPs/CIDRs) is written as an `ipset restore` file using an **atomic swap
    pattern**: `create_new populate swap with live destroy_new`. This guarantees the live `block_malicious`
    ipset is never empty during a reload.
744 4. `scripts/routing-fix.sh` watches the file's mtime every 30 seconds. On change it runs `ipset restore` and
    idempotently re-injects `FORWARD DROP` rules for both `src` and `dst` into both iptables backends (nftables
    + legacy).
745 5. `docker-compose.yml` mounts `adguard/work/userfilters/` into `routing-fix` as `/userfilters:ro` and adds `
    rule-compiler: service_completed_successfully` to its `depends_on`, eliminating the race condition where
    routing-fix could start before the file existed.
746
747 **Memory cost:** The entire 16,721-entry ipset costs only ~1.6 MiB of kernel memory negligible in the 600MB VM
    budget.
748
749 ### 11.13 Standalone Tailscale Daemon Freeze after macOS Sleep/Wake
750
751 **Problem:** When the macOS host goes to sleep and wakes up, the network interfaces and routing tables are
    rebuilt. The standalone `tailscaled` daemon (installed via Homebrew) occasionally fails to handle this
    transition and becomes completely frozen/unresponsive. In this state, any command like `tailscale down` or
    `tailscale status` hangs indefinitely, and all DNS requests sent to the local Tailscale resolver
    (`100.100.21.8`) time out, causing a total internet blackout on the host.
752
753 **Fix:** Enhanced the Tailscale verification and teardown logic in `toggle.sh` and `recover.sh`:
754 1. **Unresponsive Daemon Detection:** Instead of assuming the daemon is healthy just because the Unix socket `
    /var/run/tailscaled.socket` exists, the scripts now actively run a status check with a 5-second timeout (`
    run_with_timeout 5 tailscale status`). If the check times out (exit code 143), the daemon is classified as
    unresponsive.
755 2. **Auto-Recovery Loop:** If the daemon is unresponsive, the script now automatically attempts to restart the
    service using `brew services restart tailscale` (falling back to `sudo -n brew services restart tailscale`)
    .
756 3. **Graceful Retry:** Once restarted, the script pauses for 3 seconds for the daemon to initialize before
    proceeding with connection or teardown commands.
757

```

```

758 ### 11.14 macOS Sharing Services (AirDrop/AirPlay) Freeze on Network Transitions
759
760 **Problem:** When the gateway is turned on or off, the scripts perform network configuration cleanups which
    include flushing the routing tables (`route -n flush`), restarting the `mDNSResponder` daemon (`killall -
    HUP mDNSResponder`), and power-cycling the Wi-Fi interface (`setairportpower` or `ifconfig en0 down/up`).
    While AirDrop BLE discovery continues to function, the actual file transfers stall at "Waiting..."
    indefinitely.
761
762 **Root Cause:** The rapid tear-down and rebuild of the local routing table and Wi-Fi interface causes Apple's
    core sharing and connection daemons (`sharingd` and `rapportd`) to lose their socket bindings to the mDNS/
    Bonjour discovery interface. Instead of reconnecting gracefully, they enter a wedged state and try to route
    peer-to-peer (AWDL) IPv6/TCP transfer traffic into the default gateway (the Tailscale interface `utun`)
    instead of the direct wireless link, causing the TLS verification handshake to time out.
763
764 **Fix:** Integrated an automatic sharing services reset into both `toggle.sh` and `recover.sh`:
765 1. The script now automatically runs `sudo -n killall sharingd rapportd` at the end of both the **START** path
    and the **STOP** path (as well as inside `recover.sh`).
766 2. Killing these daemons forces macOS to immediately relaunch them, binding them fresh to the newly initialized
    network interfaces and routing tables, allowing AirDrop to work seamlessly while the gateway is active.
767
768 ### 11.15 Unlocking Mode-Locked Output Files Without `chmod` (`unlock-files.sh`)
769
770 **The one-command fix:** `bash scripts/unlock-files.sh` replaces every known locked inode in the project with
    a fresh writable one. No `chmod`. Covers `$.env` (mode `000`), `adguard/work/userfilters/compiled_rules.txt`
    , `ip_blocklist.ipset`, `cache/*.txt`, `cache/ip/*.txt`, and `adguard/work/data/filters/*.txt` (mode
    `0444`). Run it once after setup or after pulling an old repo; `toggle.sh` and `sync-rules.py` will then
    work without permission errors.
771
772 **Context:** The nullexit project has a strict project-wide policy of `no chmod from scripts` (see README 6).
    This rule exists because every prior `chmod 0444` (post-write tamper-proof lock) and `chmod 000` (post-
    toggle `$.env` lock) call from `scripts/sync-rules.py` / `toggle.sh` could leave a file permanently
    unreadable on disk if the script exited early, was killed mid-flight, or simply set a restrictive mode
    without ever being re-flipped. We've stripped every `chmod` from the codebase. But files **written by older
    versions of those scripts** are still on disk in those restrictive modes (`0444` for `compiled_rules.txt`,
    `ip_blocklist.ipset`, `data/filters/<id>.txt`; `000` for `$.env` after end-of-toggle lock). Re-running `
    toggle.sh` or `scripts/sync-rules.py` against those leftovers hits `PermissionError: [Errno 13] Permission
    denied` immediately.
773
774 **Why the stale permissions also block `mv`/`rm` of the parent folder:** The restrictive modes (`000` on `$.env`
    `, `0444` on output files) don't just block writes they prevent the kernel from unlinking the file's
    dentry during a `mv` or `rm` of the **containing directory** (`adguard/work/`). macOS enforces this at the
    VFS layer: you own the directory, but you can't delete a directory that contains entries you can't write to
    . This made the entire `adguard/work/` tree unmovable/undeletable until the locked inodes inside it were
    replaced. `unlock-files.sh` resolves this permanently by swapping each locked inode for a fresh `0644` one
    via atomic rename (`cp` + `mv`), which operates on the directory entry rather than the file contents no `
    chmod` called, no permission bypass needed.
775
776 **Problem:** You (the owner) cannot `cat`, `cp`, `rm`, `>` redirect, or any normal POSIX write against a `mode
    =000` file even though you own it the basic mode bits block ALL access for owner, group, and other. The
    script does not call `chmod` and you want a permanent fix that never relies on a `chmod` from any future run
    either.
777
778 **Solution:** Replace the locked inode with a fresh readable one. `mv` is atomic; mode bits live in the inode,
    not the directory entry. Two flavors cover every case we hit on a real machine:
779
780 **Flavor A locked file is mode `000` (owner can't even READ it):**
781 `NOPASSWD sudoers` (`/etc/sudoers.d/nullexit`) already includes `/usr/bin/python3`. So we read via `sudo python3`
    (which has the DAC override root has) and pipe the bytes around as base64 in user space, where the
    redirect happens with the user's normal umask = `0644`:
782 ````bash
783 `sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
    `decode())" > /tmp/snap.b64
784 `base64 -d < /tmp/snap.b64 > <FILE>.new
785 `mv -f <FILE>.new <FILE>
786 `ls -la <FILE>` # mode is now 0644
787 ````
788 The old `mode=000` inode is unlinked by `mv`; the new `mode=0644` inode (created by base64-decode via the
    calling user's umask) takes the path. No `chmod` ever called. The parent directory just needs to be writable
    by you (it always is, since you own it).
789
790 **Flavor B locked file is mode `0444` (you can READ but cannot WRITE; `scripts/sync-rules.py` needs to re-
    write):**
791 You own the file and can read it, you just cannot truncate/overwrite it. The simplest path is to rename it
    aside and let the writer create a fresh inode from scratch (which gets the umask-default `0644`):
792 ````bash
793 `mv <FILE> /tmp/old.<FILE>.$$
794 `python3 scripts/sync-rules.py` # now writes <FILE> cleanly at mode 0644
795 ````
796 Or apply Flavor A if you'd prefer to preserve the contents while resetting the mode.
797
798 **Why this is the canonical recovery, not a hack:**

```

```

799 - Mode bits live in the inode, not the path. `mv -f` replaces the path's inode reference; the old locked inode
      drops out of the directory entirely (dentry eviction) and loses its last link, so the kernel reclaims it.
      The new inode (whatever it is: a freshly-written one, or a base64-decoded copy) gets the path.
800 - The only prerequisite to apply Flavor B's `mv` is: you own the parent directory (so you can unlink entries
      from it) AND you have a NOPASSWD-capable binary that can read the byte stream if mode prohibits user read.
801 - This pattern generalizes. Any time a script ends up producing a "locked file that the next run can't update"
      situation, the same trick applies without a single chmod anywhere.
802
803 **Empirical verification (user machine, July 1, 2026):**
804 - `.env` was `mode=000` (owner $USER, i.e. you): Flavor A unblocked it in 3 commands, no chmod.
805 - Before: `----- 1 $USER staff 899 Jun 28 07:07 .env`
806 - After: `~rw-r--r--@ 1 $USER staff 899 Jul 1 20:39 .env`
807 - `docker compose config` immediately picked up `TS_AUTHKEY` + `WIREGUARD_PRIVATE_KEY` substitution.
808 - `compiled_rules.txt`, `ip_blocklist.ipset`, `data/filters/1782645604.txt` were `mode=0444`: Flavor B (`mv`-
      aside) unblocked all three.
809 - `scripts/sync-rules.py` then ran clean: 279587 block rules + 171 allow rules + 16710 IP entries compiled; new
      files came out at `0644`.
810 - `toggle.sh` then went end-to-end in 48s (warp / routing-fix / adguardhome / tailscale all healthy, gateway
      mesh-joined, DNS hijack verified single-server).
811
812 **When this trick is appropriate vs. inappropriate:**
813 - You own the parent directory and the file (typical desktop scenario).
814 - The lock is a leftover from a removed `chmod` call that you want off disk forever.
815 - NOPASSWD sudo includes at least one interpreter (`python3` on this system) that can be used to read mode-0
      files. If it does not, add it first: `<USER> ALL=(ALL) NOPASSWD: /usr/bin/python3`.
816 - The file is owned by another user (root, another account) and the lock is intentional respect it; do not
      bypass another user's security boundary.
817 - The parent directory is owned/writable only by another user escalate properly via sudo, or stop and ask the
      user.
818 - You do not actually own the file and there is a legitimate reason for its lock state restore the file via a
      trusted source rather than circumventing the perms.
819
820 **Reusable cookbook:**
821
822 ```bash
823 # Quick diagnostic: figure out which flavor you need.
824 WHOAMI=$(whoami)
825 ls -la <FILE>
826 stat -f 'mode=%Sp owner=%Su group=%Sg' <FILE>
827 # mode=----- or mode=---r----- : you cannot even read it Flavor A.
828 # mode=r--r--r-- but writer fails : you are blocked from write but can read Flavor B.
829 # mode=rw-r--r-- : not actually locked at all; unrelated write failure.
830
831 # Flavor A (mode=000):
832 sudo -n /usr/bin/python3 -c "import base64,sys; sys.stdout.write(base64.b64encode(open('<FILE>','rb').read()).
      decode())" > /tmp/snap.b64
833 base64 -d < /tmp/snap.b64 > <FILE>.new && mv -f <FILE>.new <FILE> && rm -f /tmp/snap.b64
834
835 # Flavor B (mode=0444, file is readable):
836 mv <FILE> /tmp/old.<FILE>.$(date +%s)
837 # (let the original writer recreate <FILE>; new file lands at mode 0644)
838
839 # Verify after either flavor:
840 ls -la <FILE> # mode should be 0644
841 <your normal validation command>
842 ```
843
844
845 ### 11.16 Gateway Breakage on Lid Close / Wi-Fi Roam Post-Wake + Post-Roam Auto-Recovery
846
847 **Context / Symptom.** Closing the lid (sleep/wake) OR losing + reconnecting Wi-Fi (e.g. in an elevator, caf,
      or roaming between APs) leaves the gateway in a partly-dead state. Symptoms range from a ~30s window of
      dead DNS to a permanent outage that only full `recover.sh` or `toggle.sh` fixes. The root cause is that
      nullexit has **no daemon watching macOS sleep/wake or network-state-change events** the existing DNS
      Watcher inside `toggle.sh` only runs in-process and is suspended along with the rest of the user shell on
      lid close, so the gateway cannot self-heal after either event.
848
849 **Diagnosis.** Two distinct failure modes share the same root gap.
850
851 * ***(A) Lid close sleep wake.** `caffeinate -i` (used by `toggle.sh`) blocks *idle* sleep but **does not
      block forced sleep from lid close**. Closing the lid hard-suspends Colima VM + every container + every
      Tailscale-quit-shells-except-tailscaled. On wake the VM clock needs ~5-30s to resync via NTP, during which
      TLS cert validation fails: Cloudflare WARP `162.159.192.1:2408` rejects the handshake gluetun's
      healthcheck (`interval: 2s, retries: 15`) eventually flips unhealthy Docker `unless-stopped` restarts the
      `warp` container (~30s gap). `mDNSResponder`'s in-memory cache still holds pre-sleep entries (stale A
      records for tailnet hosts) so the first dozen DNS queries after wake time out. `tailscaled` on the host
      often keeps its stale DERP relay preference and routes packets through a dead relay for another 30-60s.
852 * ***(B) Wi-Fi roam / loss in elevators, captive portals.** WARP is a UDP tunnel to Cloudflare's anycast edge.
      Carrier NATs and captive-portal networks silently invalidate the UDP binding on roam and on hotspot-bound
      re-association. macOS `networksetup` *should* preserve DNS settings on the same Wi-Fi service across a roam
      , but most captive-portal networks DHCP-replace DNS to the captive-portal resolver **during the captive-

```

portal dance itself**, before the user has authenticated so even DNS hijack survives a clean roam and quietly breaks on captive-portal handoff.

853

854 ****Resolution.**** A single ****launchd LaunchAgent**** (``com.nullexit.wake-recovery``) runs a long-running shell daemon (``scripts/watcher.sh``) that listens for both event sources and, on each fire, executes ``bash recover .sh --post-wake`` a **lighter-touch** recovery mode that's the opposite of the existing ``--nuclear`` semantics : it keeps the gateway live while still refreshing every stale subsystem.

855

856 **#### Apple Power Management Primer (for the unfamiliar)**

857

858 macOS exposes several relevant surfaces; the right primitive depends on what you actually need to detect. None of these are obvious and none of them have a standard splash screen in the docs.

859

860 * ****caffeinate <flags>**** creates I/O Kit power assertions via ``IOPMAssertionCreateWithName``. It does NOT block forced sleep (lid close, low-battery shutdown, sleep timer firing on a per-set Energy Settings policy). Flags:

861 * ``-i`` PreventIdleSleep (the only one ``toggle.sh`` uses; protects against the OS going idle while the user is active but a clamshell close still hard-puts it to sleep)

862 * ``-s`` PreventSystemSleep (only honored on AC power; the user may be on battery)

863 * ``-u`` DeclareUserActive (resets the idle timer every 30s by default; equivalent to keeping the cursor moving)

864 * ``-d`` PreventDisplaySleep

865 * ``-m`` PreventDiskIdleSleep

866 * There is NO ``-l`` (lid-block) flag in any modern macOS. Lid close is a kernel-firmware-level event that ignores user-space assertions.

867 * To prove any of this on live hardware: ``pmset -g assertions`` lists every active assertion with type / process / reason / timeout.

868 * ****`pmset`**** is the read-side of the same system:

869 * ``pmset -g log | grep -E 'Sleep|Wake|DarkWake'`` shows the recent timeline.

870 * ``pmset -g history`` is the per-day summary.

871 * ``pmset -g assertions`` is the live assertion list (matches ``-i -s -d -u`` to processes).

872 * ****`launchd`** does NOT have a native "on-wake" hook.** Not in any version of macOS as of Sonoma. The canonical recipe is ``StartCalendarInterval`` with a sub-minute cadence and let ``launchd`` queue missed intervals for replay-on-wake, or use a third-party daemon (Sleepwatcher). For our use case a long-running shell daemon listening to the unified log is more responsive than a polling agent.

873 * ****`com.apple.system.powermanagement.\*`** Darwin notifications** (``willSleep``, ``didWake``, ``didDim``, ``didUndim``) are the C-level signal. From a shell, they are best reached through the unified log with ``log stream --predicate`` see the watcher implementation.

874 * ****`scutil n.watch`**** is the canonical CLI for live-following network-state changes. Pairs with ``n.add State:/Network/Global/IPv4`` to get a single tap that fires on ****any**** IPv4 link/route/DNS/address change across ****all**** interfaces. This is what ``scripts/watcher.sh``'s network-change listener uses.

875 * ****`com.apple.networkChange`** Darwin notification** is the C-level equivalent but has no shell-friendly listener; use ``scutil`` instead.

876 * ****`SCNetworkReachability`**** is the Objective-C API used by SOCKS5/HTTP clients to detect route changes per target. Never a fit for shell scripts.

877

878 **#### The Fix (component-by-component)**

879

880 1. ****`recover.sh --post-wake`**** (new flag, ****non-destructive**** by design). Adding ``--post-wake`` to the existing nuclear recovery script inverts the semantics. The default mode still tears down Tailscale, resets DNS to empty, stops caffeinate, runs ``docker compose down``, power-cycles Wi-Fi. The new mode does:

881 * Skip Tailscale disconnect (we WANT it to keep the mesh connection)

882 * ****Re-hijack**** DNS to the gateway Tailscale IP read from ``gateway_ip`` (instead of resetting to empty) applies to both ``ACTIVE_SERVICE`` and ``ENO_SERVICE`` because the system resolver is scoped to ``en0``

883 * ****Refresh**** the exit-node preference with ``tailscale up --reset --ssh=true --accept-dns=false --exit-node=$TS_IP`` ``--exit-node-allow-lan-access=true`` (the exact flags ``toggle.sh`` START uses). This re-asserts DERP mapping without dropping the mesh

884 * Inspect ``docker inspect --format '{{.State.Health.Status}}`` ``nullexit-warp-1`` and ****only**** ``docker compose up -d --force-recreate warp`` if ``gluetun`` is unhealthy this single targeted recreate nudges the UDP NAT binding back to Cloudflare without disturbing Tailscale or AdGuard

885 * Run the 10.27 ``sharingd``-reset step (existing fix from ``toggle.sh`` START path) so AirDrop doesn't freeze on the new IP lease

886 * Verify the gateway with ``dig +tcp @127.0.0.1 -p 5354 google.com`` (AdGuard via TCP through Colima's SSH tunnel) and a 4-second ``tailscale ping`` to the gateway Tailscale IP, instead of the default mode's direct-to-1.1.1.1 check

887 * Crucially: ****skip `sudo -v`**** because the LaunchAgent has no TTY to prompt on; all privileged calls use ``sudo -n`` and lean on ``/etc/sudoers.d/nullexit`` NOPASSWD entries (``networksetup``, ``dnsmasq``, ``dscacheutil``, ``killall``, ``pkill``).

888 2. ****`toggle.sh`**** writes and clears ``/tmp/nullexit-gateway-active.marker`` (an ISO-8601 UTC timestamp) at the bottom of the START path and the top of the STOP path / ``cleanup_handler``. Two new helpers: ``write_gateway_active_marker`` and ``clear_gateway_active_marker``. The watcher uses this file as its ****only signal**** that the gateway is live: if ``toggle.sh`` was stopped (or never started), the watcher no-ops. This prevents waking-up-only-on-counterfactual events from churning DNS.

889 3. ****`scripts/watcher.sh`**** (long-running daemon, sibling-style source file). Two backgrounded pipelines:

890 * ****Wake listener****: ``log stream --predicate 'composedMessage CONTAINS[c] "didWake" OR composedMessage CONTAINS[c] "Wake from Sleep" OR composedMessage CONTAINS[c] "Waking from" OR composedMessage CONTAINS[c] "DarkWake"' --style compact --info`` piped through ``while read; do run_recover "WAKE: ..."; done``. ``[c]`` makes the predicate case-insensitive (the unified log writes phrases in mixed case across versions).

891 * ****Network listener****: ``(echo "n.add State:/Network/Global/IPv4"; echo "n.watch"; while true; do sleep 86400; done) | scutil 2>/dev/null`` piped through a ``case`` match on ``n.state*`` / ``SCEventUpdate*``.

```

892 * **run_recover** checks the marker, debounces via `/tmp/nullexit-watcher.last-recovery` (default 10s,
      configurable via `NULLEXIT_DEBOUNCE_SECONDS`), and shells out to `bash recover.sh --post-wake`. The 10s
      debounce prevents a single wake event (which typically emits 2-3 log lines) from triggering multiple
      recoveries.
893 * `trap ... TERM INT` `kill 0` cleanly tears down the process group on launchctl bootout` so the `sleep`
      86400 inside the scutil pipe also dies.
894 * `wait` blocks indefinitely while both pipelines feed it.
895 4. **launchd/com.nullexit.wake-recovery.plist** (LaunchAgent in the user domain runs as the console user at
      every login without needing sudo).
896 * `Label: com.nullexit.wake-recovery`
897 * `ProgramArguments: ["/bin/bash", "<absolute-path-to-your-nullexit-install>/scripts/watcher.sh"]`
898 * `RunAtLoad: true` starts immediately on launchctl load` (no need to log out and back in)
899 * `KeepAlive: { SuccessfulExit: false, Crashed: false }` **don't** restart on clean SIGTERM exit (so `launchctl bootout`
      doesn't get stuck in a relaunch loop). Also **don't** restart on hard crash: we observed
      that macOS Sonoma's sleep/wake suspend-resume path accumulates orphan watcher.sh PIDs because SIGCONT
      does not reliably clean up the prior instance's listener grandchildren, and a `KeepAlive.Crashed=true`-
      driven relaunch actively races with the still-live prior process. The script enforces single-instance on
      its own via a `flock`-style PID-file lock, so a relaunch-on-crash would be skipped by the lock and produce
      0 listener coverage until the live instance happened to die. Trade-off: a real crash leaves the user
      without auto-recovery until they run `launchctl load -w` manually acceptable because (a) gateway still
      works without auto-recovery, (b) a relaunch wouldn't fix whatever caused the crash, and (c) the gateway-
      recovery itself is observably broken (no DNS, no exit-node) if it does go down.
900
901 ### 11.17 Host-Side Traffic Leaking Past the Exit Node (with Remote Clients Routing Correctly)
902
903 **Symptom.** `toggle.sh` reports success. Remote tailnet clients (e.g. an S24 phone) correctly egress through
      Cloudflare WARP and see a Cloudflare IP on `whatismyip.com`. But on the **HOST Mac running the gateway**, `whatismyip.com`
      reports the underlying physical ISP's ASN (e.g. `campus_isp` for a university campus Wi-Fi).
      The gateway container itself is healthy; the leak is specifically on the host.
904
905 **Root causes.** Three distinct mechanisms can each produce this exact symptom on the host alone, while leaving
      remote clients unaffected. They are mutually rankable so the diagnostic picks the active one
      deterministically.
906
907 * **(A) SOCKS5 fallback lane active.** `toggle.sh`'s three Phase-B pre-flight checks (`tailscale ping` AdGuard
      via `dig +tcp` WARP internet, `toggle.sh` ~lines 750-820) sometimes flake during the first minute. When ANY
      of the three fails, the script sets `SKIP_EXIT_NODE=true` (`toggle.sh`:849) and instead activates the SOCKS5
      fallback path (`toggle.sh`:894). The SOCKS5 fallback hijacks DNS to `127.0.0.1` (so AdGuard filtering still
      works) but **modern browsers on macOS do NOT honor the system SOCKS5 proxy** they ignore `networksetup -`
`setsocksfirewallproxy`. Result: DNS gets ad-filtered but actual HTTP/S traffic exits direct through the
      campus ISP. This branch is invisible from a remote tailnet client because the client's tunnel is unaffected
      .
908
909 * **(B) IPv6 leak over campus Wi-Fi.** The Tailscale exit node and WARP tunnel are both IPv4-only (gluetun + `post-rules.txt`
      drop all IPv6 forwarded traffic; devref 10.6). But many university and enterprise APs are
      dual-stack the host happily accepts AAAA DNS responses and sends IPv6 traffic straight out `en0`. macOS
      happily encrypts that traffic through Tailscale ONLY if Tailscale has IPv6 advertised; with an IPv4-only
      exit node, IPv6 traffic skips Tailscale entirely. This branch is invisible from a phone because iOS/Android
      Tailscale apps actively sinkhole IPv6 in their VPN profile, but the macOS host's `tailscaled` does not.
910
911 * **(C) Route-table freeze and `--accept-routes` reset after `tailscaled` wake/toggle.** Running `tailscale up`
`--reset` clears the exit-node preference and reverts `--accept-routes` back to its default (`false`).
      Without explicitly passing `--accept-routes=true`, `tailscaled` will not attempt to install the default
      route through the `utun` tunnel interface even if the exit node preference is reconciled. Additionally,
      macOS occasionally freezes and fails to apply route-table changes (devref 10.26). In both cases, `netstat -rn`
      shows the default route still pointing to the physical Wi-Fi gateway (e.g. `172.17.0.1`) instead of the
      Tailscale `utun` interface, causing host traffic to exit direct while remote clients route correctly.
912
913 **Why remote clients don't see this.** Each remote phone/laptop uses its OWN Tailscale tunnel to reach the
      container's `100.x.x.x:443` over the mesh they're end-to-end encrypted regardless of how the host Mac
      itself is configured. Only the HOST's own egress sockets (HTTP requests from the host's browser, curl, etc
      .) are affected.
914
915 **Resolution.** A single script: `scripts/diagnose-host-leak.sh`. It runs the 7 checks, classifies into one of
      A / B / C / OK, prints a verdict + a ready-to-run fix command on stdout, AND writes the full report to `host-leak-diagnostic-<UTC>.txt`
      so the user can paste the file back instead of scrolling-and-copying from
      the terminal. With `--fix`, the matched remediation is applied and the two checks that could change (warp +
      ipv6) are re-verified. With `--watch` (or `--watch 30`), it runs the full baseline diagnostic then enters
      a continuous loop checking warp/IPv6/default-route every N seconds, alerting only on state changes logs to
      `host-leak-watch.log`.
916
917 `bash`
918 # Diagnose only:
919 `bash scripts/diagnose-host-leak.sh`
920
921 # Diagnose + apply matched fix + re-verify:
922 `bash scripts/diagnose-host-leak.sh --fix`
923 `bash`
924
925 **What it prints.** A 7-section report (`tailscale status`, SOCKS5 state, `cdn-cgi/trace` warp=, IPv6 leak
      probe, host default route, last `toggle.sh` output.log hints, resolved gateway Tailscale IP), followed by a

```



```

classification block (`Scenario: A | B | C | OK`) with the exact command(s) the user needs to paste into
their terminal they don't have to figure out which `tailscale up` flags match their environment.
926
927 **What it does NOT do.** It does not touch `toggle.sh` or `toggle.sh`'s pre-flight logic, does not modify `.env`
  (except `--fix` mode appends `GATEWAY_BYPASS_PING=true` when fixing Scenario A), and does not restart
  Docker. It is a read-only diagnostic with an opt-in remediation flag. Safe to run on a healthy gateway the
  worst case is a 30-line report that says "OK".
928
929 **Why cdn-cgi/trace over whatismyip.com for the verdict.** `whatismyip.com`'s ISP database routinely mislabels
  campus IPv6 ranges as residential ISPs (that's why the local ISP's ASN appears even when you're routing
  through Cloudflare). `cdn-cgi/trace` is hosted by Cloudflare ITSELF and reports `warp=on|off` plus the
  exact edge colo serving the request. It is the only public endpoint that can truthfully confirm whether the
  WARP tunnel is in use.
930
931 **Common false alarms.**
932
933 * **`warp=on` but whatismyip.com still shows the local ISP.** `whatismyip.com`'s ISP-detection database is
  unreliable on academic networks. Trust `cdn-cgi/trace`'s `warp=on` over `whatismyip.com`'s ISP string. Use
  `https://api.ipify.org` (returns just the IP, no ISP DB lookup) as a third confirmatory check.
934 * **`tailscale status` shows the host online but exit node preference is missing.** `tailscale up --reset --
  exit-node=` (empty value) clears any stale preference; `tailscale up --reset --exit-node=<100.x.x.x>` re-
  establishes it. The diagnostic prints the exact command with the resolved TS_IP filled in.
935 * **Both IPv6 leak AND route-freeze flags set simultaneously.** Scenario B outranks Scenario C fixing IPv6
  makes IPv4 the only viable egress, after which the route-freeze usually self-resolves within ~30s (or one
  more `toggle.sh` cycle).
936
937 #### 10.30.1 Host Leak Probe Continuous Sub-Second Egress Monitor
938
939 **What it is.** `scripts/host-leak-probe.sh` is a lightweight background daemon (~1.4 MB RSS, ~01% CPU) that
  polls `https://www.cloudflare.com/cdn-cgi/trace` every 300ms directly from the host via `curl` not via `
  docker compose exec`. This is a fundamentally different vantage point from the WARP Watcher (10.32): the
  WARP Watcher asks the WARP *container* whether it sees `warp=on`; the Host Leak Probe asks what a real
  browser request leaving the host's physical NIC looks like. The two blind spots it covers that the WARP
  Watcher cannot:
940
941 1. **Host-side flash-leaks** a Cloudflare anycast edge re-route, a browser reusing a stale pooled connection
  across a tunnel state change, or a split-second routing gap during a Gluetun healthcheck restart (see 10.18
  acknowledged 14s window).
942 2. **Host egress while container is healthy** the container can report `warp=on` while the host's own traffic
  exits direct (the three-scenario leak class documented in 10.30).
943
944 **Lifecycle.** Started by `toggle.sh`'s START path (`start_host_leak_probe`) immediately after `
  start_warp_watcher`. Controlled by:
945
946 | Signal path | What happens |
947 |---|---|
948 | `toggle.sh` STOP | `stop_host_leak_probe()` SIGTERM + PID file cleanup |
949 | `toggle.sh` `cleanup_handler` (error/INT/TERM/HUP) | Same |
950 | `recover.sh` (default, non `--post-wake`) | 6d block kill by PID file |
951 | `recover.sh --post-wake` | Left running the gateway is still live |
952
953 PID file: `/tmp/nullexit-host-leak-probe.pid`. Toggle with `HOST_LEAK_PROBE=false` in `.env` to disable at next
  gateway start.
954
955 **Output format.** All events are written to `output.log` (repo root) with UTC timestamps. Only state *changes*
  are logged the probe is silent during normal healthy operation.
956
957 ```
958 [09:10:26] LEAK warp=off ip=45.23.1.1 prev_warp=on prev_ip=104.28.246.50
959 [09:10:26] ROTATE warp=on ip=104.28.248.11 prev_ip=104.28.246.50
960 [09:10:26] HOST-PROBE failed/timeout (count=1)
961 ```
962
963 curl transport errors (`TLS handshake failed`, `Connection refused`, etc.) are also appended to `output.log`
  via `2>>output.log` nothing is silenced to `/dev/null`.
964
965 **Grep cheatsheet.**
966
967 ```bash
968 grep LEAK output.log          # real warp=off events from the host
969 grep ROTATE output.log        # Cloudflare anycast edge rotations (harmless)
970 grep 'HOST-PROBE' output.log   # curl failures / timeouts
971 ```
972
973 **Resource budget.** ~1.4 MB RSS, ~0% CPU at rest, ~3 HTTPS requests/second to Cloudflare. Safe to leave
  running for hours. Back off the polling interval (`bash scripts/host-leak-probe.sh 1.0`) if you observe
  probe-failure pile-ups indicating Cloudflare rate-limiting.
974
975 **Detection vs prevention.** This script detects leaks; it does not prevent them. A sub-300ms flash-leak can
  still slip between two polls undetected. If `grep LEAK output.log` shows confirmed leak events, the next
  step is a kill-switch (prevention) an `iptables` rule on the host's `en0` that drops non-Tailscale, non-

```

```

    local egress whenever the gateway is active. That is a separate engineering task not yet implemented.
976 * `ProcessType: Background` hint to App Nap not to suspend us.
977 * `StandardOutPath/StandardErrorPath: /tmp/nullexit-watcher.{out,err}.log` separates launchd-level
    diagnostics from the script's own `/tmp/nullexit-watcher.log` (which `exec >>` redirects).
978
979 ##### Install (one-time)
980
981 The plist lives in the repo at **`launchd/com.nullexit.wake-recovery.plist`** for version control. The
    installed (live) copy must land in `~/Library/LaunchAgents/` so launchd picks it up on the user session.
982
983 ```bash
984 # Copy the plist to the user-launchd location (run from repo root)
985 cp ./launchd/com.nullexit.wake-recovery.plist \
986 ~/Library/LaunchAgents/
987
988 # Substitute __NULLEXIT_HOME__ with your local install absolute path.
989 # The plist uses WorkingDirectory + a relative program arg, so this
990 # single substitution is the only place it references your machine.
991 sed -i 's|__NULLEXIT_HOME__|$(pwd)|' \
992 ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
993
994 # Validate the XML
995 plutil -lint ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
996
997 # Load + persist this user session + every future login
998 launchctl load -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
999
1000 # Confirm it actually started
1001 launchctl list | grep nullexit
1002 ```
1003
1004 To **stop** the watcher (without uninstalling):
1005
1006 ```bash
1007 launchctl bootout gui/$UID/com.nullexit.wake-recovery
1008 # or:
1009 launchctl unload -w ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1010 ```
1011
1012 To **uninstall** completely:
1013
1014 ```bash
1015 launchctl bootout gui/$UID/com.nullexit.wake-recovery 2>/dev/null || true
1016 rm ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
1017 ```
1018
1019 ##### Why this fix was preferred over alternatives
1020
1021 * **Polling with `StartCalendarInterval`** would have given ~30-60s detection latency per wake. The unified-log
    stream is sub-second.
1022 * **Sleepwatcher** would have covered wake events but not network changes. We'd still need a separate `scutil`
    listener, and introducing a third-party dependency for what amounts to ~150 lines of shell is unjustified.
1023 * **Forcing `caffeinate -l`** doesn't exist and the closest equivalent (`caffeinate -s`) only works on AC power
    . The whole point of the gateway is to stay usable on battery while traveling lid close must NOT silently
    kill the gateway.
1024 * **A kernel extension** is impossible (no entitlement) and out of scope.
1025
1026 ##### Reproduction recipe
1027
1028 Test both events in isolation to confirm the fix actually closes the gap.
1029
1030 * ***(A) Lid-only repro***: with the gateway up, run in a terminal: `pmset displaysleepnow && sleep 30 && pmset
    wake` (without physically closing the lid). Watch the wake wake-fires the watcher post-wake routine runs
    login should still work in <5s after wake. Compare to baseline pre-fix: pre-fix the gateway would be dead
    for 30-90s.
1031 * ***(B) Roam-only repro***: with the gateway up: `networksetup -setairportpower off && sleep 30 && networksetup
    -setairportpower on` (or simply walk out of Wi-Fi range and back). The captive-portal WILL repave DHCP DNS
    for a few seconds; the watcher fires on the second `State:/Network/Global/IPv4` change (after the Wi-Fi re-
    associates) and the post-wake routine re-hijacks DNS within 2-3s.
1032
1033 ##### Operative files
1034
1035 * `recover.sh` added arg parsing; wrapped destructive sections behind `POST_WAKE`; added re-hijack DNS /
    refresh exit-node / force-recreate-if-unhealthy path.
1036 * `toggle.sh` added `write_gateway_active_marker` / `clear_gateway_active_marker` called at the END of START
    and TOP of STOP / cleanup_handler.
1037 * `scripts/watcher.sh` new long-running daemon.
1038 * `launchd/com.nullexit.wake-recovery.plist` launchd LaunchAgent descriptor.
1039 * `~/Library/LaunchAgents/com.nullexit.wake-recovery.plist` the installed copy (one-time copy).
1040
1041 ##### Honest assessment (read before testing)

```


1042 This fix has two asymmetric halves:

1043

1044

1045 * **Network / roam / captive-portal path RELIABLE.** `scutil n.watch` on `State:/Network/Global/IPv{4,6}` catches every Wi-Fi roam, captive-portal DHCP-rebind, hotspot handoff, and VPN change with zero missed events. To validate, drop Wi-Fi for 30 s and re-add it; `/tmp/nullexit-watcher.log` will record a `NET:` trigger and `recover.sh --post-wake` will run within ~10 s of the network coming back.

1046 * **Lid close / wake path BEST-EFFORT** on macOS Sonoma+. Apple *suspends* LaunchAgents process-state during forced sleep and *resumes* them on wake; it does NOT re-launch them every wake. As a result: (a) `log stream` is a live subscription events emitted during the agent's suppressed window are DROPPED, not buffered, so the wake event that prompted the resume is invisible to `log stream` on resume; (b) the "fire on startup" guard runs once on `launchctl load -w` and after a `KeepAlive` crash-recovery, NOT on every successive wake; (c) `subsystem == "com.apple.powermanagement"` will catch FUTURE emissions (display dim/restore, manual sleep/wake) but not the lid-closeopen wake directly.

1047

1048 In practice the lid-wake gap is short, because `toggle.sh`'s existing DNS Watcher already polls every 5 s and re-hijacks DNS (so DNS recovers within 5 s of any roam or wake), `tailscaled` self-heals via DERP fallback within 3060 s, and `gluetun` reconnects via its healthcheck retry within 30 s. The user-visible pain on lid open is ~30 s of wonky DNS, not a permanent outage.

1049

1050 **Test the ROAM path FIRST.** If ROAM works in your testing, the lid-wake gap is acceptable. If lid-wake handling is critical, the two clean follow-ups are:

1051 1. **Sleepwatcher** (one canonical install): `brew install sleepwatcher`; drop `recover.sh --post-wake` into `~/sleep` and `~/wake` so Sleepwatcher invokes it directly on every wake without the launchd propagation problem.

1052 2. **Move the watcher to a LaunchDaemon** (root) and use `IOPMSchedulePowerEvent` via a tiny C wrapper not portable to shell.

1053

1054 A truly reliable shell-only solution to "wake events from inside a LaunchAgent" does not exist on macOS Sonoma+.

1055

1056 **Empirical lessons from deployment**

1057

1058 Two findings came up while deploying this fix end-to-end; recording so the next operator doesn't lose an evening to each.

1059

1060 1. **Bash function-call-before-definition gotcha.** While introducing the gateway-active marker helpers, the defensive `clear\_gateway\_active\_marker` call was inserted at the very top of `toggle.sh` (right after `set -e`), but the function definition lived near `PID\_FILE="/tmp/nullexit-cafeinate.pid"` ~90 lines down. Bash parses top-to-bottom and resolves names at first invocation, so the call site exited with code 127 (`clear\_gateway\_active\_marker: command not found`). The gateway stayed unreachable from the START path even though `bash -n` passed (the parser doesn't catch order-of-resolution errors). Rule: define functions BEFORE any block that calls them. Verify with `grep -n 'name()'s\*{'` followed by `grep -n 'name\s\*\$'` the latter must appear on a line number AFTER the former.

1061

1062 2. **networksetup -setairportpower off+on** is not a faithful roam reproduction. Synthetic Wi-Fi radio-cycling (`sudo networksetup -setairportpower Wi-Fi off` for 30 s, then back on) is expected to produce a `NET:` trigger in `scutil n.watch` but produced ZERO during testing because `setairportpower` powers the radio at the driver level while leaving the network SERVICE in the "up" state from scutil's perspective, so no `n.state State:/Network/Global/IPv4` event is generated. A real roam (macOS briefly loses the AP and re-associates into a different one) DOES fire the event because the link actually changes.

1063 Better reproduction recipes in priority order:

1064 * Walk between two real SSIDs via `networksetup -switchtolocation` (or actual physical station movement) guaranteed to fire the scutil event.

1065 * Force a DHCP rebind on the same SSID with `sudo ipconfig set en0 DHCP` triggers `State:/Network/Global/IPv4` to update.

1066 * Trust the existing 30-second DNS Watcher inside `toggle.sh` for the DNS path: it re-hijacks DNS automatically. The post-wake watcher adds clear value mostly for tailscale exit-node re-assertion and warp gluetun force-recreate, both of which self-heal within ~30 s anyway.

1067

1068 **11.18 Infinite Egress Routing Loops & Subnet Takeover Paradoxes**

1069

1070 **Problem:** Two distinct network loops can break host egress connectivity entirely when the exit node is active:

1071

1072 1. **The Recursive Host-VM Tunnel Loop:**

1073 - **Mechanism:** The host Mac's default route points to the Tailscale exit node (`utun\*`). The exit node container sends its encrypted tunnel packets (destined for the WARP endpoint `162.159.192.1`) back to the host via the VM NAT. Without a static route exception on the host Mac, the host Mac routes those packets right back into the exit node (`utun\*`), creating an infinite loop that blackouts all host network egress.

1074 - **ARP-Scoping Pitfall:** Simply binding the bypass route to the interface (`route add -host 162.159.192.1 -interface en0`) fails when the default route is deleted or redirected to `utun\*`. Since `162.159.192.1` is not local to the physical link, macOS fails to resolve it via ARP, dropping all tunnel packets.

1075 - **Fix:** Dynamically resolve the active physical gateway IP (e.g. `172.17.0.1`) on startup, and add the static host routes for the WARP endpoints via the gateway IP directly (`route add -host 162.159.192.1 <gateway\_ip>`). Additionally, because standalone `tailscaled` on macOS does not automatically modify the system default route via the CLI, we manually redirect the default gateway to `utun\*` using `setup\_exit\_node\_routing`.

1076

1077 2. **The Docker Compose Subnet Takeover Collision:**

```

1078 - **Mechanism:** To avoid collisions with the host's campus network, Colima's default bridge (`docker.bip`)
is moved (e.g. to `10.200.0.1/24`). However, because `172.17.0.0/16` is now free inside the VM, Docker
Compose's IPAM dynamically takes it over for the project's custom network (`nullexit_default`). Because the
containers receive IPs on `172.17.0.0/16`, the host Mac cannot send local peer discovery packets (e.g.,
Tailscale `magicsock` packets on `172.17.0.2:41641`) directly. The host routing table sends them out `en0`
to the physical Wi-Fi, returning `host is down` or `no route to host`.
1079 - **Fix:** Explicitly configure a custom default network subnet `10.200.1.0/24` in `docker-compose.yml` to
prevent Docker Compose from ever reclaiming the conflicting `172.17/172.18` ranges.
1080
1081 > See also 10.23 (The Hotspot Paradox) for the related infinite loop that occurs when the physical upstream
router is also a Tailscale exit-node client.
1082
1083 ### 11.19 In-Flight WARP Tunnel Liveness Monitor & Statistical Auto-Shutdown
1084
1085 **Why.** nullexit's core promise is that your IP always appears as Cloudflare. If the WARP tunnel silently dies
due to a UDP NAT timeout, a Cloudflare edge rotation, a Colima VM clock skew causing TLS cert rejection,
or any of a dozen other failure modes the host silently falls back to egressing through the physical ISP.
The user sees the gateway as "up" (containers running, Tailscale connected, DNS hijacked) but their real IP
is exposed. This is the exact class of failure that Ingo Blechschmidt documented in his chilling 2024
Linux kernel post-mortem: for over two years, LUKS disk encryption on suspend was silently broken because a
refactored kernel syscall returned success while doing nothing ([source](https://mathstodon.xyz/@iblech
/116769502749142438)). The lesson: **a security mechanism that fails silently is worse than no mechanism at
all** it creates a false sense of safety that prevents the user from taking corrective action.
1086
1087 **Design principle.** The WARP Watcher (`start_warp_watcher()` in `toggle.sh`) is a background daemon that
polls `cdn-cgi/trace` every 30 seconds and applies a **statistically calibrated threshold** before
triggering any safety response. This threshold distinguishes transient network blips (which self-heal
within seconds) from genuine outages (which require intervention). A single `warp=off` reading is
meaningless Docker might be slow, the network might be congested, a container healthcheck might be
restarting. But 6 consecutive off readings (30 continuous seconds of downtime) has vanishingly low
probability of being a false positive: even at an unrealistically high 10% false-positive rate per poll, P
(6 consecutive) = 0.1 0.0001%. In practice, the per-poll false-positive rate is far lower than 10%, making
the real probability astronomically small.
1088
1089 **What it does.**
1090
1091 1. **Always logs.** Every state transition (onoff, offon) is timestamped to `output.log`. The user can `grep
WARP output.log` to audit the tunnel's entire lifetime. This is the "never fail silently" mandate.
1092
1093 2. **Tracks consecutive failures.** A counter increments on each `off` reading and resets to 0 on any `on`
reading. This ensures the watcher never fires on intermittent flapping.
1094
1095 3. **Auto-shuts down at threshold.** When the counter reaches `WARP_FAIL_THRESHOLD` (default 6, configurable in
`.env`), the watcher declares a statistically significant outage and runs `recover.sh` the nuclear
recovery script that disconnects Tailscale, stops Docker containers, resets DNS to `1.1.1.1`, flushes
routes, and power-cycles Wi-Fi. This instantly reverts the host to normal (un-encrypted) internet,
guaranteeing no traffic leaks through a dead tunnel while the user thinks they're protected. A trigger file
at `/tmp/nullexit-warp-shutdown-triggered` prevents double-firing.
1096
1097 4. **Exits cleanly.** After shutdown, the watcher exits (the gateway is dead, there's nothing left to monitor).
`stop_warp_watcher()` is also called by `toggle.sh`'s STOP path and `cleanup_handler` to terminate the
daemon cleanly during normal teardown.
1098
1099 **Configuration.** Set `WARP_FAIL_THRESHOLD=3` in `.env` for a more aggressive 15s timeout, or `
WARP_FAIL_THRESHOLD=12` for a conservative 60s window. The variable is parsed from `.env` using the same `
grep` pattern as `GATEWAY_MSS` and `GATEWAY_BYPASS_PING`.
1100
1101 **Log sample.** After a real outage (or a forced test via `docker compose pause warp`):
1102
1103 ```
1104 [2026-07-03T21:15:17Z] WARP DOWN cdn-cgi/trace reports warp=off
1105 [2026-07-03T21:15:47Z] WARP SHUTDOWN 6 consecutive failures (threshold=6). Running recover.sh to kill the
gateway and restore normal internet. Your IP is NO LONGER Cloudflare.
1106 [2026-07-03T21:15:52Z] WARP SHUTDOWN recover.sh completed, watcher exiting. Your IP is now your real ISP IP.
1107 ```
1108
1109 Or, if WARP self-recovers before the threshold:
1110
1111 ```
1112 [2026-07-03T21:15:17Z] WARP DOWN cdn-cgi/trace reports warp=off
1113 [2026-07-03T21:15:32Z] WARP RECOVERED warp=on after 3 consecutive off readings (threshold was 6)
1114 ```
1115
1116 **Relationship to other monitors.** The WARP Watcher is the **innermost safety layer** it runs as a child of `
toggle.sh` and only while the gateway is active. It complements (does not replace) the `scripts/watcher.sh`
daemon (10.29, which handles sleep/wake and Wi-Fi roam) and `scripts/diagnose-host-leak.sh` (10.30, which
is a manual diagnostic tool). The WARP Watcher is the only component that actively verifies end-to-end
tunnel liveness and takes autonomous action when it fails.
1117
1118 ---
1119

```

```

1120 ## 12. Recent Updates
1121
1122 Reflects the current branch's state. Each entry one line + commit hash; read the commit message for detail.
1123
1124 - **July 10, 2026 `fix(race): suppress WARP Watcher during post-wake force-recreate`** Fixed a race condition
    where switching Wi-Fi caused the host IP to leak as the raw ISP IP (`132.x`) on every network roam. `
    recover.sh --post-wake` force-recreating the warp container (~35s) caused the background WARP Watcher to
    accumulate 6 consecutive `warp=off` readings and fire nuclear `recover.sh`, killing a gateway that was
    healing itself. Fixed by adding a `/tmp/nullexit-warp-inhibit.marker`: `recover.sh` writes it at startup in
    `--post-wake` mode (with `trap ... EXIT` for guaranteed cleanup); the WARP Watcher loop in `toggle.sh`
    checks it first and resets `consec_off=0` while it exists. See 25 for full post-mortem.
1125
1126 - **July 6, 2026 `fix: resolve edge-case vulnerabilities and system state leaks`** Addressed a suite of
    subtle but critical edge cases identified in security review:
1127 1. **DNS Watcher Interface Independence** Fixed `start_dns_watcher` in `toggle.sh` to dynamically detect the
    active interface via `get_active_service()` rather than hardcoding a `grep` for "Wi-Fi", ensuring roaming
    on Ethernet/Thunderbolt stays protected.
1128 2. **Robust Lock Verification** Upgraded the `toggle.sh` lock-file logic to use `ps -p` verification,
    preventing permanent lockouts if a crashed script's PID is recycled.
1129 3. **Fail-Closed Crypto** Changed the `crypto.sh` integrity check in `toggle.sh`/`recover.sh` from an
    executable bit check (`[ -x ]`) to a pure file-existence check (`[ -f ]`), guaranteeing execution via `bash
    scripts/crypto.sh` even if git strips the `+x` bit.
1130 4. **Strict `iptables` Pinning** Upgraded `post-rules.txt` FORWARD-chain rules to explicitly pin traffic
    between `tailscale0` and `tun0` in both directions, mathematically preventing Tailscale traffic from
    escaping through `eth0` during WARP tunnel drops.
1131 5. **Docker Local Network Isolation** Modified `docker-compose.yml` to explicitly bind the exposed WARP
    container ports (`1080`, `5354`) to `127.0.0.1`. This prevents other devices on the local bridged Wi-Fi/LAN
    from accessing the SOCKS5 proxy or DNS resolver if Colima provisions a LAN IP.
1132 6. **Routing Verification & Sudoers** Added a `netstat -nr` check (handling macOS `0/1` syntax) to verify
    the default route (`0.0.0.0/1`) actually overrides to Tailscale, and updated the `README.md` and `setup.sh`
    sudoers templates to ensure `/usr/bin/python3` is permitted so the fallback DNS proxy doesn't silently
    fail.
1133
1134 - **July 4, 2026 `cc789c5` (fix: resolve concurrency, setup variables, CI workflow, and untrack review file)**
    Added concurrency protection on `toggle.sh` using `/tmp/nullexit-toggle.lock` to prevent parallel runs.
    Initialized `TS_AUTHKEY` defensively in `setup-common.sh` to prevent unbound variable crashes under `set -u`
    . Added Go compilation (`go vet` and `go build`) steps to GitHub Actions CI workflow. Ported AdGuard Home
    container to cap logs. Fixed macOS BSD `grep` compatibility in `setup-common.sh` using `grep -oE`, resolved
    space stripping in `common.sh`, fixed quoting/PID file/routing text in `recover-linux.sh`, and removed
    redundant bypass routes block in `recover.sh`.
1135
1136 - **July 2, 2026 `8c5fae1` + `181e1e1` (feat(routing): resolve recursive host-VM tunnel loop & Docker subnet
    collision)** Two infinite routing loops fixed. (1) Host-VM tunnel loop: WARP endpoint bypass routes now
    injected via physical gateway IP (not interface scoping) to avoid ARP failures under default-route
    redirection; `setup_exit_node_routing` added to manually force the host default route to `utun*` since
    standalone `tailscaled` CLI on macOS does not update the routing table automatically. (2) Docker Compose
    subnet takeover: `docker-compose.yml` now locks the default project network to `10.200.1.0/24` to prevent
    IPAM reclaiming `172.17/172.18` after `docker.bip` is moved. `--accept-routes=true` added explicitly to all
    `tailscale up --reset` calls. Documented in 10.31.
1137
1138 - **July 4, 2026 `fix: post-wake routing loop & destructive recovery` (networking: routing)** Resolved a
    critical fatal bug where waking the Mac from sleep or roaming Wi-Fi permanently broke the internet and
    required a full system reboot to fix.
1139 1. **The Routing Loop Bug** `common.sh`'s `add_warp_bypass_routes` was previously refactored to dynamically
    look up the default route without accepting explicit interface arguments. During `post_wake`, Tailscale is
    already active, so `route get default` resolved to the Tailscale `utun` interface. This caused Cloudflare
    WARP traffic to be routed back *into* Tailscale, creating an infinite recursive routing loop that instantly
    killed the internet on every wake. Fixed by making `add_warp_bypass_routes` accept explicit target
    arguments and updating `recover.sh` to pass the correct physical interface/gateway.
1140 2. **The Destructive Recovery Bug** When users ran `recover.sh` to fix the broken internet, the script
    blindly ran `sudo route -n flush`. On macOS, this doesn't just clear VPN routes it destroys essential system
    routes including loopback (`127.0.0.1`) and multicast (`224.0.0.0`). Bouncing Wi-Fi does not automatically
    restore these internal routes, meaning macOS network services (`configd`, `mDNSResponder`) remained
    permanently wedged until a reboot. Fixed by replacing `route -n flush` with targeted deletions of specific
    gateway routes (`0.0.0.0/1`, `128.0.0.0/1`) and WARP bypass routes.
1141
1142 - **July 2, 2026 `feat: secure execution` (security: script lockdown & auth)** Fixed a major privilege
    escalation vulnerability and added native macOS authentication prompts to `Toggle Gateway.app` and `Recover
    Gateway.app`. Previous setups granted `/usr/bin/python3` blanket `NOPASSWD` sudo access, allowing malware
    to bypass macOS security. We restricted `NOPASSWD` exclusively to `/usr/bin/python3 /Users/<USER>/Developer
    /nullexit/scripts/dns-proxy.py`. Additionally, `scripts/dns-proxy.py` was locked down with `chown root:
    wheel` and `chmod 755` so attackers cannot overwrite the script and exploit the restricted sudo rule. `
    toggle.sh` was also updated to capture `warp` container logs on failure before tearing down the environment
    .
1143 - **July 3, 2026 `fix: resolve numerous system regressions`** Addressed several breakages introduced in
    recent commits:
1144 1. Fixed a truncated line syntax error in `/etc/sudoers.d/nullexit` that broke `NOPASSWD` fallback commands.
1145 2. Fixed a `[Errno 13] Permission denied` error in `sync-rules.py` cache writes by using a temporary file and
    atomic `os.replace()` to bypass inherited mode locks (0444/000). **(Later reverted see below)**
1146 3. Fixed a quoting bug in `recover.sh` where `ts_args=""` passed literal escaped quotes.

```

```

1147 4. Repaired ANSI color outputs in `diagnose-host-leak.sh` and `fix-docker-bridge-collision.sh` using `%b`
    instead of `%s` and `-e` flags.
1148 5. Fixed `toggle-linux.sh` calling a macOS-only `networksetup` command instead of `resolvectl dns`.
1149 6. Corrected the AdGuard REST API refresh POST in `sync-rules.py` to correctly target port `3000`.
1150 7. Restored the missing `START_GATEWAY=true` variable in `toggle.sh` and purged a stale port `80` mapping
    from `docker-compose.yml`.
1151 8. Verified `output.log` is untracked and safely in `.gitignore`.
1152 - **July 3, 2026 `unlock-files.sh` + revert atomic writes in `sync-rules.py`** The `.tmp` + `os.replace()`
    atomic-write pattern introduced in the commit above was a workaround for stale restrictive file permissions
    (mode `0444`/`000`) left on disk by old `chmod` calls that had already been removed from the codebase.
    Rather than keep the workaround, we created `scripts/unlock-files.sh` a one-shot script that permanently
    fixes the stale permissions by replacing each locked inode with a fresh writable one via atomic rename (`cp`
    + `mv`), operating on the directory entry instead of the file contents. No `chmod` used. All 5 `.tmp` + `os.replace()` sites in `sync-rules.py` were reverted back to direct writes. `scripts/unlock-files.sh`
    covers: `.env` (mode `000`), `adguard/work/userfilters/compiled_rules.txt`, `ip_blocklist.ipset`, `cache/*.txt`, `cache/ip/*.txt`, and `adguard/work/data/filters/*.txt` (mode `0444`). Run once after setup or
    pulling an old repo; `toggle.sh` and `sync-rules.py` then work without permission errors. Documented in
    10.28 + 3.
1153 - **July 1, 2026 `c5b92c0` (refactor: personal-leak cleanup)** eliminated every residual personal-username
    leak across source + config files. The launchd `com.nulllexit.wake-recovery.plist` now uses `WorkingDirectory` + a relative program arg with the `__NULLEXIT_HOME__` sentinel; the install recipe in
    10.29 gains a single `sed -i 's|__NULLEXIT_HOME__|$(pwd)|'` step right after the `cp`. 8 `devref.md`
    prose sites generalized to `~/colima/...`, `~/Library/LaunchAgents/...`, `$USER` for sample output, `

```

```

1187 1. **Physical Device Integrity:** You trust your physical devices are not compromised.
1188 2. **Cloudflare Integrity:** You trust Cloudflare not to correlate your WARP tunnel with exit traffic.
1189 3. **Tailscale Integrity:** You trust Tailscale's coordination server not to MITM WireGuard keys.
1190
1191 The goal: no single provider has a complete picture. Your ISP sees an encrypted tunnel. WARP sees traffic with
      no account identity. Tailscale sees node topology but not content.
1192
1193 ### OPSEC: Tor-over-WARP Architecture
1194
1195 When extreme anonymity is required, integrating the Tor network into `nullexit` provides a powerful Tor-over-
      VPN topology:
1196 * **The University/ISP** sees only an encrypted Cloudflare WireGuard tunnel, rendering them completely blind to
      the fact that you are using Tor (effectively bypassing Enterprise firewall Tor blocks).
1197 * **Cloudflare** sees an encrypted TCP stream heading to a Tor Guard Node. They cannot see your DNS lookups,
      the destination website, or the content of your traffic.
1198 * **The Tor Exit Node** sees your traffic, but not your real IP (only the Cloudflare IP).
1199
1200 #### The "Transparent Proxy" Trap (What NOT to do)
1201 It is tempting to build a system-wide "Transparent Tor Proxy" that forces all host traffic (e.g., standard
      Chrome/Safari browsers, background iCloud syncs) through Tor automatically. **This is a massive OPSEC trap
      and destroys anonymity.**
1202 Modern operating systems and browsers will instantly leak your identity through background syncing (e.g., Apple
      Mail, Google Drive) and browser fingerprinting (canvas fingerprints, WebRTC, screen resolution). The Tor
      network protects your *IP*, but if your payload contains identifying cookies or unique fingerprints, the
      Tor Exit Node (which could be run by malicious actors) will instantly tie your session to your real
      identity.
1203
1204 #### The nullexit Implementation: Isolated Procedural Proxy
1205 To safely integrate Tor without triggering the transparent proxy trap, `nullexit` implements an **Isolated Tor
      SOCKS5 Proxy** with **Procedurally Generated Ports**:
1206 1. **Isolated Container:** A lightweight `tor` Docker container runs securely inside the `warp` network
      namespace. It does not hijack system traffic.
1207 2. **Opt-In Usage:** You must consciously configure a highly-hardened browser (like the official Tor Browser or
      a dedicated Firefox profile) to use the proxy. This prevents accidental background sync leaks.
1208 3. **Procedurally Generated Ports:** Rather than hardcoding the standard proxy ports (like `127.0.0.1:9050` or
      `1080`), `toggle.sh` randomizes all internal proxy ports on every boot into the ephemeral range
      (10000-65000) and silently writes them to `/tmp/nullexit-ports.env`. This completely defeats static
      fingerprinting by malware.
1209 4. **Kernel-Level Collision Avoidance:** To prevent the randomizer from picking a port already in use by a root
      daemon or Apple service (which would crash the gateway), the script directly queries the macOS kernel
      socket table (`netstat -an -p tcp`) to verify the port is absolutely free before assigning it.
1210 5. **The Honey-Port Tripwire:** While procedural ports defeat static fingerprinting, advanced malware might
      attempt a full sequential port scan of `localhost` to find the hidden proxy. To counter this, `toggle.sh`
      spawns a "Honey-Port" (a fake random port with a netcat listener attached). If any local application
      attempts to connect to the Honey-Port, it instantly logs a critical security alert to `output.log` and
      fires a macOS desktop notification, serving as an early-warning Intrusion Detection System (IDS) for local
      malware.
1211
1212 #### Hardening: Tor Bridges and obfs4 Pluggable Transports
1213 For users operating under severe Deep Packet Inspection (DPI), `nullexit` supports configuring Tor to connect
      exclusively through private bridges using the `obfs4` pluggable transport.
1214
1215 When `TOR_USE_BRIDGES=true` is set:
1216 - **What it does:** It hides the entry IP from being matched against the public Tor consensus list, and
      disguises the traffic's protocol fingerprint from the WARP exit provider (Cloudflare).
1217 - **What it does NOT do:** It does *not* add any extra layers of anonymity beyond what Tor already provides. It
      is purely a censorship-circumvention and obfuscation mechanism.
1218 - **Limitations:** `obfs4` is highly effective against passive DPI, but it is not guaranteed to defeat active-
      probing-resistant detection by a highly sophisticated state-level adversary.
1219
1220 > **Architecture Rule:** The `nullexit` gateway deliberately does NOT bundle, hardcode, or automatically fetch
      bridge lines. Bridge lines are highly sensitive and expire. Users must obtain their own bridge lines from
      https://bridges.torproject.org and manually populate the `tor-bridges.txt` file. If bridges are enabled
      but the file is missing or empty, the Tor container will intentionally crash and fail to start rather than
      silently falling back to public guard relays.
1221
1222 **The Bootstrapping Paradox (Out-of-Band Key Exchange)**
1223 Automating the bridge acquisition process (e.g., scripting the gateway to log into the Telegram `@GetBridgesBot`
      via MTPROTO API) introduces catastrophic OPSEC flaws:
1224 1. **Identity Leaks:** Baking a Telegram session into the container explicitly links the "anonymous" gateway to
      a real-world physical phone number.
1225 2. **The Chicken & Egg Deadlock:** In heavily censored regimes, Telegram itself is usually blocked. If the
      container requires Telegram to fetch bridges to bypass the firewall, it will fail because it cannot bypass
      the firewall to reach Telegram.
1226 3. **Bridge Exhaustion & CAPTCHAs:** Automated scraping behaves identically to state-sponsored scrapers,
      triggering Tor's anti-bot CAPTCHAs which leads to silent proxy failures and exhausts the limited public
      bridge pool.
1227
1228 Instead, users should rely on a less secure layer to manually reach Telegram, obtain the bridges, and populate
      `tor-bridges.txt`. This can be done via the standard `nullexit` WARP tunnel, a mobile phone on cellular
      data, Mullvad VPN, or a secure remote VPS. *(Note: We plan on fully integrating Mullvad and VPS

```



```

architectures directly into `nullexit` in the future, but have not had the time to build it yet).* This
intentionally "air-gaps" the cryptographic key exchange from the proxy infrastructure, leaving zero API
tokens and zero network traces for an adversary to exploit.
1229
1230 ### OPSEC: Python Runtime Airgap (Isolated Mode)
1231 `nullexit` explicitly relies on the host's native system Python 3 runtime for critical components like `scripts
/dns-proxy.py`. Because this script runs with elevated privileges (`sudo -n`) to bind to the privileged UDP
/53 socket, modifying or polluting this Python environment is strictly forbidden.
1232
1233 **The Zero-Dependency Rule & Isolated Mode (-I)**
1234 You must **NEVER** install third-party `pip` packages (e.g., `requests`, `pyyaml`) into the system Python
environment that `nullexit` uses.
1235 - All Python scripts in this repository must be written using *only* the Python Standard Library (`socket`, `
urllib`, `json`, `ssl`).
1236 - Third-party packages introduce a massive supply-chain attack vector. A compromised `pip` package installed
globally could allow local malware to hook into the Python runtime and exploit the `sudo` privilege of the
DNS proxy to gain full root access.
1237 - To enforce strict immunity against user-installed malicious packages, `toggle.sh` explicitly invokes the
proxy using **Python's Isolated Mode** (`python3 -I`). This flag forces the interpreter to completely
ignore `PYTHONPATH`, `PYTHONHOME`, and the user's `site-packages` directory, executing exclusively from the
read-only, Apple-signed system framework.
1238
1239 ### Recommended Upgrades
1240
1241 | Layer | Default | Upgraded |
1242 |---|---|---|
1243 | VPN Exit | Cloudflare WARP (US) | Mullvad (Swedish, proven no-logs, anonymous payment) |
1244 | DNS Resolver | AdGuard via WARP | AdGuard via Mullvad DoH (Cloudflare-blind) |
1245 | Coordination | Tailscale (US, OAuth) | Headscale (self-hosted, no third-party) |
1246 | Auth | OAuth / persistent keys | Ephemeral auth keys (no identity persistence) |
1247 | Content | WireGuard asymmetric | WireGuard + PSKs (coordination server blind) |
1248
1249 #### Mullvad (Replacing WARP)
1250 Swedish-based. Police raided their offices in 2023 and left empty-handed no logs exist to seize. Accounts are
random numbers. Payment by cash or Monero. Replace `VPN_SERVICE_PROVIDER=custom` with `VPN_SERVICE_PROVIDER
=mullvad` in `docker-compose.yml`.
1251
1252 #### Mullvad DoH Upstreams
1253 Point AdGuard's upstream resolvers to `https://adblock.dns.mullvad.net/dns-query`. Cloudflare WARP only sees
encrypted HTTPS to Mullvad's IPs completely blind to your DNS queries.
1254
1255 #### Headscale
1256 Self-hosted Tailscale coordination server. Eliminates Tailscale the company, keeping all node topology under
your control.
1257
1258 #### Ephemeral Auth Keys
1259 Generate in the Tailscale admin panel. Used once to authenticate; node disappears from the admin panel after
disconnect. Breaks persistent identity linkage.
1260
1261 #### Pre-Shared Keys (PSKs)
1262 Add a WireGuard PSK between peer nodes. Adds a symmetric encryption layer that Tailscale's coordination server
has no knowledge of, blinding it from reading transit content.
1263
1264 ### Structural Limitations
1265 - **Traffic Analysis / Metadata:** Passive fiber tapping (UPSTREAM) can observe timestamps, data volumes, and
IP ranges without reading content. Defeating traffic analysis requires mixing networks (Tor) or continuous
dummy packet padding both heavily degrade performance.
1266 - **Targeted State-Level Adversaries:** Consumer-grade VPNs are not a complete shield against a nation-state
actively targeting you. This stack is designed to defeat bulk passive surveillance and corporate dragnet
collection.
1267
1268 ---
1269
1270 ## 14. Per-Device Access Control
1271
1272 Because every mesh device routes internet-bound traffic through the `warp` container's `FORWARD` chain, they
all appear with their stable `100.x.x.x` Tailscale IP as the `src` address. This gives two powerful
enforcement surfaces:
1273
1274 ### IP & Port Level (`scripts/routing-fix.sh`)
1275 Write raw `iptables` rules targeting specific devices. The 30-second idempotent loop auto-survives Gluetun
reconnects:
1276
1277 ```bash
1278 # Block a specific device from a target subnet
1279 iptables -I FORWARD -s 100.x.x.x -d 203.0.113.0/24 -j DROP
1280
1281 # Restrict a device to only HTTP/HTTPS
1282 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 3000 -j DROP
1283 iptables -I FORWARD -s 100.x.x.x -p tcp ! --dport 443 -j DROP
1284

```

```

1285
1286 # Block a device from internet egress entirely (mesh only)
1287 iptables -I FORWARD -s 100.x.x.x -j DROP
1288
1289 # Time-based (e.g., cut off 10 PM 7 AM)
1290 iptables -I FORWARD -s 100.x.x.x -m time --timestart 22:00 --timestop 07:00 -j DROP
1291 ```
1292
1293 ### DNS Level (AdGuard REST API)
1294 AdGuard Home supports per-client rules keyed by Tailscale IP:
1295
1296 ```bash
1297 curl -X POST http://localhost:80/control/clients/add \
1298   -H "Content-Type: application/json" \
1299   -d '{
1300     "name": "kids-ipad",
1301     "ids": ["100.x.x.x"],
1302     "blocked_services": ["youtube", "tiktok", "instagram"],
1303     "safesearch_enabled": true
1304   }'
1305 ```
1306
1307 ---
1308
1309 ## 15. Packaging nullexit as a macOS Application (.dmg)
1310
1311 nullexit can be packaged into a native `.app` bundle, but this introduces nested virtualization requirements.
1312
1313 nullexit uses Colima (Apple Virtualization Framework `vz`) to run a Linux VM hosting Docker. Installing the
1314   `.app` inside a virtualized macOS environment (Parallels, cloud Mac) means running a Linux VM inside a
1315   macOS VM requiring hardware-level nested virtualization.
1316
1317 ### Hardware Support
1318 - Supported: Apple Silicon M3/M4 (macOS Sequoia 15+), Intel Macs with VMX passthrough.
1319 - Unsupported: M1, M2, A18 Pro. Colima crashes silently; the terminal (`setup.sh`) surfaces crash logs.
1320
1321 ### Build Steps
1322 ```bash
1323 # 1. Verify nested virtualization support
1324 sysctl -a | grep hv_nested_virt_supported # expect: 1
1325
1326 # 2. Compile the AppleScript launcher
1327 osacompile -o "Nullexit.app" "Toggle Gateway.applescript"
1328
1329 # 3. Embed scripts into app resources
1330 mkdir -p "Nullexit.app/Contents/Resources/scripts"
1331 cp toggle.sh setup.sh recover.sh docker-compose.yml "Nullexit.app/Contents/Resources/"
1332
1333 # 4. Package into DMG
1334 hdiutil create -volname "Nullexit Installer" -srcfolder "./Nullexit.app" -ov -format UDZO Nullexit.dmg
1335
1336 # 5. Bypass Gatekeeper (unsigned app)
1337 xattr -cr /Applications/Nullexit.app
1338 ```
1339
1340 Without a paid Apple Developer certificate ($99/yr), users see an "App is damaged" Gatekeeper warning and need
1341 to run step 5.
1342
1343 ---
1344
1345 ## 16. Moonlight/Sunshine + Tailscale Race Condition
1346
1347 When streaming from a remote Windows host running Sunshine over a Tailscale mesh (e.g., while the client is on
1348 a cellular hotspot), a race condition can occur on boot:
1349
1350 1. Both Tailscale and Sunshine are set to start automatically on boot.
1351 2. Tailscale takes a few seconds to fully initialize its `100.x.x.x` virtual adapter and establish a connection
1352 3. Sunshine starts too fast. It launches before Tailscale is fully ready. Since Sunshine only binds to
1353    network interfaces that are active at the exact moment of startup, it misses the Tailscale adapter entirely
1354 4. Moonlight clients (even when configured with the correct Tailscale IP) will fail to connect because Sunshine
1355    isn't listening on that interface.
1356
1357 The "Magic Toggle" Symptom:
1358 If the user manually enables or disables Wi-Fi on the host PC, it triggers a global Windows network change
1359 event. Sunshine listens for these events, re-enumerates all network adapters, and says, "Oh, there's a
1360 Tailscale adapter here!" and finally binds to it. Suddenly, Moonlight can connect.
1361
1362 The Permanent Fix:

```



```

1355 On the Windows host, open `services.msc`, locate the Sunshine Service, and change its Startup type from Automatic to Automatic (Delayed Start). This forces Windows to wait a minute or two after booting
    before launching Sunshine, ensuring Tailscale's virtual adapter is fully initialized first.
1356
1357 ---
1358
1359 ## 17. Future Work
1360
1361 - Direct P2P on VM Hosts: Non-Linux hosts run Docker inside a VM, making true UDP hole-punching impossible.
    The only working solution is a native Linux host (Raspberry Pi, Intel NUC) where Docker runs without VM
    hypervisor translation.
1362 - Native Linux Deployment: Benchmark on a Raspberry Pi native container footprint is ~75MB without macOS
    hypervisor overhead.
1363 - Mesh-Wide Filesystem (SFTP over Tailscale): Any mesh device passively exposes its filesystem over SFTP.
    Android via Termux + OpenSSH + wakelock; iOS is client-only due to background process limits.
1364 - Post-Quantum Cryptography (PQC): Tailscale uses Curve25519, theoretically vulnerable to "harvest now,
    decrypt later" quantum attacks. A future iteration could replace the Tailscale container with raw WireGuard
    + [Rosenpass](https://rosenpass.eu/) to negotiate post-quantum PSKs.
1365
1366 ## 18. Geo-IP Blocking
1367
1368 To block traffic to and from specific countries, the system dynamically downloads country-specific IP ranges
    from [ipdeny.com](http://www.ipdeny.com/ipblocks/data/countries/) and drops them via iptables `FORWARD`
    rules.
1369
1370 To add a new country to the blocklist:
1371 1. Find the 2-letter ISO country code (e.g., `cn` for China, `ru` for Russia, `il` for Israel, `kp` for North
    Korea).
1372 2. Open your `.env` file.
1373 3. Update the `BLOCKED_COUNTRIES` variable: e.g., `BLOCKED_COUNTRIES="kp il cn ru"`.
1374 4. Run `toggle.sh` to restart the gateway and apply the new environment variables to the routing-fix container.
1375
1376 > [!NOTE]
1377 > Limitations of Geo-IP Blocking: IP-based blocking is highly effective at blocking servers, direct apps,
    and raw infrastructure physically located and registered in a specific country (e.g., `www.gov.il`).
    However, it will not block high-profile websites (e.g., `jpost.com`) that route their traffic through
    global Content Delivery Networks (CDNs) like Google Cloud, Cloudflare, or Fastly. Because the CDN's IP
    addresses are registered in the US or globally, the network traffic physically never routes to the blocked
    country, allowing it to bypass the Geo-IP firewall.
1378
1379 ## 19. Incident Post-Mortems
1380
1381 ### Incident: The Overnight Silent IP Leak (July 2026)
1382
1383 The Problem: The host leaked traffic over its raw, unencrypted `eth0` IP continuously for roughly 8 hours,
    completely bypassing the VPN. The `toggle.sh` state and the container liveness checks all reported the
    gateway as healthy and active.
1384
1385 The Investigation:
1386 1. Sleep/Wake Checks: We initially suspected macOS sleep cycles broke the routing table. A check of `pmset
    -g log` confirmed the laptop literally never slept (0 sleep events recorded since boot), ruling out all
    sleep/wake code paths.
1387 2. Keepalive Expiry: We discovered `docker-compose.yml` was missing `
    WIREGUARD_PERSISTENT_KEEPALIVE_INTERVAL`. Without a keepalive, standard router NAT tables (which typically
    garbage collect UDP after 30 to 120 seconds of silence) quietly expired the WireGuard port mapping
    overnight.
1388 3. Container State Masking: Because WireGuard is stateless, the `warp` container didn't immediately crash.
    It stayed "Up", which meant Docker's healthchecks and our `toggle.sh` liveness checks never noticed the
    tunnel inside it was totally dead.
1389
1390 The Solution:
1391 1. Added `WIREGUARD_PERSISTENT_KEEPALIVE_INTERVAL=25s` (the WireGuard standard to beat standard 30s NAT
    timeouts) to keep the UDP tunnel permanently pinned open through the router.
1392 2. Injected a Fail-Closed Kill-Switch into `routing-fix.sh`: A 30-second loop now actively verifies tunnel
    health by running `curl` over `tun0` to Cloudflare's trace endpoint. If it fails 3 consecutive times, it
    immediately rewrites the default route to `blackhole`, severing all traffic instead of letting it leak
    through `eth0`.
1393 3. Added a listener in `watcher.sh` that detects this fail-closed event and instantly pushes a native macOS UI
    notification to the desktop, alerting the user to manually intervene.
1394
1395 > [!NOTE]
1396 > Why dual failure systems?
1397 > The system now relies on two independent failure handlers that cover each other's blind spots:
1398 > 1. Automated Self-Healing (WARP Watcher + `recover.sh`): Triggered when the `warp` container logs `warp=
    off`. This happens when the server actively kicks the client. Because the container knows it is
    disconnected, it triggers `recover.sh` to safely reboot Docker and reconnect.
1399 > 2. Fail-Closed Kill-Switch (`routing-fix.sh` + `watcher.sh`): Triggered when a silent network failure
    occurs (e.g., NAT timeouts). The container incorrectly believes it is connected (`warp=on`), so auto-
    recovery never fires. Instead of auto-recovering (which risks falling back to raw Wi-Fi mid-process while
    the user isn't looking), this kill-switch actively blackholes the internet and forces a manual intervention
    via a desktop notification.

```

```

1400
1401 ## 20. Design Decisions: Exit Node IP Rotation
1402
1403 A common question for privacy architectures is whether the system should aggressively rotate its public exit IP
1404 (e.g., every 5 minutes, like a Tor circuit or a proxy rotator).
1405
1406 `nullexit` uses Cloudflare WARP via Gluetun. WARP provides anonymity in a crowd rather than aggressive
1407 rotation. Thousands of users share the same Cloudflare Edge IP simultaneously, making your traffic
1408 indistinguishable from the herd. The IP may occasionally rotate on its own (which `host-leak-probe.sh` logs
1409 as a `ROTATE` event), but it remains generally stable.
1410
1411 Why aggressive IP rotation was intentionally excluded:
1412 1. Broken TCP Connections: Every rotation instantly drops all long-lived TCP sessions (SSH, large downloads
1413 , WebSockets, gaming, Zoom).
1414 2. CAPTCHA & 2FA Hell: Modern web security (e.g., Cloudflare, Akamai) aggressively flags IP-hopping as bot
1415 behavior. Aggressive rotation guarantees the user will be bombarded with "Verify you are human" CAPTCHAs
1416 and "New Login Detected" 2FA emails constantly.
1417 3. WireGuard Architecture: WireGuard is stateless and does not natively support IP hopping on the fly. To
1418 rotate an IP, the VPN client must actively drop the tunnel, request a new endpoint, and re-handshake. This
1419 introduces latency gaps where the kill-switch would repeatedly trigger and blackhole the user's internet.
1420
1421 Verdict: For a daily-driver secure gateway, shared-IP crowding (WARP) is superior to aggressive rotation.
1422 As an added benefit, Cloudflare WARP IPs are actually highly static (tied to the persistent WireGuard
1423 public key generated for your device identity). Because your IP stays static and is part of Cloudflare's
1424 own trusted network, it builds a high "trust score," virtually eliminating CAPTCHA loops while still hiding
1425 you within the massive crowd of other users in that data center. All mesh devices routing through your
1426 gateway will reliably share this exact same trusted IP.
1427
1428 Note: The only exception where aggressive IP rotation is genuinely required is for specialized offensive
1429 security tasks like high-volume web scraping or dark web research where avoiding IP bans or active tracking
1430 overrides the need for TCP stability.)*
1431
1432 ## 21. Cloudflare WARP Privacy & Logging
1433
1434 Because `nullexit` uses Cloudflare WARP (via Gluetun) as its upstream exit node, the system inherits Cloudflare
1435 's consumer privacy policy.
1436
1437 What Cloudflare DOES NOT log (Strict No-Logs Policy):
1438 * Browsing History: They do not log the domains or URLs you visit.
1439 * Traffic Destinations: They do not log the IP addresses of the servers you connect to.
1440 * Content: They do not inspect or log the payload/data of your traffic.
1441 * Data Brokering: They explicitly guarantee they will never sell or rent your data to advertisers.
1442
1443 What Cloudflare DOES log (Minimal Telemetry):
1444 * Data Transfer Volume: Total bandwidth used (required to enforce 10GB/mo quotas if you are not using a
1445 paid WARP+ key).
1446 * Performance Metrics: Anonymized connection speeds and latency to optimize their network routing.
1447 * Aggregate Volume: Total traffic volumes to popular destinations in aggregate (e.g., "datacenter X sent 50
1448 TB to Netflix"), stripped of all user IDs.
1449 * Device ID (Public Key): They store the persistent WireGuard public key generated by your container to
1450 authorize the connection.
1451
1452 Privacy Verdict:
1453 Cloudflare WARP provides excellent historical privacy. Because they do not log your destinations, they
1454 cannot comply with historical subpoenas asking what websites you visited yesterday. However, it is not
1455 absolute anonymity. They still know your real ISP IP address while you are actively connected. For daily-
1456 driver privacy against ISPs, hackers, and corporate trackers, it is top-tier. For nation-state evasion, use
1457 Tor.
1458
1459 ---
1460
1461 ## 22. Battery & Power Measurement Quirks
1462
1463 When trying to optimize this framework (or any daemon) for battery life, developers often look for a way to
1464 programmatically measure the exact Wattage consumed by a specific process (e.g., "How many Watts is `toggle
1465 .sh` using?").
1466
1467 This is a hardware impossibility on macOS and Linux.
1468
1469 ### 22.1 Why Activity Monitor is Lying to You
1470 Your machine's logic board only has physical power sensors for the entire CPU package, the GPU, and the RAM.
1471 It knows the CPU is pulling 5W total, but it has no physical hardware capability to know whether Docker is
1472 using 3W and Chrome is using 2W.
1473
1474 The "Energy Impact" score you see in macOS Activity Monitor is a synthetic, heuristic score calculated by `
1475 powerd`. It penalizes apps for waking up the CPU (idle wakes), doing disk I/O, or keeping the screen awake.
1476 It is a relative score, not actual Wattage.
1477
1478 ### 22.2 The Proper A/B Testing Methodology
1479 If you want to truly know how many Watt-hours or mAh your background daemons (`routing-fix.sh`, `host-leak-
1480 probe.sh`, etc.) are costing you, you must perform a hardware-level A/B drain test:

```

```

1450
1451 1. Unplug the laptop, run the gateway, and note your exact battery mAh using:
1452     ``bash
1453     ioreg -l | grep "AppleRawCurrentCapacity"
1454     ``
1455 2. Leave the laptop idle for exactly 1 hour.
1456 3. Run the command again to see exactly how many mAh were drained.
1457 4. Turn the gateway off and repeat the test for another hour.
1458
1459 The delta in mAh drained between the two tests is the true, exact cost of running the software. When optimizing
    polling loops (e.g., changing `sleep 5` to `sleep 30`), this is the only reliable way to measure the
    actual battery life returned to the user.
1460
1461 ---
1462
1463 ## 23. Architectural Limitations: Host Lockdown Mode
1464
1465 A common privacy feature request is a "Host Lockdown Mode" (Mode 3): A mode where the Docker exit node remains
    perfectly functional for remote devices, but the host machine itself is completely blocked from accessing
    the internet (to prevent even encrypted host traffic from leaking metadata).
1466
1467 **Why this is impossible on macOS:**
1468 On Linux (e.g., a Raspberry Pi), Docker runs natively. The host's traffic traverses the `OUTPUT` iptables chain
    , while Docker traffic traverses the `FORWARD` chain. We could easily drop all `OUTPUT` traffic to kill the
    host's internet, and Docker would continue routing traffic perfectly.
1469
1470 However, Docker on macOS runs inside a Linux Virtual Machine (via `qemu` or Apple `Virtualization.framework`).
    This VM runs as a standard macOS user-space application. If you configure the macOS firewall (`pf`) to drop
    all host outbound traffic, it will indiscriminately drop the VM application's traffic as well, instantly
    killing the Cloudflare WARP tunnel and the exit node.
1471
1472 Writing complex macOS `pf` rules to "allow the VM app, allow Cloudflare WARP IPs, allow Tailscale DERP IPs, but
    block everything else" is highly fragile and heavily discouraged. Since the current `toggle.sh` default
    mode already fully encrypts the host's traffic via the WARP tunnel, the host is already protected from ISP
    leaks.
1473
1474 ### 23.1 July 4, 2026: Multi-stage Docker Builds for Go Ports & Teardown Hang Fix
1475 **Symptom:** The Python implementations of `logger.py` and `sync-rules.py` were slow and required a heavy
    python runtime inside Alpine containers. Also, `routing-fix` container teardown was hanging for 30s during
    `toggle.sh` shutdown.
1476 **Root Cause:**
1477 - Python is slower than Go and installing it dynamically via `apk add` added overhead and complexity to the
    containers.
1478 - Docker containers ignore `SIGTERM` if the init process doesn't handle it, causing a full 30s timeout on
    shutdown.
1479 **Resolution:**
1480 - Ported `logger` and `sync-rules` to Go using Goroutines for massive concurrent speedups.
1481 - Implemented **Multi-stage Docker builds** (using `golang:1.22-alpine` as builder) to compile the static
    binaries inside Docker. The final containers are raw Alpine with zero dependencies.
1482 - Added `--build` to `docker compose up -d` in `toggle.sh` to ensure updates are consistently applied on boot.
1483 - Added `stop_grace_period: 1s` to `routing-fix` in `docker-compose.yml` which eliminates the 30-second
    teardown hang instantly.
1484 - Implemented a cryptographic integrity checker (`scripts/crypto.sh`) using HMAC-SHA256 and `NULLEXIT_SEED` in
    `.env` to prevent tampering of bash scripts.
1485
1486 ### 23.2 July 4, 2026: Consolidation of Core Bash Scripts (Refactoring)
1487 **Symptom:** Over 200 lines of bash logic were directly duplicated across `toggle.sh` and `recover.sh` (e.g. `
    restart_tailscaled_daemon`, `stop_sleep_prevention`, WARP bypass routes). This caused drift when one script
    was updated without the other.
1488 **Root Cause:**
1489 - Historical separation of responsibilities allowed `recover.sh` to grow complex alongside `toggle.sh`.
1490 **Resolution:**
1491 - Massively refactored both scripts by extracting all duplicated networking logic, PID lifecycle management,
    and environmental configuration down to `scripts/common.sh`.
1492 - Specifically, the `stop_sleep_prevention`, `stop_dns_watcher`, `stop_host_leak_probe`, and `stop_warp_watcher`
    functions were collapsed into a single `stop_pidfile_daemon()` abstraction.
1493 - Proxy disable loops were abstracted to `disable_all_proxies()`.
1494 - The `162.159.192.1/193.1` WARP edge IPs are now dynamically resolved from `.env` instead of hardcoded.
1495
1496 ---
1497
1498 ### 23.3 July 4, 2026: The "Network Unreachable" Host Leak & Post-Wake Container Destructions
1499 **Symptom:**
1500 1. Running `docker compose config` with dummy keys corrupted the user's `.env` file by appending invalid
    WireGuard keys. This caused the `warp` container to become unhealthy on boot, which triggered a full
    teardown of the gateway by `toggle.sh`.
1501 2. After fixing `.env`, `toggle.sh` successfully booted the gateway, but the host's internet started bypassing
    the tunnel entirely (a massive host leak). The Cloudflare trace returned `warp=off` and exposed the host's
    physical ISP IP.
1502 3. Running `recover.sh --post-wake` would violently force-recreate all gateway containers and drop the host's
    internet connection.

```

```

1503
1504 **Root Cause:**
1505 - **Bug 1 (The Host Leak):** In `toggle.sh`, the logic to find the Tailscale `utun` interface used `ifconfig |
grep -B4 "inet 100."`. Due to the 4 lines of before-context, this regex was accidentally capturing the
interface *above* the actual Tailscale interface (e.g. grabbing `utun4` instead of `utun5`). Because `utun4`
had no IP address, the subsequent `sudo route add -net 0.0.0.0/1 -interface utun4` silently failed with "
Network is unreachable", leaving the host's default route exposed.
1506 - **Bug 2 (The Post-Wake Destructive):** The health check in `recover.sh --post-wake` relied on `docker
compose exec -T warp curl -sf https://www.cloudflare.com/cdn-cgi/trace`. However, the newer `qmcgaw/glutun`
Alpine-based Docker image no longer ships with `curl` pre-installed. The health check failed with `
executable file not found in $PATH`, tricking `recover.sh` into thinking the tunnel was completely dead and
needed to be forcefully recreated via `docker compose up -d --force-recreate`.
1507 - **Bug 3 (Restart Flags):** Previous refactors incorrectly permitted `toggle.sh restart` instead of strictly
enforcing `toggle.sh --restart`.
1508
1509 **Resolution:**
1510 - Cleaned the `.env` file of duplicate dummy entries.
1511 - Replaced the brittle `grep -B4` interface parsing in `toggle.sh` with a strict AWK parser: `awk '/^[a-z0
-9]+/{iface=$1} /inet 100\./{print iface; exit}' | tr -d ':'`. This mathematically isolates the exact
interface possessing the Tailscale `100.x.x.x` IP address, preventing the `0.0.0.0/1` route from silently
failing.
1512 - Changed the health check in `recover.sh` to use `wget -qO- --timeout=3` which is natively installed in the
Glutun image. This stopped the unnecessary destructive recreations of the containers during post-wake
events.
1513 - Reverted the `restart` alias in `toggle.sh` to strictly require `--restart`.
1514
1515 ### 23.4 July 4-5, 2026: Tailscale Data-Plane Loop (The DERP Relay Deadlock)
1516 **Symptom:**
1517 After bypassing the Tailscale control plane (`192.200.0.0/16`) to fix the "offline" mesh status, the Mac
appeared online. However, external peer devices (like the user's phone on cellular) could not successfully
ping or SSH into the Mac. `tailscale netcheck` reported `Nearest DERP: unknown (no response to latency
probes)`.
1518
1519 **Root Cause:**
1520 While bypassing the control plane kept the daemon connected to the coordination servers, Tailscale's actual
peer-to-peer data plane relies on STUN (for NAT traversal) and DERP relays. These relays span ~80
dynamically allocated IP addresses across multiple cloud providers. Because we were forcing `0.0.0.0/1`
through the `utun*` exit node, the Mac's own outbound connection attempts to DERP relays were being routed
back into the tunnel. To prevent an infinite loop, Tailscale's daemon dropped its own packets when it saw
them returning on the TUN interface. This effectively left the daemon deaf and blind to peer connections.
1521
1522 **Resolution:**
1523 - Modified `add_warp_bypass_routes` in `scripts/common.sh` to execute a live API fetch (`curl -s https://login.
tailscale.com/derpmap/default`) during startup.
1524 - The script parses out all active DERP relay IPv4 addresses (~80 IPs) and adds a physical host bypass route
for every single one of them.
1525 - These IPs are written to a temporary file (`/tmp/nullexit-derp-ips.txt`) so they can be cleanly un-routed by
`remove_warp_bypass_routes` when the gateway shuts down. Peer connectivity (ping, SSH, SFTP) was fully
restored.
1526
1527 ### 23.5 July 5, 2026: Hardcoded WARP Endpoints in docker-compose
1528 **Symptom:**
1529 The bash scripts allowed overriding the default Cloudflare WARP IP endpoints via `.env` variables (`
WARP_ENDPOINT_1` and `WARP_ENDPOINT_2`) to establish bypass routes. However, `docker-compose.yml`
statically hardcoded `162.159.192.1` for the `warp` container's `WIREGUARD_ENDPOINT_IP`.
1530
1531 **Root Cause & Risk:**
1532 If a user set a custom endpoint in `.env`, the script would successfully bypass the custom IP on the host level
. However, Glutun would still attempt to connect to the hardcoded `162.159.192.1`. Since `162.159.192.1`
was no longer explicitly bypassed, its packets would be sucked into the `utun*` exit node, causing an
infinite WireGuard routing loop that instantly breaks the gateway.
1533
1534 **Resolution:**
1535 Modified `docker-compose.yml` to dynamically read the environment variable via `${WARP_ENDPOINT_1
:-162.159.192.1}`, ensuring both the host routing scripts and the container use the exact same endpoint.
1536
1537 ### 23.6 July 6, 2026: Packet Filter (pfctl) Defaulting Local LAN Rules to TCP-Only
1538 **Symptom:**
1539 Devices on the exact same local Wi-Fi network as the gateway were experiencing ~500ms latency to the gateway
instead of the expected ~80ms.
1540
1541 **Root Cause:**
1542 When writing the `pf.conf` rules to whitelist local LAN traffic (`pass out quick on $ext_if to {
192.168.0.0/16, ... }`), the rule omitted an explicit protocol. The macOS `pfctl` compiler implicitly
applies state tracking to all `pass` rules. However, because state tracking (`keep state`) natively
evaluates TCP sessions, `pfctl` automatically appended the `flags S/SA` modifier to the compiled rule in
the kernel. This silently restricted the whitelist to **TCP only**, causing all local UDP traffic to be
dropped by the default-deny kill-switch. Since Tailscale relies on UDP port 41641 for direct P2P
connections, the local devices were unable to hole-punch and were forced to fall back to routing traffic
through a remote Tailscale DERP relay, massively inflating latency.
1543

```

```

1544 **Resolution:**
1545 Split the single implicit rule into explicit `proto tcp`, `proto udp`, and `proto icmp` rules in `scripts/pf.conf` to guarantee the macOS compiler allows all three core transport protocols on the local subnet without mutating the rule into a TCP-only lock.
1546 ### 23.7 July 8, 2026: Network Watcher Event Mismatch, Sudo-in-Background Crash, and Loop-Prevention
1547 **Symptom:**
1548 1. Switching Wi-Fi or waking the Mac from sleep did not trigger a post-wake recovery (`recover.sh --post-wake`) automatically.
1549 2. The network connection would eventually get completely sinkholed/blocked. If the user tried to restart or wait, it did not self-heal.
1550 3. During the recovery attempt, checking the public IP on the host would return the unencrypted campus/ISP IP (starting with `132.`) instead of the encrypted tunnel IP.
1551 **Root Cause:**
1552 - **Bug 1 (Network Watcher Mismatch):** In `scripts/watcher.sh`, the network change listener watched `State:/Network/Global/IPv4` changes via `scutil n.watch` but matched them against `*.state|*SCEventUpdate`. On macOS, the actual output is `changedKey [0] = State:/Network/Global/IPv4`. Because the pattern did not match, all network switches were silently ignored.
1553 - **Bug 2 (Sudo Caching Background Crash):** When the background WARP Watcher eventually detected the broken tunnel (after 6 failures/30s), it triggered the default nuclear `recover.sh` (with `POST_WAKE=false`). Because there was no active TTY in the background context, the `sudo -v` caching check aborted instantly with a terminal required error. Due to `set -e`, `recover.sh` died immediately before resetting DNS, disabling the packet-filter killswitch, or stopping containers, leaving the host network completely sinkholed.
1554 - **Bug 3 (Post-Wake Infinite Loop):** Once the watcher patterns were fixed, `recover.sh --post-wake` triggered successfully on network changes. However, because the script deleted the default routing entries at the start, spent ~15 seconds rebuilding 88 DERP relay bypass routes, and re-added the default routes at the end, this execution time exceeded the watcher's 10-second debounce threshold. Furthermore, the watcher recorded the `start` timestamp in the debounce file. Consequently, the route changes made by the script itself triggered the watcher again immediately after completion, trapping the system in an infinite loop of recovery runs. During this loop, the VPN routes were constantly deleted, resulting in the host leaking traffic through the real ISP interface (an IP starting with `132.`) for 15s out of every cycle.
1555 **Resolution:**
1556 - **Fixed Watcher Matching:** Updated `scripts/watcher.sh` to match `*changedKey*` and `*State:/Network/Global/IPv4*` to correctly catch network changes.
1557 - **Bypassed Background Sudo:** Added `--auto`/`--non-interactive` flags to `recover.sh` and added a TTY check `[ -t 0 ]` to skip interactive `sudo -v` authentication when running headlessly in the background. Updated the WARP Watcher call in `toggle.sh` to pass `--auto`.
1558 - **Prevented Infinite Loops:** Modified `scripts/watcher.sh` to write the **completion** timestamp of `recover.sh` to `DEBOUNCE_FILE` (instead of only the start timestamp). This ensures all buffered network config events generated during route reconstruction are evaluated against the end-of-run timestamp and successfully debounced, breaking the infinite loop.
1559 - **Centralized & Timestamped Logs:** Redirected `watcher.sh` stdout/stderr from `/tmp/nullexit-watcher.log` to the repository's main `output.log` and updated `scripts/common.sh`'s `step`, `ok`, `warn`, `fail`, and `die` helpers to prefix logs with a local `[YYYY-MM-DD HH:MM:SS]` timestamp.
1560 ### 23.8 July 11, 2026: Tor Container Hardening & obfs4proxy Restoration
1561 **Symptom:**
1562 1. The Tor container entered a fatal crash loop upon boot, throwing `exec: "obfs4proxy": executable file not found in $PATH` when `TOR_USE_BRIDGES=true`.
1563 2. When the user pasted Tor bridge lines containing comments (`#`) or blank lines into `tor-bridges.txt`, the container failed to validate the file properly and crashed.
1564 **Root Cause:**
1565 - **Bug 1 (Missing Pluggable Transports):** The Tor Dockerfile was built on `debian:bullseye-slim`, which had silently dropped or failed to resolve the `obfs4proxy` package in its latest apt repositories.
1566 - **Bug 2 (Fragile Validation):** The `entrypoint.sh` bridge validation check merely checked `[ -s tor-bridges.txt ]` (file size > 0). It did not verify if the file actually contained valid, uncommented bridge lines. Passing empty lines or comments tricked the validation script into appending invalid `Bridge` directives to the `torrc`, causing the Tor daemon to instantly exit.
1567 **Resolution:**
1568 - **Upgraded Base Image:** Shifted the Tor Dockerfile base image to `debian:bookworm-slim`, restoring access to the `obfs4proxy` pluggable transport package.
1569 - **Robust Bridge Validation:** Replaced the fragile `[ -s ]` check with a strict `grep -vE '^\s*(#|)$'` command that strips comments and empty lines. If no valid lines remain, the container safely falls back to standard public guard relays instead of crashing, ensuring Tor remains available even with a misconfigured bridge file.
1570 - **Image Pinning:** Explicitly pinned `image: nullexit-tor:v1.0.0` in `docker-compose.yml` to prevent arbitrary local builds from drifting to `:latest`.
1571 ### 23.9 July 12, 2026: Tor ControlPort Dynamic Password Authentication & User Config
1572 **Symptom:**
1573 Querying the Tor Control Port (e.g., via the `/sweep` script or `check_circuits.py`) returned empty responses or connection errors. Logs showed that Tor automatically closed its ControlPort:
1574 `You have a ControlPort set to accept unauthenticated connections from a non-local address. ... That's so bad that I'm closing your ControlPort for you.`
1575 **Root Cause:**

```

```

1583 Docker port mapping (especially under Colima on macOS) requires container sockets to bind to `0.0.0.0` to be
    reachable from the host. However, when Tor's ControlPort is bound to `0.0.0.0` (non-local) with no
    authentication, Tor closes it by default as a safety precaution. Sourcing SOCKS5/ControlPort to `127.0.0.1`
    inside the container was not an option as it broke the Docker bridge routing path.
1584
1585 **Resolution:**
1586 - **Dynamic Password Authentication**: Updated the Tor container's `entrypoint.sh` to read `TOR_PASSWORD` and
    dynamically generate its HMAC hash via `tor --hash-password`, adding `HashedControlPassword <hash>` to `
    torrc`. This secures the ControlPort on `0.0.0.0` so Tor keeps it open.
1587 - **Unified Credentials in `.env`**: Exposed `ADGUARD_USER=admin` and `ADGUARD_PASSWORD=nullexit` in `.env` to
    serve as the unified gateway credentials. Sourced `TOR_PASSWORD` directly from `ADGUARD_PASSWORD` in `
    docker-compose.yml`.
1588 - **Client Authentication**: Updated `scripts/check_circuits.py` to fetch `TOR_PASSWORD` (defaulting to `
    ADGUARD_PASSWORD` or `nullexit`) and authenticate with the ControlPort using `AUTHENTICATE "<password>"` to
    regain access.
1589
1590 ---
1591
1592 ## 24. Censorship-Resistant Transport & Egress Compartmentalization
1593
1594 ### 24.1 Why this is needed
1595 WARP can be blocked from deployment countries with strict internet controls. The most likely cause isn't that "
    encrypted traffic is blocked" in general it's that WireGuard has a recognizable handshake signature, and `
    WIREGUARD_ENDPOINT_IP=162.159.192.1` on `WIREGUARD_ENDPOINT_PORT=2408` is a fixed, easily-enumerable target
    (Cloudflare's known WARP range). That combination is exactly what IP/ASN-based and DPI-based blocking is
    good at catching.
1596
1597 ### 24.2 Important correction: Gluetun's `SHADOWSOCKS=on` is the wrong direction
1598 Gluetun's built-in Shadowsocks option runs a Shadowsocks **server** inside the already-connected VPN tunnel, so
    other LAN devices can reach out through Gluetun's **existing** connection via the SS protocol. It is not a
    way to make Gluetun itself connect **outbound** through a Shadowsocks server Gluetun only speaks OpenVPN
    or WireGuard as its actual transport. There is no `VPN_TYPE=shadowsocks`. Confirmed against Gluetun's own
    maintainer/community discussion: people have asked for a "Shadowsocks-client" mode specifically to chain
    through Gluetun, and the standing answer is "run a separate Shadowsocks client container."
1599
1600 ### 24.3 Diagnose before building anything
1601 The right fix depends entirely on what's actually being blocked:
1602 - Point Gluetun at a **different provider/IP/ASN** (any other Gluetun-supported WireGuard/OpenVPN provider, or
    a self-hosted WireGuard server) and test. If that gets through, the block is Cloudflare/WARP-specific, not
    a general WireGuard block **no new component needed**, just swap `VPN_SERVICE_PROVIDER` / the endpoint.
1603 - If WireGuard fails regardless of endpoint, the block is protocol-level (DPI fingerprinting the handshake)
    proceed to obfuscation.
1604
1605 ### 24.4 Path A: Swap WARP for a different Gluetun-native transport
1606 Simplest fix, only applies if 24.3 shows the block is Cloudflare-specific. Change `VPN_SERVICE_PROVIDER` / `
    VPN_TYPE` / endpoint in `docker-compose.yml` and in the now-centralized `WARP_ENDPOINT_*` values in `.env`.
    Nothing else in the stack needs to change.
1607
1608 ### 24.5 Path B: Add an obfuscation hop
1609 Needed if WireGuard itself is being fingerprinted/blocked. Two ways to slot it in:
1610
1611 **B1 Wrap (recommended):** Run a Shadowsocks (or obfs4/v2ray) client as a new sidecar container. Point Gluetun
    's `WIREGUARD_ENDPOINT_IP` at `127.0.0.1:<local-port>` instead of Cloudflare directly; the sidecar relays
    that traffic to a Shadowsocks server you run yourself abroad. Keeps Gluetun's kill-switch, healthchecks,
    and the entire rest of the stack (Tailscale, AdGuard, ipset, `routing-fix.sh`) untouched, since none of
    them know the transport changed underneath `warp`.
1612
1613 **B2 Replace:** Swap the `warp` service entirely for a Shadowsocks-client + `tun2socks` container that owns
    the network namespace instead of Gluetun. More invasive loses Gluetun's built-in kill-switch/healthcheck,
    which would need to be rebuilt by hand (see Section 24.9 for the pluggable architecture to support this
    natively).
1614
1615 **Recommendation: B1.** Smaller surface area, preserves the self-healing architecture already built and
    documented.
1616
1617 ### 24.6 Fingerprinting caveat
1618 Plain Shadowsocks can still be caught by active-probing DPI in more aggressive censorship environments. If
    plain SS gets blocked too, the next step up is an obfuscation plugin (`v2ray-plugin`, `Cloak`) or a newer
    protocol designed against active probing (VLESS+Reality, Hysteria2). Test plain SS first don't over-build
    before confirming it's needed.
1619
1620 ### 24.7 AmneziaWG (The High-Speed Alternative)
1621 If you want to maintain the bare-metal speeds of WireGuard while bypassing DPI (e.g. in Egypt or Russia), the
    best alternative to Shadowsocks/V2Ray is AmneziaWG. AmneziaWG is a fork of WireGuard that pads the
    handshake packets with randomized "junk" data, entirely destroying the 148-byte DPI signature that
    firewalls look for.
1622 - **Pros:** Because it runs entirely as UDP without TCP-over-TCP overhead, it completely avoids "TCP meltdown"
    latency spikes associated with Shadowsocks/V2Ray wrappers.
1623 - **Cons:** You cannot use Cloudflare WARP. Furthermore, it requires completely ripping Gluetun out of the `
    nullexit` stack and building a custom Docker container, meaning you lose Gluetun's built-in kill-switches
    and health-check logic.

```



```

1624
1625 ### 24.8 Infrastructure Costs & Privacy
1626 To use either Shadowsocks (Path B) or AmneziaWG, you cannot use the free Cloudflare WARP infrastructure,
        because Cloudflare servers only speak standard WireGuard. The software for both custom protocols is 100%
        free and open-source, but you must host the remote server infrastructure yourself.
1627 - **Free Tier (Oracle Cloud):** You can host your remote server on Oracle Cloud's "Always Free" tier for $0.
        However, this routes your highly sensitive, censorship-evading traffic directly through Oracle's massive
        corporate data broker. If privacy is the core objective, handing all your metadata to Oracle defeats the
        purpose.
1628 - **Paid Tier (Hetzner / Mullvad):** The recommended path is to rent a standard ~$4/month Virtual Private
        Server (VPS) in a free country (e.g., Hetzner in Germany, subject to strict EU GDPR privacy laws) and self-
        host the server. Alternatively, Mullvad VPN (5/month) provides native v2ray/Shadowsocks bridges built into
        their network, allowing you to bypass censorship with a strict zero-logs policy. It is highly recommended
        to pay with untrackable payment methods to maintain complete anonymity, which these providers typically
        support without requiring local residency.
1629
1630 ### 24.9 The Future: Egress Compartmentalization (Pluggable Architecture)
1631 To natively support invasive replacement paths like **Path B2** or **AmneziaWG** (Section 24.7) without
        breaking the core `nullexit` routing and monitoring logic, the architecture must be refactored into a
        modular, "pluggable" design.
1632
1633 1. **The Egress Plugin System:** Currently, WARP-specific logic (like `cdn-cgi/trace` health checks and
        hardcoded interface names) is scattered across `toggle.sh`, `scripts/routing-fix.sh`, and `scripts/host-
        leak-probe.sh`. This should be abstracted into a `scripts/plugins/` directory, where each egress method (`
        warp.sh`, `v2ray.sh`) implements standardized functions: `get_egress_interface()`, `check_health()`, and `
        get_bypass_ips()`.
1634 2. **Dynamic Docker Compose:** Based on an `EGRESS_TYPE` variable in `.env`, dynamic compose file generation
        would spin up only the required egress containers (e.g., skipping the `warp` container when `EGRESS_TYPE=
        v2ray` is set).
1635 3. **Agnostic Routing & Watchers:** `routing-fix.sh` would dynamically read the `EGRESS_INTERFACE` from the
        active plugin to apply `iptables` rules, and the background watchers in `toggle.sh` would become an
        agnostic `Egress Watcher` that invokes `check_health()` periodically.
1636
1637 This compartmentalization transforms `nullexit` into a universal, censorship-resistant gateway where the entire
        egress layer can be swapped by changing a single `.env` variable and running `. /toggle.sh --restart`.
1638
1639 ---
1640
1641 ## 25. Incident Post-Mortem: WARP Watcher Race vs Post-Wake (July 10, 2026)
1642
1643 ### Symptom
1644 After switching Wi-Fi networks, the host IP appeared as the raw ISP IP (`132.x.x.x`) instead of a Cloudflare/
        WARP IP. The gateway appeared to complete post-wake recovery successfully in the logs, but nuclear shutdown
        fired immediately after and killed everything.
1645
1646 ### Root Cause
1647 A fatal race condition between two concurrent processes:
1648
1649 1. **Wi-Fi roam** `watcher.sh` fires `recover.sh --post-wake`
1650 2. **Post-wake** finds the `warp` container missing/dead calls `docker compose up -d --force-recreate` (~35
        seconds)
1651 3. **In parallel**, the WARP Watcher (running as a child of `toggle.sh`) polls the warp container every 5s
        during failures
1652 4. While the container is being recreated it returns `warp=off` for those 35 seconds 7 consecutive failures
        **WARP SHUTDOWN fires**, calling nuclear `recover.sh`
1653 5. Nuclear `recover.sh` tears down Tailscale, resets DNS to 1.1.1.1, stops all containers **gateway dead, IP
        exposed**
1654 6. The post-wake `docker compose up` finishes at ~35s mark, container healthy but the gateway is already gone
1655
1656 The evidence in `output.log`:
1657 ```
1658 [2026-07-10T06:07:27Z] NET: ... bash recover.sh --post-wake
1659 [2026-07-10T06:07:47Z] WARP DOWN cdn-cgi/trace reports warp=off watcher starts counting
1660 ...
1661 add net 0.0.0.0: gateway utun5 post-wake successfully re-routes at 02:08:22
1662 ...
1663 [2026-07-10T06:09:04Z] WARP SHUTDOWN 6 consecutive failures watcher fires nuclear
1664 ```
1665
1666 ### Fix (July 10, 2026)
1667 Added a **WARP Watcher inhibit marker** (`/tmp/nullexit-warp-inhibit.marker`):
1668
1669 - `recover.sh --post-wake` writes the marker at startup **before** any container operations
1670 - The WARP Watcher loop checks for the marker at the top of every iteration; if present it resets `consec_off
        =0` and sleeps 5s (skips the poll entirely)
1671 - `recover.sh` removes the marker at the very end (on both success and failure via `trap cleanup_inhibit EXIT`)
1672 - This guarantees the watcher never counts failures during the intentional container-down window of a post-wake
        force-recreate
1673
1674 The marker file path is hardcoded to `/tmp/nullexit-warp-inhibit.marker` in both `toggle.sh` and `recover.sh`.
1675

```

```

1676 ## 26. Incident Post-Mortem: Startup Pre-Flight Check Tailscale Race (July 10, 2026)
1677
1678 ### Symptom
1679 Immediately after startup via `toggle.sh`, the pre-flight connectivity check `[1/3] Gateway reachable via
    Tailscale` returned `FAIL`, forcing `toggle.sh` to skip the full exit-node activation and fall back to
    SOCKS5/local DNS proxy mode.
1680
1681 ### Root Cause
1682 A timing race condition during startup. In `toggle.sh`, the host joins the mesh using `tailscale up --reset` (
    Phase A) right before performing the pre-flight check. Because `tailscale up` resets and reconfigures the
    host's `tailscaled` daemon, the local daemon must fetch the latest netmap asynchronously from the
    coordination servers.
1683 This network map propagation and route establishment takes a few seconds. Running `tailscale ping` immediately
    after `tailscale up` returns (with no delay) causes the check to fail because the local daemon has not yet
    discovered the gateway node `100.100.21.8`.
1684
1685 ### Fix (July 10, 2026)
1686 Upgraded pre-flight Check 1 in both `toggle.sh` and `scripts/toggle-linux.sh` to use a 5-attempt retry loop
    with a 1-second delay between checks. This allows up to 5 seconds for local tailnet routing paths to settle
    , ensuring the gateway is correctly reached and a pong is received before deciding whether to enable exit-
    node routing.
1687
1688 ## 27. Incident Post-Mortem: Concurrency Collision between Nuclear Teardown and Post-Wake (July 10, 2026)
1689
1690 ### Symptom
1691 When the background WARP Watcher triggers a nuclear shutdown (due to a real or simulated tunnel failure), the
    gateway is completely torn down. However, the network configuration changes made during the teardown
    trigger a `State:/Network/Global/IPv4` network change event. The LaunchAgent-based network watcher (`
    watcher.sh`) intercepts this event and concurrently spawns `recover.sh --post-wake` to heal/restore the
    gateway. This causes `docker compose down` (from nuclear teardown) and `docker compose up` (from post-wake)
    to run in parallel, resulting in container errors like `dependency failed to start: container warp exited
    (0)` and the gateway becoming wedged.
1692
1693 ### Root Cause
1694 A lack of coordination/locking between the lifecycle control scripts. While `toggle.sh` used `/tmp/nullexit-
    toggle.lock` to prevent concurrent `toggle.sh` runs, `recover.sh` (both nuclear and `--post-wake` modes)
    did not utilize or check any lock files.
1695
1696 ### Fix (July 10, 2026)
1697 Implemented a shared lifecycle lock file (`/tmp/nullexit-toggle.lock`) across both scripts:
1698 - **`recover.sh` (all modes)**: Checks `/tmp/nullexit-toggle.lock` at startup. If the lock is held by another
    running lifecycle script (`toggle.sh` or `recover.sh`):
1699   - In `--post-wake` mode, it logs a skip message to `output.log` and exits cleanly (`exit 0`). This safely
    prevents the auto-recovery agent from recreating containers during a teardown.
1700   - In nuclear recovery mode, it exits with an error.
1701 - **`recover.sh` Lock Cleanup**: Cleans up the lock file upon completion using `trap cleanup_recover EXIT` (
    which checks if the lock file belongs to the current PID).
1702 - **`toggle.sh` update**: The lock check in `toggle.sh` was updated to look for both `toggle.sh` and `recover.
    sh` in the running processes to enforce proper exclusion.
1703
1704 ## 28. Incident Post-Mortem: Infinite Auto-Recovery Loop after WARP Watcher Nuclear Shutdown (July 10, 2026)
1705
1706 ### Symptom
1707 When the background WARP Watcher triggered a nuclear shutdown (due to a tunnel outage), it successfully
    executed `recover.sh` (nuclear). However, immediately after the teardown completed, the system started
    spawning `recover.sh --post-wake` and rebuilding the containers again. This created an infinite loop of
    tearing down and auto-restarting the gateway, keeping the host's internet route permanently wedged.
1708
1709 ### Root Cause
1710 An active-state marker file leak. The LaunchAgent-based network watcher (`watcher.sh`) only fires `recover.sh
    --post-wake` when `/tmp/nullexit-gateway-active.marker` is present on disk. While `toggle.sh` (STOP path)
    correctly deleted this marker, the nuclear `recover.sh` script did not. When the WARP Watcher ran `recover.
    sh` (nuclear), the containers were stopped but the active-state marker was left on disk. The network
    changes from the teardown then triggered `watcher.sh`, which saw the marker, believed the gateway was still
    supposed to be active, and triggered `--post-wake` to start it back up.
1711
1712 ### Fix (July 10, 2026)
1713 Modified the start of `recover.sh` to explicitly delete `/tmp/nullexit-gateway-active.marker` if `POST_WAKE` is
    `false` (nuclear mode). This ensures that any subsequent network configuration changes made during the
    teardown are correctly ignored by `watcher.sh` because the marker file is gone.
1714
1715 ## 29. Incident Post-Mortem: Pre-Flight Check False Negatives due to VPN Firewall STUN Blocks (July 10, 2026)
1716
1717 ### Symptom
1718 Immediately after a cold boot or full restart of the gateway, the pre-flight connectivity check `[1/3] Gateway
    reachable via Tailscale` failed, causing `toggle.sh` to skip the full exit-node configuration and fall back
    to SOCKS5/local DNS proxy mode. However, manually running `tailscale ping 100.100.21.8` immediately after
    startup succeeded in ms.
1719
1720 ### Root Cause

```

1721 An asynchronous Tailscale handshake delay caused by firewall restrictions. Inside the container network namespace, the `warp` container (gluetun) implements a strict firewall that blocks all outbound UDP traffic to the public internet except to the designated WireGuard endpoint. Because of this, the `tailscale` container's STUN requests to public STUN servers are blocked, causing the local daemon to report `UDP is blocked, trying HTTPS`.

1722 Without UDP STUN capability, Tailscale is forced to fall back to a slower DERP-assisted handshake (relayed via HTTPS/TCP). This negotiation and direct-path punch-through takes roughly 30 to 35 seconds to establish on cold boots (after a full `tailscale up --reset`). Since the pre-flight check only waited 15 seconds (15 attempts with 1-second sleeps), the check timed out and aborted before the path completed.

1723

1724 **### Fix (July 10, 2026)**

1725 Increased the pre-flight ping retry limit from 15 to 45 attempts (45 seconds) in both `toggle.sh` and `scripts/toggle-linux.sh`. This provides sufficient time for the DERP-assisted handshake to complete on cold boots, while still passing immediately (in 1-2 seconds) on warm starts or once the path is established.

1726

1727 **## 30. Incident Post-Mortem: Docker Image Build DNS Failures on Immediate Restart (July 10, 2026)**

1728

1729 **### Symptom**

1730 When executing `toggle.sh --restart`, the script successfully stops the gateway but then immediately fails during startup during `docker compose build` with DNS resolution errors:

1731 `failed to do request: Head "https://registry-1.docker.io/...": dial tcp: lookup registry-1.docker.io on 192.168.5.1:53: no such host`.

1732

1733 **### Root Cause**

1734 A race condition between physical link negotiation and virtual machine startup. The STOP path of the restart command invokes `cleanup\_network\_state` which power-cycles the host's physical Wi-Fi interface (`en0`) to flush stale routing states. Immediately after, `toggle.sh` starts the gateway boot sequence. Because `toggle.sh` did not verify the status of the physical interface, it proceeded to boot Colima and build Docker containers while `en0` was still negotiating a DHCP lease. Consequently, the host (and by extension, the VM bridge) had no active internet gateway/DNS path, causing Docker's registry check to fail.

1735

1736 **### Fix (July 10, 2026)**

1737 1. Created a unified `wait\_for\_dhcp\_settle` helper function in `scripts/common.sh` that polls `ipconfig getpacket` and `ifconfig` until a valid IP and router are assigned. To handle typical macOS Wi-Fi reassociation and DHCP negotiation latency (which commonly takes 15-20 seconds after a power-cycle), this loop was configured to poll up to 60 times (30 seconds total).

1738 2. Injected a call to `wait\_for\_dhcp\_settle` at the beginning of the `START` action in `toggle.sh` (right before checking Colima status) to block container builds until the host's physical network is online.

1739 3. Refactored `recover.sh` to reuse the unified `wait\_for\_dhcp\_settle` function.

1740

1741 **## 31. Incident Post-Mortem: PF Kill-Switch Bricks Host Internet in SOCKS5 Fallback Mode (July 10, 2026)**

1742

1743 **### Symptom**

1744 When the pre-flight checks fail (e.g. because host Tailscale was unauthenticated/logged out), the script falls back to SOCKS5 proxy mode but the host immediately loses all internet connectivity and tailscaled reports `You are logged out... connect: no route to host`. Even running `sudo tailscale up` to log in fails because the connection to the control plane times out.

1745

1746 **### Root Cause**

1747 An architectural mismatch. The macOS Packet Filter (PF) Kill-Switch works at the IP level: it blocks all outbound traffic on the physical interface (`en0`) except to explicitly whitelisted VPN endpoints, expecting all other traffic to go through the virtual VPN interface (`utun\*`).

1748 In SOCKS5 fallback mode, the exit-node (`utun\*`) is NOT enabled; instead, only specific applications (like browsers) use the localhost SOCKS5 proxy, while all other host traffic (including the `tailscaled` daemon's UDP packets) continues to go through `en0`. Because the script still enabled the PF Kill-Switch during SOCKS5 fallback, it blocked all non-SOCKS5 traffic on `en0`, completely bricking the host's internet and locking the Tailscale daemon out of the control plane.

1749

1750 **### Fix (July 10, 2026)**

1751 Modified `toggle.sh` to remove the `enable\_killswitch` call from the SOCKS5 fallback path. The firewall kill-switch is now strictly reserved for exit-node VPN mode, ensuring SOCKS5 fallback leaves the host's direct internet route open for system daemons to authenticate.

1752

1753 **## 32. Incident Post-Mortem: Host Tailscale Logouts due to Aggressive reset Flags (July 10, 2026)**

1754

1755 **### Symptom**

1756 Every time the user runs `toggle.sh` or `recover.sh` which disconnects/reconnects the host client, the host client randomly gets logged out, forcing them to re-run `sudo tailscale up` to authenticate.

1757

1758 **### Root Cause**

1759 An overly aggressive configuration reset. The host-side `tailscale up` calls (used in `disconnect\_tailscale\_host` and the startup/enable exit node paths) all passed the `--reset` flag. On macOS, `--reset` resets the entire configuration state and authentication token cache of the daemon. Since the script does not pass a `TS\_AUTHKEY` to the host's commands (which are meant for the user's private interactive client), the `--reset` flag effectively logged the user out of their tailnet, requiring interactive re-login.

1760

1761 **### Fix (July 10, 2026)**

1762 Removed the `--reset` flag from all host-side `tailscale up` invocations in `toggle.sh` and `scripts/common.sh`. This ensures that exit-node state transitions (enabling/disabling) are handled safely while preserving the host client's authenticated session.

```

1763 ## 33. Incident Post-Mortem: Discarded Docker Exec Errors in logs (July 10, 2026)
1764
1765 ### Symptom
1766 When the `warp` container is stopped, dead, or failing to connect to its endpoints, the `toggle.sh` and `
1767 recover.sh` connectivity health checks fail, but no details are printed to `output.log`. The logs only show
generic `FAIL` outputs, making it hard to see if the failure was a network timeout, Docker daemon error,
or a missing binary.
1768
1769 ### Root Cause
1770 Overly broad error silencing. The `docker compose exec` commands in the health checks (Check 3 in `toggle.sh`
and the traffic check in `recover.sh`) both redirected stderr to `/dev/null` (`&>/dev/null` and `2>/dev/
null` respectively), completely throwing away any diagnostic error messages produced by Docker or `wget`.
1771
1772 ### Fix (July 10, 2026)
1773 Redirected stderr of the `docker compose exec` check commands to `output.log` (`2>> output.log`). This ensures
that any command execution failures or connection timeout details are logged.
1774
1775 ## 34. Architectural Design: How nullexit Mimics Commercial VPNs during Roaming
1776
1777 The `nullexit` roaming/network recovery subsystem replicates the exact engineering mechanics used by commercial
, premium VPN clients (like Mullvad or NordVPN) to transition across Wi-Fi networks safely and seamlessly.
1778
1779 ### The 4-Step Transition Cycle
1780 When your Mac disconnects from one Wi-Fi and associates with another, `nullexit` coordinates the following
events:
1781
1782 1. Firewall Lock (Kill-Switch): The macOS Packet Filter (PF) anchor `com.apple/nullexit` remains locked on
the physical interface (`en0`), blocking all direct outbound IPv4 and IPv6 traffic. This guarantees that
your real ISP IP or unencrypted DNS requests never leak onto the new network while the connection is
rebuilding.
1783 2. System Event Interception: A launchd LaunchAgent (`watcher.sh`) listens to the macOS System
Configuration framework for the `State:/Network/Global/IPv4` key changes, spawning `recover.sh --post-wake`
in under 500ms when a change is detected.
1784 3. Bypass Routing: The script dynamically resolves the new physical network's IP gateway (e.g.
`192.168.137.1`), cleans up stale bypass routes from the previous network, and adds static bypass routes
pointing to Cloudflare's WARP endpoints and the Tailscale control plane directly via the new gateway. This
allows the VPN client to negotiate its handshake outside of the tunnel.
1785 4. Hole-Punching / NAT Re-negotiation: The script runs a target check on the `warp` container's tunnel. If
the WireGuard connection is wedged by the network switch, it force-recreates the container to establish a
fresh NAT binding to Cloudflare, restoring connection in ~15 seconds.
1786
1787 ## 35. Incident Post-Mortem: macOS Tailscale Flag Requirement Abort (July 10, 2026)
1788
1789 ### Symptom
1790 When the `toggle.sh` stop trap or the roaming watcher triggered, the gateway failed to shut down or disconnect
properly. The `output.log` showed the error: `Error: changing settings via 'tailscale up' requires
mentioning all non-default flags. To proceed, either re-run your command with --reset...`
1791
1792 ### Root Cause
1793 In a previous attempt to stop host-side logouts (Section 32), we removed the `--reset` flag from all `tailscale
up` invocations. However, if the Tailscale daemon on macOS is already running with non-default flags (like
`--ssh` or `--accept-routes`), invoking `tailscale up` with only a subset of flags (like `--exit-node=`)
triggers a strict safety check in the Tailscale CLI that aborts the command completely. Because the command
aborted, exit-node routing remained stuck.
1794
1795 ### Fix (July 10, 2026)
1796 Migrated all specific preference changes in `toggle.sh` and `scripts/common.sh` from `tailscale up` to a two-
step sequence:
1797 1. `tailscale up` (with no flags) to safely ensure the interface is brought up using the current authenticated
state.
1798 2. `tailscale set --exit-node=...` to modify the specific target preference cleanly without triggering the
missing flags error or requiring `--reset`.
1799
1800 ## 36. Incident Post-Mortem: macOS SNAT Endpoint Poisoning & The P2P Blackhole (July 10, 2026)
1801
1802 ### Symptom
1803 When a Tailscale client (like an S24) connected to the exact same Wi-Fi network as the macOS gateway, it
completely lost internet connectivity. However, if the S24 connected to a Windows PC's hotspot on the same
network, the tunnel worked flawlessly.
1804
1805 ### Root Cause
1806 Claude's critique correctly dismantled the port collision theory: the gateway daemon wasn't failing to bind,
and macOS wasn't load-balancing ports. The real culprit was macOS SNAT Source Port Mangling poisoning
Tailscale's path discovery.
1807
1808 Here is the exact step-by-step of the "infinite loop" blackhole:
1809 1. The P2P Handshake: The S24 and the Mac are on the same Wi-Fi (`192.168.1.x`). The S24 discovers the Mac's
local IP and sends a UDP hole-punch packet to `192.168.1.5:41641`. Colima successfully forwards this to
the gateway container.

```

```

1810 2. **The Bypass Rule:** The gateway replies. Because `routing-fix.sh` forces all Tailscale UDP traffic out `
eth0` to bypass WARP, the reply goes to the macOS host.
1811 3. **macOS SNAT Mangling:** macOS routes the reply out `en0` to the S24. Because the packet crosses from the
Colima VM bridge to the Wi-Fi interface, macOS applies Source NAT (SNAT). Critically, macOS **randomizes
the source port** (e.g., changes it from `41641` to `58291`).
1812 4. **Endpoint Poisoning:** The S24 receives the authenticated Tailscale packet from `192.168.1.5:58291`.
Tailscale's `magicsock` is designed to handle NAT dynamically, so it assumes the Mac has roamed to port
`58291`! The S24 updates its endpoint and starts sending all its encrypted internet traffic to
`192.168.1.5:58291`.
1813 5. **The Blackhole:** macOS has no port forward for `58291`. It drops 100% of the S24's outbound data packets.
The connection is completely dead.
1814
1815 ### Why did the Hotspot work?
1816 When the S24 connected to the PC hotspot, it was behind the PC's NAT. The S24 sent the packet, PC NATted it,
and the Mac replied (mangled to `58291`). When the reply hit the PC, the PC's strict NAT dropped it because
the source port (`58291`) didn't match the destination port it originally sent to (`41641`). Because the
mangled packet was dropped, the S24 never poisoned its endpoint! It safely gave up on P2P, fell back to the
100% reliable TCP DERP relay, and the internet worked perfectly.
1817
1818 ### Fix & Resolution (July 10, 2026)
1819
1820 **Phase 1 Port disambiguation:** Changed the gateway container's Tailscale listen port from the default
`41641` to `41642` across `docker-compose.yml`, `post-rules.txt`, and `routing-fix.sh`. This prevents any
bind conflict with the macOS host's native Tailscale daemon.
1821
1822 **Phase 2 RFC1918 DROP rules:** Updated `routing-fix.sh`'s `add_tailscale_p2p_bypass()` to explicitly drop all
outgoing Tailscale UDP packets destined for private subnets (`192.168.0.0/16`, `10.0.0.0/8`,
`172.16.0.0/12`) when `TAILSCALE_ALLOW_LAN_P2P=false`. This surgically kills the local P2P hole-punch
attempt before it can trigger the macOS gvproxy SNAT mangling. The S24 receives a clean failure and
immediately locks onto the public DERP relay with 0% packet loss.
1823
1824 **Phase 3 Automatic network detection:** Because `TAILSCALE_ALLOW_LAN_P2P=false` breaks P2P on trusted
hotspots that genuinely don't have AP isolation, a `.env` toggle and a full auto-detection system were
implemented:
1825
1826 - **`.env` toggle:** `TAILSCALE_ALLOW_LAN_P2P=true/false` static fallback, used only if auto-detection cannot
run.
1827 - **`watcher.sh` auto-detection (`detect_lan_p2p_mode`):** On every network change, `watcher.sh` runs two
checks:
1828 1. **WPA2-Enterprise detection:** `airport -I` is used to read the current Wi-Fi `link auth` field. If the
auth type does not contain `psk` (e.g., it is plain `wpa2` = 802.1x), the network is classified as
enterprise and P2P is forced off.
1829 2. **AP Isolation probe:** For WPA2-Personal networks, a short `ping` is sent to the default gateway. If the
gateway responds (it always should on a real home/hotspot network), P2P is allowed. If not, AP isolation is
inferred and P2P is forced off.
1830 - The result (`true` or `false`) is written to `.lan_p2p_detected` in the repo root.
1831 - **`routing-fix.sh` dynamic reading:** `add_tailscale_p2p_bypass()` re-reads `.lan_p2p_detected` on every 30-
second loop iteration and atomically adds or removes the RFC1918 DROP rules. **No container restart is
needed when switching networks.**
1832
1833 ### Known Unknowns (Pending Verification)
1834 - The gvproxy SNAT source port mangling theory is mechanistically sound and backed by observed behavior (
gvproxy's UDP proxy changelog has documented edge cases in this exact code path), but has not yet been
confirmed via `tcpdump -i en0 udp portrange 40000-60000` while triggering a P2P handshake from the phone. A
packet capture cross-referenced with `tailscale ping` endpoint reports on the phone side would confirm the
theory definitively.
1835 - Claude's recommendation: run `tcpdump -i en0 udp port 41642 or portrange 40000-60000` on the Mac while
triggering the handshake from the phone to see whether the reply leaves with a different source port than
`41642`.
1836 - **Client-Side VPN Cycling (Roaming Recovery):** An observation (July 10, 2026) suggests that when a hung DNS
probe state is triggered by roaming between Access Points (APs) on a WPA2-Enterprise network, simply
toggling the VPN connection (Tailscale) off and on locally on the client phone immediately resolves the
issue. You do not need to cycle the phone's physical Wi-Fi. This reinforces the theory that roaming on
Enterprise networks leaves a stale/poisoned P2P endpoint in the phone's Tailscale magicsock state, which
gets instantly flushed upon a local VPN restart. (Status: Possible but not formally verified).
1837
1838 ## 37. TODO
1839
1840 * **Verify Wi-Fi Roaming:** Investigate and test whether switching Wi-Fi networks now successfully and smoothly
recovers the gateway end-to-end without any tailscale flag errors or dead tunnels. (Pending user
verification).
1841 * **Fix Diagnostic Script Check 5/8:** ~Investigate why `diagnose-host-leak.sh` check 5/8 (`Host default route
`) sometimes reports "default route goes via physical Wi-Fi Tailscale route assertion failed" on macOS
even though the `warp=on` egress checks confirm that traffic is successfully tunneling.~ **Closed** fixed
by switching check 5 from `netstat -rn | grep default` to `route -n get 1.1.1.1`. macOS Tailscale uses
interface-scoped routes and does not replace the DHCP default route entry. See 36 Known Unknowns.
1842 * **Expose Tor Control Port (9051):** ~Consider mapping the Tor Control port to localhost and adding `nyx` (
the Tor terminal status monitor) to allow users to inspect their entry, middle, and exit node IPs in real-
time.~ **Closed** mapped the Tor Control Port to a procedurally generated, localhost-bound port `
TOR_CONTROL_PORT`, configured `ControlPort` and disabled cookie authentication in `docker/tor/entrypoint.sh
`, and integrated it into the `/sweep` verification protocol.

```

```

1843 * **Host-Only Tor Mode (Pluggable Egress):** Implement an `EGRESS_TYPE=tor_host_only` mode for users in heavily
      censored (DPI-restricted) environments. This mode would skip booting `warp` and `tailscale`, boot `
      adguardhome` and `tor` (using Telegram `obfs4` bridges), set AdGuard's upstream to Tor's `DNSPort`, and use
      the `pf` kill-switch to force all host Mac traffic strictly through the Tor network. This provides a 100%
      free, DPI-resistant evasion path without requiring an external VPS.
1844 * **The Technical Realities (What You Need to Watch Out For):**
1845 * **The UDP Problem:** Tor only transports TCP traffic. It does not support UDP (aside from DNS requests
      passed directly to its `DNSPort`). macOS is incredibly chatty over UDP (FaceTime, mDNS, QUIC protocols). Your
      pf rules defined in `scripts/pf.conf` must be configured to aggressively and silently DROP all host UDP
      traffic (except for local DNS queries to AdGuard). If you don't drop it cleanly, macOS services might hang
      indefinitely while waiting for timeouts.
1846 * **The "Daily Driver" Friction:** Routing a whole host OS through Tor is brutal on performance. Background
      daemons (iCloud sync, macOS software updates, Spotlight indexing) will blindly attempt to download
      gigabytes of data over Tor, which could saturate the circuits or time out completely.
1847 * **Bridge Rot:** State firewalls actively hunt and block obfs4 bridges. When a user's bridges burn out,
      they will lose all internet access due to the pf kill-switch. You will need to ensure the user has a
      frictionless way to update the `tor-bridges.txt` file and restart the Tor container without exposing their
      IP.
1848 * **Exit Node Discrimination:** Because all traffic will exit from known Tor nodes, the user will face an
      onslaught of Cloudflare CAPTCHAs, and many banking or streaming services will flat-out reject the
      connections.
1849
1850 ## 38. Observation: AdGuard Returns Two Different Sinkhole IPs (Unverified)
1851
1852 > **Status: Observed but not formally verified. Needs a deeper audit of the rule-compiler sources to confirm.**
1853
1854 ### Observation (July 10, 2026)
1855 When querying AdGuard Home for blocked ad domains, some domains resolve to `0.0.0.0` and others resolve to
      `127.0.0.1`. Both are blocked, both cause connection failure, but the different responses were unexpected.
1856
1857 ...
1858 doubleclick.net          0.0.0.0    (blocked)
1859 ads.google.com           127.0.0.1  (blocked)
1860 pagead2.googlesyndication.com 0.0.0.0    (blocked)
1861 scorecardresearch.com    127.0.0.1  (blocked)
1862 ...
1863
1864 ### Likely Explanation (Unverified)
1865 There are three historical schools of thought on the "correct" DNS sinkhole response, and AdGuard faithfully
      preserves whichever the source blocklist used:
1866
1867 | Sinkhole IP | Convention | Origin |
1868 |---|---|---|
1869 | `0.0.0.0` | AdBlock-style lists (`\|\|domain`) | Modern DNS-level blocking; AdGuard's native format. Fails
      fast OS doesn't attempt a TCP connection. Works for both IPv4 and IPv6 without a separate rule. |
1870 | `127.0.0.1` | Hosts-file-style lists (`127.0.0.1 domain`) | 90s-era `/etc/hosts` tradition. Steven Black's
      hosts list (500k+ domains) still uses this. |
1871 | `NXDOMAIN` | Purist / Pi-hole default | Returns "domain doesn't exist." Semantically most accurate but
      requires browsers to handle NXDOMAIN gracefully. |
1872
1873 The dual-response behavior in this project is because `rule-compiler` pulls from both AdBlock-format sources (
      Hagezi, uBlock origin lists) and hosts-format sources (Spamhaus DROP, Steven Black, Feodo Tracker), and
      AdGuard returns whatever sinkhole IP the matching rule specifies.
1874
1875 ### To Verify
1876 Run `docker compose exec tailscale cat /etc/hosts` to confirm no hosts-file rules are being injected at the
      container level, and check which specific AdGuard rule matched each domain via the AdGuard query log at `
      http://100.100.21.8:3000/#logs`.
1877
1878 ## 39. Observation: AGY CLI Appears to Generate Native macOS Notification Pop-ups (Unverified)
1879
1880 > **Status: Observed but source unverified. macOS security logs ruled out system-level origin.**
1881
1882 ### Observation (July 10, 2026)
1883 While running a DNS blocklist verification via the Antigravity CLI (`agy`), a command was proposed that
      included `malware.wicar.org` (a known-safe DNS blocklist test domain) in a `dig` loop. Shortly after, a
      macOS-style notification popup appeared describing a "malicious" or "blocked" event. The AGY CLI terminal
      session then closed.
1884
1885 ### Investigation
1886 - **macOS XProtect / Gatekeeper logs:** Empty. Zero events in the 30-minute window around the incident.
1887 - **macOS Endpoint Security / System Policy logs:** Empty.
1888 - **Broad `log show` search for "malicious", "malware", "blocked":** Empty.
1889 - **Conclusion:** macOS itself did not generate the notification. No system security subsystem fired.
1890
1891 ### Likely Explanation (Unverified)
1892 The notification appears to have originated from **AGY CLI's own internal safety system**. AGY CLI is a native
      macOS application (not just a terminal process) and has access to the macOS Notification Center API (
      `UNUserNotificationCenter`). When its guardrails detected a domain containing the word "malware" (
      `malware.wicar.org`) in a proposed shell command, it likely:
1893 1. Blocked the command from executing

```



```

1894 2. Posted a native macOS-style notification via the Notification Center
1895
1896 This is surprising behavior for a CLI tool most terminal programs don't send GUI notifications but AGY CLI
1897 appears to bridge both worlds: it runs in a terminal but is packaged as a native app with full system API
1898 access. The notification would be indistinguishable from a system security alert at a glance.
1899
1900 ### Notes
1901 - `malware.wicar.org` is the DNS/AV equivalent of the EICAR test file a deliberately benign domain that
1902 triggers blocklist/AV systems to verify they work. It was included in the test set to confirm the AdGuard
1903 blocklist was catching it. AGY's safety system is not aware of this distinction.
1904 - For future DNS blocklist testing, use only ad/tracking domains (e.g., `doubleclick.net`, `adnxs.com`) to
1905 avoid triggering AGY's guardrails.
1906
1907 ## 40. Apple Silently Removed the `airport` Binary in macOS Sonoma 14.4 (March 2024)
1908
1909 ### What happened
1910 `watcher.sh`'s `detect_lan_p2p_mode()` was written to use the `airport` CLI tool at:
1911 ```
1912 /System/Library/PrivateFrameworks/Apple80211.framework/Versions/Current/Resources/airport
1913 ```
1914 This path returned `exit 127: no such file or directory`. The function silently fell back to `reason="no-wifi"`
1915 even while the Mac was actively connected to WPA2 Enterprise Wi-Fi.
1916
1917 ### Why Apple removed it
1918 `airport` was never a real public binary it lived inside a private framework and was always technically
1919 unsupported. Apple deprecated it quietly years ago and finally removed it in macOS Sonoma 14.4 (March
1920 2024). The removal is part of a broader privacy clampdown on Wi-Fi metadata:
1921 - Starting in macOS 14.5, BSSIDs and other Wi-Fi association details are redacted (`<redacted>`) in
1922 most tools, including the official replacement `wdutil`
1923 - Apple's official recommendation is to use the CoreWLAN framework (Swift/Objective-C) with proper
1924 entitlements useless for a bash script
1925 - The official CLI replacement `wdutil` requires `sudo` for most useful output also useless for a background
1926 LaunchAgent
1927
1928 ### Fix
1929 Replaced `airport -I | awk '/link auth/'` with:
1930 ```bash
1931 system_profiler SPAirPortDataType | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF
1932 ; exit}'
1933 ```
1934 This returns human-readable strings like `WPA2 Enterprise`, `WPA2 Personal`, or `None` for the currently
1935 associated network without `sudo`, and without any removed private binary.
1936
1937 **Confirmed working output on this machine:**
1938 ```
1939 P2P detect: security='Enterprise' allow=false (reason: 802.1x-enterprise)
1940 ```
1941
1942 ### Risk: `system_profiler` may not survive forever either
1943 `system_profiler SPAirPortDataType` still works as of macOS Sequoia (15.x) without sudo and without redaction.
1944 However, given Apple's trajectory on Wi-Fi privacy, this could be locked down in a future release. If `
1945 security` comes back empty on a future macOS version while the Mac is actively on Wi-Fi, the function will
1946 silently fall back to `allow=false` (safe default), but the detection will be broken again. The long-term
1947 fix would be a small Swift helper using CoreWLAN that we compile once and bundle in the repo.
1948
1949 ## 41. Fixed: Watcher.sh Suppressed Exit Code Bug
1950
1951 ### Observation
1952 The `scripts/watcher.sh` script is a long-running daemon that invokes `recover.sh --post-wake` when it detects
1953 system wake or network roam events. Previously, it contained the following bug:
1954 ```bash
1955 bash "$RECOVER" --post-wake
1956 local exit_code=$?
1957 ...
1958 return $exit_code || true
1959 ```
1960 The `|| true` mistakenly swallowed the actual `$exit_code` returned by `recover.sh`, causing `run_recover()` to
1961 always return `0` (success), masking any underlying failures in the wake recovery process.
1962
1963 ### Fix
1964 The `|| true` statement was removed from the return line:
1965 ```bash
1966 return $exit_code
1967 ```
1968
1969 Since `watcher.sh` explicitly omits `set -e` (to prevent pipe breakages from killing the daemon), removing `||
1970 true` safely allows the underlying exit code to propagate without risking daemon termination. The watcher
1971 daemon was then reloaded (`launchctl kickstart -k`) to pick up the script changes in memory.
1972
1973 ## 42. Network Readiness Race Condition on `--restart` (Fixed)
1974
1975 **Problem:** When running `./toggle.sh --restart`, the script tears down the gateway and immediately begins the
1976 startup sequence. On systems with Wi-Fi (or any wireless interface), the OS takes a moment to re-associate

```

```

    with the network after the teardown's DNS/routing changes. If the startup sequence ran before the OS had a
    default route, `get_active_service` would return nothing, routing bypass targets would be wrong, and DNS/
    WARP setup would silently fail resulting in a partially configured gateway that self-heals only once the `
    watcher.sh` re-runs.
1953
1954 **Root cause:** The original code had no gate before the first network-dependent call (`ACTIVE_SERVICE=$(
    get_active_service)` at the top of the start branch). This was a pure timing race.
1955
1956 **First fix (Wi-Fi specific):** A polling loop on `ipconfig getifaddr en0` was added to wait for an IPv4
    address on `en0` before proceeding. This worked but was brittle it only handled Wi-Fi and would fail for
    Ethernet, USB-C adapters, or iPhone USB tethering.
1957
1958 **Generalized fix:** Replaced the `en0` check with `route get default | grep 'gateway:'`. This detects a valid
    default gateway on any interface, covering all connection types. The loop polls every 2 seconds with a
    60-second hard timeout (then proceeds with a warning rather than hanging forever). The output prints the
    detected gateway IP so failures are easy to diagnose in logs.
1959
1960 ```bash
1961 # In toggle.sh, before ACTIVE_SERVICE=$(get_active_service)
1962 while true; do
1963     if route get default 2>/dev/null | grep -q 'gateway: '; then
1964         _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
1965         echo " Network ready (default gateway: $_gw)."
1966         break
1967     fi
1968     ...
1969 done
1970 ```
1971
1972 **Why not ping?** Pinging an IP (e.g. `1.1.1.1`) during restart is unreliable because DNS may still be hijacked
    from the previous session, and the exit node routing may not yet be cleared, causing the ping to loop back
    through the tunnel. `route get default` is a pure local kernel query no packets sent.
1973
1974 ## 43. Low Gateway Throughput (DERP Relay & MTU Fragmentation) Fixed
1975
1976 **Problem:** The host Mac's internet throughput through the gateway dropped significantly (e.g., from 34Mbps
    raw to 2.1Mbps via gateway). This was caused by a combination of two bottlenecks:
1977
1978 **1. Tailscale DERP Relay Bottleneck:**
1979 Because the host Mac and the Docker container operate behind the same public IP, standard hole-punching for
    Tailscale P2P failed, forcing traffic through Tailscale's NYC DERP relay. Normally, `routing-fix.sh`
    explicitly drops container Tailscale UDP packets destined for local subnets (when `TAILSCALE_ALLOW_LAN_P2P=
    false` in `.env` or auto-detected). We nuke these P2P connections on purpose to prevent the severe macOS `
    gvproxy` SNAT endpoint poisoning issue we faced earlier (see `devref.md 36`). However, this strict policy
    unintentionally blocked direct communication between the container and the Mac host itself via the Docker
    bridge network.
1980
1981 **Fix:** Tailscale's control plane registers the Mac host using its physical IP (e.g., `172.17.52.223`), not
    the Docker bridge IP. If this physical IP falls within the dropped local subnets (like `172.16.0.0/12`),
    the P2P handshake is dropped. To solve this natively, `scripts/common.sh` now features a `write_host_ips()`
    function that actively grabs all physical IPv4 addresses on the Mac and writes them to `.host_ips`. `
    routing-fix.sh` dynamically reads this file every 30 seconds and injects priority `RETURN` rules for those
    exact physical IPs. This guarantees direct container-to-host P2P over the local bridge (bypassing DERP)
    while continuing to block external LAN devices (like phones) to prevent the 36 SNAT poisoning bug.
1982
1983 **2. MTU Double-Encapsulation Fragmentation:**
1984 The host's Tailscale interface (`utun5`) defaulted to an MTU of 1280. The WARP tunnel inside the container also
    uses an MTU of 1280. When a full 1280-byte Tailscale packet from the host entered the container and was
    encapsulated by WARP (adding ~60 bytes of overhead), the resulting packet exceeded 1280 bytes, leading to
    severe fragmentation and performance degradation.
1985
1986 **Fix:** In `toggle.sh`'s `setup_exit_node_routing` function, the host's Tailscale `utun` interface MTU is
    explicitly lowered to `1200` (configurable via `HOST_MTU` in `.env`). This ensures that host-originated
    packets can comfortably fit inside the container's WARP tunnel encapsulation without fragmenting.
1987
1988 ## 44. Architectural Limitation: Chrome Remote Desktop Performance (UDP NAT)
1989
1990 ### Observation (July 11, 2026)
1991 When the user connects to their host Mac using Chrome Remote Desktop (CRD) while `nullexit` is active, the
    connection suffers from massive latency, lag, and degraded video quality. AdGuard Home's `blocked.log`
    shows zero blocked DNS requests for Google, WebRTC, or `chromoting` endpoints, indicating this is not a DNS
    sinkhole issue.
1992
1993 ### Root Cause
1994 This is a fundamental limitation of tunneling all host traffic through a strict commercial NAT (Cloudflare WARP
    ).
1995 1. Chrome Remote Desktop attempts to use **STUN (UDP hole-punching)** to establish a lightning-fast, direct
    peer-to-peer WebRTC connection between the client device and the host Mac.
1996 2. Because all non-Tailscale traffic is forcefully routed through the `warp` container, the STUN packets exit
    the network from a Cloudflare IP.

```

1997 3. Cloudflare's enterprise NAT infrastructure strictly drops unsolicited inbound UDP packets. The UDP hole-punch fails.

1998 4. Because a direct P2P connection cannot be established, CRD falls back to **Relay Mode** (TURN), bouncing all video data back and forth through Google's cloud servers instead of sending it directly over the local or mesh network. This relaying introduces massive latency.

1999

2000 **### Workaround / Fix**

2001 This lag is the accepted "tax" for maintaining absolute cryptographic anonymity; the only way to restore CRD speed would be to leak its UDP traffic outside the tunnel (violating zero-trust).

2002 To achieve low-latency remote access, users should bypass CRD entirely and use macOS's built-in **Screen Sharing (VNC)** directly over the Tailscale IP. Tailscale's sophisticated custom DERP/WireGuard architecture successfully UDP hole-punches through strict NATs, providing a direct, encrypted, high-speed connection without relying on external cloud relays.

2003

2004 **## 45. Architectural Decision: Dynamic IP Fetch vs. MagicDNS**

2005

2006 **### Observation**

2007 The ``nullexit`` gateway uses a dynamically fetched IP address (cached in ``gateway_ip``) to configure host routing and the ``pf`` kill-switch, rather than utilizing Tailscale's user-friendly MagicDNS hostname (e.g., ``nullexit-gateway``).

2008

2009 **### Why MagicDNS is Not Used**

2010 1. **The "Chicken and Egg" Bootstrapping Problem:**

2011 macOS's Packet Filter (``pf``) and low-level routing commands (``route add``) operate strictly at Layer 3 and require raw IP addresses. If ``toggle.sh`` or ``recover.sh`` attempted to use ``nullexit-gateway``, the host Mac would need to perform a DNS lookup to resolve it. However, the very first step of the gateway's initialization is to completely hijack the host's DNS and point it **at the gateway itself**. The Mac cannot resolve the gateway's MagicDNS name if it needs the gateway's IP to know where to send the DNS query!

2012 2. **Dynamic Self-Healing (Avoiding Stale Cache):**

2013 To solve the bootstrapping problem without causing bugs if the node is deleted and re-authenticated on the Tailscale Admin Console, ``toggle.sh`` explicitly runs ``docker compose exec ... tailscale ip -4`` during boot to fetch the absolute freshest IP dynamically. It saves this to ``gateway_ip`` so that fast-roaming scripts like ``recover.sh`` can instantly retrieve the routing target without incurring the latency of querying Docker.

2014

2015 **## 46. Tor Container Kill-Switch Inheritance & Fail-Closed Egress Verification**

2016

2017 **### Context**

2018 Unlike the host's SOCKS5 fallback path (whose kill-switch interaction is documented in 31), the ``tor`` container's connection status and fail-closed behavior under VPN tunnel drops is not governed by macOS ``pf`` rules. Instead, it relies on network namespace inheritance inside Docker Compose.

2019

2020 **### Mechanics**

2021 1. **Shared Network Namespace:** The ``tor`` service is configured with ``network_mode: service:warp`` in ``docker-compose.yml``. This shares the network namespace of the ``warp`` (GlueTun) container.

2022 2. **Inherited Egress Rules:** All socket communication within the namespace is bound to the same networking stack. GlueTun manages this stack by redirecting traffic through ``tun0`` (the WireGuard/WARP tunnel) and applying ``iptables`` rules that explicitly drop outbound traffic originating on ``eth0`` (the physical/bridge interface), except for permitted local subnets or the VPN server's endpoint.

2023 3. **Tunnel Fail-Closed:** If the WARP connection drops or the ``tun0`` interface is brought down, the routing table entries for ``tun0`` become invalid or disappear. Because the ``iptables`` rules in the shared namespace continue to drop any outgoing internet traffic attempting to exit via ``eth0``, the ``tor`` container cannot leak raw traffic to the host's underlying networks. Egress fails closed.

2024

2025 **### Verification / Manual Test Case**

2026 To manually verify that the Tor container fails closed and does not leak traffic when the tunnel dies:

2027

2028 1. **Verify active path:** Ensure the gateway is active (``toggle.sh start``) and fetch the Tor port (``$TOR SOCKS_PORT`` in ``/tmp/nullexit-ports.env``). Verify Tor traffic routes correctly:

2029 ````bash`

2030 `curl --socks5-hostname 127.0.0.1:$TOR SOCKS_PORT https://check.torproject.org/api/ip`

2031 `````

2032 `*(This should output a Tor exit node IP).*`

2033

2034 2. **Simulate tunnel drop:** Bring down the virtual interface inside the shared network namespace:

2035 ````bash`

2036 `docker compose exec warp ip link set dev tun0 down`

2037 `````

2038

2039 3. **Re-run the egress check:**

2040 ````bash`

2041 `curl --socks5-hostname 127.0.0.1:$TOR SOCKS_PORT https://check.torproject.org/api/ip`

2042 `````

2043 *** Expected Result:** The request must hang and eventually time out, or immediately fail with a proxy error (e.g., ``curl: (97) SOCKS5: connection failed`` or ``curl: (7) Failed to connect``).

2044 *** Leaked Egress Check:** Check the host's Wi-Fi router or firewall logs. No outbound packets from the Tor container should route directly over the bridge network to the internet.

2045

2046 **## 47. Architectural Decision: Transparent .onion Routing via RFC 2544 Subnet (198.18.0.0/15)**

2047

2048 **### Context**

```

2049 To enable seamless dark web browsing across the entire mesh, the gateway requires a mechanism to dynamically
2050 map `.onion` domains to IP addresses that the host operating system and network layers can route.
2051
2052 ### IP Address Conflict & Solution
2053 1. **Initial Conflict:** Initially, the Tor virtual address network was configured to map `.onion` sites within
the `10.192.0.0/10` range. However, the Colima Docker daemon dynamically allocated `10.200.1.0/24` to the
containers. Because this Docker subnet mathematically falls within the `10.192.0.0/10` block, a routing
loop occurred: all host packets destined for the Docker containers on ports like `9050` or `9051` were
intercepted by the transparent proxy NAT rules and dropped, causing proxy failures.
2054 2. **Tailscale Private IP Exclusion:** Shifting the subnet to another private range like `10.123.0.0/16`
resolved the port conflict, but led to connection timeouts. On macOS, Tailscale's Exit Node routing
actively excludes RFC 1918 private subnets (e.g. `10.0.0.0/8`, `172.16.0.0/12`, `192.168.0.0/16`) to
maintain accessibility to local LAN resources (like printers/routers). Thus, packets targeting `10.123.x.x`
were dropped locally by Tailscale before reaching the gateway tunnel.
2055 3. **The Benchmarking Subnet Solution:** To bypass these private IP exclusions without manual static routing
tables, Tor's virtual network was changed to `198.18.0.0/15` (reserved by RFC 2544 for benchmarking).
Since this is not a private RFC 1918 range, Tailscale exit-node routing automatically encapsulates and
routes all traffic destined for `198.18.0.0/15` through the tunnel to the gateway.
2056 4. **Transparent NAT Redirection:** The `routing-fix` sidecar applies NAT rules in the shared network namespace
:
2057 ```bash
2058 iptables -t nat -I PREROUTING -d 198.18.0.0/15 -p tcp -j REDIRECT --to-ports 9040
2059 ```
2060 This intercepts all incoming TCP traffic targeting the mapped `.onion` range and transparently proxies it
through Tor's transparent port (`9040`).
2061
2062 ### Exit Node Exclusions
2063 The gateway supports the ability to exclude specific countries from being used as exit nodes (e.g. to avoid
certain jurisdictions or nodes with poor network latency). This is configured globally via the `
TOR_EXCLUDE_EXIT_NODES` environment variable in `.env`, which gets dynamically appended to Tor's `
ExcludeExitNodes` and strictly enforced with `StrictNodes 1`.

```

5 diagrams.md

```

1 # nullexit Flow Diagrams & Architecture
2
3 ---
4
5 ## 1. System Architecture What's Running & Where
6
7 ```mermaid
8 graph TB
9     subgraph INTERNET[" Internet / Cloudflare Edge"]
10         CF["Cloudflare WARP\n(Exit Point)\n\nwarp=on"]
11     end
12
13     subgraph TAILNET[" Tailscale Mesh (100.x.x.x)"]
14         PHONE[" Phone / Laptop\n(Tailnet Client)"]
15     end
16
17     subgraph MACHOST[" macOS Host"]
18         direction TB
19         PFWALL["macOS pf Firewall\n(Kill-Switch & TCP MSS Clamping)"]
20         TSDAEMON["tailscaled\n(brew service)\n\nhost Tailscale"]
21         HOSTDNS["Host DNS\nGateway IP\n(hijacked by toggle.sh)"]
22         HOSTSOCKS["Host SOCKS5\n127.0.0.1:1080\n(fallback only)"]
23         HOSTTOR["Host Tor SOCKS5\n127.0.0.1:$TOR SOCKS_PORT\n(randomized)"]
24         HOSTTORCTRL["Host Tor Control\n127.0.0.1:$TOR_CONTROL_PORT\n(randomized)"]
25
26         subgraph MONITORS[" Background Daemons (toggle.sh children)"]
27             WARPWATCH["WARP Watcher\n(every 30s, via docker exec)\n\noutput.log"]
28             LEAKPROBE["Host Leak Probe\n(every 300ms, via curl from HOST)\n\noutput.log"]
29             DNSWATCHER["DNS Watcher\n(re-hijacks if drift detected)\n\noutput.log"]
30             CAFFEINATE["caffeinate -i\n(sleep prevention)"]
31         end
32     end
33
34     subgraph COLIMAVM[" Colima VM (Linux, 600MB RAM)"]
35         subgraph NETNS["warp container network namespace (shared by ALL containers)"]
36             GLUETUN["warp / Gluetun\nWireGuard tun0\n(owns the namespace)"]
37             TS["tailscale container\nAdvertises exit node\ntailscale0 interface"]
38             SOCKS["tailscale\nSOCKS5\nport 1080"]
39             ADGUARD["adguardhome\nDNS sinkhole\nport 5353"]
40             ROUTINGFIX["routing-fix sidecar\nGeo-IP Blocking & Go Logger\n(writes blocked.log)"]
41             TOR["tor container\nSOCKS5: port 9050\nControl: port 9051\nTransparent Proxy: port 9040\n\nDNSPort: port 5353"]
42         end
43     end
44 end

```

```

42     end
43
44     subgraph LAUNCHD[" launchd (always-on)"]
45         WATCHER["scripts/watcher.sh\n(sleep/wake + Wi-Fi roam\n+ LAN P2P auto-detection)"]
46     end
47 end
48
49 subgraph RECOVERY[" Recovery"]
50     RECOVER["recover.sh\n(nuclear reset)"]
51     RECOVERPOST["recover.sh --post-wake\n(light refresh)"]
52 end
53
54 %% Traffic flow
55 PHONE -->|"WireGuard / Tailscale mesh"| TSDAEMON
56 TSDAEMON -->|"exit-node route tun0"| TS
57 TS -->|"FORWARD tun0 (MASQUERADE)"| GLUETUN
58 GLUETUN -->|"WireGuard UDP (macOS Host Egress)"| PFFIREWALL
59 PFFIREWALL -->|"TCP MSS Clamped / Kill-Switch check"| CF
60 TS -->|"198.18.0.0/15 PREROUTING redirect"| TOR
61 TOR -->|"internal path / exit nodes via tun0"| GLUETUN
62
63 %% DNS
64 HOSTDNS -->|"UDP:53 TCP:5354"| ADGUARD
65 ADGUARD -->|"DNS upstream via tun0"| CF
66 ADGUARD -->|"onion queries localhost:5353"| TOR
67
68 %% Monitoring
69 WARPWATCH -->|"docker exec warp wget cdn-cgi/trace"| GLUETUN
70 LEAKPROBE -->|"curl cdn-cgi/trace (HOST NIC)"| CF
71 DNSWATCHER -->|"networksetup -getdnsservers"| HOSTDNS
72
73 %% Logging
74 ROUTINGFIX -->|"writes"| BLOCKEDLOG["blocked.log\n(Host-side)"]
75 HOSTTOR -->|"port mapping"| TOR
76 HOSTTORCTRL -->|"port mapping"| TOR
77
78 %% Recovery triggers
79 WARPWATCH -->|"6 consecutive warp=off"| RECOVER
80 WATCHER -->|"sleep/wake / roam"| RECOVERPOST
81 RECOVERPOST -->|"if gateway unhealthy"| RECOVER
82 WATCHER -->|"writes .lan_p2p_detected"| ROUTINGFIX
83
84 style MONITORS fill:#1a1a2e,color:#e0e0ff
85 style COLIMAVM fill:#0d2137,color:#e0e0ff
86 style NETNS fill:#0a1628,color:#e0e0ff
87 style RECOVERY fill:#2d0a0a,color:#ffe0e0
88 style INTERNET fill:#0a2d1a,color:#e0ffe0
89 style TAILNET fill:#1a2d0a,color:#e8ffe0
90 style LAUNCHD fill:#2d1a2d,color:#ffe0ff
91 ...
92
93 ---
94
95 ## 2. toggle.sh Full START Flow
96
97 ```mermaid
98 flowchart TD
99     START(["./toggle.sh"])
100     CHECK{"is_gateway_active?\n(containers running OR\nDNS 1.1.1.1)"}
101
102     START --> CHECK
103     CHECK -->|"YES already ON"| STOPFLOW[" See STOP Flow"]
104     CHECK -->|"NO start it"| S1
105
106     S1["1. Reset DNS 1.1.1.1\n(prevent deadlocks)"]
107     S2["2. tailscale down\n(prevent exit-node routing during boot)"]
108     S3["3. Check Colima VM\n(colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged)"]
109     S4["4. Configure VM swap\n(400MB, prevent OOM)"]
110     S5["5. Clean corrupted AdGuard config\n(prevent container crash loop)"]
111     S6["6. docker compose up -d --build\n(compile Go threat rules & boot containers)"]
112     S7["7. Poll tailscale status\n(wait for 'offers exit node'\nup to 60s)"]
113     TSREADY{"container\nTailscale ready?"}
114     ABORT([" ABORT\n cleanup_handler"])
115     S8["8. Resolve gateway Tailscale IP\n(.gateway_ip or docker exec)"]
116     S9["9. Verify tailscaled running\n(auto-start if needed)"]
117     S10["10. tailscale up --reset\n--exit-node=\n--accept-routes=true\n--ssh=true --accept-dns=false"]
118
119     PF1{"Pre-flight 1/3\n tailscale ping gateway"}
120     PF2{"Pre-flight 2/3\n dig +tcp AdGuard DNS"}
121     PF3{"Pre-flight 3/3\n WARP internet check"}
122     ALLPASS{"All 3 pass?"}

```

```

123 EXITPATH[" EXIT NODE PATH\n tailscale up --exit-node=\n force_dns_to_gateway (3 retry)\n SOCKS5 disabled"]
124 SOCKSPATH[" SOCKS5 FALLBACK PATH\n macOS SOCKS5 127.0.0.1:1080\n DNS proxy 127.0.0.1:53\n No exit node (
125 SKIP_EXIT_NODE=true)"]
126
127 DAEMONS["Start background daemons\n caffeinate -i (sleep prevention)\n DNS Watcher\n WARP Watcher\n Host
128 Leak Probe\n(all write output.log)"]
129
130 RULECOMPILE["Rule compiler status\n(log rule/IP counts)"]
131 WRITEMARKER["write_gateway_active_marker\n(/tmp/nullexit-gateway-active.marker)"]
132 DONE([" Gateway UP"])
133
134 S1 --> S2 --> S3 --> S4 --> S5 --> S6 --> S7
135 S7 --> TSREADY
136 TSREADY -->|"NO (40 NoState)"| ABORT
137 TSREADY -->|"YES"| S8
138 S8 --> S9 --> S10
139 S10 --> PF1 --> PF2 --> PF3
140 PF1 & PF2 & PF3 --> ALLPASS
141 ALLPASS -->|"YES"| EXITPATH
142 ALLPASS -->|"NO"| SOCKSPATH
143 EXITPATH --> DAEMONS
144 SOCKSPATH --> DAEMONS
145 DAEMONS --> RULECOMPILE --> WRITEMARKER --> DONE
146
147 style ABORT fill:#5c1010,color:#fff
148 style EXITPATH fill:#0a3d1a,color:#c0ffc0
149 style SOCKSPATH fill:#3d2d00,color:#ffe0a0
150 style DAEMONS fill:#0a1f3d,color:#c0d8ff
151 style DONE fill:#0a3d1a,color:#fff
152
153 ---
154
155 ## 3. toggle.sh STOP Flow
156
157 ```mermaid
158 flowchart TD
159     STOP(["./toggle.sh\n(gateway already ON)"])
160
161     ST1["1. tailscale down\n(disconnect from mesh)"]
162     ST2["2. docker compose down -t 5\n(stop all containers)"]
163     ST3["3. Reset DNS 1.1.1.1\n(networksetup on all services)"]
164     ST4["4. stop_local_dns_proxy\n(kill Python UDP proxy)"]
165     ST5["5. stop_pidfile_daemon\n(kill caffeinate, rm PID)"]
166     ST6["6. stop_pidfile_daemon\n(kill DNS Watcher, rm PID)"]
167     ST7["7. stop_pidfile_daemon\n(kill WARP Watcher, rm PID)"]
168     ST8["8. stop_pidfile_daemon\n(kill Leak Probe, rm PID)"]
169     ST9["9. clear_gateway_active_marker\n(rm /tmp/nullexit-gateway-active.marker)"]
170     ST10["10. cleanup_network_state\n disable_all_proxies\n Flush DNS cache (dscacheutil)\n Flush stale routes\n n Power-cycle Wi-Fi (offon)"]
171
172     DONE([" Gateway DOWN\nInternet restored"])
173
174     STOP --> ST1 --> ST2 --> ST3 --> ST4
175     ST4 --> ST5 --> ST6 --> ST7 --> ST8 --> ST9 --> ST10 --> DONE
176
177     style DONE fill:#0a3d1a,color:#fff
178
179 ---
180
181 ## 4. Monitoring Layer The 4 Background Daemons
182
183 ```mermaid
184 graph LR
185     subgraph DAEMONS["Background Daemons (all start with gateway, stop on teardown)"]
186         direction TB
187
188         subgraph D1[" Sleep Prevention"]
189             CAFF["caffeinate -i\nPID: /tmp/nullexit-caffeinate.pid\n\nKeeps Mac awake while\ngateway is active"]
190         end
191
192         subgraph D2[" DNS Watcher"]
193             DNSW["Polls networksetup every ~30s\nPID: /tmp/nullexit-dns-watcher.pid\n\nIf DNS drifts from gateway IP\n re-hijacks automatically\n output.log"]
194         end
195
196         subgraph D3[" WARP Watcher"]
197

```



```

198     WARPW["Polls every 30s via:\ndocker compose exec warp wget cdn-cgi/trace\nPID: /tmp/nullexit-warp-
watcher.pid\n\nCounts consecutive warp=off\n WARP_FAIL_THRESHOLD (default 6=30s)\n runs recover.sh (nuclear
)\n output.log"]
199     end
200
201     subgraph D4[" Host Leak Probe"]
202         LEAKP["Polls every 300ms via:\ncurl cdn-cgi/trace (HOST NIC directly)\nPID: /tmp/nullexit-host-leak
-probe.pid\n\nLogs ONLY on state change:\n LEAK / ROTATE / HOST-PROBE failed\nAll errors  output.log\n(
HOST_LEAK_PROBE=true in .env)"]
203     end
204 end
205
206 subgraph BLIND["What each monitor can/can't see"]
207     B1["WARP Watcher sees:\n Container WARP tunnel state\n Host NIC traffic\n Flash leaks < 5s"]
208     B2["Host Leak Probe sees:\n Host NIC egress (browser path)\n Sub-second transitions\n Container
internals"]
209 end
210
211 D3 -->|"covers"| B1
212 D4 -->|"covers"| B2
213
214 style D1 fill:#1a2d0a
215 style D2 fill:#0a1a2d
216 style D3 fill:#2d0a2d
217 style D4 fill:#2d1a00
218 style BLIND fill:#1a1a1a,color:#aaa
219 ---
220
221 ---
222
223 ## 5. Traffic Flow Data Path (Normal Operation)
224
225 ```mermaid
226 sequenceDiagram
227     participant PHONE as Phone<br/>(Tailnet client)
228     participant TS_HOST as tailscaled<br/>(macOS host)
229     participant TS_CONTAINER as tailscale<br/>(container)
230     participant GLUETUN as Gluetun/warp<br/>(container)
231     participant CF as Cloudflare<br/>WARP Edge
232     participant INTERNET as Internet
233
234     Note over PHONE,INTERNET: Normal EXIT NODE path (all 3 pre-flights passed)
235
236     PHONE->>TS_HOST: WireGuard packet<br/>(encrypted, UDP 41641)
237     TS_HOST->>TS_CONTAINER: route via utun* tailscale0
238     TS_CONTAINER->>GLUETUN: FORWARD chain<br/>(iptables MASQUERADE on tun0)
239     GLUETUN->>TS_HOST: WireGuard tunnel (tun0) egresses macOS Host
240     Note over TS_HOST: macOS pf Firewall applies<br/>Kill-Switch & TCP MSS Clamping
241     TS_HOST->>CF: re-encrypted UDP packet (MSS 1160)
242     CF->>INTERNET: Plain HTTPS from<br/>Cloudflare IP
243
244     Note over PHONE,INTERNET: DNS Resolution path
245     PHONE->>TS_HOST: DNS query
246     TS_HOST->>TS_CONTAINER: UDP:53 TCP:5354 AdGuard :5335
247     TS_CONTAINER->>GLUETUN: AdGuard upstream DNS via tun0
248     GLUETUN->>CF: Encrypted DNS query
249     CF->>GLUETUN: DNS response
250     GLUETUN-->>TS_CONTAINER: response
251     TS_CONTAINER-->>TS_HOST: response
252     TS_HOST-->>PHONE: DNS answer
253 ---
254
255 ---
256
257 ## 6. recover.sh Decision Tree
258
259 ```mermaid
260 flowchart TD
261     INVOKE(["recover.sh called"])
262     MODE["--post-wake\nflag?"]
263
264     subgraph POSTWAKE[" --post-wake (light refresh, gateway stays UP)"]
265         PW1["Re-hijack DNS gateway IP"]
266         PW2["Flush DNS cache"]
267         PW3["Refresh tailscale exit-node"]
268         PW4["Force-recreate warp container\n(if Gluetun healthcheck failed)"]
269         PW5["Leave: containers, Wi-Fi,\ncaffeinate, DNS watcher,\nHost Leak Probe untouched"]
270     end
271
272     subgraph NUCLEAR[" Default (nuclear gateway torn down)"]
273         N0["Sudo keep-alive loop (background)"]

```

```

274     N1["1. tailscale down\n(disconnect from mesh)"]
275     N2["2. Reset DNS 1.1.1.1\n(ALL network services)"]
276     N3["3. disable_all_proxies\n(all interfaces)"]
277     N4["4. Flush DNS cache\n(dscacheutil -flushcache)"]
278     N5["5. Flush stale routes\n(WARP endpoint routes)"]
279     N6a["6a. docker compose down -t 30\n(stop all containers)"]
280     N6b["6b. stop_pidfile_daemon\n(kill caffeinate)"]
281     N6c["6c. stop_pidfile_daemon\n(kill DNS Watcher)"]
282     N6d["6d. stop_pidfile_daemon\n(kill Leak Probe)"]
283     N7["7. Power-cycle Wi-Fi\n(networksetup offslepp 2on)"]
284     N8["8. Verify internet working\ncurl ifconfig.io"]
285 end
286
287 DONE_PW([" Gateway refreshed\n(still running)"])
288 DONE_NUC([" Internet restored\n(gateway fully stopped)"])
289
290 INVOKE --> MODE
291 MODE -->|"yes"| PW1
292 PW1 --> PW2 --> PW3 --> PW4 --> PW5 --> DONE_PW
293 MODE -->|"no"| NO
294 NO --> N1 --> N2 --> N3 --> N4 --> N5
295 N5 --> N6a --> N6b --> N6c --> N6d --> N7 --> N8 --> DONE_NUC
296
297 style POSTWAKE fill:#0a2d1a,color:#c0ffc0
298 style NUCLEAR fill:#2d0505,color:#ffc0c0
299 style DONE_PW fill:#0a3d1a,color:#fff
300 style DONE_NUC fill:#3d1a00,color:#ffe0c0
301 ...
302
303 ---
304
305 ## 7. Failure Paths & Self-Healing
306
307 ``mermaid
308 flowchart LR
309     subgraph EVENTS["Failure / Event"]
310         E1["WARP tunnel drops\n(warp=off)"]
311         E2["Mac wakes from sleep\nor Wi-Fi roams"]
312         E3["DNS drifts from\ngateway IP"]
313         E4["toggle.sh crashes /\nCtrl-C / SIGTERM"]
314         E5["Host-side flash leak\ndetected (HOST NIC)"]
315         E6["Silent NAT timeout\n(ping stops working)"]
316     end
317
318     subgraph DETECTORS["Detector"]
319         D1["WARP Watcher\n(30s poll, docker exec)"]
320         D2["scripts/watcher.sh\n(launchd, always on)"]
321         D3["DNS Watcher\n(30s poll, networksetup)"]
322         D4["cleanup_handler\n(ERR/INT/TERM/HUP trap)"]
323         D5["Host Leak Probe\n(300ms poll, HOST curl)"]
324         D6["routing-fix.sh\n(30s curl over tun0)"]
325     end
326
327     subgraph ACTIONS["Response"]
328         A1["threshold consecutive off\nrecover.sh (nuclear)"]
329         A2["recover.sh --post-wake\n(gentle refresh)"]
330         A3["Re-hijack DNS\n(networksetup setdnsservers)"]
331         A4["Stop all daemons\nReset DNS 1.1.1.1\nTear down Tailscale"]
332         A5["Log LEAK to output.log"]
333         A6["Blackhole internet traffic\n+ Drop TUNNEL_FAILED_CLOSED.marker"]
334         A7["Push macOS Desktop Notification\n(via watcher.sh listener)"]
335     end
336
337     E1 --> D1 --> A1
338     E2 --> D2 --> A2
339     E3 --> D3 --> A3
340     E4 --> D4 --> A4
341     E5 --> D5 --> A5
342     E6 --> D6 --> A6
343     A6 -. ->|"marker detected"| D2
344     D2 --> A7
345
346     style EVENTS fill:#2d1010,color:#ffcccc
347     style DETECTORS fill:#0a1a2d,color:#cce0ff
348     style ACTIONS fill:#0a2d0a,color:#ccffcc
349 ...
350
351 ---
352
353 ## 8. output.log Who Writes What
354

```

```

355 All events converge into a single `output.log` at the repo root.
356
357 | Writer | Event Types | Format |
358 |-----|-----|-----|
359 | `toggle.sh` | All startup/shutdown steps, errors, pre-flight results | Plain text with timestamps |
360 | `recover.sh` | Every recovery step (nuclear or post-wake) | Plain text |
361 | **WARP Watcher** | `WARP DOWN`, `WARP RECOVERED`, `WARP SHUTDOWN` | `[UTC timestamp]` prefix |
362 | **Host Leak Probe** | `LEAK`, `ROTATE`, `HOST-PROBE failed/timeout` | `[HH:MM:SS.mmm]` prefix |
363 | **DNS Watcher** | DNS re-hijack events | Appended inline |
364 | `docker compose` | Container logs on failure (last 100 lines of warp) | Dumped on ERR path |
365 | `routing-fix.sh` | (via `nullexit-logger` inside container) | Structured |
366
367 **Grep cheatsheet:**
368 ```bash
369 grep LEAK output.log          # Host-side warp=off events (real leaks)
370 grep ROTATE output.log        # Cloudflare edge IP rotations (harmless)
371 grep 'HOST-PROBE' output.log   # curl failures from probe
372 grep 'WARP DOWN' output.log    # Container-side tunnel drops
373 grep 'WARP SHUTDOWN' output.log # Auto-recovery triggered
374 grep 'WARP RECOVERED' output.log # Self-healed before threshold
375 ```

```

6 toggle.sh

```

1  #!/bin/bash
2  # toggle.sh nullexit gateway toggle script for macOS
3  # Automatically handles Docker containers, Colima, DNS hijacking, and Tailscale exit node routing.
4
5  set -e
6
7  SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
8
9  # Enforce Cryptographic Script Integrity
10 if [ -f "scripts/crypto.sh" ]; then
11     if ! bash scripts/crypto.sh --verify; then
12         exit 1
13     fi
14 fi
15
16 # Define log file
17 LOG_FILE="$PWD/output.log"
18
19 # Rotate log file if it exceeds 50MB (52428800 bytes)
20 if [ -f "$LOG_FILE" ]; then
21     log_size=$(stat -f%z "$LOG_FILE" 2>/dev/null || stat -c%s "$LOG_FILE" 2>/dev/null || echo 0)
22     if [ "$log_size" -gt 52428800 ]; then
23         mv "$LOG_FILE" "${LOG_FILE}.old" 2>/dev/null || rm -f "$LOG_FILE"
24         echo "[$(date '+%Y-%m-%d %H:%M:%S')] Log file exceeded 50MB; rotated to output.log.old" > "$LOG_FILE"
25     fi
26 fi
27
28 # --- MAIN EXECUTION LOGGING ---
29 echo -e "\n===== " >> "$LOG_FILE"
30 echo "[$(date '+%Y-%m-%d %H:%M:%S')] toggle.sh invoked" >> "$LOG_FILE"
31 echo "===== " >> "$LOG_FILE"
32
33 # (Previously this script chmod 600'd .env at start to "unlock" it for docker compose,
34 # and chmod 000'd it on exit. That pattern broke docker compose on macOS/Colima when
35 # the umask produced 0600 the script removed its own ability to read .env before it could even proceed.)
36
37 # --- PROCEDURAL PORT GENERATION ---
38 # To prevent static port fingerprinting by malware, we randomize exposed ports
39 # on every boot and persist them to a hidden environment file.
40 PORTS_FILE="/tmp/nullexit-ports.env"
41 if [[ "$1" == "" || "$1" == "--restart" ]]; then
42
43     # Collision-avoidance function
44     GENERATED_PORTS=""
45     get_free_port() {
46         local port
47         while true; do
48             port=$(jot -r 1 10000 65000)
49             # 1. Prevent self-collision within this loop
50             if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
51                 continue
52             fi

```

```

53 # 2. Prevent collision with ANY process on the Mac (root daemons, terminal apps, etc)
54 # We use netstat instead of lsof because netstat reads the kernel socket table
55 # directly and doesn't require sudo to see ports bound by other users/root.
56 if ! netstat -an -p tcp | awk '{print $4}' | grep -q "\.$port$"; then
57     echo "$port"
58     return
59 fi
60 done
61 }
62
63 export TOR SOCKS_PORT=$(get_free_port)
64 GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
65
66 export TOR_CONTROL_PORT=$(get_free_port)
67 GENERATED_PORTS="$GENERATED_PORTS $TOR_CONTROL_PORT"
68
69 export SOCKS_PROXY_PORT=$(get_free_port)
70 GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
71
72 export DNS_PROXY_PORT=$(get_free_port)
73
74 echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
75 echo "export TOR_CONTROL_PORT=$TOR_CONTROL_PORT" >> "$PORTS_FILE"
76 echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
77 echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
78 ADGUARD_PASS=$(grep -E "^ADGUARD_PASSWORD=" .env 2>/dev/null | cut -d'=' -f2- | tr -d '\n')
79 echo "export TOR_PASSWORD=${ADGUARD_PASS:-nullexit}" >> "$PORTS_FILE"
80 elif [ -f "$PORTS_FILE" ]; then
81     # On STOP, source the existing ports so docker-compose down matches
82     source "$PORTS_FILE"
83 else
84     # Fallbacks if file is lost
85     export TOR SOCKS_PORT=9050
86     export TOR_CONTROL_PORT=9051
87     export SOCKS_PROXY_PORT=1080
88     export DNS_PROXY_PORT=5354
89 fi
90
91 # Start execution timer
92 TOGGLE_START_TIME=$SECONDS
93
94 if [[ "$1" == "--restart" ]]; then
95     echo "Executing Gateway Restart Sequence..."
96     # We must source common.sh here temporarily to check is_gateway_active before we proceed
97     source "$SCRIPT_DIR/scripts/common.sh"
98     if is_gateway_active; then
99         echo "Gateway is currently running. Stopping it first..."
100         bash "$0"
101         echo "Gateway stopped. Now starting it..."
102     else
103         echo "Gateway is already stopped. Starting it..."
104     fi
105     bash "$0"
106     exit 0
107 fi
108
109 # Global flag to track if the script completed successfully
110 SUCCESS_RUN=false
111
112 # Global variable to track the currently running background command PID
113 CURRENT_BG_PID=""
114
115 # Source common bash functions
116 source "$SCRIPT_DIR/scripts/common.sh"
117
118 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
119 LOCK_FILE="/tmp/nullexit-toggle.lock"
120 if [ -f "$LOCK_FILE" ]; then
121     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
122     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
123         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
124             die "Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running."
125         fi
126     fi
127 fi
128 echo "$$" > "$LOCK_FILE"
129
130 # Helper function to run a GUI command as the logged-in console user
131 # (Prevents permission failures if the script is run with sudo)
132 run_gui_cmd() {
133     local console_user

```

```

134 console_user=$(stat -f '%Su' /dev/console 2>> output.log || echo "$SUDO_USER")
135 if [ -z "$console_user" ] || [ "$console_user" = "root" ]; then
136     console_user=$(logname 2>> output.log || echo "$USER")
137 fi
138
139 if [ -n "$console_user" ] && [ "$console_user" != "root" ] && [ "$EUID" -eq 0 ]; then
140     sudo -u "$console_user" "$@"
141 else
142     "$@"
143 fi
144 }
145
146 # Active-state marker: written at end of START path so external watchers
147 # (see launchd/com.nullexit.wake-recovery.plist + scripts/watcher.sh +
148 # devref 10.29) know the gateway is currently up. They only fire
149 # recover.sh --post-wake when this file is present. Cleared at the top
150 # of the STOP path and inside cleanup_handler on any error/signal path
151 # so a half-broken gateway never re-triggers post-wake refreshes.
152 write_gateway_active_marker() {
153     date -u +%FT%TZ > /tmp/nullexit-gateway-active.marker
154     # Set the watcher debounce timestamp to now to prevent a redundant post-startup recovery run.
155     date +%s > /tmp/nullexit-watcher.last-recovery 2>/dev/null || true
156 }
157
158 clear_gateway_active_marker() {
159     rm -f /tmp/nullexit-gateway-active.marker
160     rm -f "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/TUNNEL_FAILED_CLOSED.marker"
161 }
162
163 # Defensive: clear any stale marker from a prior crashed/aborted run. If a
164 # previous toggle.sh wrote the marker but was OOM-killed in the middle of the
165 # START block (before the END-of-START write) or if the user rebooted mid
166 # session this prevents the watcher from firing post-wake against a
167 # non-running gateway on every wake event. Placed AFTER the function
168 # definitions so bash resolves the call at parse time.
169 clear_gateway_active_marker
170
171 PID_FILE="$PID_CAFFEINATE"
172
173 # Start macOS caffeine to prevent sleep while gateway is running
174 start_sleep_prevention() {
175     if ! command -v caffeine >/dev/null 2>&1; then
176         echo " [Warning] caffeine tool not found. Sleep prevention unavailable."
177         return 1
178     fi
179
180     # Stop any existing sleep prevention process first to avoid duplicates
181     stop_pidfile_daemon "/tmp/nullexit-caffeinate.pid" "system sleep prevention"
182
183     echo " Enabling system sleep prevention (caffeinate)..."
184     # Run caffeine wrapped in a bash trap to prevent idle system sleep.
185     # Critically, this traps macOS shutdown signals (SIGTERM) and automatically
186     # flushes hijacked DNS back to normal right before the Mac powers off.
187     # We do not use recover.sh here because it is a heavy recovery script (managing
188     # containers, restarting tailscaled, etc.) which takes too long and gets
189     # terminated by macOS before it can reset the DNS. Instead, we surgically and
190     # instantly restore normal DNS and proxy settings here.
191     nohup bash -c "
192     trap '
193         echo \"[Shutdown Trap] Reverting network settings...\" >> \"\$PWD/output.log\" 2>&1
194         kill \$! 2>/dev/null || true
195         sudo -n networksetup -setdnsservers \"\$ACTIVE_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
196         sudo -n networksetup -setdnsservers \"\$ENO_SERVICE\" 1.1.1.1 >> \"\$PWD/output.log\" 2>&1 || true
197         sudo -n networksetup -setsearchdomains \"\$ACTIVE_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
198         sudo -n networksetup -setsearchdomains \"\$ENO_SERVICE\" \"Empty\" >> \"\$PWD/output.log\" 2>&1 || true
199         sudo -n networksetup -setsocksfirewallproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 ||
200 true
201         sudo -n networksetup -setsocksfirewallproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
202         sudo -n networksetup -setwebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
203         sudo -n networksetup -setsecurewebproxystate \"\$ACTIVE_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
204         sudo -n networksetup -setsecurewebproxystate \"\$ENO_SERVICE\" off >> \"\$PWD/output.log\" 2>&1 || true
205         if command -v tailscale >/dev/null 2>&1; then
206             tailscale up >> \"\$PWD/output.log\" 2>&1 || true
207             tailscale set --accept-dns=false --exit-node= >> \"\$PWD/output.log\" 2>&1 || true
208             TS_DOWN_ARGS=""
209             if [ \"\$KILL_SWITCH\" = \"true\" ]; then TS_DOWN_ARGS="--accept-risk=lose-ssh\"; fi
210             tailscale down \$TS_DOWN_ARGS >> \"\$PWD/output.log\" 2>&1 &
211         fi
212         sleep 1
213         echo \"[Shutdown Trap] Cleanup complete.\" >> \"\$PWD/output.log\" 2>&1

```

```

214     exit 0
215     ' SIGTERM SIGINT SIGHUP
216     caffeinate -i &
217     wait \${!}
218     " >> output.log 2>&1 &
219     local caffe_pid=${!}
220     echo "$caffe_pid" > "$PID_FILE"
221     echo " Sleep prevention active (PID $caffe_pid). Your Mac won't sleep while the gateway is running."
222 }
223
224
225
226 DNS_WATCHER_PID_FILE="$PID_DNS_WATCHER"
227 WARP_WATCHER_PID_FILE="/tmp/nullexit-warp-watcher.pid"
228
229 start_dns_watcher() {
230     local target_ip=${1}
231     stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
232     echo " Starting background DNS Watcher for seamless Wi-Fi roaming..."
233     nohup bash -c "
234         source \"\$SCRIPT_DIR/scripts/common.sh\"
235         trap 'exit 0' SIGTERM SIGINT SIGHUP
236         while true; do
237             ACTIVE_IF=$(get_active_service)
238             if [ -n \"\$ACTIVE_IF\" ]; then
239                 CURRENT_DNS=$(networksetup -getdnsservers \"\$ACTIVE_IF\" 2>/dev/null)
240                 if [ \"\$CURRENT_DNS\" != \"\$target_ip\" ]; then
241                     networksetup -setdnsservers \"\$ACTIVE_IF\" \"\$target_ip\" >/dev/null 2>&1
242                 fi
243             fi
244             sleep 30
245         done
246     " >> output.log 2>&1 &
247     echo $! > "$DNS_WATCHER_PID_FILE"
248 }
249
250
251
252 # Background WARP tunnel liveness monitor. Polls cdn-cgi/trace every 30s
253 # while healthy, but accelerates to polling every 5s if a failure is detected.
254 # Always logs state transitions to output.log. When warp=off persists for
255 # WARP_FAIL_THRESHOLD consecutive polls (default 6 = 30s of downtime), the watcher
256 # triggers a nuclear recovery: runs recover.sh to tear down the entire
257 # gateway disconnecting Tailscale, stopping containers, resetting DNS,
258 # and power-cycling Wi-Fi. Note: This minimizes exposure window, but
259 # only the pf kill switch guarantees absolutely zero leakage on drop.
260 #
261 # The threshold (default 6) is statistically chosen: even at an
262 # unrealistically high 10% false-positive rate per poll, P(6 consecutive
263 # false positives) = 0.1^6 = 0.0001%. Real-world false-positive rates
264 # are far lower, making 30s of continuous downtime overwhelmingly likely
265 # to be a genuine outage.
266 #
267 # Configurable via .env: WARP_FAIL_THRESHOLD=3 (15s, more aggressive)
268 start_warp_watcher() {
269     stop_warp_watcher
270     # Parse threshold from .env (same pattern as GATEWAY_MSS, GATEWAY_BYPASS_PING, etc.)
271     local threshold
272     threshold=$(read_env_var WARP_FAIL_THRESHOLD)
273     if [ -z "$threshold" ]; then
274         threshold="{WARP_FAIL_THRESHOLD:-6}"
275     fi
276     local trigger_file="/tmp/nullexit-warp-shutdown-triggered"
277     rm -f "$trigger_file"
278     echo " Starting background WARP Watcher (polling every 30s [accelerating to 5s on failure], shutdown after $
279         {threshold} consecutive failures)..."
280     nohup bash -c "
281         trap 'exit 0' SIGTERM SIGINT SIGHUP
282         last_state='on'
283         consec_off=0
284         threshold='$threshold'
285         trigger_file='$trigger_file'
286         out_log='$PWD/output.log'
287         recover_bin='$PWD/recover.sh'
288         inhibit_file='/tmp/nullexit-warp-inhibit.marker'
289         while true; do
290             # Inhibit check
291             # recover.sh --post-wake writes this marker while force-recreating the
292             # warp container. During that window, the container is intentionally
293             # down don't count it as a failure or we'll fire nuclear recover.sh
294             # and kill a gateway that was in the process of healing itself.

```



```

294     if [ -f \"$inhibit_file\" ]; then
295         consec_off=0
296         sleep 5
297         continue
298     fi
299
300     state='off'
301     if docker compose exec -T warp wget -q0- --timeout=5 \
302         https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null \
303         | grep -q 'warp=on'; then
304         state='on'
305     fi
306
307     # State-transition logging
308     if [ \"$state\" != \"$last_state\" ]; then
309         if [ \"$state\" = 'off' ]; then
310             echo \"[\$(date -u +%FT%TZ)] WARP DOWN   cdn-cgi/trace reports warp=off\" >> \"$out_log\"
311         fi
312         last_state=\"$state\"
313     fi
314
315     # Consecutive-failure tracking + auto-shutdown
316     if [ \"$state\" = 'off' ]; then
317         consec_off=$((consec_off + 1))
318         if [ \"$consec_off\" -ge \"$threshold\" ] && [ ! -f \"$trigger_file\" ]; then
319             touch \"$trigger_file\"
320             echo \"[\$(date -u +%FT%TZ)] WARP SHUTDOWN  \${consec_off} consecutive failures (threshold=\${threshold})
. Running recover.sh to kill the gateway and restore normal internet. Your IP is NO LONGER Cloudflare.\" >>
\"$out_log\"
321             # Nuclear recovery: tear down the entire gateway.
322             # recover.sh resets DNS 1.1.1.1, disconnects Tailscale,
323             # stops Docker containers, flushes routes, power-cycles Wi-Fi.
324             bash \"$recover_bin\" --auto >> \"$out_log\" 2>&1 || true
325             echo \"[\$(date -u +%FT%TZ)] WARP SHUTDOWN  recover.sh completed, watcher exiting. Your IP is now
your real ISP IP.\" >> \"$out_log\"
326             exit 0
327         fi
328     else
329         if [ \"$consec_off\" -gt 0 ]; then
330             echo \"[\$(date -u +%FT%TZ)] WARP RECOVERED  warp=on after \${consec_off} consecutive off readings (
threshold was \${threshold})\" >> \"$out_log\"
331             fi
332             consec_off=0
333         fi
334
335         if [ \"$state\" = \"off\" ]; then
336             sleep 5
337         else
338             sleep 30
339         fi
340     done
341     \" >> output.log 2>&1 &
342     echo $! > \"$WARP_WATCHER_PID_FILE\"
343 }
344
345 stop_warp_watcher() {
346     if [ -f \"$WARP_WATCHER_PID_FILE\" ]; then
347         local wp
348         wp=$(cat \"$WARP_WATCHER_PID_FILE\")
349         if [ -n \"$wp\" ] && kill -0 \"$wp\" 2>/dev/null; then
350             echo \" Stopping background WARP Watcher (PID $wp)...\"
351             kill \"$wp\" 2>/dev/null || true
352         fi
353         rm -f \"$WARP_WATCHER_PID_FILE\"
354     fi
355 }
356
357
358
359
360
361 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
362 cleanup_handler() {
363     local exit_code=$?
364     local trigger_type=\"$1\"
365     local line_no=\"$2\"
366
367     if [ \"$SUCCESS_RUN\" = \"false\" ]; then
368         echo -e \"\n[Self-Correction] Script interrupted or failed (\${trigger_type}). Restoring host DNS to 1.1.1.1...
\"
369         if [ -n \"$ACTIVE_SERVICE\" ]; then

```

```

370     reset_dns
371 fi
372
373 if [ -n "$CURRENT_BG_PID" ]; then
374     echo "Terminating active command (PID $CURRENT_BG_PID)..."
375     kill -15 "$CURRENT_BG_PID" 2>> output.log || true
376 fi
377
378 # Disconnect host Tailscale and close the GUI application on failure
379 disconnect_tailscale_host
380
381 # Stop local DNS proxy if running
382 stop_local_dns_proxy
383
384 # Stop sleep prevention
385 stop_pidfile_daemon "/tmp/nullexit-caffeinate.pid" "system sleep prevention"
386 stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
387 stop_warp_watcher
388 clear_gateway_active_marker
389
390 # Capture warp logs on failure for debugging before teardown
391 if [ "$trigger_type" = "ERR" ]; then
392     echo -e "\n--- WARP FAILURE LOGS ---" >> output.log
393     docker compose logs warp --tail=100 >> output.log 2>&1 || true
394     echo "-----" >> output.log
395 fi
396
397 # Best-effort container teardown on failure
398 docker compose down --remove-orphans -t 30 2>> output.log || true
399
400 # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
401 if [ -n "$ACTIVE_SERVICE" ]; then
402     cleanup_network_state
403 fi
404
405
406 if [ "$trigger_type" = "ERR" ]; then
407     echo -e "\n=====
408     echo "ERROR: Script failed at line $line_no with exit code $exit_code."
409     echo "=====
410     echo "Troubleshooting steps:"
411     echo "1. Run 'colima status' to verify the VM is active."
412     echo "2. Run 'docker compose ps' to check container health."
413     echo "3. Run 'docker compose logs' to view application logs."
414     echo "4. Run 'tailscale status' to check host Tailscale status."
415     echo "=====
416 fi
417 rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"
418 fi
419 }
420
421 trap 'cleanup_handler ERR $LINENO' ERR
422 trap 'cleanup_handler INT' INT
423 trap 'cleanup_handler TERM' TERM
424 trap 'cleanup_handler HUP' HUP
425
426 # NOPASSWD sudo
427 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
428 # needed (networksetup, dnsmasq, dscacheutil, killall, pkill, socat).
429 # All privileged calls below use `sudo -n` which will run without prompting.
430 # No credential caching loop is needed.
431
432 # Add Homebrew and standard paths
433 setup_standard_path
434
435 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
436 # Uses a Python DNS proxy (handles TCP wire format correctly).
437 # Python3 is built into macOS no external dependencies.
438 DNS_PROXY_BIN=""
439 if command -v python3 >> output.log 2>&1; then
440     DNS_PROXY_BIN="python3 -I"
441 elif command -v python >> output.log 2>&1; then
442     DNS_PROXY_BIN="python -I"
443 fi
444
445 # Path to the DNS proxy script (sibling of this script)
446 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/scripts/dns-proxy.py"
447
448 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
449 # (`brew install tailscale` + `brew services start tailscale`) NOT the
450 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox

```

```

451 # to launch, no menu-bar click to perform, and no scutil Network Service to
452 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
453 # if you ran the install with sudo) and the control socket is always available.
454 TS_BIN=""
455 if command -v tailscale >> output.log 2>&1; then
456     TS_BIN="tailscale"
457 fi
458
459 # Baseline: disable Tailscale DNS management at the daemon level.
460 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
461 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
462 # re-enable `--accept-dns=true` and nullify this `set`.
463 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass
464 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
465 #
466 # This initial `set` is still useful as a baseline for non-reset operations
467 # (e.g. if the user runs `tailscale up` manually without `--reset`) and covers
468 # the brief window between this line and the first `--reset` call.
469 #
470 # Runs once at script start while tailscaled is still connected (if it was
471 # left running from a previous toggle), so the pref takes effect before any
472 # disconnection or reconnection logic.
473 if [ -n "$TS_BIN" ]; then
474     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
475 fi
476
477 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
478 # *entire* teardown: no scutil tunnel stop, no Tailscale.app GUI quit, no pkill.
479 # tailscaled keeps running (as a LaunchAgent / LaunchDaemon), which is intentional:
480 # the next `tailscale up` then reconnects in seconds rather than waiting for a
481 # slow .app launch + Network Extension activation. The `tailscale down` call also
482 # retains the user's auth state, so we don't force a browser re-login every cycle.
483 #
484 # Defined early (before the if/else branches and referenced by cleanup_handler's
485 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
486
487 # Wait for Network Readiness
488 # On --restart, the gateway tears down while the host network interface may
489 # still be reconnecting (Wi-Fi, Ethernet, USB tethering, etc.). If we proceed
490 # too early, get_active_service returns nothing, routing targets the wrong
491 # interface, and DNS/WARP setup silently fails.
492 #
493 # We use `route get default` to detect a valid default gateway rather than
494 # checking a specific interface (e.g. en0) this generalizes across all
495 # connection types: Wi-Fi, Ethernet, USB-C adapters, iPhone tethering, etc.
496 # The moment the OS has any routable default gateway, we proceed.
497 _net_wait_secs=0
498 _net_timeout=60
499 echo "Waiting for network connectivity before starting..."
500 while true; do
501     if route get default 2>/dev/null | grep -q 'gateway: '; then
502         _gw=$(route get default 2>/dev/null | awk '/gateway:/{print $2}')
503         echo "    Network ready (default gateway: $_gw)."
504         break
505     fi
506     if [ "$_net_wait_secs" -ge "$_net_timeout" ]; then
507         echo "    [Warning] No default gateway after ${_net_timeout}s proceeding anyway."
508         break
509     fi
510     sleep 2
511     _net_wait_secs=$(( _net_wait_secs + 2 ))
512 done
513
514 ACTIVE_SERVICE=$(get_active_service)
515
516 # Resolve the service name for en0 (usually "Wi-Fi") macOS scutil DNS resolver
517 # is commonly scoped to en0, so per-service DNS changes on other interfaces are
518 # ignored unless en0's service is also updated.
519 ENO_SERVICE=$(get_en0_service)
520
521 # Helper to add host-side bypass routes for the WARP WireGuard endpoints.
522 # This prevents an infinite recursive routing loop (tunnel loop) where
523 # the container's WireGuard packets exit the VM, reach the host, match the
524 # default route (the exit node), and get routed back into the tunnel.
525 # We route via the gateway IP (not interface) to ensure packets are routed
526 # properly even when the host's default route changes to the Tailscale interface.
527
528 # Helper to manually override the host's default route to point through the
529 # Tailscale utun* interface. Standalone tailscaled on macOS (Homebrew version)
530 # lacks the platform-specific integration to override the default gateway

```

```

532 # automatically when --exit-node is used.
533 setup_exit_node_routing() {
534     local ts_iface
535     ts_iface=$(ifconfig | awk '/^[a-z0-9]+:/{iface=$1} /inet 100\./{print iface; exit}' | tr -d ':')
536
537     if [ -n "$ts_iface" ]; then
538         echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
539         local host_mtu
540         host_mtu=$(read_env_var HOST_MTU)
541         host_mtu=${host_mtu:-1200}
542
543         # Lower the host Tailscale MTU (defaults to 1200) to prevent fragmentation when passing
544         # through the container's WARP tunnel (which also has an MTU of 1280).
545         sudo -n ifconfig "$ts_iface" mtu "$host_mtu" >> output.log 2>&1 || true
546
547         # DO NOT delete the physical default route! It crashes Colima.
548         # Use the /1 trick to mathematically override the default route.
549         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
550         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
551         sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
552         sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
553
554         if netstat -nr | grep -E -q "^(0\.\0\.\0/1|0/1)[[:space:]]+.*$ts_iface"; then
555             echo " [] Default route successfully overridden via $ts_iface."
556         else
557             echo " [!] Failed to verify exit node routing on $ts_iface."
558         fi
559     else
560         echo "[Warning] Could not detect Tailscale utun interface for host routing."
561     fi
562 }
563
564 # Helper to restore host DNS to a clean state (used on script start, on failure,
565 # and after a successful STOP). Without this, a successful toggle-off leaves the
566 # host's resolver pinned to a now-dead gateway IP every lookup stalls for macOS's
567 # DNS timeout (~5s) before falling through to whatever next server is configured.
568 # With it, we always leave the host in `1.1.1.1 / no search domain`.
569 #
570 # We set DNS on BOTH the active service AND the en0 service because macOS's scutil DNS
571 # resolver is commonly scoped to en0 (Wi-Fi). A per-service change on e.g.
572 # "USB 10/100 LAN" is ignored by the system resolver unless the en0 service is also updated.
573 reset_dns() {
574     if [ -n "$ACTIVE_SERVICE" ]; then
575         for dns_svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
576             networksetup -setsearchdomains "$dns_svc" "Empty" 2>> output.log || true
577             networksetup -setdnsservers "$dns_svc" 1.1.1.1 2>> output.log || true
578             done
579         fi
580     }
581
582 # Helper to nuke leftover network state after teardown (proxy settings, routing
583 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
584 # problem where stale state kills internet after the gateway stops.
585 cleanup_network_state() {
586     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."
587
588     # 1. Disable any leftover proxy settings (needs root on many macOS versions)
589     for svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
590         disable_all_proxies "$svc"
591     done
592     echo " Proxies disabled."
593
594     # 2. Flush DNS cache
595     if command -v dscacheutil >> output.log 2>&1; then
596         sudo -n dscacheutil -flushcache 2>> output.log || true
597     fi
598     sudo -n killall -HUP mDNSResponder 2>> output.log || true
599     echo " DNS cache flushed."
600
601     # 3. Flush stale routing table entries
602     if command -v route >> output.log 2>&1; then
603         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
604         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
605     fi
606     remove_warp_bypass_routes
607     disable_killswitch
608     echo " Routing table and firewall flushed."
609
610     # 4. Power-cycle Wi-Fi to clear any lingering interface state
611     WIFI_PORT=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print $2
    }')

```

```

612 if [ -n "$WIFI_PORT" ]; then
613     sudo -n networksetup -setairportpower "$WIFI_PORT" off 2>> output.log || true
614     sleep 2
615     sudo -n networksetup -setairportpower "$WIFI_PORT" on 2>> output.log || true
616     echo " Wi-Fi power-cycled (interface $WIFI_PORT)."
617     sleep 3
618 else
619     echo " Wi-Fi interface not detected; bouncing en0..."
620     sudo -n ifconfig en0 down 2>> output.log || true
621     sudo -n ifconfig en0 up 2>> output.log || true
622 fi
623
624 # Force restart sharing services to prevent AirDrop / AirPlay discovery freezes
625 # after network interface/DNS changes.
626 echo " Resetting macOS sharing services (AirDrop/AirPlay)..."
627 reset_sharing_services
628
629 echo "Network state cleanup complete."
630 }
631
632 # SOCKS5 Proxy (WARP Tunnel)
633 # When Tailscale's exit node can't route through WARP (due to userspace
634 # forwarding), we use a SOCKS5 proxy running inside the warp container's
635 # network namespace. Connections created by this proxy go through the
636 # kernel routing table (table 200 -> tun0 -> WARP), unlike Tailscale's
637 # userspace exit node which bypasses it.
638 #
639 # The SOCKS5 proxy is exposed on localhost:${SOCKS_PROXY_PORT} via Docker port mapping.
640 # `networksetup -setsocksfirewallproxy` on macOS routes system TCP traffic
641 # through the proxy, which then goes through WARP.
642 #
643 # IMPORTANT: CLI tools like curl do NOT respect macOS system proxy settings.
644 # They must use --socks5-hostname or the ALL_PROXY env variable explicitly.
645
646 SOCKS_PROXY_PORT=${SOCKS_PROXY_PORT}
647
648 enable_socks_proxy() {
649     local svc="$1"
650     if [ -z "$svc" ]; then
651         svc="$ACTIVE_SERVICE"
652     fi
653
654     echo -n " Enabling system-wide SOCKS5 proxy on $svc (localhost:${SOCKS_PROXY_PORT})... "
655     sudo -n networksetup -setsocksfirewallproxy "$svc" 127.0.0.1 $SOCKS_PROXY_PORT 2>> output.log || true
656     sudo -n networksetup -setsocksfirewallproxystate "$svc" on 2>> output.log || true
657
658     # Also set on en0 service for macOS resolver consistency
659     if [ "$svc" != "$ENO_SERVICE" ]; then
660         sudo -n networksetup -setsocksfirewallproxy "$ENO_SERVICE" 127.0.0.1 $SOCKS_PROXY_PORT 2>> output.log ||
661             true
662         sudo -n networksetup -setsocksfirewallproxystate "$ENO_SERVICE" on 2>> output.log || true
663     fi
664
665     # IMPORTANT: Exclude local LAN and mDNS traffic from the proxy so P2P works natively
666     # without source-IP masking from the Colima VM bridge.
667     sudo -n networksetup -setproxybypassdomains "$svc" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
668     172.16.0.0/12 "*.local" 2>> output.log || true
669     if [ "$svc" != "$ENO_SERVICE" ]; then
670         sudo -n networksetup -setproxybypassdomains "$ENO_SERVICE" 127.0.0.1 localhost 192.168.0.0/16 10.0.0.0/8
671         172.16.0.0/12 "*.local" 2>> output.log || true
672     fi
673
674     echo "done."
675     echo " All TCP traffic now routed through gateway -> WARP tunnel -> internet."
676     echo " (CLI tools: export ALL_PROXY=socks5://127.0.0.1:${SOCKS_PROXY_PORT})"
677 }
678
679 disable_socks_proxy() {
680     for svc in "$ACTIVE_SERVICE" "$ENO_SERVICE"; do
681         sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
682     done
683     echo " SOCKS5 proxy disabled."
684 }
685
686 # Local DNS Proxy (Python)
687 # When the tailnet data plane cannot establish, we use a Python DNS proxy.
688 # It listens on UDP:53, receives DNS queries, forwards them over TCP to
689 # AdGuard at 127.0.0.1:${DNS_PROXY_PORT} (Docker port mapping), and returns the response.
690 # The Python script handles the DNS-over-TCP wire format (2-byte length prefix)
691 # correctly unlike socat which just forwards raw bytes.
692 #

```

```

690 # Python3 is built into macOS no external dependencies.
691 DNS_PROXY_PID=""
692
693 # Kill any leftover DNS proxy process
694 stop_local_dns_proxy() {
695     if [ -n "$DNS_PROXY_PID" ]; then
696         kill "$DNS_PROXY_PID" 2>/dev/null || true
697         wait "$DNS_PROXY_PID" 2>/dev/null || true
698         DNS_PROXY_PID=""
699     fi
700     # Clean up any stale Python DNS proxy processes
701     sudo -n pkill -f "dns-proxy.py" 2>/dev/null || true
702 }
703
704 # Start local DNS proxy (Python)
705 start_local_dns_proxy() {
706     if [ -z "$DNS_PROXY_BIN" ]; then
707         echo " Python not found. Python3 is built into macOS this shouldn't happen."
708         return 1
709     fi
710
711     if [ ! -f "$DNS_PROXY_SCRIPT" ]; then
712         echo " DNS proxy script not found at $DNS_PROXY_SCRIPT"
713         return 1
714     fi
715
716     # Kill any leftover process first
717     stop_local_dns_proxy
718
719     echo -n " Starting local DNS proxy via Python (UDP:53 TCP:${DNS_PROXY_PORT})... "
720     sudo -n bash -c "TARGET_PORT=$DNS_PROXY_PORT \"$DNS_PROXY_BIN\" \"$DNS_PROXY_SCRIPT\" &
721     DNS_PROXY_PID=$!
722     disown "$DNS_PROXY_PID" 2>/dev/null || true
723
724     # Give it a moment to bind
725     sleep 0.5
726
727     if kill -0 "$DNS_PROXY_PID" 2>/dev/null; then
728         echo "started (PID $DNS_PROXY_PID)."

```



```

771 # investigate which is the correct failure mode.
772 force_dns_to_gateway() {
773     local target_ip="$1"
774     if [ -z "$target_ip" ] || [ -z "$ACTIVE_SERVICE" ]; then
775         echo " [force_dns] ERROR: target_ip or ACTIVE_SERVICE is empty" >&2
776         return 1
777     fi
778
779     local attempt
780     for attempt in 1 2 3; do
781         echo -n " [force_dns] Attempt $attempt/3: setting DNS to $target_ip on \"\$ACTIVE_SERVICE\" + \"
782             \$ENO_SERVICE\"... "
783
784         # Apply to BOTH the active service AND the en0 service the scutil DNS resolver
785         # is scoped to en0 (Wi-Fi) and ignores per-service changes that don't include it.
786         # Fail fast on ACTIVE_SERVICE; best-effort on ENO_SERVICE (it may not exist).
787         networksetup -setsearchdomains "$ACTIVE_SERVICE" "ts.net" || true
788         if ! networksetup -setdnsservers "$ACTIVE_SERVICE" "$target_ip"; then
789             echo "FAILED (networksetup error check permissions)"
790             sleep 1
791             continue
792         fi
793         networksetup -setsearchdomains "$ENO_SERVICE" "ts.net" || true
794         networksetup -setdnsservers "$ENO_SERVICE" "$target_ip" || true
795
796         local entries
797         entries=$(networksetup -getdnsservers "$ACTIVE_SERVICE" 2>>> output.log \
798             | awk '/^(25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\. (25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)
799             \. (25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)\. (25[0-5]|2[0-4][0-9]|[01]?[0-9][0-9]?)$/ {print}')
800         if echo "$entries" | grep -qFx "$target_ip"; then
801             echo "VERIFIED"
802             return 0
803         fi
804
805         local current
806         current=$(echo "$entries" | tr '\n' ' ' | sed 's/ $//')
807         echo "NOT YET (current DNS: [{current:-none}])"
808
809         if [ "$attempt" = "3" ]; then sleep 2; else sleep 1; fi
810     done
811
812     echo " [force_dns] FAILED after 3 attempts"
813     return 1
814 }
815
816 # Ensure DNS is set to 1.1.1.1 immediately at script start to prevent any DNS deadlocks/hangs
817 # during status checks, container startup, VM boot, or teardown. We *also* clear any leftover
818 # `ts.net` search domain from a previous `tailscale up --accept-dns=true` run, otherwise macOS
819 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
820
821 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
822 reset_dns
823
824
825
826 echo "Checking Gateway Status..."
827 HIJACK_HOST=$(read_env_var GATEWAY_HIJACK_HOST | tr '[:upper:]' '[:lower:]')
828
829 if is_gateway_active; then
830     echo -e "\n=====
831     echo "Gateway is RUNNING. Stopping it now..."
832     echo -e "=====
833
834 # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
835 if [[ "$HIJACK_HOST" != "false" ]]; then
836     disconnect_tailscale_host
837     echo ""
838 fi
839
840 echo "Stopping Docker containers..."
841 docker compose down --remove-orphans -t 30
842
843 # Only stop Colima if the user explicitly opted in (false by default) to prevent breaking their other Docker
844 # dev projects
845 STOP_COLIMA=$(read_env_var STOP_COLIMA_ON_EXIT | tr '[:upper:]' '[:lower:]')
846 if [[ "$STOP_COLIMA" == "true" ]]; then
847     echo -e "\nStopping Colima VM to free up host RAM and battery..."
848     run_with_timeout 30 colima stop >> output.log 2>&1 || echo "Warning: Failed to stop Colima gracefully."
849 else

```

```

849     echo -e "\nLeaving Colima running. (Set STOP_COLIMA_ON_EXIT=true in .env to change this)"
850 fi
851
852 if [[ "$HIJACK_HOST" != "false" ]]; then
853     # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
854     # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
855     # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
856     echo -e "\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
857     reset_dns
858
859     # 3. Stop local DNS proxy if running
860     stop_local_dns_proxy
861
862     # Stop background daemons
863     stop_pidfile_daemon "/tmp/nullexit-cafeinate.pid" "system sleep prevention"
864     stop_pidfile_daemon "/tmp/nullexit-dns-watcher.pid" "background DNS Watcher"
865     stop_warp_watcher
866     clear_gateway_active_marker
867
868     # 4. Nuke leftover network state so internet actually works after teardown
869     cleanup_network_state
870 fi
871
872 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
873 echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
874 else
875     START_GATEWAY=true
876     echo "[$(date +%Y-%m-%d %H:%M:%S)] Action: STARTUP GATEWAY" >> "$LOG_FILE"
877     echo -e "\n===== "
878     echo "Gateway is STOPPED. Starting it now..."
879     echo -e "===== \n"
880
881     # 1. Prevent host exit-node deadlock during VM / Container startup
882     if [[ "$HIJACK_HOST" != "false" ]]; then
883         disconnect_tailscale_host
884     fi
885
886     # 2. Wait for physical DHCP lease to settle (crucial for restarts/wake-up/roaming)
887     wait_for_dhcp_settle
888
889     # 3. Boot Colima VM if it is not already running
890     echo -e "\nChecking Colima VM status..."
891     if ! run_with_timeout 15 colima status >> output.log 2>&1; then
892         colima_network_mode="bridged"
893         security_type=""
894         if [[ "$OSTYPE" == "darwin*" ]]; then
895             security_type=$(system_profiler SPAirPortDataType 2>/dev/null | awk '/Current Network Information:/{found
896             =1} found && /Security:/{print $NF; exit}')
897             if echo "$security_type" | grep -qi "Enterprise"; then
898                 echo -e "\n [!] WPA2-Enterprise Wi-Fi detected! Bridged mode will cause MAC-spoofing deauthentication.
899                 "
900                 echo "      Falling back to '--network-mode shared' for Colima."
901                 colima_network_mode="shared"
902             fi
903         fi
904         echo "Colima is not running. Starting Colima (600MB RAM allocation, vz VM, network address,
905         $colima_network_mode)..."
906         run_with_timeout 120 colima start --memory 0.6 --vm-type vz --network-address --network-mode "
907         $colima_network_mode"
908         echo "$colima_network_mode" > /tmp/nullexit_colima_mode.txt
909     else
910         echo "Colima is already running."
911     fi
912
913     # 3b. Auto-detect and fix Docker Subnet Collisions (e.g. if the user travels to a 172.17.x.x Wi-Fi)
914     if [ -f "scripts/fix-docker-bridge-collision.sh" ]; then
915         if bash scripts/fix-docker-bridge-collision.sh --dry | grep -q "collision detected"; then
916             echo -e "\n[!] WARNING: Docker Subnet Collision Detected with your current Wi-Fi!"
917             echo -e "      Auto-fixing Colima config to prevent internet jitter..."
918             bash scripts/fix-docker-bridge-collision.sh --skip-toggle || true
919         fi
920     fi
921
922     # 3c. Configure swap file inside the VM to prevent OOM on low-memory limits
923     # We also set vm.swappiness=10 so the Linux kernel strictly prefers physical RAM
924     # and avoids unnecessarily wearing out the SSD with proactive background swapping.
925     if ! run_with_timeout 15 colima ssh -- grep -q 'swapfile' /proc/swaps >> output.log 2>&1; then
926         echo "Configuring 400MB swap file inside the VM to prevent OOM..."
927         run_with_timeout 30 colima ssh -- sudo sh -c "if [ ! -f /swapfile ]; then dd if=/dev/zero of=/swapfile bs=1
928         M count=400 status=none && mkswap /swapfile; fi && swapon /swapfile && sysctl vm.swappiness=10" >> output.
929         log 2>&1 || echo "Warning: Failed to enable swap file inside the VM."

```

```

924 fi
925
926 # 4. Clean up corrupted AdGuardHome configurations
927 if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then
928     rm -f "adguard/conf/AdGuardHome.yaml"
929     echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
930 fi
931
932 # 4b. Auto-heal stale Docker socket from hard reboots
933 if ! run_with_timeout 15 docker ps >/dev/null 2>&1; then
934     echo "Docker daemon is unresponsive (likely a stale socket from a crash). Auto-healing Colima..."
935     run_with_timeout 120 colima restart >> output.log 2>&1
936 fi
937
938 # 4c. Inject TCP MSS clamp from .env into post-rules.txt before starting
939 if [ -f .env ] && grep -q "^GATEWAY_MSS=" .env; then
940     GATEWAY_MSS=$(read_env_var GATEWAY_MSS)
941     if [ "$GATEWAY_MSS" =~ ^[0-9]+$ ]; then
942         # macOS sed syntax
943         sed -i '' "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null || \
944         # Linux sed fallback
945         sed -i "s/--set-mss [0-9]*/--set-mss ${GATEWAY_MSS}/g" post-rules.txt 2>/dev/null
946     fi
947 fi
948
949 # 5. Start compose services
950 write_host_ips
951
952 # Procedurally generated ports have already been exported at the top of the script.
953 echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
954
955 # --- HONEY-PORT TRIPWIRE ---
956 # Generate a random trap port. If any malware does a sequential port scan on localhost,
957 # it will inevitably hit this port. When it does, we instantly fire a macOS alert.
958 HONEY_PORT=$(get_free_port)
959 (
960     if nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then
961         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
962             just pinged the local Honey-Port ($HONEY_PORT)!" >> "$LOG_FILE"
963         osascript -e 'display notification "An unknown application is aggressively scanning your local ports.
964             Malware port-scan suspected." with title "nullexit OPSEC Alert" sound name "Basso" 2>/dev/null || true
965         fi
966     ) &
967     disown %1 2>/dev/null || true
968     echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
969
970     echo -e "\nStarting Docker containers..."
971     docker compose up -d --build
972
973     # Log the output of the rule compiler for debugging before removing it
974     docker compose logs rule-compiler >> output.log 2>&1
975     # Clean up the one-off rule compiler container so it doesn't clutter the Docker UI
976     docker compose rm -s -f rule-compiler >> output.log 2>&1
977
978     RULE_COUNT=$(grep "! Total Block Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
979     NATIVE_COUNT=$(grep "! Native AdGuard Rules:" adguard/work/userfilters/compiled_rules.txt 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
980
981     if [ "$RULE_COUNT" != "0" ] && [ "$RULE_COUNT" != "" ]; then
982         if [ "$NATIVE_COUNT" != "0" ] && [ "$NATIVE_COUNT" != "" ]; then
983             TOTAL_COUNT=$((RULE_COUNT + NATIVE_COUNT))
984             # Format with commas for readability (e.g. 441,578)
985             if command -v printf &> /dev/null; then
986                 FORMATTED_RULE=$(printf "%'d" "$RULE_COUNT")
987                 FORMATTED_TOTAL=$(printf "%'d" "$TOTAL_COUNT")
988                 echo " DNS: Compiled $FORMATTED_RULE unique custom rules (Total active in AdGuard: ~$FORMATTED_TOTAL
989                     rules)."
990             else
991                 echo " DNS: Compiled $RULE_COUNT unique custom rules (Total active in AdGuard: ~$TOTAL_COUNT rules)."
992             fi
993         fi
994         echo " DNS: Compiled and loaded $RULE_COUNT optimized rules."
995     fi
996 fi
997
998 # Report IP blocklist compilation results
999 IP_COUNT=$(grep "^# Entries:" adguard/work/userfilters/ip_blocklist.ipset 2>/dev/null | awk -F': ' '{print $2}' || echo "0")
1000 if [ "$IP_COUNT" != "0" ] && [ "$IP_COUNT" != "" ]; then
1001     if command -v printf &> /dev/null; then

```

```

999     FORMATTED_IP=$(printf "%d" "$IP_COUNT")
1000     echo " IP: Loaded $FORMATTED_IP threat intelligence IPs/CIDRs into kernel firewall."
1001     else
1002     echo " IP: Loaded $IP_COUNT threat intelligence IPs/CIDRs into kernel firewall."
1003     fi
1004 fi
1005
1006 # 6. Wait for the gateway container's Tailscale connection to be ready
1007 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
1008 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
1009
1010 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
1011 echo " (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
1012 connected=false
1013 consecutive_failures=0
1014 for i in {1..60}; do
1015     # Run tailscale status and capture its output so the user can see what's happening.
1016     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
1017     ts_output=$(run_with_timeout 5 docker compose exec -T tailscale tailscale status 2>&1 || true)
1018
1019     if echo "$ts_output" | grep -q "offers exit node"; then
1020         connected=true
1021         break
1022     fi
1023
1024     # Track consecutive failures to detect a stuck state.
1025     # If we see 40+ consecutive 'NoState' responses, give up early
1026     # the daemon is alive but can't connect to the coordination server.
1027     if echo "$ts_output" | grep -q "NoState"; then
1028         consecutive_failures=$((consecutive_failures + 1))
1029         if [ "$consecutive_failures" -ge 40 ]; then
1030             echo ""
1031             echo " [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
1032             echo " Tailscale daemon is running but can't reach the coordination server."
1033             break
1034         fi
1035     else
1036         consecutive_failures=0
1037     fi
1038
1039     # Print a condensed status every 10 seconds so the user can see progress
1040     if [ $((i % 10)) -eq 0 ]; then
1041         ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
1042         echo " [attempt $i/60] $ts_short"
1043     elif [ $((i % 5)) -eq 0 ]; then
1044         echo -n "$[i]"
1045     else
1046         echo -n "."
1047     fi
1048     sleep 1
1049 done
1050 echo ""
1051
1052 if [ "$connected" = "true" ]; then
1053     echo "Gateway container is online on the Tailscale mesh."
1054 elif [ "$BYPASS_PING" = "true" ]; then
1055     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
1056 else
1057     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
1058     echo " The container was seen in the following states during the wait:" >&2
1059     echo "$ts_output" | sed 's/~ / /' >&2
1060     echo "" >&2
1061     echo " This could mean:" >&2
1062     echo " 1. Tailscale auth key has expired check TS_AUTHKEY in .env" >&2
1063     echo " 2. Network issue inside the container check: docker compose logs tailscale" >&2
1064     echo " 3. gluetun VPN tunnel not connecting check: docker compose logs warp | tail -20" >&2
1065     echo "" >&2
1066     echo " Diagnose manually:" >&2
1067     echo " docker compose exec -T tailscale tailscale status" >&2
1068     echo " docker compose exec -T tailscale tailscale netcheck" >&2
1069     cleanup_handler ERR $LINENO
1070     exit 1
1071 fi
1072
1073 # 6b. Resolve the gateway's Tailscale IP.
1074 # Dynamically query the container for the IP (handles IP changes after re-auth)
1075 # and cache it in .gateway_ip for fast retrieval by recover.sh during Wi-Fi roaming.
1076 #
1077 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
1078 # returns (\r) into captured output. If $TS_IP ends with \r, then
1079 # `networksetup -setdnsservers` silently rejects the malformed IP and

```

```

1080 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
1081 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
1082 # the first field of the first line.
1083 TS_IP=""
1084 if [ "$connected" = "true" ]; then
1085     TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
1086         awk 'NR==1{print $1; exit}' || true)
1087     if [ -n "$TS_IP" ]; then
1088         echo "Resolved gateway Tailscale IP dynamically: $TS_IP"
1089         echo "$TS_IP" > .gateway_ip
1090     else
1091         echo " [!] Failed to resolve gateway IP dynamically. Trying fallback cache..." >&2
1092         TS_IP=$(read_adguard_ip || true)
1093     fi
1094     TS_IP=$(read_adguard_ip || true)
1095 fi
1096
1097 # 7. Connect host Mac to Tailscale mesh.
1098 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
1099 # system LaunchDaemon via 'brew services start tailscale'), the previous five-
1100 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
1101 # no menu bar item to click, and no Accessibility permission required.
1102 #
1103 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
1104 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
1105 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
1106 # are impossible without the host on the mesh).
1107 #
1108 # We connect in two phases:
1109 # Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
1110 # Phase B Set exit node only after pre-flight checks confirm the gateway works
1111 HOST_ON_MESH=false
1112 SKIP_EXIT_NODE=false
1113 if [ "$HIJACK_HOST" = "false" ]; then
1114     echo -e "\n[Info] HIJACK_HOST is false (Headless Mode). Host networking will remain untouched."
1115     SKIP_EXIT_NODE=true
1116 elif [ -n "$TS_BIN" ]; then
1117     echo -n "Verifying tailscaled is reachable"
1118     daemon_ready=false
1119     status_exit=0
1120     for i in {1..10}; do
1121         # Test if the daemon responds to status check.
1122         # If status returns non-zero, check if it was a normal "stopped/disconnected" state
1123         # (exit code 1 is normal for disconnected). If the command exited quickly (not killed by timeout),
1124         # and the error is just "Tailscale is stopped", then the daemon is alive and ready.
1125         # But if it timed out (exit code 143) or is completely unresponsive, it's wedged.
1126         status_exit=0
1127         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
1128             daemon_ready=true
1129             break
1130         else
1131             status_exit=$?
1132             if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
1133                 daemon_ready=true
1134                 break
1135             fi
1136         fi
1137         echo -n "."
1138         sleep 1
1139     done
1140     echo ""
1141
1142     # If the daemon was unresponsive or socket check failed, attempt auto-restart
1143     if [ "$daemon_ready" != "true" ]; then
1144         if [ "$status_exit" -eq 143 ]; then
1145             warn "tailscaled daemon is wedged/unresponsive. Restarting it..."
1146         else
1147             warn "tailscaled daemon is not running. Attempting to start it..."
1148         fi
1149         restart_tailscaled_daemon
1150
1151         # Re-verify
1152         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
1153             daemon_ready=true
1154         fi
1155     fi
1156
1157     if [ "$daemon_ready" = "true" ]; then
1158         echo "tailscaled is responsive (running as a system service)."
1159     fi

```

```

1160 # Phase A: Join mesh without exit node
1161 echo "Connecting host to Tailscale mesh (no exit node yet)..."
1162 if $TS_BIN up; then
1163     $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node= || true
1164     HOST_ON_MESH=true
1165     echo "Host is on Tailscale mesh."
1166 else
1167     warn "tailscale up failed even though the daemon is running."
1168     echo " This usually means the host hasn't authenticated with your tailnet yet."
1169     echo " Run this command in a terminal to authenticate:"
1170     echo ""
1171     echo "         sudo tailscale up"
1172     echo ""
1173     echo " A browser will open. Log in to your Tailscale account, then re-run toggle.sh."
1174 fi
1175 else
1176     fail "tailscaled could NOT be started automatically."
1177     echo " Run this in a terminal to start and authenticate Tailscale:"
1178     echo ""
1179     echo "         sudo brew services start tailscale"
1180     echo "         sudo tailscale up"
1181     echo ""
1182     echo " Then re-run toggle.sh."
1183 fi
1184 else
1185     echo -e "\n[Warning] tailscale CLI not found on PATH. Skipping host Tailscale configuration..."
1186 fi
1187
1188 # Phase B: Pre-flight checks + enable exit node only if safe
1189 if [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" != "false" ] && [ -n "$TS_IP" ]; then
1190     echo -e "\n"
1191     echo "Pre-flight connectivity check"
1192     echo ""
1193
1194     # Check 1: Can we reach the gateway via Tailscale (uses tailscale ping, not ICMP)?
1195     # NOTE: tailscale ping exits 1 when the connection goes through a DERP relay
1196     # ("direct connection not established") even though the pong was received.
1197     # We check for 'pong' in the output (stderr discarded) to accept relayed connections.
1198     echo -n " [1/3] Gateway reachable via Tailscale... "
1199     ping_ok=false
1200     # The host tailscaled needs a few seconds to propagate the new network map and
1201     # establish a connection route after 'tailscale up --reset'. We retry up to 45 times.
1202     for attempt in {1..45}; do
1203         if $TS_BIN ping --until-direct=false -c 2 --timeout 2s "$TS_IP" 2>/dev/null | grep -q "pong"; then
1204             ping_ok=true
1205             break
1206         fi
1207         sleep 1
1208     done
1209     if [ "$ping_ok" = "true" ]; then
1210         echo "PASS"
1211     else
1212         echo "FAIL"
1213         SKIP_EXIT_NODE=true
1214     fi
1215
1216     # Check 2: Can we reach AdGuard via localhost (exposed port ${DNS_PROXY_PORT})?
1217     # Colima's SSH tunnel only forwards TCP (not UDP), so we use +tcp.
1218     echo -n " [2/3] AdGuard DNS via localhost... "
1219     if command -v dig &>/dev/null; then
1220         if dig +tcp @127.0.0.1 -p ${DNS_PROXY_PORT} google.com +short +timeout=5 &>/dev/null; then
1221             echo "PASS"
1222         else
1223             echo "FAIL"
1224             SKIP_EXIT_NODE=true
1225         fi
1226     elif command -v nslookup &>/dev/null; then
1227         if nslookup -port=${DNS_PROXY_PORT} google.com 127.0.0.1 &>/dev/null; then
1228             echo "PASS"
1229         else
1230             echo "FAIL"
1231             SKIP_EXIT_NODE=true
1232         fi
1233     else
1234         echo "SKIP (dig/nslookup not available)"
1235     fi
1236
1237     # Check 3: Does the WARP container have external internet?
1238     echo -n " [3/3] WARP container internet... "
1239     tmp_err=$(mktemp)

```



```

1240 if docker compose exec -T warp wget -q0- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace >/dev/null 2>
1241 "$tmp_err"; then
1242     echo "PASS"
1243 else
1244     echo "FAIL"
1245     if [ -s "$tmp_err" ]; then
1246         err_msg=$(cat "$tmp_err" | tr -d '\r')
1247         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Check 3] WARP container check failed: $err_msg" >> output.log
1248     fi
1249     SKIP_EXIT_NODE=true
1250 fi
1251 rm -f "$tmp_err"
1252
1253 # Act based on check results
1254 if [ "$SKIP_EXIT_NODE" != "true" ]; then
1255     echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
1256     $TS_BIN up || true
1257     if $TS_BIN set --ssh=true --accept-dns=false --accept-routes=true --exit-node="$TS_IP" --exit-node-allow-
lan-access=true; then
1258         echo "Exit node enabled."
1259         add_warp_bypass_routes
1260         setup_exit_node_routing
1261         enable_killswitch
1262     else
1263         echo "[Warning] Failed to set exit node."
1264         SKIP_EXIT_NODE=true
1265     fi
1266 fi
1267
1268 if [ "$SKIP_EXIT_NODE" = "true" ]; then
1269     warn "Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED."
1270     echo "Host stays on Tailscale mesh but your internet routes through your normal connection."
1271     echo "Troubleshoot with:"
1272     echo "docker compose logs warp | tail -20"
1273     echo "docker compose exec -T warp wget -q0- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
1274     echo ""
1275     echo "Falling back to SOCKS5 proxy for traffic routing..."
1276 fi
1277 elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
1278     echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
1279     SKIP_EXIT_NODE=true
1280 fi
1281
1282 # 8. Apply DNS Hijacking.
1283 # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
1284 # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
1285 # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
1286 if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
1287     echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."
1288     echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
1289     if force_dns_to_gateway "$TS_IP"; then
1290         echo "DNS hijack verified: single server = $TS_IP."
1291     else
1292         echo "[Warning] DNS hijack could not be verified."
1293         echo "If DNS is broken, run manually:"
1294         echo "networksetup -setdnsservers \"$ACTIVE_SERVICE\" \"$TS_IP"
1295         echo "networksetup -setsearchdomains \"$ACTIVE_SERVICE\" ts.net"
1296     fi
1297 elif [ -n "$TS_IP" ] && [ "$HIJACK_HOST" != "false" ]; then
1298     echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
1299     echo "Trying local DNS proxy for ad-blocking..."
1300     if start_local_dns_proxy; then
1301         echo "Ad-blocking active via local DNS proxy through AdGuard."
1302     else
1303         echo "Local DNS proxy unavailable. DNS stays at 1.1.1.1."
1304         echo "AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
1305     fi
1306
1307 # Enable SOCKS5 proxy for traffic routing through WARP
1308 # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
1309 echo -e "\nEnabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
1310 echo "(SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
1311 enable_socks_proxy "$ACTIVE_SERVICE"
1312
1313 # Verify the SOCKS5 proxy works through WARP
1314 echo -n "Verifying SOCKS5 proxy routes through WARP..."
1315 if curl --socks5-hostname 127.0.0.1:$SOCKS_PROXY_PORT --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
trace 2>/dev/null | grep -q "warp=on"; then
1316     echo "ok (warp=on)."
1317     echo "All TCP traffic now encrypted through Cloudflare WARP tunnel."
1318 else

```

```

1318     echo "FAILED (proxy not routing through WARP)."
1319     echo "  Disabling SOCKS5 proxy  internet stays on direct connection."
1320     disable_socks_proxy
1321     echo "  SOCKS proxy available at localhost:${SOCKS_PROXY_PORT} for manual configuration."
1322   fi
1323 else
1324     echo -e "\n[Warning] Could not resolve gateway Tailscale IP."
1325 fi
1326
1327 # 9. Verify host connectivity
1328 if [ "$HOST_ON_MESH" = "true" ] && [ -n "$TS_BIN" ]; then
1329     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
1330         echo "Host is online on the Tailscale mesh."
1331     else
1332         echo "[Warning] Host is not responding on Tailscale."
1333     fi
1334 fi
1335
1336 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
1337 echo -e "\nGateway has been successfully STARTED in ${ELAPSED} seconds."
1338
1339 # Prevent system from going to sleep while gateway is running
1340 start_sleep_prevention
1341
1342 if [ [ "$HIJACK_HOST" != "false" ] ]; then
1343     if [ -n "$TS_IP" ] && [ "$TS_IP" != "1.1.1.1" ]; then
1344         start_dns_watcher "$TS_IP"
1345     fi
1346
1347     # Start WARP liveness monitor (logs to output.log on warp flip)
1348     start_warp_watcher
1349 fi
1350
1351 # Reset sharing services after network configuration to prevent AirDrop freezes
1352 echo "Resetting macOS sharing services (AirDrop/AirPlay)..."
1353 reset_sharing_services
1354
1355 # Tell external watchers (post-wake / network-change) the gateway is up.
1356 write_gateway_active_marker
1357
1358 # Local Network Surveillance Check
1359 echo -e "\n"
1360 echo "Local Network Surveillance Check"
1361 echo ""
1362 # Count populated ARP cache entries to detect network scanning or high congestion
1363 # Using '-an' avoids hanging on DNS reverse resolution when the network state is in transition.
1364 ARP_COUNT=$(arp -an 2>/dev/null | grep -iv 'incomplete' | wc -l | tr -d ' ')
1365 if [ "$ARP_COUNT" -gt 15 ]; then
1366     warn "High ARP activity detected ($ARP_COUNT devices in cache)."
1367     warn "This Wi-Fi network may be heavily congested or actively scanned (e.g., arp-scan)."
1368     echo -e "  [] Your traffic remains fully encrypted and invisible.\n"
1369 else
1370     echo -e "  [] Local network looks quiet ($ARP_COUNT devices). No aggressive scanning detected.\n"
1371 fi
1372 fi
1373
1374 SUCCESS_RUN=true
1375
1376 # Final DNS state summary
1377 if [ -n "$ACTIVE_SERVICE" ]; then
1378     FINAL_DNS=$(networksetup -getdnsservers "$ACTIVE_SERVICE" 2>> output.log || true)
1379     echo -e "\n"
1380     echo -e "DNS STATE: $FINAL_DNS"
1381     echo -e ""
1382 fi
1383
1384 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
1385     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
1386         echo -e "\n[Warning] One or more blacklist URLs failed to download (404/Offline)."
1387         echo "  The gateway gracefully fell back to yesterday's cached blacklist."
1388         echo "  (Run 'docker logs rule-compiler' later to investigate which link died.)"
1389     fi
1390 fi
1391
1392 rm -f "${LOCK_FILE:-/tmp/nullexit-toggle.lock}"
1393 if [ -t 0 ]; then
1394     read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
1395     if [ [ "${USER_CHOICE}" == "r" ] || "${USER_CHOICE}" == "R" ]; then
1396         echo "Reversing gateway state..."
1397         exec bash "$0"
1398     fi

```

```

1399 fi
1400
1401 echo -e "\nYou can close this terminal window now."

```

7 recover.sh

```

1 #!/bin/bash
2 # recover.sh nullexit KILL-SWITCH recovery script for macOS
3 # Fixes internet when toggle.sh leaves you stranded.
4 #
5 # This is a NUCLEAR option: it undoes EVERYTHING toggle.sh does by:
6 # 1. Disconnecting host Tailscale from the mesh (exit-node off)
7 # 2. Resetting DNS to default on ALL network services
8 # 3. Disabling rogue proxy settings (SOCKS, web, secure web)
9 # 4. Flushing macOS DNS cache
10 # 5. Flushing stale routing table entries
11 # 6. Stopping gateway Docker containers
12 # 7. Power-cycling Wi-Fi
13 # 8. Verifying internet is working
14 #
15 # Usage: double-click Recover-Gateway.applescript
16 #         OR ./recover.sh
17 #         OR ./recover.sh --post-wake    (lighter-touch refresh, keeps gateway live)
18 #
19 set -e
20
21 # Resolve script directory (used for path-relative refs)
22 # recover.sh lives at the repo root; SCRIPT_DIR lets us launch toggle.sh
23 # (and reference any other repo-root file) from the same directory no
24 # matter where the user invoked recover.sh from.
25 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
26 cd "$SCRIPT_DIR" || exit 1
27
28 # Enforce Cryptographic Script Integrity
29 if [ -f "scripts/crypto.sh" ]; then
30     if ! bash scripts/crypto.sh --verify; then
31         exit 1
32     fi
33 fi
34
35 # Argument parsing
36 # --post-wake: lightweight refresh after sleep/wake or Wi-Fi roam.
37 #             Keeps the gateway live while HUPing DNS caches, refreshing the
38 #             Tailscale exit-node leak, re-hijacking host DNS, and force-
39 #             recreating the warp container if its gluetun healthcheck failed.
40 #             Does NOT tear down Docker, Tailscale, sleep prevention, or Wi-Fi.
41 #             Called from scripts/watcher.sh on every wake / network-change event
42 #             while /tmp/nullexit-gateway-active.marker is present (i.e. the
43 #             gateway is currently up). See devref.md 10.29 for the why.
44 POST_WAKE=false
45 AUTO_MODE=false
46 for arg in "$@"; do
47     if [ "$arg" = "--post-wake" ]; then
48         POST_WAKE=true
49     elif [ "$arg" = "--auto" ] || [ "$arg" = "--non-interactive" ]; then
50         AUTO_MODE=true
51     fi
52 done
53
54 rm -f "$SCRIPT_DIR/TUNNEL_FAILED_CLOSED.marker"
55
56 # Prevent concurrent execution of lifecycle scripts (toggle.sh / recover.sh)
57 LOCK_FILE="/tmp/nullexit-toggle.lock"
58 if [ -f "$LOCK_FILE" ]; then
59     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
60     if [ -n "$LOCK_PID" ] && [ "$LOCK_PID" != "$$" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
61         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -q -E "toggle\.sh|recover\.sh"; then
62             if [ "$POST_WAKE" = "true" ]; then
63                 echo "[$(date -u +%FT%TZ)] POST-WAKE skipped: another lifecycle script (PID $LOCK_PID) is running." >>
64                 output.log
65                 exit 0
66             else
67                 echo "Error: Another instance of toggle.sh or recover.sh (PID $LOCK_PID) is already running." >&2
68                 exit 1
69             fi
70         fi
71     fi
72 fi

```

```

70 fi
71 fi
72 echo "$$" > "$LOCK_FILE"
73
74 # Clear the active-state marker in nuclear recovery mode since the gateway is being
75 # completely torn down. This prevents scripts/watcher.sh from triggering post-wake
76 # recoveries in response to network changes caused by the teardown itself.
77 if [ "$POST_WAKE" = "false" ]; then
78     rm -f "/tmp/nullexit-gateway-active.marker"
79 fi
80
81 # WARP Watcher inhibit marker: written immediately in --post-wake mode so the
82 # background WARP Watcher (in toggle.sh) skips failure counting while we are
83 # intentionally tearing down / recreating the warp container. Without this, the
84 # watcher accumulates 6 consecutive warp=off readings during force-recreate (~35s)
85 # and fires nuclear recover.sh, killing a gateway that would have been healthy.
86 WARP_INHIBIT_FILE="/tmp/nullexit-warp-inhibit.marker"
87 if [ "$POST_WAKE" = "true" ]; then
88     touch "$WARP_INHIBIT_FILE"
89     echo "[(date -u +%FT%TZ)] POST-WAKE started WARP Watcher inhibited to prevent spurious shutdown during warp
90         recreate." >> output.log
91 fi
92
93 # Ensure the inhibit marker and lock file are always removed when the script exits (success or error)
94 cleanup_recover() {
95     rm -f "$WARP_INHIBIT_FILE"
96     if [ -f "$LOCK_FILE" ] && [ "$(cat "$LOCK_FILE" 2>/dev/null)" = "$$" ]; then
97         rm -f "$LOCK_FILE"
98     fi
99 }
100 trap cleanup_recover EXIT
101
102 # Source common formatting and helper functions
103 source "$SCRIPT_DIR/scripts/common.sh"
104
105 # Always add Homebrew path
106 setup_standard_path
107
108 echo ""
109 if [ "$POST_WAKE" = "true" ]; then
110     echo -e "${YELLOW}${BOLD}${NC}"
111     echo -e "${YELLOW}${BOLD} Nullexit POST-WAKE / POST-ROAM LIGHT ${NC}"
112     echo -e "${YELLOW}${BOLD} (keeps the gateway live, refreshes) ${NC}"
113     echo -e "${YELLOW}${BOLD}${NC}"
114     echo ""
115     echo "Re-hijacking DNS, refreshing Tailscale exit-node, force-recreating"
116     echo "warp if unhealthy, and resetting sharing services. The gateway stays"
117     echo "up. docker compose / Wi-Fi / caffeinate are NOT touched."
118     echo ""
119 else
120     echo -e "${RED}${BOLD}${NC}"
121     echo -e "${RED}${BOLD} Nullexit Gateway RECOVERY MODE ${NC}"
122     echo -e "${RED}${BOLD}${NC}"
123     echo ""
124     echo "This script will tear down the gateway and restore normal internet."
125     echo ""
126 fi
127
128 # Cache sudo credentials upfront (default mode ONLY)
129 # Skip in --post-wake: the LaunchAgent-launched watcher has no TTY, so `sudo -v`
130 # would block forever waiting for a password. All post-wake commands use `sudo -n`
131 # which is non-blocking and relies on /etc/sudoers.d/nullexit NOPASSWD entries.
132 SUDO_KEEPER_PID=""
133 if [ "$POST_WAKE" = "false" ] && [ "$AUTO_MODE" = "false" ] && [ -t 0 ]; then
134     echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
135     sudo -v
136     (
137         while true; do
138             sudo -n true 2>/dev/null
139             sleep 60
140         done
141     ) &
142     SUDO_KEEPER_PID=$!
143     trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
144 fi
145
146 # Resolve physical default interface and gateway upfront
147 PHYSICAL_IFACE=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
148     getline; print $2; exit}')
149 [ -z "$PHYSICAL_IFACE" ] && PHYSICAL_IFACE="en0"

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```

149 if [ "$POST_WAKE" = "true" ]; then
150     wait_for_dhcp_settle
151 else
152     # Quick resolution in non-post-wake mode (non-blocking)
153     PHYSICAL_GW=$(ipconfig getpacket "$PHYSICAL_IFACE" 2>> output.log | awk -F'[{}]' '/router/{print $2}')
154 fi
155
156 # 1. Disconnect host Tailscale (default) / Refresh exit-node (post-wake)
157
158 if [ "$POST_WAKE" = "true" ]; then
159     step "Refreshing host Tailscale exit-node routing (post-wake)"
160     if command -v tailscale >> output.log 2>&1; then
161         # Re-issue the SAME exit-node up command toggle.sh START uses. This refreshes
162         # the DERP relay mapping and re-asserts the exit-node preference without
163         # dropping the host's mesh connection (which `tailscale down` would).
164         TS_IP=""
165         for src in .gateway_ip; do
166             [ -f "$src" ] || continue
167             TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
168             [ -n "$TS_IP" ] && break
169         done
170
171         if [ -n "$TS_IP" ]; then
172             step "Re-establishing exit node $TS_IP via 'tailscale set --exit-node'"
173             # `tailscale set` only mutates the named preference (exit-node here).
174             # Crucially it does NOT call --reset, which would re-apply ALL flags and
175             # trigger a sub-second route-table re-evaluation cycle each post-wake.
176             # On already-connected nodes, the lighter `set` keeps the existing tunnel
177             # alive while only changing the exit-node preference.
178             if run_with_timeout 15 tailscale set --exit-node="$TS_IP" \
179                 --exit-node-allow-lan-access=true \
180                 >> output.log 2>&1; then
181                 ok "Exit-node $TS_IP re-asserted via 'tailscale set'"
182             else
183                 warn "tailscale set --exit-node=$TS_IP didn't respond; falling back to mesh-only"
184                 run_with_timeout 10 tailscale set --exit-node= \
185                     >> output.log 2>&1 || true
186             fi
187         else
188             warn ".gateway_ip missing cannot re-assert exit node"
189             run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= \
190                 >> output.log 2>&1 || true
191         fi
192     else
193         warn "tailscale CLI not found"
194     fi
195 else
196     step "Disconnecting host Tailscale from mesh"
197     disconnect_tailscale_host "tailscale"
198 fi
199
200 # 2. Reset DNS to default on ALL network services (default) / Re-hijack gateway DNS (post-wake)
201 if [ "$POST_WAKE" = "true" ]; then
202     step "Re-hijacking host DNS to gateway IP (post-wake / post-roam)"
203     TS_IP=""
204     for src in .gateway_ip; do
205         [ -f "$src" ] || continue
206         TS_IP=$(cat "$src" 2>> output.log | tr -d '\r' | awk 'NR==1{print $1; exit}' || true)
207         [ -n "$TS_IP" ] && break
208     done
209
210     if [ -n "$TS_IP" ]; then
211         # Detect the active network service (using helper in common.sh)
212         ACTIVE_SVC=$(get_active_service)
213         ENO_SVC=$(get_en0_service)
214
215         networksetup -setsearchdomains "$ACTIVE_SVC" "ts.net" 2>> output.log || true
216         networksetup -setdnsservers "$ACTIVE_SVC" "$TS_IP" 2>> output.log || true
217         networksetup -setsearchdomains "$ENO_SVC" "ts.net" 2>> output.log || true
218         networksetup -setdnsservers "$ENO_SVC" "$TS_IP" 2>> output.log || true
219
220         HITS=$(networksetup -getdnsservers "$ACTIVE_SVC" 2>> output.log | grep -Fx "$TS_IP" || true)
221         if [ -n "$HITS" ]; then
222             ok "DNS re-hijacked to $TS_IP on services: $ACTIVE_SVC, $ENO_SVC"
223         else
224             warn "networksetup accepted but DNS read-back didn't include $TS_IP"
225         fi
226     else
227         warn ".gateway_ip missing cannot re-hijack DNS (user must run toggle.sh START)"
228     fi
229 else

```

```

230 step "Resetting DNS to default on all network services (empty = DHCP)"
231
232 # Get ALL network services (handles spaces in names, skips disabled ones)
233 SERVICES=$(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "^$" ||
234     true)
235
236 if [ -z "$SERVICES" ]; then
237     # Fallback to common names
238     SERVICES="Wi-Fi Ethernet Thunderbolt Ethernet USB 10/100 LAN"
239 fi
240
241 RESET_COUNT=0
242 while IFS= read -r service; do
243     # Strip leading asterisk (disabled services)
244     service_clean=$(echo "$service" | sed 's/^\*//')
245     [ -z "$service_clean" ] && continue
246
247     # Check if this service has a hardware port (skip VPN/service-only entries)
248     if networksetup -listallhardwareports 2>> output.log | grep -A1 "Port: $service_clean$" | grep -q "Device:"
249     || [ "$service_clean" = "Wi-Fi" ]; then
250     (
251         networksetup -setsearchdomains "$service_clean" "Empty" 2>> output.log
252         # Set to "empty" so macOS uses DHCP-assigned DNS (not hardcoded 1.1.1.1)
253         sudo networksetup -setdnsservers "$service_clean" "empty" 2>> output.log
254     ) && RESET_COUNT=$((RESET_COUNT + 1)) || true
255 fi
256 done <<< "$SERVICES"
257
258 ok "DNS reset on $RESET_COUNT service(s)"
259 fi
260
261 # 3. Disable rogue proxy settings (default only gateway uses proxy in non-exit-node path)
262 if [ "$POST_WAKE" = "false" ]; then
263     step "Disabling proxy settings (SOCKS, web, secure web)"
264     for svc in $(networksetup -listallnetworkservices 2>> output.log | grep -v "^An asterisk" | grep -v "^$" ||
265         echo "Wi-Fi"); do
266         svc_clean=$(echo "$svc" | sed 's/^\*//')
267         [ -z "$svc_clean" ] && continue
268         disable_all_proxies "$svc_clean"
269     done
270     ok "Proxy settings cleared"
271 else
272     step "Proxy settings: leaving untouched (post-wake keeps gateway SOCKS wiring alive)"
273 fi
274
275 # 4. Flush macOS DNS cache (always safe and useful in both modes)
276 step "Flushing DNS cache"
277 if command -v dscacheutil >> output.log 2>&1; then
278     flush_dns_cache
279     ok "DNS cache flushed (mDNSResponder HUP)"
280 else
281     warn "dscacheutil not found"
282 fi
283
284 # 5. Clear stale routing table (always safe in both modes)
285 step "Flushing stale routing table entries"
286 # Resolve physical default gateway IP before flushing
287 # We cannot trust `route get default` here because Tailscale may have already hijacked
288 # the default route to utunX. We must read the actual DHCP router assignment from the hardware.
289 # PHYSICAL_IFACE and PHYSICAL_GW resolved upfront
290
291 # Instead of full route -n flush (which destroys macOS loopback/multicast and wedges the OS until reboot),
292 # we cleanly delete the Tailscale overrides and Cloudflare WARP bypass routes in ALL modes.
293 # This ensures tailscaled isn't blocked from reaching its control plane when waking from sleep.
294 sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
295 sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
296 remove_warp_bypass_routes
297 ok "Routing overrides cleared"
298
299 if [ "$POST_WAKE" = "false" ]; then
300     disable_killswitch
301     ok "Routing table flush completed via targeted deletes and firewall disabled"
302 else
303     ok "Routing overrides cleared (post-wake mode to preserve Colima bridge100)"
304 fi
305
306 # In post-wake mode we don't bounce Wi-Fi, so we MUST manually restore the physical default route
307 if [ "$POST_WAKE" = "true" ] && [ -n "$PHYSICAL_GW" ]; then
308     # Ensure physical gateway is set just in case it was dropped
309     sudo -n route add default "$PHYSICAL_GW" >> output.log 2>&1 || true
310 fi

```



```

308
309 # CONDITIONAL BYPASS ROUTE RESTORATION:
310 # If this was a post-wake recovery and the exit node is active, the route flush
311 # just wiped the static bypass routes for the WARP endpoints. We must re-add
312 # them immediately to prevent a routing loop when Tailscale starts sending traffic.
313 if [ "$POST_WAKE" = "true" ]; then
314     if [ -z "${ACTIVE_IF:-}" ]; then
315         ACTIVE_IF=$(route get default 2>> output.log | awk '/interface:/{print $2; exit}')
316         if [ -z "$ACTIVE_IF" ] || [[ "$ACTIVE_IF" =~ ^tun ]] || [[ "$ACTIVE_IF" =~ ^tun ]]; then
317             for i in en0 en1 en2 en3; do
318                 if ifconfig "$i" 2>> output.log | grep -q 'status: active'; then
319                     ACTIVE_IF="$i"; break
320                 fi
321             done
322         fi
323         [ -z "$ACTIVE_IF" ] && ACTIVE_IF="en0"
324     fi
325
326     echo "Re-adding host bypass routes for Cloudflare WARP endpoints..."
327     remove_warp_bypass_routes
328     if [ -n "${PHYSICAL_GW:-}" ]; then
329         add_warp_bypass_routes "$PHYSICAL_GW"
330     else
331         add_warp_bypass_routes "$ACTIVE_IF"
332     fi
333
334     # Also re-apply the default route pointing to the Tailscale interface
335     ts_iface=$(ifconfig 2>> output.log | grep -B4 "inet 100." | grep -E '[a-z0-9]+' | cut -d: -f1 | head -n 1)
336     if [ -n "$ts_iface" ]; then
337         echo "Re-routing default gateway to $ts_iface using 0.0.0.0/1 split..."
338         # DO NOT delete the physical default route! It crashes Colima.
339         # Use the /1 trick to mathematically override the default route.
340         sudo -n route delete -net 0.0.0.0/1 >> output.log 2>&1 || true
341         sudo -n route delete -net 128.0.0.0/1 >> output.log 2>&1 || true
342         sudo -n route add -net 0.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
343         sudo -n route add -net 128.0.0.0/1 -interface "$ts_iface" >> output.log 2>&1 || true
344         enable_killswitch
345     fi
346 fi
347
348 # 6. Stop gateway Docker containers (default) / Force-recreate warp if unhealthy (post-wake)
349 if [ "$POST_WAKE" = "true" ]; then
350     # 6a. Check for Roaming Network Mismatch (Bridged vs Enterprise Wi-Fi)
351     if [ -f "/tmp/nullexit_colima_mode.txt" ] && [ -f "$SCRIPT_DIR/../../lan_p2p_detected" ]; then
352         current_colima_mode=$(cat /tmp/nullexit_colima_mode.txt)
353         p2p_allowed=$(cat "$SCRIPT_DIR/../../lan_p2p_detected")
354         required_colima_mode="bridged"
355         [ "$p2p_allowed" == "false" ] && required_colima_mode="shared"
356
357         if [ "$current_colima_mode" != "$required_colima_mode" ]; then
358             warn "Network roaming mismatch detected!"
359             warn "Colima is running in '$current_colima_mode' mode but new network requires '$required_colima_mode'."
360             warn "Executing full gateway restart to protect against MAC-spoofing deauthentication..."
361             echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Roam] Mismatch ($current_colima_mode vs $required_colima_mode).
362             Triggering toggle.sh --restart." >> output.log
363             nohup bash "$SCRIPT_DIR/../../toggle.sh" --restart >/dev/null 2>&1 &
364             exit 0
365         fi
366     fi
367
368     step "Checking Colima VM state"
369     colima_status=$(colima status 2>&1)
370     if echo "$colima_status" | grep -qi "running"; then
371         ok "Colima VM is running"
372         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima status check passed." >> output.log
373     else
374         warn "Colima VM is NOT running or unresponsive"
375         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Colima is unhealthy or dead! Output: $colima_status"
376         >> output.log
377         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Attempting to restart Colima..." >> output.log
378         colima restart >> output.log 2>&1 || warn "Failed to restart Colima"
379     fi
380
381     step "Checking warp container health (gluetun UDP tunnel)"
382     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Inspecting warp container state..." >> output.log
383     # The warp container is the most failure-prone link in the chain at wake/roam:
384     # its UDP NAT binding to Cloudflare's edge (162.159.192.1:2408) silently dies
385     # during a Wi-Fi roam. We verify the tunnel is actually passing traffic via curl.
386     WARP_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
387     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container status: $WARP_STATE" >> output.log

```

```

387 if [ "$WARP_STATE" = "missing" ]; then
388     warn "warp container is missing leaving gateway containers alone (full relaunch requires \"\$SCRIPT_DIR/
toggle.sh\")"
389     echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container missing from Docker engine. Requires
full toggle.sh launch." >> output.log
390 else
391     tmp_err=$(mktemp)
392     if docker compose exec -T warp wget -q0- --timeout=3 https://www.cloudflare.com/cdn-cgi/trace 2>"$tmp_err"
| grep -q "warp=on"; then
393         ok "warp container is healthy and actively tunneling traffic (no recreate needed)"
394         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container passed live traffic healthcheck (
Cloudflare trace successful)." >> output.log
395     else
396         warn "warp container tunnel is dead (Docker status: $WARP_STATE) force-recreating all gateway containers
to restore network namespace"
397         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container is DEAD or STUCK. Forcing recreation
of gateway stack..." >> output.log
398         if [ -s "$tmp_err" ]; then
399             err_msg=$(cat "$tmp_err" | tr -d '\r')
400             echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] Healthcheck failure details: $err_msg" >> output.
log
401         fi
402         docker compose up -d --force-recreate >> output.log 2>&1 || warn "force-recreate failed"
403
404         NEW_STATE=$(docker inspect --format '{{.State.Health.Status}}' warp 2>> output.log || echo "missing")
405         ok "warp container health after recreate: $NEW_STATE"
406         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [Docker/Colima] warp container health after recreation: $NEW_STATE"
>> output.log
407         fi
408         rm -f "$tmp_err"
409     fi
410 else
411     step "Stopping gateway Docker containers"
412     if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then
413         if [ -f "docker-compose.yml" ]; then
414             docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
415         else
416             warn "docker-compose.yml not found in current directory"
417         fi
418     else
419         warn "Docker not available skipping"
420     fi
421 fi
422
423 # 6b. Stopping sleep prevention (caffeinate) default only
424 if [ "$POST_WAKE" = "false" ]; then
425     step "Stopping sleep prevention"
426     stop_pidfile_daemon "$PID_CAFFEINATE" "system sleep prevention"
427 else
428     step "Sleep prevention: leaving untouched (caffeinate already re-acquired by parent shell)"
429 fi
430
431 # 6c. Stopping DNS Watcher default only
432 if [ "$POST_WAKE" = "false" ]; then
433     step "Stopping DNS Watcher"
434     stop_pidfile_daemon "$PID_DNS_WATCHER" "background DNS Watcher"
435 else
436     step "DNS Watcher: leaving untouched (it keeps re-hijacking DNS on every roam)"
437 fi
438
439 # 7. Power-cycle Wi-Fi default only
440 if [ "$POST_WAKE" = "false" ]; then
441     step "Power-cycling Wi-Fi"
442     bounce_wifi_interfaces
443 else
444     step "Wi-Fi: leaving untouched (post-wake keeps the current Wi-Fi association)"
445 fi
446
447 # 8. Verify (default checks direct internet; post-wake verifies the gateway)
448 if [ "$POST_WAKE" = "true" ]; then
449     step "Verifying gateway is healthy after post-wake refresh"
450
451     # Wait a moment for tailscale + warp healthcheck to settle
452     sleep 3
453
454     GATEWAY_OK=false
455     if command -v dig >> output.log 2>&1; then
456         # Source procedural ports if available
457         if [ -f "/tmp/nullexit-ports.env" ]; then
458             source "/tmp/nullexit-ports.env"
459         fi

```

```

460 DNS_PORT=${DNS_PROXY_PORT:-5354}
461
462 # AdGuard is exposed at localhost:${DNS_PORT}. If this resolves, the warp tunnel
463 # is functioning properly because AdGuard routes upstream to warp.
464 if dig +tcp @127.0.0.1 -p "$DNS_PORT" google.com +short +timeout=5 >> output.log 2>&1; then
465     ok "Gateway DNS (AdGuard via localhost:${DNS_PORT}) works"
466     GATEWAY_OK=true
467 else
468     warn "Gateway DNS not responding yet"
469 fi
470 fi
471
472 # Direct host gateway check via Tailscale ping (relayed pong counts as success)
473 if command -v tailscale >> output.log 2>&1; then
474     TS_IP=""
475     [ -f .gateway_ip ] && TS_IP=$(read_adguard_ip || true)
476     if [ -n "$TS_IP" ] && tailscale ping --until-direct=false -c 1 --timeout 4s "$TS_IP" 2>> output.log | grep
477         -q pong; then
478         ok "Gateway $TS_IP reachable via Tailscale"
479         GATEWAY_OK=true
480     else
481         warn "Tailscale ping to gateway did not pong exit-node may still be re-establishing"
482     fi
483 fi
484
485 if [ "$GATEWAY_OK" = "true" ]; then
486     echo -e " ${GREEN}${BOLD} Gateway is healthy after post-wake refresh.${NC}"
487 else
488     echo -e " ${YELLOW}${BOLD} Gateway recovery is in-progress.${NC}"
489     echo " The route may need another 30s before queries succeed."
490     echo " If it stays broken for >60s, run: bash \"${SCRIPT_DIR}/toggle.sh\""
491 fi
492 else
493     step "Verifying internet connectivity"
494
495     # Wait a moment for the network to settle
496     sleep 2
497
498     INTERNET_OK=false
499
500     # Try DNS resolution first
501     if host -W 3 google.com 1.1.1.1 >> output.log 2>&1; then
502         ok "DNS resolution works (google.com via 1.1.1.1)"
503         INTERNET_OK=true
504     elif nslookup google.com 1.1.1.1 >> output.log 2>&1; then
505         ok "DNS resolution works (nslookup google.com via 1.1.1.1)"
506         INTERNET_OK=true
507     else
508         warn "DNS resolution check failed will still try ping/curl"
509     fi
510
511     # Try actual HTTP connectivity
512     if command -v curl >> output.log 2>&1; then
513         if curl -sf --max-time 5 https://1.1.1.1 >> output.log 2>&1; then
514             ok "Internet reachable via HTTP"
515             INTERNET_OK=true
516         elif curl -sf --max-time 5 http://1.1.1.1 >> output.log 2>&1; then
517             ok "Internet reachable via HTTP (plain)"
518             INTERNET_OK=true
519         else
520             warn "HTTP check failed"
521         fi
522     elif command -v ping >> output.log 2>&1; then
523         if ping -c 1 -W 3 1.1.1.1 >> output.log 2>&1; then
524             ok "Internet reachable via ping"
525             INTERNET_OK=true
526         else
527             warn "Ping check failed"
528         fi
529     fi
530 fi
531
532 # Summary
533 echo ""
534 echo -e " ${BOLD}${NC}"
535 if [ "$POST_WAKE" = "true" ]; then
536     echo -e " ${BOLD}POST-WAKE SUMMARY${NC}"
537 else
538     echo -e " ${BOLD}RECOVERY SUMMARY${NC}"
539 fi
540 echo -e " ${BOLD}${NC}"

```

```

540
541 # Show final DNS state
542 FINAL_DNS=$(networksetup -getdnsservers "Wi-Fi" 2>/dev/null || echo "N/A")
543 echo "   DNS (Wi-Fi):   $FINAL_DNS"
544
545 FINAL_DNS2=$(networksetup -getdnsservers "Ethernet" 2>/dev/null || true)
546 if [[ -z "$FINAL_DNS2" || "$FINAL_DNS2" == *"not a recognized"* || "$FINAL_DNS2" == *"Error"* ]]; then
547     FINAL_DNS2="N/A (Not configured)"
548 fi
549 echo "   DNS (Ethernet): $FINAL_DNS2"
550
551 echo ""
552
553 if [ "$POST_WAKE" = "false" ]; then
554     if [ "$INTERNET_OK" = "true" ]; then
555         echo -e "   ${GREEN}${BOLD} Internet is working.${NC}"
556     else
557         echo -e "   ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
558         echo "       This might mean:"
559         echo "       - Your Wi-Fi is disconnected from the router"
560         echo "       - Tailscale is still interfering (try restarting tailscaled:"
561         echo "         brew services restart tailscale)"
562         echo "       - You need to re-connect to Wi-Fi in the menu bar"
563         echo "       - Try toggling Wi-Fi off/on in the menu bar"
564         echo ""
565         echo "       Or try manually:"
566         echo "       networksetup -setdnsservers Wi-Fi empty"
567         echo "       tailscale down"
568         echo "       sudo route delete -net 0.0.0.0/1"
569         echo "       sudo route delete -net 128.0.0.0/1"
570         echo "       brew services restart tailscale"
571     fi
572 fi
573
574 # Reset sharing services to prevent AirDrop freezes after interface changes
575 # This is the SAME step in BOTH modes (post-wake inherits the previous fix from 10.27).
576 echo -e "   Resetting macOS sharing services (AirDrop/AirPlay)..."
577 reset_sharing_services
578
579 echo ""
580 if [ "$POST_WAKE" = "true" ]; then
581     echo -e "${YELLOW}${BOLD}Post-wake refresh complete.${NC}"
582     # Remove the WARP Watcher inhibit marker now that the gateway is fully healthy.
583     # The watcher will resume normal liveness monitoring on its next 5s poll.
584     rm -f "$WARP_INHIBIT_FILE"
585     echo "[$(date -u +%FT%TZ)] POST-WAKE complete  WARP Watcher inhibit released." >> output.log
586 else
587     echo -e "${GREEN}${BOLD}Recovery complete.${NC}"
588 fi
589 echo ""

```

8 setup.sh

```

1 #!/bin/bash
2 # setup.sh nullexit one-shot setup script (macOS)
3 # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4 set -euo pipefail
5
6 # Source common formatting and helper functions
7 source "$(dirname "$0")/scripts/common.sh"
8
9 # Run common installation logic
10 source "$(dirname "$0")/scripts/setup-common.sh"
11
12 # Compile macOS Shortcuts
13 echo -e "\n${BOLD}Compiling macOS Desktop Shortcuts...${NC}"
14 if command -v osascript &> /dev/null; then
15     osascript -o "Toggle Gateway.app" "Toggle-Gateway.applescript" >/dev/null 2>&1
16     osascript -o "Recover Gateway.app" "Recover-Gateway.applescript" >/dev/null 2>&1
17     echo "   Created 'Toggle Gateway.app' and 'Recover Gateway.app'"
18 fi
19
20 # Done
21 echo ""
22 echo -e "${GREEN}${BOLD}${NC}"
23 echo -e "${GREEN}${BOLD}           Setup complete!           ${NC}"

```

```

24 echo -e "${GREEN}${BOLD}${NC}"
25 echo ""
26 echo -e "${BOLD}One manual step remaining  approve the exit node:${NC}"
27 echo "   https://login.tailscale.com/admin/machines"
28 echo "   Find your gateway  '...'  'Edit route settings'  enable Exit Node"
29 echo ""
30
31 echo -e "${YELLOW}${BOLD}Sudoers / Background Execution Requirements:${NC}"
32 echo "   To allow silent, non-interactive execution of the firewall and network routes (especially during sleep/
wake recovery), you must configure passwordless sudo."
33 echo "   Run this exact command in your terminal:"
34 echo "       echo \"\$USER ALL=(root) NOPASSWD: /sbin/pfctl, /usr/sbin/networksetup, /usr/bin/dscacheutil, /usr/
bin/killall, /usr/bin/pkill, /bin/kill, /sbin/route, /sbin/ifconfig, /usr/bin/true, /opt/homebrew/bin/brew,
/usr/local/bin/brew, /usr/bin/python3 -I \$PWD/scripts/dns-proxy.py, /opt/homebrew/bin/python3 -I \$PWD/
scripts/dns-proxy.py, /usr/local/bin/python3 -I \$PWD/scripts/dns-proxy.py\" | sudo tee /etc/sudoers.d/
nullexit"
35 echo ""
36
37 if [[ -n "${TS_IP:-}" ]]; then
38     echo -e "${BOLD}AdGuard Home dashboard:${NC}   http://${TS_IP}:3000   (username: admin)"
39     echo ""
40 fi
41
42 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
43 echo "   macOS:   double-click the 'Toggle Gateway' app icon!"
44 echo "   Windows: double-click Toggle-Gateway.bat"
45 echo ""
46
47 if [[ "$OSTYPE" == "darwin*" ]]; then
48     # LuLu localhost filtering check
49     if [ -d "/Applications/LuLu.app" ] || systemextensionsctl list 2>/dev/null | grep -q "com.objective-see.
lulu"; then
50         echo -e "${YELLOW}${BOLD}LuLu Firewall Detected:${NC}"
51         echo "   LuLu's localhost (loopback) filtering is OFF by default. To properly protect the"
52         echo "   local nullexit proxy ports from malicious local processes, you must manually enable"
53         echo "   'Allow Loopback' (or block it selectively) in LuLu's Preferences -> Rules."
54         echo ""
55     fi
56
57     echo -e "${YELLOW}${BOLD}Note on macOS Tailscale Sandboxing:${NC}"
58     echo "   The Tailscale GUI app (tailscale-app) installed on macOS is sandboxed."
59     echo "   While you cannot run a Tailscale SSH server on this Mac host, you can"
60     echo "   still SSH outbound to other devices on the mesh using their MagicDNS"
61     echo "   addresses directly (e.g. ssh user@device.tailnet-name.ts.net)."
62     echo ""
63 fi

```

9 scripts/toggle-linux.sh

```

1 #!/bin/bash
2 # toggle-linux.sh  nullexit gateway toggle script for Linux
3 # Automatically handles Docker containers, Colima, DNS hijacking, and Tailscale exit node routing.
4
5 set -e
6
7 # --- PROCEDURAL PORT GENERATION ---
8 # To prevent static port fingerprinting by malware, we randomize exposed ports
9 # on every boot and persist them to a hidden environment file.
10 PORTS_FILE="/tmp/nullexit-ports.env"
11 if [[ "$1" == "" || "$1" == "--restart" || "$1" == "restart" ]]; then
12
13     # Collision-avoidance function
14     GENERATED_PORTS=""
15     get_free_port() {
16         local port
17         while true; do
18             # Linux uses shuf or $RANDOM instead of jot
19             port=$(shuf -i 10000-65000 -n 1 2>/dev/null || echo $((10000 + RANDOM % 55000)))
20             # 1. Prevent self-collision within this loop
21             if [[ "$GENERATED_PORTS" == *"$port"* ]]; then
22                 continue
23             fi
24             # 2. Prevent collision with ANY process on the machine (root daemons, terminal apps, etc)
25             # We use ss to read the kernel socket table directly.
26             if ! ss -ltn | awk '{print $4}' | grep -q ":$port$"; then
27                 echo "$port"

```

```

28         return
29     fi
30     done
31 }
32
33 export TOR SOCKS_PORT=$(get_free_port)
34 GENERATED_PORTS="$GENERATED_PORTS $TOR SOCKS_PORT"
35
36 export TOR_CONTROL_PORT=$(get_free_port)
37 GENERATED_PORTS="$GENERATED_PORTS $TOR_CONTROL_PORT"
38
39 export SOCKS_PROXY_PORT=$(get_free_port)
40 GENERATED_PORTS="$GENERATED_PORTS $SOCKS_PROXY_PORT"
41
42 export DNS_PROXY_PORT=$(get_free_port)
43
44 echo "export TOR SOCKS_PORT=$TOR SOCKS_PORT" > "$PORTS_FILE"
45 echo "export TOR_CONTROL_PORT=$TOR_CONTROL_PORT" >> "$PORTS_FILE"
46 echo "export SOCKS_PROXY_PORT=$SOCKS_PROXY_PORT" >> "$PORTS_FILE"
47 echo "export DNS_PROXY_PORT=$DNS_PROXY_PORT" >> "$PORTS_FILE"
48 elif [ -f "$PORTS_FILE" ]; then
49     # On STOP, source the existing ports so docker-compose down matches
50     source "$PORTS_FILE"
51 else
52     # Fallbacks if file is lost
53     export TOR SOCKS_PORT=9050
54     export TOR_CONTROL_PORT=9051
55     export SOCKS_PROXY_PORT=1080
56     export DNS_PROXY_PORT=5354
57 fi
58
59 # Start execution timer
60 TOGGLE_START_TIME=$SECONDS
61
62 if [[ "$1" == "restart" ]] || [[ "$1" == "--restart" ]]; then
63     echo "Executing Gateway Restart Sequence..."
64     # We must source common.sh here temporarily to check is_gateway_active before we proceed
65     source "$(dirname "${BASH_SOURCE[0]}")/common.sh"
66     if is_gateway_active; then
67         echo "Gateway is currently running. Stopping it first..."
68         bash "$0"
69         echo "Gateway stopped. Now starting it..."
70     else
71         echo "Gateway is already stopped. Starting it..."
72     fi
73     bash "$0"
74     exit 0
75 fi
76
77 # Global flag to track if the script completed successfully
78 SUCCESS_RUN=false
79
80 # Global variable to track the currently running background command PID
81 CURRENT_BG_PID=""
82
83 # Source common bash functions
84 source "$(dirname "${BASH_SOURCE[0]}")/common.sh"
85
86
87 PID_FILE="/tmp/nullexit-caffeinate.pid"
88
89 # Start macOS caffeinate to prevent sleep while gateway is running
90 start_sleep_prevention() {
91     if ! command -v systemd-inhibit >/dev/null 2>&1; then
92         echo " [Warning] systemd-inhibit tool not found. Sleep prevention unavailable."
93         return 1
94     fi
95
96     # Stop any existing sleep prevention process first to avoid duplicates
97     stop_sleep_prevention
98
99     echo " Enabling system sleep prevention (systemd-inhibit)..."
100     # Run systemd-inhibit wrapped in a bash trap to prevent idle system sleep.
101     # Critically, this traps shutdown signals (SIGTERM) and automatically
102     # flushes hijacked DNS back to normal right before the PC powers off.
103     # We do not use recover-linux.sh here because it is a heavy recovery script
104     # which takes too long and gets terminated before it can reset the DNS. Instead,
105     # we surgically and instantly restore normal DNS settings here.
106     nohup bash -c "
107         trap '
108             echo \"[Shutdown Trap] Reverting network settings...\\" >> \"\$PWD/output.log\" 2>&1

```

```

109     kill \${ 2>/dev/null || true
110     sudo -n resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"\$PWD/output.log\" 2>&1 || true
111     sudo -n resolvectl revert \"\$ACTIVE_SERVICE\" >> \"\$PWD/output.log\" 2>&1 || true
112     resolvectl domain \"\$ACTIVE_SERVICE\" \"\" >> \"\$PWD/output.log\" 2>&1 || true
113     resolvectl revert \"\$ACTIVE_SERVICE\" >> \"\$PWD/output.log\" 2>&1 || true
114     if command -v tailscale >/dev/null 2>&1; then
115         tailscale up --accept-dns=false --exit-node= >> \"\$PWD/output.log\" 2>&1 &
116         TS_DOWN_ARGS=\"\"
117         if [ \"\$KILL_SWITCH\" = \"true\" ]; then TS_DOWN_ARGS=\"--accept-risk=lose-ssh\"; fi
118         tailscale down \"\$TS_DOWN_ARGS >> \"\$PWD/output.log\" 2>&1 &
119     fi
120     sleep 1
121     echo \"[Shutdown Trap] Cleanup complete.\" >> \"\$PWD/output.log\" 2>&1
122     exit 0
123     ' SIGTERM SIGINT SIGHUP
124     systemd-inhibit --what=sleep --who=nullexit --why=\"gateway running\" --mode=block sleep infinity &
125     wait \$!
126     \" >> output.log 2>&1 &
127     local caffe_pid=$!
128     echo \"\$caffe_pid\" > \"\$PID_FILE\"
129     echo \" Sleep prevention active (PID \$caffe_pid). Your PC won't sleep while the gateway is running.\"
130 }
131
132 # Stop sleep prevention when gateway is stopped
133 stop_sleep_prevention() {
134     if [ -f \"\$PID_FILE\" ]; then
135         local caffe_pid
136         caffe_pid=$(cat \"\$PID_FILE\")
137         if [ -n \"\$caffe_pid\" ] && kill -0 \"\$caffe_pid\" 2>/dev/null && ps -p \"\$caffe_pid\" -o command= 2>/dev/null |
138             grep -q -E \"systemd-inhibit|recover-linux.sh\"; then
139             echo \" Stopping system sleep prevention (systemd-inhibit wrapper PID \$caffe_pid)...\"
140             kill \"\$caffe_pid\" 2>/dev/null || true
141         fi
142         rm -f \"\$PID_FILE\"
143     fi
144 }
145
146 DNS_WATCHER_PID_FILE=\"/tmp/nullexit-dns-watcher.pid\"
147
148 start_dns_watcher() {
149     local target_ip=$1
150     stop_dns_watcher
151     echo \" Starting background DNS Watcher for seamless Wi-Fi roaming...\"
152     nohup bash -c \"
153         trap 'exit 0' SIGTERM SIGINT SIGHUP
154         while true; do
155             if [ -n \"\$ACTIVE_SERVICE\" ]; then
156                 CURRENT_DNS=$(resolvectl dns \"\$ACTIVE_SERVICE\" 2>/dev/null | grep -oE
157                 '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head -1)
158                 if [ \"\$CURRENT_DNS\" != \"\$target_ip\" ]; then
159                     resolvectl dns \"\$ACTIVE_SERVICE\" \"\$target_ip\" >/dev/null 2>&1 || true
160                 fi
161             fi
162             sleep 30
163         done
164     \" >> output.log 2>&1 &
165     echo $! > \"\$DNS_WATCHER_PID_FILE\"
166 }
167
168 stop_dns_watcher() {
169     if [ -f \"\$DNS_WATCHER_PID_FILE\" ]; then
170         local watcher_pid
171         watcher_pid=$(cat \"\$DNS_WATCHER_PID_FILE\")
172         if [ -n \"\$watcher_pid\" ] && kill -0 \"\$watcher_pid\" 2>/dev/null; then
173             echo \" Stopping background DNS Watcher (PID \$watcher_pid)...\"
174             kill -9 \"\$watcher_pid\" 2>/dev/null || true
175         fi
176         rm -f \"\$DNS_WATCHER_PID_FILE\"
177     fi
178 }
179
180 # Cleanup handler to restore DNS to 1.1.1.1 on error or user interrupt (Ctrl+C / SIGTERM / SIGHUP)
181 cleanup_handler() {
182     local exit_code=$?
183     local trigger_type=\"$1\"
184     local line_no=\"$2\"
185     if [ \"\$SUCCESS_RUN\" = \"false\" ]; then
186         echo -e \"\n[Self-Correction] Script interrupted or failed ($trigger_type). Restoring host DNS to 1.1.1.1...\"
187     fi
188     if [ -n \"\$ACTIVE_SERVICE\" ]; then

```



```

187     reset_dns
188 fi
189
190 if [ -n "$CURRENT_BG_PID" ]; then
191     echo "Terminating active command (PID $CURRENT_BG_PID)..."
192     kill -15 "$CURRENT_BG_PID" 2>> output.log || true
193 fi
194
195 # Disconnect host Tailscale and close the GUI application on failure
196 disconnect_tailscale_host
197
198 # Stop local DNS proxy if running
199 stop_local_dns_proxy
200
201 # Stop sleep prevention
202 stop_sleep_prevention
203 stop_dns_watcher
204
205 # Nuke leftover network state (proxies, routes, DNS cache, Wi-Fi)
206 if [ -n "$ACTIVE_SERVICE" ]; then
207     cleanup_network_state
208 fi
209
210
211 if [ "$trigger_type" = "ERR" ]; then
212     echo -e "\n=====
213     echo "ERROR: Script failed at line $line_no with exit code $exit_code."
214     echo "=====
215     echo "Troubleshooting steps:"
216     echo "1. Run 'colima status' to verify the VM is active."
217     echo "2. Run 'docker compose ps' to check container health."
218     echo "3. Run 'docker compose logs' to view application logs."
219     echo "4. Run 'tailscale status' to check host Tailscale status."
220     echo "=====
221 fi
222 fi
223 }
224
225 trap 'cleanup_handler ERR $LINENO' ERR
226 trap 'cleanup_handler INT' INT
227 trap 'cleanup_handler TERM' TERM
228 trap 'cleanup_handler HUP' HUP
229
230 # NOPASSWD sudo
231 # The user has NOPASSWD in /etc/sudoers.d/nullexit for the specific commands
232 # needed (networksetup, dnsmasq, dscacheutil, killall, pkill, socat).
233 # All privileged calls below use `sudo -n` which will run without prompting.
234 # No credential caching loop is needed.
235
236 # Add Homebrew and standard paths
237 export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
238
239 # Check for local DNS proxy tools (used when tailnet data plane is unavailable)
240 # Uses a Python DNS proxy (handles TCP wire format correctly).
241 # Python3 is built into macOS no external dependencies.
242 DNS_PROXY_BIN=""
243 if command -v python3 >/dev/null 2>&1; then
244     DNS_PROXY_BIN="python3 -I"
245 elif command -v python >/dev/null 2>&1; then
246     DNS_PROXY_BIN="python -I"
247 fi
248
249 # Path to the DNS proxy script (sibling of this script)
250 DNS_PROXY_SCRIPT="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/dns-proxy.py"
251
252 # Resolve Tailscale CLI on host. We rely on the standalone Homebrew formula
253 # (`brew install tailscale` + `systemctl start tailscaled`) NOT the
254 # Tailscale.app GUI so there is no .app bundle or Network Extension sandbox
255 # to launch, no menu-bar click to perform, and no scutil Network Service to
256 # start manually. tailscaled runs as a per-user LaunchAgent (or LaunchDaemon
257 # if you ran the install with sudo) and the control socket is always available.
258 TS_BIN=""
259 if command -v tailscale >> output.log 2>&1; then
260     TS_BIN="tailscale"
261 fi
262
263 # Baseline: disable Tailscale DNS management at the daemon level.
264 # `tailscale set` writes a persistent preference. However, `tailscale up --reset`
265 # (used in steps 7 and 9) RESETS all unspecified flags to defaults, which would
266 # re-enable `--accept-dns=true` and nullify this `set`.
267 # Therefore, EVERY `tailscale up --reset` call in this file MUST also pass

```

```

268 # `--accept-dns=false` explicitly. See steps 7 and 9 for the fix.
269 #
270 # This initial `set` is still useful as a baseline for non-reset operations
271 # (e.g. if the user runs `tailscale up` manually without --reset) and covers
272 # the brief window between this line and the first `--reset` call.
273 #
274 # Runs once at script start while tailscaled is still connected (if it was
275 # left running from a previous toggle), so the pref takes effect before any
276 # disconnection or reconnection logic.
277 if [ -n "$TS_BIN" ]; then
278     $TS_BIN set --accept-dns=false >> output.log 2>&1 || true
279 fi
280
281 # Disconnect host Tailscale from the mesh. With the standalone daemon, this is the
282 # *entire* teardown no need to mess with system network managers.
283 # tailscaled keeps running (as a systemd service), which is intentional:
284 # the next `tailscale up` then reconnects in seconds rather than waiting for a
285 # slow daemon launch. The `tailscale down` call also retains the user's
286 # auth state, so we don't force a browser re-login every cycle.
287 #
288 # Defined early (before the if/else branches and referenced by cleanup_handler's
289 # ERR trap) so that any failure before the main logic can still tear down Tailscale.
290 restart_tailscaled_daemon() {
291     echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
292     echo " [Tailscale Recovery] Attempting to restart tailscaled service via systemctl..."
293     if run_with_timeout 15 sudo -n systemctl restart tailscaled >> output.log 2>&1; then
294         echo " [Tailscale Recovery] Successfully restarted tailscaled service."
295     else
296         echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
297     fi
298     sleep 3
299 }
300 disconnect_tailscale_host() {
301     if [ -n "$TS_BIN" ]; then
302         echo "Disconnecting host Tailscale from mesh (tailscaled stays running as system service)..."
303         local ts_args=""
304         if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
305         if ! run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1; then
306             restart_tailscaled_daemon
307             run_with_timeout 10 $TS_BIN down $ts_args >> output.log 2>&1 || true
308         fi
309     fi
310 }
311
312 # Function to get the active network service name (e.g., "Wi-Fi" or "USB 10/100 LAN")
313 get_active_service() {
314     # Get the interface used for default internet routing
315     local iface
316     iface=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
317     if [ -z "$iface" ]; then
318         echo ""
319         return 1
320     fi
321     echo "$iface"
322 }
323
324 ACTIVE_SERVICE=$(get_active_service)
325
326 # Resolve the service name for en0 (usually "Wi-Fi") macOS scutil DNS resolver
327 # is commonly scoped to en0, so per-service DNS changes on other interfaces are
328 # ignored unless en0's service is also updated.
329 # Helper to restore host DNS to a clean state (used on script start, on failure,
330 # and after a successful STOP). Without this, a successful toggle-off leaves the
331 # host's resolver pinned to a now-dead gateway IP every lookup stalls for macOS's
332 # DNS timeout (~5s) before falling through to whatever next server is configured.
333 # With it, we always leave the host in `1.1.1.1 / no search domain`.
334 reset_dns() {
335     if [ -n "$ACTIVE_SERVICE" ]; then
336         resolvectl domain "$ACTIVE_SERVICE" "" 2>/dev/null || true
337         resolvectl revert "$ACTIVE_SERVICE" 2>/dev/null || true
338     fi
339 }
340
341 # Helper to nuke leftover network state after teardown (proxy settings, routing
342 # table, DNS cache, Wi-Fi). Prevents the "had to write a whole recovery script"
343 # problem where stale state kills internet after the gateway stops.
344 cleanup_network_state() {
345     echo -e "\nCleaning up network state (proxies, routes, DNS cache, Wi-Fi)..."
346
347     # 1. Disable any leftover proxy settings (needs root on many macOS versions)
348     # (Skipped for Linux since we don't set global SOCKS proxy)

```

```

349 echo " Proxies disabled."
350
351 # 2. Flush DNS cache
352 if command -v resolvectl >> output.log 2>&1; then
353     sudo -n resolvectl flush-caches 2>> output.log || true
354 elif command -v systemd-resolve >> output.log 2>&1; then
355     sudo -n systemd-resolve --flush-caches 2>> output.log || true
356 fi
357 echo " DNS cache flushed."
358
359 # 3. Flush stale routing table entries
360 if command -v ip >> output.log 2>&1; then
361     sudo -n ip route flush cache >> output.log 2>&1 || true
362 fi
363 echo " Routing table flushed."
364
365 # 4. Power-cycle Wi-Fi to clear any lingering interface state
366 echo " Restarting network interface..."
367 sudo -n ip link set dev "$ACTIVE_SERVICE" down 2>> output.log || true
368 sleep 1
369 sudo -n ip link set dev "$ACTIVE_SERVICE" up 2>> output.log || true
370 echo " Interface power-cycled ($ACTIVE_SERVICE)."
371 sleep 3
372
373 echo "Network state cleanup complete."
374 }
375
376 # SOCKS5 Proxy (WARP Tunnel)
377 # When Tailscale's exit node can't route through WARP (due to userspace
378 # forwarding), we use a SOCKS5 proxy running inside the warp container's
379 # network namespace. Connections created by this proxy go through the
380 # kernel routing table (table 200 -> tun0 -> WARP), unlike Tailscale's
381 # userspace exit node which bypasses it.
382 #
383 # The SOCKS5 proxy is exposed on localhost:${SOCKS_PROXY_PORT} via Docker port mapping.
384 # We route system TCP traffic through the proxy, which then goes through WARP.
385 #
386 # IMPORTANT: CLI tools like curl do NOT respect macOS system proxy settings.
387 # They must use --socks5-hostname or the ALL_PROXY env variable explicitly.
388
389 enable_socks_proxy() {
390     local svc="$1"
391     echo " (Linux global SOCKS proxy configuration is desktop-environment specific, skipping.)"
392     echo " SOCKS5 proxy is active and listening on 127.0.0.1:${SOCKS_PROXY_PORT}"
393 }
394
395 disable_socks_proxy() {
396     local svc="$1"
397     echo " (Linux global SOCKS proxy configuration skipped.)"
398 }
399
400 # Local DNS Proxy (Python)
401 # When the tailnet data plane cannot establish, we use a Python DNS proxy.
402 # It listens on UDP:53, receives DNS queries, forwards them over TCP to
403 # AdGuard at 127.0.0.1:${DNS_PROXY_PORT} (Docker port mapping), and returns the response.
404 # The Python script handles the DNS-over-TCP wire format (2-byte length prefix)
405 # correctly unlike socat which just forwards raw bytes.
406 #
407 # Python3 is built into macOS no external dependencies.
408 DNS_PROXY_PID=""
409
410 # Kill any leftover DNS proxy process
411 stop_local_dns_proxy() {
412     if [ -n "$DNS_PROXY_PID" ]; then
413         kill "$DNS_PROXY_PID" 2>/dev/null || true
414         wait "$DNS_PROXY_PID" 2>/dev/null || true
415         DNS_PROXY_PID=""
416     fi
417     # Clean up any stale Python DNS proxy processes
418     sudo -n pkill -f "dns-proxy.py" 2>/dev/null || true
419 }
420
421 # Start local DNS proxy (Python)
422 start_local_dns_proxy() {
423     if [ -z "$DNS_PROXY_BIN" ]; then
424         echo " Python not found. Python3 is built into macOS this shouldn't happen."
425         return 1
426     fi
427
428     if [ ! -f "$DNS_PROXY_SCRIPT" ]; then
429         echo " DNS proxy script not found at $DNS_PROXY_SCRIPT"

```

```

430     return 1
431 fi
432
433 # Kill any leftover process first
434 stop_local_dns_proxy
435
436 echo -n " Starting local DNS proxy via Python (UDP:53 TCP:${DNS_PROXY_PORT})... "
437 sudo -n bash -c "TARGET_PORT=${DNS_PROXY_PORT} \"${DNS_PROXY_BIN}\" \"${DNS_PROXY_SCRIPT}\"" &
438 DNS_PROXY_PID=$!
439 disown "${DNS_PROXY_PID}" 2>/dev/null || true
440
441 # Give it a moment to bind
442 sleep 0.5
443
444 if kill -0 "${DNS_PROXY_PID}" 2>/dev/null; then
445     echo "started (PID ${DNS_PROXY_PID})."
446
447     # Hijack host DNS to localhost
448     echo -n " Hijacking host DNS to 127.0.0.1 for ad-blocking... "
449     resolvectl domain "${ACTIVE_SERVICE}" "" 2>/dev/null || true
450     resolvectl domain "${ACTIVE_SERVICE}" "" 2>/dev/null || true
451     if ! sudo -n resolvectl dns "${ACTIVE_SERVICE}" 127.0.0.1; then
452         echo "FAILED (resolvectl error check permissions)."

```

```

511 # Ensure DNS is set to 1.1.1.1 immediately at script start to prevent any DNS deadlocks/hangs
512 # during status checks, container startup, VM boot, or teardown. We *also* clear any leftover
513 # `ts.net` search domain from a previous `tailscale up --accept-dns=true` run, otherwise macOS
514 # would prepend it to every lookup and most public DNS queries would resolve to NXDOMAIN.
515 # Capture current DNS before resetting so we can check if it was hijacked
516 INITIAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | head
    -1 || true)
517
518 echo "Initializing DNS to 1.1.1.1 to ensure reliable internet access..."
519 reset_dns
520
521 # Local Network Surveillance Check
522 echo -e "\n"
523 echo "Local Network Surveillance Check"
524 echo ""
525 # Count populated ARP cache entries to detect network scanning or high congestion
526 if command -v ip >/dev/null 2>&1; then
527     ARP_COUNT=$(ip neigh | grep -iv 'incomplete' | wc -l | tr -d ' ')
528 else
529     # Using '-an' avoids hanging on DNS reverse resolution when the network state is in transition.
530     ARP_COUNT=$(arp -an 2>/dev/null | grep -iv 'incomplete' | wc -l | tr -d ' ')
531 fi
532 if [ "$ARP_COUNT" -gt 15 ]; then
533     echo -e " \033[1;33m[!] WARNING: High ARP activity detected ($ARP_COUNT devices in cache).\033[0m"
534     echo -e " \033[1;33m[!] This Wi-Fi network may be heavily congested or actively scanned (e.g., arp-scan)
535     \033[0m"
536     echo -e " [] Your traffic remains fully encrypted and invisible.\n"
537 else
538     echo -e " [] Local network looks quiet ($ARP_COUNT devices). No aggressive scanning detected.\n"
539 fi
540
541 echo "Checking Gateway Status..."
542 if is_gateway_active; then
543     echo -e "\n=====
544     echo "Gateway is RUNNING. Stopping it now..."
545     echo -e "=====
546
547 # 1. Disconnect host Tailscale cleanly first before stopping the exit-node container
548 disconnect_tailscale_host
549 echo ""
550
551 echo "Stopping Docker containers..."
552 docker compose down -t 5
553
554 echo -e "\nDocker daemon handles container teardown automatically."
555
556 # The host's DNS was hijacked to the gateway IP during ENABLE; now that the
557 # gateway is down, restore DNS to 1.1.1.1 immediately so subsequent lookups
558 # don't stall for the macOS DNS timeout before the 1.1.1.1 fallback engages.
559 echo -e "\nRestoring host DNS to 1.1.1.1 (gateway is gone)... "
560 reset_dns
561
562 # 3. Stop local DNS proxy if running
563 stop_local_dns_proxy
564
565 # Stop sleep prevention
566 stop_sleep_prevention
567 stop_dns_watcher
568
569 # 4. Nuke leftover network state so internet actually works after teardown
570 cleanup_network_state
571
572 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
573 echo -e "\nGateway has been successfully STOPPED in ${ELAPSED} seconds."
574 else
575     echo -e "\n=====
576     echo "Gateway is STOPPED. Starting it now..."
577     echo -e "=====
578
579 # 1. Prevent host exit-node deadlock during VM / Container startup
580 disconnect_tailscale_host
581
582 echo -e "\nUsing native Linux Docker..."
583
584 # 4. Clean up corrupted AdGuardHome configurations
585 if [ -f "adguard/conf/AdGuardHome.yaml" ] && [ ! -s "adguard/conf/AdGuardHome.yaml" ]; then
586     rm -f "adguard/conf/AdGuardHome.yaml"
587     echo "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop."
588 fi
589

```

```

590
591
592 # 5. Start compose services
593 write_host_ips
594 # Procedurally generated ports have already been exported at the top of the script.
595 echo " [Security] Procedurally randomized proxy ports generated to evade fingerprinting."
596
597 # --- HONEY-PORT TRIPWIRE ---
598 # Generate a random trap port. If any malware does a sequential port scan on localhost,
599 # it will inevitably hit this port. When it does, we instantly fire a desktop alert.
600 HONEY_PORT=$(get_free_port)
601 (
602     if nc -l -p "$HONEY_PORT" 127.0.0.1 >/dev/null 2>&1 || nc -l localhost "$HONEY_PORT" >/dev/null 2>&1; then
603         echo "[$(date '+%Y-%m-%d %H:%M:%S')] [CRITICAL SECURITY ALERT] PORT SCAN DETECTED! An unknown application
        just pinged the local Honey-Port ($HONEY_PORT)!" >> "$PWD/output.log"
604         if command -v notify-send >/dev/null 2>&1; then
605             notify-send -u critical "nullexit OPSEC Alert" "An unknown application is aggressively scanning your
        local ports. Malware port-scan suspected."
606         fi
607     fi
608 ) &
609 disown %1 2>/dev/null || true
610 echo " [Security] Honey-Port Tripwire armed on random port to detect malware port-scans."
611
612 echo -e "\nStarting Docker containers..."
613 docker compose up -d
614
615 # Clean up the one-off rule compiler container so it doesn't clutter the Docker UI
616 docker compose rm -s -f rule-compiler >> output.log 2>&1
617
618 # 6. Wait for the gateway container's Tailscale connection to be ready
619 BYPASS_PING=$(read_env_var GATEWAY_BYPASS_PING | tr '[:upper:]' '[:lower:]')
620 USE_EXIT_NODE=$(read_env_var GATEWAY_USE_EXIT_NODE | tr '[:upper:]' '[:lower:]')
621
622 echo "Waiting for gateway container's Tailscale to connect to the tailnet..."
623 echo " (This can take 30-60s Tailscale goes through: NoState Starting Running Online)"
624 connected=false
625 consecutive_failures=0
626 for i in {1..60}; do
627     # Run tailscale status and capture its output so the user can see what's happening.
628     # Previously this was hidden behind 2>> output.log, making it impossible to debug hangs.
629     ts_output=$(run_with_timeout 5 docker compose exec -T tailscale tailscale status 2>&1 || true)
630
631     if echo "$ts_output" | grep -q "offers exit node"; then
632         connected=true
633         break
634     fi
635
636     # Track consecutive failures to detect a stuck state.
637     # If we see 40+ consecutive 'NoState' responses, give up early
638     # the daemon is alive but can't connect to the coordination server.
639     if echo "$ts_output" | grep -q "NoState"; then
640         consecutive_failures=$((consecutive_failures + 1))
641         if [ "$consecutive_failures" -ge 40 ]; then
642             echo ""
643             echo " [attempt $i/60] Stuck in 'NoState' for $consecutive_failures checks."
644             echo " Tailscale daemon is running but can't reach the coordination server."
645             break
646         fi
647     else
648         consecutive_failures=0
649     fi
650
651     # Print a condensed status every 10 seconds so the user can see progress
652     if [ $((i % 10)) -eq 0 ]; then
653         ts_short=$(echo "$ts_output" | head -1 | tr -d '\r\n')
654         echo " [attempt $i/60] $ts_short"
655     elif [ $((i % 5)) -eq 0 ]; then
656         echo -n "[i]"
657     else
658         echo -n "."
659     fi
660     sleep 1
661 done
662 echo ""
663
664 if [ "$connected" = "true" ]; then
665     echo "Gateway container is online on the Tailscale mesh."
666 elif [ "$BYPASS_PING" = "true" ]; then
667     echo "[Warning] Gateway container check timed out. Proceeding anyway (GATEWAY_BYPASS_PING is true)..."
668 else

```

```

669     echo "ERROR: Gateway container failed to initialize Tailscale (timed out after 60s)." >&2
670     echo "  The container was seen in the following states during the wait:" >&2
671     echo "$ts_output" | sed 's/-/ /' >&2
672     echo "" >&2
673     echo "  This could mean:" >&2
674     echo "    1. Tailscale auth key has expired  check TS_AUTHKEY in .env" >&2
675     echo "    2. Network issue inside the container  check: docker compose logs tailscale" >&2
676     echo "    3. gluetun VPN tunnel not connecting  check: docker compose logs warp | tail -20" >&2
677     echo "" >&2
678     echo "  Diagnose manually:" >&2
679     echo "    docker compose exec -T tailscale tailscale status" >&2
680     echo "    docker compose exec -T tailscale tailscale netcheck" >&2
681     cleanup_handler ERR $LINENO
682     exit 1
683 fi
684
685 # 6b. Resolve the gateway's Tailscale IP.
686 # Dynamically query the container for the IP (handles IP changes after re-auth)
687 # and cache it in .gateway_ip for fast retrieval by recover.sh during Wi-Fi roaming.
688 #
689 # CRITICAL: `docker compose exec -T` on macOS/Colima injects carriage
690 # returns (\r) into captured output. If $TS_IP ends with \r, then
691 # `networksetup -setdnsservers` silently rejects the malformed IP and
692 # DNS stays at 1.1.1.1 the primary bug this project was hitting.
693 # `tr -d '\r'` strips the CR; `awk 'NR==1{print $1; exit}'` takes only
694 # the first field of the first line.
695 TS_IP=""
696 if [ "$connected" = "true" ]; then
697     TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale tailscale ip -4 2>> output.log | tr -d '\r' |
698         awk 'NR==1{print $1; exit}' || true)
699     if [ -n "$TS_IP" ]; then
700         echo "Resolved gateway Tailscale IP dynamically: $TS_IP"
701         echo "$TS_IP" > .gateway_ip
702     else
703         echo "    [!] Failed to resolve gateway IP dynamically. Trying fallback cache..." >&2
704         TS_IP=$(read_adguard_ip || true)
705     fi
706 else
707     TS_IP=$(read_adguard_ip || true)
708 fi
709
710 # 7. Connect host Mac to Tailscale mesh.
711 # With the standalone tailscaled daemon (registered as a per-user LaunchAgent or
712 # system LaunchDaemon via 'systemctl start tailscaled'), the previous five-
713 # phase flow collapses into two trivial CLI calls. There is no .app GUI to launch,
714 # no menu bar item to click, and no Accessibility permission required.
715 #
716 # CRITICAL: If tailscaled is not running, `tailscale up` will silently fail.
717 # We auto-start it before proceeding, and SKIP the rest of the Tailscale setup
718 # if it can't be started (DNS hijack to a 100.x.x.x IP and exit-node routing
719 # are impossible without the host on the mesh).
720 #
721 # We connect in two phases:
722 #   Phase A Join mesh WITHOUT exit node (so pre-flight checks can run)
723 #   Phase B Set exit node only after pre-flight checks confirm the gateway works
724 HOST_ON_MESH=false
725 SKIP_EXIT_NODE=false
726 if [ -n "$TS_BIN" ]; then
727     echo -n "Verifying tailscaled is reachable"
728     daemon_ready=false
729     for i in {1..10}; do
730         # `tailscale status` returns exit code 1 when the daemon is alive but
731         # disconnected ("Tailscale is stopped."). We check the daemon socket
732         # directly as a fallback if the socket exists, tailscaled is running
733         # even if the host isn't connected to the mesh yet.
734         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1 || [ -S /var/run/tailscaled.socket ]; then
735             daemon_ready=true
736             break
737         fi
738         echo -n "."
739         sleep 1
740     done
741     echo ""
742     if [ "$daemon_ready" != "true" ]; then
743         echo -e "\033[0;33m[!] tailscaled daemon is not running. Attempting to start it...\033[0m"
744
745         # Try system-wide daemon (with timeout sudo may prompt for password)
746         if [ "$daemon_ready" != "true" ]; then
747             if run_with_timeout 10 sudo systemctl start tailscaled 2>> output.log; then
748                 echo "  Started tailscaled as system LaunchDaemon."

```



```

749     for i in {1..15}; do
750         if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
751             daemon_ready=true
752             break
753         fi
754         sleep 1
755     done
756 fi
757 fi
758 fi
759
760 if [ "$daemon_ready" = "true" ]; then
761     echo "tailscaled is responsive (running as a system service)."

```

```

830     else
831         echo "FAIL"
832         SKIP_EXIT_NODE=true
833     fi
834 else
835     echo "SKIP (dig/nslookup not available)"
836 fi
837
838 # Check 3: Does the WARP container have external internet?
839 echo -n " [3/3] WARP container internet... "
840 if docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace &>/dev/null;
841 then
842     echo "PASS"
843 else
844     echo "FAIL"
845     SKIP_EXIT_NODE=true
846 fi
847
848 # Act based on check results
849 if [ "$SKIP_EXIT_NODE" != "true" ]; then
850     echo -e "\nAll checks passed. Enabling exit node $TS_IP..."
851     if $TS_BIN up --reset --ssh=true --accept-dns=false --exit-node="$TS_IP" --exit-node-allow-lan-access=
852     true; then
853         echo "Exit node enabled."
854     else
855         echo "[Warning] Failed to set exit node."
856         SKIP_EXIT_NODE=true
857     fi
858 fi
859
860 if [ "$SKIP_EXIT_NODE" = "true" ]; then
861     echo -e "\n033[0;33m[!] Pre-flight checks failed EXIT NODE + DNS HIJACK SKIPPED.\033[0m"
862     echo " Host stays on Tailscale mesh but your internet routes through your normal connection."
863     echo " Troubleshoot with:"
864     echo "     docker compose logs warp | tail -20"
865     echo "     docker compose exec -T warp wget -qO- --timeout=5 https://www.cloudflare.com/cdn-cgi/trace"
866     echo ""
867     echo " Falling back to SOCKS5 proxy for traffic routing..."
868 fi
869 elif [ "$HOST_ON_MESH" = "true" ] && [ "$USE_EXIT_NODE" = "false" ]; then
870     echo "USE_EXIT_NODE is false. Skipping exit node and DNS hijack."
871     SKIP_EXIT_NODE=true
872 fi
873
874 # 8. Apply DNS Hijacking.
875 # If exit node is enabled: hijack DNS to gateway's Tailscale IP (routes through tailnet).
876 # Otherwise, leave DNS at 1.1.1.1 (already set by reset at the top).
877 # AdGuard is also available at localhost:${DNS_PROXY_PORT} for manual configuration in apps.
878 if [ -n "$TS_IP" ] && [ "$HOST_ON_MESH" = "true" ] && [ "$SKIP_EXIT_NODE" != "true" ]; then
879     echo -e "\nHijacking host DNS to point ONLY at AdGuard Home ($TS_IP)..."
880     echo "Setting MagicDNS search domain 'ts.net' so tailnet hostnames resolve..."
881     if force_dns_to_gateway "$TS_IP"; then
882         echo "DNS hijack verified: single server = $TS_IP."
883         start_dns_watcher "$TS_IP"
884     else
885         echo "[Warning] DNS hijack could not be verified."
886         echo " If DNS is broken, run manually:"
887         echo "     networksetup -setdnsservers \"$ACTIVE_SERVICE\" $TS_IP"
888         echo "     networksetup -setsearchdomains \"$ACTIVE_SERVICE\" ts.net"
889     fi
890 fi
891 elif [ -n "$TS_IP" ]; then
892     echo -e "\n[Info] Exit node not enabled (pre-flight checks failed)."
893     echo " Trying local DNS proxy for ad-blocking..."
894     if start_local_dns_proxy; then
895         echo " Ad-blocking active via local DNS proxy through AdGuard."
896     else
897         echo " Local DNS proxy unavailable. DNS stays at 1.1.1.1."
898         echo " AdGuard available at http://localhost:80 (port ${DNS_PROXY_PORT} for DNS via TCP)."
899     fi
900 fi
901
902 # Enable SOCKS5 proxy for traffic routing through WARP
903 # (enabled whenever exit node is skipped, regardless of HOST_ON_MESH)
904 echo -e "\n Enabling SOCKS5 proxy for TCP traffic routing through gateway WARP tunnel..."
905 echo " (SOCKS5 proxy runs inside gateway container, routes through tun0 -> WARP)"
906 enable_socks_proxy "$ACTIVE_SERVICE"
907
908 # Verify the SOCKS5 proxy works through WARP
909 echo -n " Verifying SOCKS5 proxy routes through WARP... "
910 if curl --socks5-hostname 127.0.0.1:$SOCKS_PROXY_PORT --max-time 10 -s https://www.cloudflare.com/cdn-cgi/
911 trace 2>/dev/null | grep -q "warp=on"; then
912     echo "ok (warp=on)."

```

```

908     echo " All TCP traffic now encrypted through Cloudflare WARP tunnel."
909 else
910     echo "FAILED (proxy not routing through WARP)."
```

```

911     echo " Disabling SOCKS5 proxy internet stays on direct connection."
912     disable_socks_proxy
913     echo " SOCKS proxy available at localhost:$SOCKS_PROXY_PORT for manual configuration."
914 fi
915 else
916     echo -e "\n[Warning] Could not resolve gateway Tailscale IP."
917 fi
918
919 # 9. Verify host connectivity
920 if [ "$HOST_ON_MESH" = "true" ] && [ -n "$TS_BIN" ]; then
921     if run_with_timeout 5 $TS_BIN status >> output.log 2>&1; then
922         echo "Host is online on the Tailscale mesh."
923     else
924         echo "[Warning] Host is not responding on Tailscale."
925     fi
926 fi
927
928 ELAPSED=$(( SECONDS - TOGGLE_START_TIME ))
929 echo -e "\nGateway has been successfully STARTED in ${ELAPSED} seconds."
930
931 # Prevent system from going to sleep while gateway is running
932 start_sleep_prevention
933 fi
934
935 SUCCESS_RUN=true
936
937 # Final DNS state summary
938 if [ -n "$ACTIVE_SERVICE" ]; then
939     FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>/dev/null | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr '
940 ' ' || true)
941     echo -e "\n"
942     echo -e "DNS STATE: $FINAL_DNS"
943     echo -e ""
944 fi
945
946 if [ "$START_GATEWAY" = "true" ] && command -v docker >/dev/null 2>&1; then
947     if docker compose logs rule-compiler 2>/dev/null | grep -q "Warning: Failed to fetch"; then
948         echo -e "\n[Warning] One or more blocklist URLs failed to download (404/Offline)."
```

```

949         echo " The gateway gracefully fell back to yesterday's cached blocklist."
950         echo " (Run 'docker logs rule-compiler' later to investigate which link died.)"
951     fi
952 fi
953
954 echo -e "\n"
955 read -rp "Press [r] and Enter to instantly reverse state, or just press Enter to exit: " USER_CHOICE
956 if [[ "${USER_CHOICE}" == "r" || "${USER_CHOICE}" == "R" ]]; then
957     echo "Reversing gateway state..."
958     exec bash "$0"
959 fi
960
961 echo -e "\nYou can close this terminal window now."

```

10 scripts/recover-linux.sh

```

1 #!/bin/bash
2 # recover.sh nullexit KILL-SWITCH recovery script for Linux
3 # Fixes internet when toggle.sh leaves you stranded.
4 #
5 # This is a NUCLEAR option: it undoes EVERYTHING toggle.sh does by:
6 # 1. Disconnecting host Tailscale from the mesh (exit-node off)
7 # 2. Resetting DNS to default on ALL network services
8 # 3. Disabling rogue proxy settings (SOCKS, web, secure web)
9 # 4. Flushing macOS DNS cache
10 # 5. Flushing stale routing table entries
11 # 6. Stopping gateway Docker containers
12 # 7. Power-cycling Wi-Fi
13 # 8. Verifying internet is working
14 #
15 # Usage: double-click Recover-Gateway.applescript
16 # OR ./recover.sh
17
18 set -e
19

```

```

20 # Source common formatting and helper functions
21 source "${dirname "$0"}/common.sh"
22
23 # Always add Homebrew path
24 export PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:$PATH"
25
26 echo ""
27 echo -e "${RED}${BOLD}${NC}"
28 echo -e "${RED}${BOLD}      Nullexit Gateway  RECOVERY MODE      ${NC}"
29 echo -e "${RED}${BOLD}${NC}"
30 echo ""
31 echo "This script will tear down the gateway and restore normal internet."
32 echo ""
33
34 # Cache sudo credentials upfront
35 # Prevents the script from hanging halfway through when it needs to run
36 # privileged commands (route flush, Wi-Fi power cycle, etc.).
37 echo -e "${YELLOW}${BOLD}Authentication required for network recovery...${NC}"
38 sudo -v
39 (
40     while true; do
41         sudo -n true 2>/dev/null
42         sleep 60
43     done
44 ) &
45 SUDO_KEEPER_PID=$!
46 trap 'kill $SUDO_KEEPER_PID 2>/dev/null' EXIT
47
48 # 1. Disconnect host Tailscale
49 step "Disconnecting host Tailscale from mesh"
50 if command -v tailscale >> output.log 2>&1; then
51     # Check if actually connected first (with timeout tailscaled could be wedged)
52     if run_with_timeout 5 tailscale status >> output.log 2>&1; then
53         # Also explicitly reset any exit-node so it doesn't linger.
54         if run_with_timeout 10 tailscale up --reset --ssh=true --accept-dns=false --exit-node= >> output.log 2>&1;
55             then
56                 ok "Exit-node preference cleared"
57             else
58                 warn "tailscale up --reset didn't respond (tailscaled may be wedged)"
59             fi
60             ts_args=""
61             if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
62             if run_with_timeout 10 tailscale down $ts_args >> output.log 2>&1; then
63                 ok "Tailscale disconnected"
64             else
65                 warn "tailscale down didn't respond"
66             fi
67             warn "tailscaled not reachable skipping (timeout after 5s)"
68         fi
69     else
70         warn "tailscale CLI not found skipping"
71     fi
72
73 # 2. Reset DNS to default on ALL network services
74 step "Resetting DNS to default on active interface"
75
76 ACTIVE_SERVICE=$(ip route get 1.1.1.1 2>/dev/null | awk '{print $5; exit}')
77 if [ -n "$ACTIVE_SERVICE" ]; then
78     sudo resolvectl revert "$ACTIVE_SERVICE" 2>> output.log || true
79     ok "DNS reset on $ACTIVE_SERVICE"
80 else
81     warn "Could not determine active network interface"
82 fi
83
84 # 3. Disable rogue proxy settings
85 step "Disabling proxy settings (SOCKS, web, secure web)"
86 echo " (Linux global SOCKS proxy configuration skipped.)"
87 ok "Proxy settings cleared"
88
89
90 # 4. Flush macOS DNS cache
91 step "Flushing DNS cache"
92 if command -v resolvectl >> output.log 2>&1; then
93     sudo resolvectl flush-caches 2>> output.log || true
94     ok "DNS cache flushed"
95 else
96     warn "resolvectl not found"
97 fi
98
99 # 5. Clear stale routing table

```

```

100 step "Flushing stale routing table entries"
101 sudo ip route flush cache >> output.log 2>&1 || true
102 ok "Routing table flushed"
103
104 # 6. Stop gateway Docker containers
105 step "Stopping gateway Docker containers"
106 if command -v docker >> output.log 2>&1 && docker info >> output.log 2>&1; then
107     if [ -f "docker-compose.yml" ]; then
108         docker compose down -t 5 2>> output.log && ok "Containers stopped" || warn "No containers were running"
109     else
110         warn "docker-compose.yml not found in current directory"
111     fi
112 else
113     warn "Docker not available skipping"
114 fi
115
116 # 6b. Stopping sleep prevention (systemd-inhibit)
117 step "Stopping sleep prevention"
118 PID_FILE="/tmp/nullexit-cafeinate.pid"
119 if [ -f "$PID_FILE" ]; then
120     INHIBIT_PID=$(cat "$PID_FILE")
121     if [ -n "$INHIBIT_PID" ] && kill -0 "$INHIBIT_PID" 2>/dev/null; then
122         kill "$INHIBIT_PID" 2>/dev/null || true
123         ok "Sleep prevention stopped (PID $INHIBIT_PID)"
124     else
125         warn "Sleep prevention process not running, cleaning up stale PID file"
126     fi
127     rm -f "$PID_FILE"
128 else
129     ok "Sleep prevention was not active"
130 fi
131
132 # 7. Power-cycle Wi-Fi
133 step "Power-cycling Wi-Fi"
134 if command -v nmcli &>/dev/null; then
135     sudo nmcli radio wifi off 2>> output.log || true
136     sleep 2
137     sudo nmcli radio wifi on 2>> output.log || true
138     ok "Wi-Fi bounced via nmcli"
139     sleep 3
140 else
141     warn "nmcli not found. Falling back to ip link..."
142     WIFI_IFACE=$(ip link | grep -E '[0-9]+: (wl[a-zA-Z0-9]+|wlan[0-9]+):' | awk -F': ' '{print $2}')
143     if [ -n "$WIFI_IFACE" ]; then
144         sudo ip link set "$WIFI_IFACE" down 2>> output.log || true
145         sleep 2
146         sudo ip link set "$WIFI_IFACE" up 2>> output.log || true
147         ok "Wi-Fi bounced via ip link (interface $WIFI_IFACE)"
148         sleep 3
149     else
150         warn "No Wi-Fi interface detected."
151     fi
152 fi
153
154 # 8. Verify internet connectivity
155 step "Verifying internet connectivity"
156
157 # Wait a moment for the network to settle
158 sleep 2
159
160 INTERNET_OK=false
161
162 # Try DNS resolution first
163 if host -W 3 google.com 1.1.1.1 >> output.log 2>&1; then
164     ok "DNS resolution works (google.com via 1.1.1.1)"
165     INTERNET_OK=true
166 elif nslookup google.com 1.1.1.1 >> output.log 2>&1; then
167     ok "DNS resolution works (nslookup google.com via 1.1.1.1)"
168     INTERNET_OK=true
169 else
170     warn "DNS resolution check failed will still try ping/curl"
171 fi
172
173 # Try actual HTTP connectivity
174 if command -v curl >> output.log 2>&1; then
175     if curl -sf --max-time 5 https://1.1.1.1 >> output.log 2>&1; then
176         ok "Internet reachable via HTTP"
177         INTERNET_OK=true
178     elif curl -sf --max-time 5 http://1.1.1.1 >> output.log 2>&1; then
179         ok "Internet reachable via HTTP (plain)"
180         INTERNET_OK=true

```

```

181 else
182     warn "HTTP check failed"
183 fi
184 elif command -v ping >> output.log 2>&1; then
185     if ping -c 1 -W 3 1.1.1.1 >> output.log 2>&1; then
186         ok "Internet reachable via ping"
187         INTERNET_OK=true
188     else
189         warn "Ping check failed"
190     fi
191 fi
192
193 # Summary
194 echo ""
195 echo -e "${BOLD}${NC}"
196 echo -e "${BOLD}RECOVERY SUMMARY${NC}"
197 echo -e "${BOLD}${NC}"
198
199 # Show final DNS state
200 FINAL_DNS=$(resolvectl dns "$ACTIVE_SERVICE" 2>> output.log | grep -oE '[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+' | tr '
201 ' ' ' || echo "N/A")
202 echo "  DNS ($ACTIVE_SERVICE):  $FINAL_DNS"
203
204 echo ""
205
206 if [ "$INTERNET_OK" = "true" ]; then
207     echo -e "  ${GREEN}${BOLD} Internet is working.${NC}"
208 else
209     echo -e "  ${YELLOW}${BOLD} Could not verify internet connectivity.${NC}"
210     echo "    This might mean:"
211     echo "    - Your Wi-Fi is disconnected from the router"
212     echo "    - Tailscale is still interfering (try restarting tailscaled:"
213     echo "      systemctl restart tailscaled)"
214     echo "    - You need to re-connect to Wi-Fi in the menu bar"
215     echo "    - Try toggling Wi-Fi off/on in the menu bar"
216     echo ""
217     echo "    Or try manually:"
218     echo "      resolvectl revert $ACTIVE_SERVICE"
219     extra_args=""
220     if is_kill_switch_enabled; then extra_args="--accept-risk=lose-ssh"; fi
221     echo "      tailscale down $extra_args"
222     echo "      sudo ip route flush cache"
223     echo "      systemctl restart tailscaled"
224 fi
225
226 echo ""
227 echo -e "${GREEN}${BOLD}Recovery complete.${NC}"
228 echo ""

```

11 scripts/setup-linux.sh

```

1 #!/bin/bash
2 # scripts/setup-linux.sh nullexit one-shot setup script (Linux)
3 # Configures Cloudflare WARP + Tailscale + AdGuard Home from scratch.
4 set -euo pipefail
5
6 # Source common formatting and helper functions
7 source "${dirname "$0"}/common.sh"
8
9 # Run common installation logic
10 source "${dirname "$0"}/setup-common.sh"
11
12 # Compile Linux Desktop Shortcuts
13 echo -e "\n${BOLD}Creating Linux Desktop Shortcuts...${NC}"
14 DIR="$(pwd)"
15 cat <<EOF > "Toggle Gateway.desktop"
16 [Desktop Entry]
17 Version=1.0
18 Name=Toggle Gateway
19 Comment=Start or Stop Nullexit Gateway
20 Exec=bash -c "cd '$DIR' && ./scripts/toggle-linux.sh; read -p 'Press Enter to close...' dummy"
21 Icon=utilities-terminal
22 Terminal=true
23 Type=Application
24 Categories=Network;Security;
25 EOF

```

```

26
27 cat <<EOF > "Recover Gateway.desktop"
28 [Desktop Entry]
29 Version=1.0
30 Name=Recover Gateway
31 Comment=Emergency DNS and Network Recovery
32 Exec=bash -c "cd '$DIR' && ./scripts/recover-linux.sh; read -p 'Press Enter to close...' dummy"
33 Icon=utilities-terminal
34 Terminal=true
35 Type=Application
36 Categories=Network;Security;
37 EOF
38 echo "    Created 'Toggle Gateway.desktop' and 'Recover Gateway.desktop'"
39
40 # Done
41 echo ""
42 echo -e "${GREEN}${BOLD}${NC}"
43 echo -e "${GREEN}${BOLD}                Setup complete!                ${NC}"
44 echo -e "${GREEN}${BOLD}${NC}"
45 echo ""
46 echo -e "${BOLD}One manual step remaining  approve the exit node:${NC}"
47 echo "    https://login.tailscale.com/admin/machines"
48 echo "    Find your gateway '...' 'Edit route settings'  enable Exit Node"
49 echo ""
50
51 if [[ -n "${TS_IP:-}" ]]; then
52     echo -e "${BOLD}AdGuard Home dashboard:${NC}  http://${TS_IP}:3000  (username: admin)"
53     echo ""
54 fi
55
56 echo -e "${BOLD}To toggle the gateway on/off:${NC}"
57 echo "    Linux: double-click the 'Toggle Gateway' desktop icon!"
58 echo "    (or run ./scripts/toggle-linux.sh in terminal)"
59 echo ""

```

12 docker-compose.yml

```

1 services:
2   warp:
3     image: qmcgaw/gluetun:v3.41.1
4     container_name: warp
5     hostname: nullexit-gateway
6     cap_add:
7       - NET_ADMIN
8     devices:
9       - /dev/net/tun:/dev/net/tun
10    ports:
11      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/udp" # AdGuard DNS (5335 is used by SSH on host, use
12        procedural port instead)
13      - "127.0.0.1:${DNS_PROXY_PORT:-5354}:5335/tcp" # AdGuard DNS (TCP)
14      - "41642:41642/udp" # Tailscale WireGuard (direct connection to host)
15      - "127.0.0.1:${SOCKS_PROXY_PORT:-1080}:1080/tcp" # SOCKS5 proxy (routed through WARP tunnel)
16      - "127.0.0.1:${TOR SOCKS PORT:-9050}:9050/tcp" # Tor SOCKS5 proxy (procedurally generated)
17      - "127.0.0.1:${TOR_CONTROL_PORT:-9051}:9051/tcp" # Tor Control port (procedurally generated)
18    environment:
19      - TZ=America/New_York
20      - VPN_SERVICE_PROVIDER=custom
21      - VPN_TYPE=wireguard
22      - WIREGUARD_PRIVATE_KEY=${WIREGUARD_PRIVATE_KEY}
23      - WIREGUARD_PUBLIC_KEY=${WIREGUARD_PUBLIC_KEY}
24      - WIREGUARD_ADDRESSES=${WIREGUARD_ADDRESSES}
25      - WIREGUARD_ENDPOINT_IP=${WARP_ENDPOINT_1:-162.159.192.1}
26      - WIREGUARD_ENDPOINT_PORT=2408
27      # Set to 25s (WireGuard default) to beat standard 30s hardware NAT/firewall UDP timeouts
28      - WIREGUARD_PERSISTENT_KEEPALIVE_INTERVAL=25s
29      # MTU Fix (Critical): Prevent packet fragmentation from double-encryption
30      - WIREGUARD_MTU=1280
31      # Give WARP more time to establish the connection before internal restart (default is 6s which is too
32      short)
33      - HEALTH_VPN_DURATION_INITIAL=30s
34      # Use plaintext DNS resolvers instead of DNS-over-TLS (DoT) to prevent startup failures on networks where
35      port 853 is blocked
36      - DNS_UPSTREAM_RESOLVER_TYPE=plain
37      - DNS_UPSTREAM_PLAIN_ADDRESSES=1.1.1.1:53,8.8.8.8:53
38      # Allow local network traffic to bypass WARP so Tailscale can establish fast, direct peer-to-peer
39      connections (no DERP relays) on the LAN

```



```

36 - FIREWALL_OUTBOUND_SUBNETS=192.168.0.0/16,172.16.0.0/12,10.0.0.0/8
37 # Built-in DNS blocking for ads, malicious domains, and tracking
38 # NOTE: This is the massive "Hagezi-style" built-in blocklist.
39 # You can turn these to 'on' for maximum security if you have spare RAM
40 # or are running this on a cloud exit node with adequate autonomous resources (requires approx 1-2GB of
41 RAM).
42 # Disabled intentionally: AdGuard Home provides the DNS filtering layer instead.
43 # If AdGuard fails, there is no fallback filtering, but the healthcheck
44 # dependency chain ensures Tailscale won't come up if AdGuard is down.
45 - BLOCK_MALICIOUS=off
46 - BLOCK_SURVEILLANCE=off
47 - BLOCK_ADS=off
48 dns:
49 - 1.1.1.1
50 sysctls:
51 - net.ipv4.ip_forward=1
52 volumes:
53 - ./post-rules.txt:/iptables/post-rules.txt:ro
54 restart: unless-stopped
55 healthcheck:
56   test: ["CMD-SHELL", "/gluetun-entrypoint healthcheck"]
57   interval: 2s
58   timeout: 5s
59   retries: 15
60   start_period: 60s
61 logging:
62   driver: "json-file"
63   options:
64     max-size: "10m"
65     max-file: "3"
66 tailscale:
67   image: tailscale/tailscale:v1.98.4
68   container_name: tailscale
69   network_mode: service:warp
70   depends_on:
71     warp:
72       condition: service_healthy
73     adguardhome:
74       condition: service_healthy
75   cap_add:
76     - NET_ADMIN
77     - NET_RAW
78     - SYS_MODULE
79   sysctls:
80     - net.ipv4.ip_forward=1
81     - net.ipv6.conf.all.forwarding=1
82   environment:
83     - TZ=America/New_York
84     - TS_EXTRA_ARGS=--advertise-exit-node
85     - PORT=41642
86     - TS_USERSPACE=false
87     - TS_AUTHKEY=${TS_AUTHKEY}
88     - TS_STATE_DIR=/var/lib/tailscale
89     # (Note: For headless/cloud deployments, you can bypass this footgun by
90     # using Tailscale Ephemeral Keys which auto-clean themselves).
91     - TS_AUTH_ONCE=true
92     # MTU Optimization: Force Tailscale packets to be smaller than WARP's 1280 MTU
93     # to prevent WireGuard-in-WireGuard fragmentation, completely eliminating bufferbloat
94     - TS_DEBUG_MTU=1200
95     # Enable Tailscale's built-in SOCKS5 proxy to route out of the WARP tunnel
96     - TS_SOCKS5_SERVER=:1080
97   devices:
98     - /dev/net/tun:/dev/net/tun
99   volumes:
100     - tailscale_state:/var/lib/tailscale
101   restart: unless-stopped
102   logging:
103     driver: "json-file"
104     options:
105       max-size: "10m"
106       max-file: "3"
107
108 routing-fix:
109   image: nullexit-routing-fix:v1.0.0
110   build:
111     context: ./scripts
112     dockerfile: Dockerfile.routing-fix
113   container_name: routing-fix
114   network_mode: service:warp
115   # NOTE: Deliberately NOT sharing tailscale's PID namespace the shared

```

```

116 # namespace caused routing-fix to be killed whenever tailscale restarted
117 # (e.g. during gluetun health-check flaps), which dropped all routing
118 # table changes and left traffic leaking through eth0 instead of tun0.
119 cap_add:
120   - NET_ADMIN
121   - SYSLOG
122 depends_on:
123   warp:
124     condition: service_healthy
125   rule-compiler:
126     condition: service_completed_successfully
127 environment:
128   - TZ=America/New_York
129   - WARP_ENDPOINT=${WARP_ENDPOINT_1:-162.159.192.1}
130   - BLOCKED_COUNTRIES=${BLOCKED_COUNTRIES}
131   - TAILSCALE_ALLOW_LAN_P2P=${TAILSCALE_ALLOW_LAN_P2P:-false}
132 volumes:
133   - ./scripts/routing-fix.sh:/routing-fix.sh:ro
134   # SECURITY: ./adguard/work/ is bind-mounted into this container.
135   # Never write to it from the host or another container. The
136   # single allowed writer is `rule-compiler` container manual
137   # edits corrupt the AdGuard cache + BoltDB session store.
138   - ./adguard/work/userfilters:/userfilters:ro
139   - ./adguard/work:/adguard_work:ro
140   - ./app
141 restart: unless-stopped
142 stop_grace_period: 1s
143 logging:
144   driver: "json-file"
145   options:
146     max-size: "1m"
147     max-file: "1"
148
149 adguardhome:
150   # WARNING: AdGuard is pre-configured with hardcoded credentials (admin/nullexit).
151   # This is fine for a local Mac running behind a NAT, but if you are deploying
152   # this on a public cloud VPS with port 80/3000 exposed, change this immediately!
153   image: adguard/adguardhome:v0.107.77
154   container_name: adguardhome
155   network_mode: service:warp
156   depends_on:
157     warp:
158       condition: service_healthy
159     rule-compiler:
160       condition: service_completed_successfully
161   environment:
162     - TZ=America/New_York
163   healthcheck:
164     test: ["CMD-SHELL", "nc -z 127.0.0.1 5335 || exit 1"]
165     interval: 2s
166     timeout: 5s
167     retries: 10
168   volumes:
169     # SECURITY: ./adguard/work/ is bind-mounted into this container.
170     # Never write to it from the host or another container. The
171     # single allowed writer is `rule-compiler` container manual
172     # edits corrupt the AdGuard cache + BoltDB session store.
173     - ./adguard/work:/opt/adguardhome/work
174     - ./adguard/conf:/opt/adguardhome/conf
175   restart: unless-stopped
176   stop_grace_period: 2s
177   logging:
178     driver: "json-file"
179     options:
180       max-size: "10m"
181       max-file: "3"
182
183 rule-compiler:
184   image: nullexit-rule-compiler:v1.0.0
185   build:
186     context: ./scripts
187     dockerfile: Dockerfile.rule-compiler
188   container_name: rule-compiler
189   volumes:
190     - ./app
191
192 tor:
193   build: ./docker/tor
194   image: nullexit-tor:v1.0.0
195   container_name: tor
196   network_mode: service:warp

```

```

197     depends_on:
198         warp:
199             condition: service_healthy
200     environment:
201         - TZ=America/New_York
202         - TOR_USE_BRIDGES=${TOR_USE_BRIDGES:-false}
203         - TOR_BRIDGE_LINES_FILE=${TOR_BRIDGE_LINES_FILE:-./tor-bridges.txt}
204         - TOR_PASSWORD=${ADGUARD_PASSWORD:-nullexit}
205         - TOR_EXCLUDE_EXIT_NODES=${TOR_EXCLUDE_EXIT_NODES:-{us}}
206     working_dir: /nullexit
207     volumes:
208         - ./:/nullexit:ro
209         - ./output.log:/output.log:rw
210     restart: unless-stopped
211     logging:
212         driver: "json-file"
213         options:
214             max-size: "10m"
215             max-file: "3"
216
217 volumes:
218     tailscale_state:
219
220 networks:
221     default:
222         ipam:
223             config:
224                 - subnet: 10.200.1.0/24

```

13 scripts/common.sh

```

1  #!/bin/bash
2  # common.sh shared bash utilities for nullexit scripts
3
4  # Formatting
5  RED='\033[0;31m'
6  GREEN='\033[0;32m'
7  YELLOW='\033[1;33m'
8  BLUE='\033[0;34m'
9  BOLD='\033[1m'
10 NC='\033[0m'
11
12 # Constants & Environment
13 export MARKER_FILE="/tmp/nullexit-gateway-active.marker"
14 export PID_CAFFEINATE="/tmp/nullexit-caffeinate.pid"
15 export PID_DNS_WATCHER="/tmp/nullexit-dns-watcher.pid"
16 export DEFAULT_WARP_ENDPOINT_1="162.159.192.1"
17 export DEFAULT_WARP_ENDPOINT_2="162.159.193.1"
18
19 setup_standard_path() {
20     export PATH="/opt/homebrew/bin:/opt/homebrew/sbin:/usr/local/bin:$PATH"
21 }
22
23 step() { echo -e "\n[ $(date '+%Y-%m-%d %H:%M:%S') ] ${BLUE}${BOLD} ${*}${NC}"; }
24 ok() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${GREEN} ${*}${NC}"; }
25 warn() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${YELLOW} ${*}${NC}"; }
26 fail() { echo -e "[ $(date '+%Y-%m-%d %H:%M:%S') ] ${RED} ${*}${NC}"; }
27 die() { echo -e "\n[ $(date '+%Y-%m-%d %H:%M:%S') ] ${RED} ${*}${NC}\n"; exit 1; }
28
29 # Helper Functions
30
31 # Pure-bash timeout (no dependency on GNU coreutils' timeout)
32 # Runs a command with a safety cutoff so a wedged daemon can't hang the script.
33 run_with_timeout() {
34     local timeout_sec="$1"
35     shift
36     if [ $# -eq 0 ]; then return 1; fi
37
38     "$@" &
39     local cmd_pid=$!
40     CURRENT_BG_PID="$cmd_pid"
41
42     (
43         sleep "$timeout_sec"
44         if kill -0 "$cmd_pid" 2>> output.log; then
45             echo -e "\n[Timeout] Command '$*' exceeded $timeout_sec seconds. Terminating..." >&2

```

```

46     kill -15 "$cmd_pid" 2>> output.log || true
47     sleep 2
48     if kill -0 "$cmd_pid" 2>> output.log; then
49         kill -9 "$cmd_pid" 2>> output.log || true
50     fi
51 fi
52 ) &
53 local watcher_pid=$!
54
55 # Disable set -e temporarily to capture exit status of wait
56 set +e
57 wait "$cmd_pid" 2>> output.log
58 local exit_status=$?
59 set -e
60
61 # Kill the watcher since the command finished
62 kill "$watcher_pid" 2>> output.log && wait "$watcher_pid" 2>> output.log || true
63
64 CURRENT_BG_PID=""
65 return "$exit_status"
66 }
67
68 # Check if the gateway is active (either containers are running, or host DNS is hijacked)
69 is_gateway_active() {
70     # 1. Check if containers are running (suppress stderr to avoid errors if docker is down)
71     if run_with_timeout 15 docker compose ps --status running 2>/dev/null | grep -q 'warp'; then
72         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (warp container is running)" >> "$LOG_FILE"
73         return 0
74     fi
75
76     # 2. Check if host DNS was hijacked (not 1.1.1.1 and not default/empty)
77     local current_dns=""
78     local active_svc=""
79     if [[ "$OSTYPE" == "darwin"* ]]; then
80         active_svc=$(get_active_service)
81         current_dns=$(networksetup -getdnsservers "$active_svc" 2>> output.log || true)
82     else
83         current_dns=$(resolvectl dns 2>/dev/null | awk '/Global|Link/ {for(i=4;i<=NF;i++) print $i}' | head -n1)
84     fi
85
86     if [[ -n "$current_dns" && "$current_dns" != "1.1.1.1" && ! "$current_dns" =~ "There aren't any DNS Servers"
87         ]]; then
88         echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (DNS was hijacked: $current_dns)" >> "
89             $LOG_FILE"
90         return 0
91     fi
92
93     # 3. Check if SOCKS proxy is enabled (macOS only)
94     if [[ "$OSTYPE" == "darwin"* ]]; then
95         local socks_proxy
96         socks_proxy=$(networksetup -getsocksfirewallproxy "$active_svc" 2>> output.log || true)
97         if echo "$socks_proxy" | grep -q "Enabled: Yes"; then
98             echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: TRUE (SOCKS proxy enabled)" >> "$LOG_FILE"
99             return 0
100         fi
101     fi
102
103     echo "[$(date '+%Y-%m-%d %H:%M:%S')] is_gateway_active: FALSE (Containers down, DNS clean, SOCKS disabled)"
104     >> "$LOG_FILE"
105     return 1
106 }
107
108 # Reads an environment variable from a file (default: .env), strips quotes and whitespace
109 read_env_var() {
110     local var_name="$1"
111     local env_file="$2:-.env"
112     grep -E "^${var_name}=" "$env_file" 2>/dev/null | cut -d '=' -f2- | tr -d '"\`\' ' | sed -e 's/^[[:space:]]*//'
113     -e 's/[[:space:]]*$//' || echo ""
114 }
115
116 # Load KILL_SWITCH variable from .env (defaults to false if not found/empty)
117 export KILL_SWITCH
118 KILL_SWITCH=$(read_env_var "KILL_SWITCH" 2>/dev/null | tr '[:upper:]' '[:lower:]')
119 if [ -z "$KILL_SWITCH" ]; then
120     KILL_SWITCH="false"
121 fi
122
123 # Function to get the active network service name (e.g., "Wi-Fi" or "USB 10/100 LAN")
124 get_active_service() {
125     local iface
126     iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
127 }

```

```

123
124 # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
125 if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
126     for i in en0 en1 en2 en3; do
127         if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "
128             inet "; then
129                 iface="$i"
130                 break
131             fi
132         done
133     fi
134 # Default fallback if still empty or not enX
135 if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
136     iface="en0"
137 fi
138
139 # Map interface (e.g., en0) to service name (e.g., Wi-Fi)
140 local service
141 service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B 1 "Device: $iface" | head -n 1 | sed
142     -E 's/^\([0-9\*]+\)\s*//' )
143 if [ -n "$service" ]; then
144     echo "$service"
145 else
146     echo "Wi-Fi"
147 fi
148 }
149
150 get_en0_service() {
151     local service
152     service=$(networksetup -listnetworkserviceorder 2>> output.log | grep -B1 "Device: en0" | head -1 | sed -E 's
153         /\([0-9\*]+\)\s*//' || true)
154     if [ -z "$service" ]; then
155         echo "Wi-Fi"
156     else
157         echo "$service"
158     fi
159 }
160
161 get_warp_endpoint_1() { read_env_var "WARP_ENDPOINT_1" ".env" | grep -Eo '[0-9\.]+' || echo "
162     $DEFAULT_WARP_ENDPOINT_1"; }
163 get_warp_endpoint_2() { read_env_var "WARP_ENDPOINT_2" ".env" | grep -Eo '[0-9\.]+' || echo "
164     $DEFAULT_WARP_ENDPOINT_2"; }
165
166 restart_tailscaled_daemon() {
167     echo " [Tailscale Recovery] tailscaled daemon appears to be wedged/unresponsive."
168     echo " [Tailscale Recovery] Attempting to restart tailscaled service..."
169
170     if run_with_timeout 15 brew services restart tailscale >> output.log 2>&1; then
171         echo " [Tailscale Recovery] Successfully restarted tailscaled (user service)."
172     elif run_with_timeout 15 sudo -n brew services restart tailscale >> output.log 2>&1; then
173         echo " [Tailscale Recovery] Successfully restarted tailscaled (system service)."
174     else
175         echo " [Tailscale Recovery] WARNING: Failed to restart tailscaled. Manual intervention may be needed."
176     fi
177     sleep 3
178 }
179
180 disconnect_tailscale_host() {
181     local ts_bin="${1:-tailscale}"
182     if command -v "$ts_bin" >> output.log 2>&1; then
183         echo " Disconnecting host Tailscale from mesh..."
184
185         # Check if actually connected first (with timeout tailscaled could be wedged)
186         local status_ok=false
187         if run_with_timeout 5 "$ts_bin" status >> output.log 2>&1; then
188             status_ok=true
189         else
190             local status_exit=$?
191             # If it was a quick exit code 1 (disconnected), it is still reachable
192             if [ "$status_exit" -ne 143 ] && [ -S /var/run/tailscaled.socket ]; then
193                 status_ok=true
194             fi
195         fi
196
197         if [ "$status_ok" = "false" ]; then
198             restart_tailscaled_daemon
199         fi
200
201         # Explicitly reset any exit-node so it doesn't linger.

```

```

199     "$ts_bin" up >> output.log 2>&1 || true
200     if run_with_timeout 10 "$ts_bin" set --ssh=true --accept-dns=false --exit-node= >> output.log 2>&1; then
201         echo " [] Exit-node preference cleared."
202     else
203         echo " [!] tailscale up didn't respond (tailscaled may be wedged)"
204     fi
205
206     local ts_args=""
207     if is_kill_switch_enabled; then ts_args="--accept-risk=lose-ssh"; fi
208     if run_with_timeout 10 "$ts_bin" down $ts_args >> output.log 2>&1; then
209         echo " [] Tailscale disconnected."
210     else
211         echo " [!] tailscale down didn't respond"
212     fi
213 else
214     echo " [!] tailscale CLI not found skipping"
215 fi
216 }
217
218 stop_pidfile_daemon() {
219     local pid_file="$1"
220     local label="$2"
221     if [ -f "$pid_file" ]; then
222         local pid
223         pid=$(cat "$pid_file")
224         if kill -0 "$pid" 2>/dev/null; then
225             echo " Stopping $label (PID $pid)..."
226             sudo -n kill "$pid" 2>> output.log || kill "$pid" 2>> output.log || true
227             sleep 0.5
228             if kill -0 "$pid" 2>/dev/null; then
229                 sudo -n kill -9 "$pid" 2>> output.log || kill -9 "$pid" 2>> output.log || true
230             fi
231         fi
232         rm -f "$pid_file"
233     fi
234 }
235
236 disable_all_proxies() {
237     local svc="$1"
238     if [ -n "$svc" ]; then
239         sudo -n networksetup -setsocksfirewallproxystate "$svc" off 2>> output.log || true
240         sudo -n networksetup -setwebproxystate "$svc" off 2>> output.log || true
241         sudo -n networksetup -setsecurewebproxystate "$svc" off 2>> output.log || true
242     fi
243 }
244
245 flush_dns_cache() {
246     if [[ "$OSTYPE" == "darwin"* ]]; then
247         sudo -n dscacheutil -flushcache 2>> output.log || true
248         sudo -n killall -HUP mDNSResponder 2>> output.log || true
249     else
250         resolvectl flush-caches 2>> output.log || true
251     fi
252 }
253
254 wait_for_dhcp_settle() {
255     # Resolve physical interface (Wi-Fi or Ethernet)
256     local physical_iface
257     physical_iface=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: (Wi-Fi|Ethernet)/{
258         getline; print $2; exit}')
259     [ -z "$physical_iface" ] && physical_iface="en0"
260
261     echo -n "Waiting for physical network interface ($physical_iface) DHCP lease to settle..."
262     local settled=false
263     for attempt in {1..60}; do
264         PHYSICAL_GW=$(ipconfig getpacket "$physical_iface" 2>> output.log | awk -F'[{]' '{print $2}')
265         local local_ip
266         local_ip=$(ifconfig "$physical_iface" 2>/dev/null | awk '/inet/{print $2}')
267
268         if [ -n "$PHYSICAL_GW" ] && [[ "$PHYSICAL_GW" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
269             [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
270             [[ "$local_ip" != "169.254.*" ]]; then
271             echo " settled (Interface IP: $local_ip, Router IP: $PHYSICAL_GW)."
272             settled=true
273             break
274         fi
275
276         if [ "$attempt" -gt 15 ] && [ -n "$local_ip" ] && [[ "$local_ip" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]] && \
277             [[ "$local_ip" != "169.254.*" ]]; then
278             local fallback_gw
279             fallback_gw=$(route get default 2>> output.log | awk '/gateway/{print $2}')

```

```

278     if [ -n "$fallback_gw" ] && [[ "$fallback_gw" =~ ^[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+$ ]]; then
279         PHYSICAL_GW="$fallback_gw"
280     fi
281     echo " static/settled detected (Interface IP: $local_ip, Gateway IP: ${PHYSICAL_GW:-unknown})."
282     settled=true
283     break
284 fi
285 echo -n "."
286 sleep 0.5
287 done
288 if [ "$settled" = "false" ]; then
289     echo " timed out or no active link. Proceeding anyway."
290 fi
291 }
292
293 reset_sharing_services() {
294     if [[ "$OSTYPE" == "darwin"* ]]; then
295         sudo -n killall sharingd rapportd 2>> output.log || true
296     fi
297 }
298
299 read_adguard_ip() {
300     if [ -f ".gateway_ip" ]; then
301         tr -d '\r' < ".gateway_ip" | awk 'NR==1{print $1;exit}'
302     fi
303 }
304
305 is_kill_switch_enabled() {
306     [ "$KILL_SWITCH" = "true" ]
307 }
308
309 add_warp_bypass_routes() {
310     local target="${1:-}"
311     local ep1
312     ep1=$(get_warp_endpoint_1)
313     local ep2
314     ep2=$(get_warp_endpoint_2)
315
316     if [[ "$OSTYPE" == "darwin"* ]]; then
317         local routing_arg=""
318         local msg_via=""
319         if [ -n "$target" ]; then
320             if [[ "$target" =~ ^en[0-9]+$ || "$target" == "bridge"* ]]; then
321                 routing_arg="-interface $target"
322                 msg_via="interface $target"
323             else
324                 routing_arg="$target"
325                 msg_via="gateway $target"
326             fi
327         else
328             local gateway_ip
329             gateway_ip=$(route get default 2>> output.log | awk '/gateway:/ {print $2}')
330             if [ -z "$gateway_ip" ]; then
331                 local iface
332                 iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
333                 if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
334                     iface="en0"
335                 fi
336                 routing_arg="-interface $iface"
337                 msg_via="interface $iface"
338             else
339                 routing_arg="$gateway_ip"
340                 msg_via="gateway $gateway_ip"
341             fi
342         fi
343
344         echo -e "\nAdding host bypass routes for Cloudflare WARP endpoints via $msg_via..."
345         sudo -n route delete -host "$ep1" 2>/dev/null || true
346         sudo -n route delete -host "$ep2" 2>/dev/null || true
347         sudo -n route add -host "$ep1" $routing_arg >> output.log 2>&1 || true
348         sudo -n route add -host "$ep2" $routing_arg >> output.log 2>&1 || true
349
350         echo -e "Adding host bypass routes for Tailscale control plane via $msg_via..."
351         sudo -n route delete -net 192.200.0.0/24 2>/dev/null || true
352         sudo -n route add -net 192.200.0.0/24 $routing_arg >> output.log 2>&1 || true
353
354         echo -e "Adding host bypass routes for Tailscale DERP relays via $msg_via..."
355         local derp_ips
356         derp_ips=$(curl -s --connect-timeout 5 https://login.tailscale.com/derpmap/default | grep -oE '"IPv4"[[:space:]]*:[[:space:]]*"[0-9.]+"' | cut -d'"' -f4 | grep -E '^([0-9]+\.[0-9]+\.[0-9]+\.[0-9])+$' || true)
357         if [ -n "$derp_ips" ]; then

```



```

358     echo "$derp_ips" > /tmp/nullexit-derp-ips.txt
359     for ip in $derp_ips; do
360         sudo -n route delete -host "$ip" 2>/dev/null || true
361         sudo -n route add -host "$ip" $routing_arg >> output.log 2>&1 || true
362     done
363     fi
364 else
365     local gateway_ip="${target}"
366     if [ -z "$gateway_ip" ]; then
367         gateway_ip=$(ip route show default 2>/dev/null | awk '/default/ {print $3}')
368     fi
369     if [ -n "$gateway_ip" ]; then
370         sudo ip route add "$ep1" via "$gateway_ip" >> output.log 2>&1 || true
371         sudo ip route add "$ep2" via "$gateway_ip" >> output.log 2>&1 || true
372     fi
373 fi
374 }
375
376 remove_warp_bypass_routes() {
377     local ep1
378     ep1=$(get_warp_endpoint_1)
379     local ep2
380     ep2=$(get_warp_endpoint_2)
381
382     if [[ "$OSTYPE" == "darwin"* ]]; then
383         echo -e "\nRemoving host bypass routes for Cloudflare WARP endpoints..."
384         sudo -n route delete -host "$ep1" >> output.log 2>&1 || true
385         sudo -n route delete -host "$ep2" >> output.log 2>&1 || true
386         echo -e "Removing host bypass routes for Tailscale control plane..."
387         sudo -n route delete -net 192.200.0.0/24 >> output.log 2>&1 || true
388         if [ -f /tmp/nullexit-derp-ips.txt ]; then
389             echo -e "Removing host bypass routes for Tailscale DERP relays..."
390             while read -r ip; do
391                 sudo -n route delete -host "$ip" >> output.log 2>&1 || true
392             done < /tmp/nullexit-derp-ips.txt
393             rm -f /tmp/nullexit-derp-ips.txt
394         fi
395     else
396         sudo ip route del "$ep1" >> output.log 2>&1 || true
397         sudo ip route del "$ep2" >> output.log 2>&1 || true
398     fi
399 }
400
401 bounce_wifi_interfaces() {
402     if [[ "$OSTYPE" == "darwin"* ]]; then
403         local wifi_port
404         wifi_port=$(networksetup -listallhardwareports 2>> output.log | awk '/Hardware Port: Wi-Fi/{getline; print $2}')
405         if [ -n "$wifi_port" ]; then
406             sudo -n networksetup -setairportpower "$wifi_port" off 2>> output.log || true
407             sleep 2
408             sudo -n networksetup -setairportpower "$wifi_port" on 2>> output.log || true
409             echo "Wi-Fi bounced (interface $wifi_port)"
410             sleep 3
411         else
412             echo "Could not detect Wi-Fi interface. Bouncing fallback interfaces..."
413             set +e
414             for iface in en0 en1 en2 en3 en4 en5; do
415                 if ifconfig "$iface" >> output.log 2>&1; then
416                     sudo -n ifconfig "$iface" down 2>> output.log || true
417                     sudo -n ifconfig "$iface" up 2>> output.log || true
418                 fi
419             done
420             set -e
421             echo "Fallback interfaces bounced"
422         fi
423     fi
424 }
425
426 get_active_interface() {
427     local iface
428     iface=$(route get default 2>> output.log | awk '/interface:/ {print $2}')
429
430     # If the default interface is empty or a VPN tunnel (like utunX), fallback to the active physical interface
431     if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
432         for i in en0 en1 en2 en3; do
433             if ifconfig "$i" 2>> output.log | grep -q "status: active" && ifconfig "$i" 2>> output.log | grep -q "inet "; then
434                 iface="$i"
435                 break
436             fi

```

```

437     done
438 fi
439
440 # Default fallback if still empty or not enX
441 if [[ -z "$iface" || ! "$iface" =~ ^en[0-9]+$ ]]; then
442     iface="en0"
443 fi
444
445 echo "$iface"
446 }
447
448 enable_killswitch() {
449     if [[ "$OSTYPE" == "darwin"* ]]; then
450         if ! is_kill_switch_enabled; then
451             return 0
452         fi
453
454         local ext_if
455         ext_if=$(get_active_interface)
456         local ep1
457         ep1=$(get_warp_endpoint_1)
458         local ep2
459         ep2=$(get_warp_endpoint_2)
460         local pf_conf_path
461         pf_conf_path="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/pf.conf"
462
463         echo -e "\nEnabling macOS Packet Filter (PF) Kill-Switch on $ext_if..."
464
465         # Enable PF and capture the reference token
466         local token_out
467         token_out=$(sudo -n pfctl -E 2>> output.log || true)
468         local token
469         token=$(echo "$token_out" | awk '/Token/{print $NF}')
470         if [ -n "$token" ]; then
471             echo "$token" > /tmp/nullexit-pf-token.txt
472             echo " [] PF enabled with token $token."
473         else
474             echo " [!] WARNING: Could not capture PF reference token (may already be enabled or sudo failed)."
475         fi
476
477         # Load the rules into the anchor
478         if sudo -n pfctl -D "ext_if=$ext_if" -a com.apple/nullexit -f "$pf_conf_path" >> output.log 2>&1; then
479             echo " [] Ruleset loaded into anchor com.apple/nullexit."
480         else
481             echo " [!] ERROR: Failed to load ruleset into anchor."
482         fi
483
484         # Populate tables in the anchor
485         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T flush >> output.log 2>&1 || true
486         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$ep1" >> output.log 2>&1 || true
487         sudo -n pfctl -a com.apple/nullexit -t vpn_endpoints -T add "$ep2" >> output.log 2>&1 || true
488
489         if [ -f /tmp/nullexit-derp-ips.txt ]; then
490             local derp_ips
491             derp_ips=$(tr '\n' ' ' < /tmp/nullexit-derp-ips.txt)
492             if [ -n "$derp_ips" ]; then
493                 sudo -n pfctl -a com.apple/nullexit -t derp_relays -T flush >> output.log 2>&1 || true
494                 sudo -n pfctl -a com.apple/nullexit -t derp_relays -T add $derp_ips >> output.log 2>&1 || true
495             fi
496         fi
497         echo " [] Kill-switch tables populated."
498     fi
499 }
500
501 disable_killswitch() {
502     if [[ "$OSTYPE" == "darwin"* ]]; then
503         if ! is_kill_switch_enabled; then
504             return 0
505         fi
506         echo -e "\nDisabling macOS PF Kill-Switch..."
507
508         # Flush rules from anchor com.apple/nullexit
509         sudo -n pfctl -a com.apple/nullexit -F all >> output.log 2>&1 || true
510
511         # Release the enable reference if token is present
512         if [ -f /tmp/nullexit-pf-token.txt ]; then
513             local token
514             token=$(cat /tmp/nullexit-pf-token.txt)
515             if [ -n "$token" ]; then
516                 sudo -n pfctl -X "$token" >> output.log 2>&1 || true
517                 echo " [] Released PF enable reference token $token."

```

```

518     fi
519     rm -f /tmp/nullexit-pf-token.txt
520 else
521     echo " [] Rules flushed from anchor com.apple/nullexit."
522 fi
523 fi
524 }
525
526 write_host_ips() {
527     # Gather all active IPv4 addresses on the host and write to .host_ips
528     # routing-fix.sh will read this file to explicitly whitelist them for direct Tailscale P2P
529     local repo_dir
530     repo_dir="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
531
532     if [[ "$OSTYPE" == "darwin"* ]]; then
533         ifconfig | awk '/inet /{print $2}' | grep -v '127.0.0.1' > "$repo_dir/.host_ips" 2>/dev/null || true
534     else
535         ip -4 addr show 2>/dev/null | awk '/inet /{print $2}' | cut -d/ -f1 | grep -v '127.0.0.1' > "$repo_dir/.
536         host_ips" 2>/dev/null || true
537     fi
538 }

```

14 scripts/watcher.sh

```

1  #!/bin/bash
2  # scripts/watcher.sh nullexit post-wake / post-roam recovery watcher
3  #
4  # Long-running daemon. Launched by launchd as a LaunchAgent at
5  # ~/Library/LaunchAgents/com.nullexit.wake-recovery.plist
6  # (label com.nullexit.wake-recovery).
7  #
8  # Two independent listeners run as backgrounded pipelines:
9  #
10 # (1) WAKE listener `log stream` live-follows the macOS unified log for
11 # power-management events (lid closewake, manual wake, DarkWake from
12 # clamshell). Each event triggers run_recover (with a global debounce so
13 # a single wake that emits 2-3 log lines only causes ONE post-wake run).
14 #
15 # (2) NETWORK listener `scutil n.watch` on `State:/Network/Global/IPv4`
16 # fires on every Wi-Fi roam, ethernet swap, hotspot join, captive-portal
17 # re-bind, VPN up/down, etc. Same debounce.
18 #
19 # Both listeners call run_recover, which:
20 # - checks /tmp/nullexit-gateway-active.marker (only act when toggle.sh left
21 # the gateway up)
22 # - debounces (10s default) so multiple events don't pile up
23 # - shells out to `bash <repo>/recover.sh --post-wake` directly (no GUI auth prompt needed due to NOPASSWD
24 # sudoers)
25 #
26 # See devref.md 10.29 for the full Apple power management primer and the
27 # rationale behind using `log stream` + `scutil n.watch` from a shell daemon.
28 #
29 # NOTE: errexit (`set -e`) is intentionally NOT enabled. This is a long-running
30 # daemon, not a discrete-step script, and errexit actively breaks the reconnect
31 # loops below: any failed pipe (log stream on rotation, scutil on state churn)
32 # would kill the outer subshell before `echo "reconnecting..."; sleep 5;` runs,
33 # AND the parent's final `wait` would propagate a non-zero child exit and kill
34 # the entire watcher. Every error path is already wrapped in `|| true` /
35 # `2>/dev/null` fallbacks.
36 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
37 source "$SCRIPT_DIR/common.sh"
38 setup_standard_path
39
40 LOG="$SCRIPT_DIR/./output.log"
41 # Resolve our own directory; recover.sh lives one level up at the repo root.
42 # This makes the daemon install-agnostic no need to template the path
43 # in launchd, no need to keep a deploy-specific hardcoded location in sync.
44 RECOVER="$SCRIPT_DIR/./recover.sh"
45 MARKER="$MARKER_FILE"
46 DEBOUNCE_FILE="/tmp/nullexit-watcher.last-recovery"
47 DEBOUNCE_SECONDS="${NULLEXIT_DEBOUNCE_SECONDS:-10}"
48
49 exec >> "$LOG" 2>&1
50
51 # Single-instance lock

```

```

51 # Without this, repeated lid-closewake cycles under macOS Sonoma+ accumulate
52 # orphan watcher.sh processes: launchd suspends on sleep and SIGCONT-resumes
53 # on wake, and KeepAlive.Crashed=true caused relaunch on any non-zero exit;
54 # the listener grandchildren (log stream | while ... and (echo n.add ...;
55 # sleep 86400) | scutil | while ...) stay alive because pipe producers don't
56 # reliably respond to plain SIGTERM if the consumer is in a half-closed state.
57 #
58 # We solve it with a PID-file lock (POSIX-portable; macOS does NOT ship `flock`).
59 # The trap below removes the lock file on every exit so a stale lock never
60 # outlives a crashed watcher. On launch: read the existing PID, check it is
61 # alive AND is actually a `scripts/watcher.sh` process; if so, exit 0; else
62 # overwrite with our PID and proceed. The check-then-write window has a TOCTOU
63 # race but is microseconds wide and only matters for hand-launched duplicate
64 # test invocations; production is single-launch via launchctl load -w.
65 LOCK_FILE="/tmp/nullexit-watcher.lock"
66 LOCK_PID=""
67 if [ -e "$LOCK_FILE" ]; then
68     LOCK_PID=$(cat "$LOCK_FILE" 2>/dev/null || echo "")
69     if [ -n "$LOCK_PID" ] && kill -0 "$LOCK_PID" 2>/dev/null; then
70         # Verify the holder is actually a scripts/watcher.sh process (not some
71         # unrelated process that got the recycled PID). The exact-launchd pattern
72         # `bash <abs-path>/scripts/watcher.sh` is hard to accidentally match with
73         # a `grep scripts/watcher.sh .` invocation because we anchor on `^bash`
74         # followed by whitespace and end-of-line on the script path.
75         if ps -p "$LOCK_PID" -o args= 2>/dev/null | grep -qE "(^|[:space:])bash[:space:]+.*scripts/watcher\\.sh$"; then
76             echo "[$(date -u +%FT%TZ)] watcher.sh: another instance holds $LOCK_FILE (pid=$LOCK_PID), exiting"
77             exit 0
78         fi
79     fi
80 fi
81 echo $$ > "$LOCK_FILE"
82
83 echo "[$(date -u +%FT%TZ)] watcher.sh start (pid=$$ ppid=$PPID user=$(id -un) lock=acquired)"
84
85 # Treat launchd-resume as a wake signal. Apple's launchd SUSPENDS LaunchAgents
86 # during system sleep and restores them on wake; during that suspended gap,
87 # macOS's `log stream` cannot catch the wake event that prompted the resume.
88 # The next thing that happens after wake IS our own startup, so we always
89 # fire one post-wake run unconditionally here. The marker check + debounce
90 # in run_recover() make this idempotent: if the gateway isn't up or we just
91 # debounced, it no-ops. Without this, the FIRST wake-after-install goes
92 # unseen and the watcher appears broken until the SECOND wake.
93 # Helper: run recover.sh --post-wake if the gateway is active
94 run_recover() {
95     local why="$1"
96     if [ ! -f "$MARKER" ]; then
97         echo "[$(date -u +%FT%TZ)] $why no $MARKER, skip"
98         return 0
99     fi
100     local now last
101     now=$(date +%s)
102     last=$(cat "$DEBOUNCE_FILE" 2>/dev/null || echo 0)
103     if [ "$(now - last)" -lt "$DEBOUNCE_SECONDS" ]; then
104         echo "[$(date -u +%FT%TZ)] $why debounced ($(now - last)s since last, threshold ${DEBOUNCE_SECONDS}s)"
105         return 0
106     fi
107     echo "$now" > "$DEBOUNCE_FILE"
108     echo "[$(date -u +%FT%TZ)] $why bash $RECOVER --post-wake"
109     bash "$RECOVER" --post-wake
110     local exit_code=?
111     echo "$(date +%s)" > "$DEBOUNCE_FILE"
112     echo "[$(date -u +%FT%TZ)] $why exit=$exit_code (now=$(date +%s))"
113     return $exit_code
114 }
115
116 if [ -f "$MARKER" ]; then
117     run_recover "initial post-wake (launchd resume = wake signal)"
118 fi
119
120 # LAN P2P Auto-Detection
121 # Detects whether the current network allows safe Tailscale LAN P2P by:
122 # 1. Checking WPA2-Enterprise (802.1x) via airport -I always forces P2P=false
123 #    because enterprise networks almost universally have AP Client Isolation.
124 # 2. AP Isolation probe sends a broadcast ping and checks if ANY LAN host responds.
125 #    If no LAN hosts respond on a non-empty network, AP Isolation is active.
126 # Writes 'true' or 'false' to .lan_p2p_detected in the repo root.
127 # routing-fix.sh reads this file dynamically every 30s loop iteration.
128 P2P_OVERRIDE_FILE="$SCRIPT_DIR/../../lan_p2p_detected"
129
130

```

```

131 detect_lan_p2p_mode() {
132     local reason="unknown" allow="false"
133
134     # Step 1: Detect Wi-Fi security type via system_profiler (airport binary removed in macOS 14.4+)
135     # SPAirPortDataType lists all known networks; the first "Security:" line is the current association.
136     local security
137     security=$(system_profiler SPAirPortDataType 2>/dev/null \
138         | awk '/Current Network Information:/{found=1} found && /Security:/{print $NF; exit}')
139
140     if [ -z "$security" ]; then
141         # Not on Wi-Fi or system_profiler unavailable fail safe
142         reason="no-wifi" allow="false"
143     elif echo "$security" | grep -qi "Enterprise"; then
144         # WPA2/WPA3 Enterprise = 802.1x always AP isolated
145         reason="802.1x-enterprise" allow="false"
146     elif echo "$security" | grep -qi "Personal\|None\|Open"; then
147         # WPA2/WPA3 Personal or open probe for AP isolation
148         local gw
149         gw=$(route -n get 1.1.1.1 2>/dev/null | awk '/gateway:/{print $2}')
150         if [ -z "$gw" ]; then
151             reason="no-default-route" allow="false"
152         else
153             local responses
154             responses=$(ping -c 3 -t 1 -q "$gw" 2>/dev/null | awk '/packets transmitted/{print $4}')
155             if [ "${responses:-0}" -gt 0 ]; then
156                 reason="wpa-personal-lan-reachable" allow="true"
157             else
158                 reason="wpa-personal-ap-isolated" allow="false"
159             fi
160         fi
161     else
162         # Unknown security type fail safe
163         reason="unknown-security-type" allow="false"
164     fi
165
166     echo "[$(date -u +%FT%TZ)] P2P detect: security='${security:-none}' allow=${allow} (reason: ${reason})"
167     echo "$allow" > "$P2P_OVERRIDE_FILE"
168     write_host_ips
169 }
170
171 # Run once on startup
172 detect_lan_p2p_mode
173
174 # Listener 1: WAKE events via unified log
175 # Predicate picks up:
176 # * "didWake" regular kernel wake events (lid open, RTC alarm)
177 # * "Wake from Sleep" powerd-driven normal wake
178 # * "DarkWake" clamshell-display wake that didn't fully wake the
179 # Mac (relevant when lid opens during display sleep)
180 # * "Waking from" generic catch-all for variations in newer macOS
181 # `[c]` makes the match case-insensitive (the unified log uses mixed-case
182 # phrases). --info reduces noise by dropping debug/trace log levels.
183 # Subsystem-based predicate catches every powermanagement emission (willSleep,
184 # didWake, didDim, didUndim, DarkWake, etc.) regardless of message phrasing
185 # across kernel versions. Phrasing-based predicates ('didWake', 'Wake from Sleep')
186 # are fragile across macOS releases. We accept the extra log noise because the
187 # run_recover() debounce + marker check filter out anything that isn't a real
188 # gateway-relevant event.
189 (
190     # Reconnect loop: macOS rotates the unified log buffer periodically; when
191     # that happens, this `log stream` subscriber's pipe EOFs, the inner
192     # `while read` consumer terminates, and without this wrapper the listener
193     # would be silently dead for the rest of the watcher's lifetime.
194     while ;; do
195         log stream --predicate \
196             'subsystem == "com.apple.powermanagement"' \
197             --style compact --info 2>/dev/null \
198             | while IFS= read -r line; do
199                 [ -z "$line" ] && continue
200                 run_recover "WAKE: $(echo "$line" | head -c 100)"
201             done
202         echo "[$(date -u +%FT%TZ)] log stream subshell exited; reconnecting in 5s"
203         sleep 5
204     done
205 ) &
206
207 # Listener 2: NETWORK state changes via scutil
208 # `scutil n.watch` on a state key prints SCEventUpdate lines whenever the key
209 # changes. We watch Global/IPv4 because that's the umbrella that catches link,
210 # address, route, and DNS changes for ALL interfaces in one namespace.
211 # The pipe is kept alive by an `sleep 86400` loop so scutil never sees EOF

```

```

212 # from us (which would silently terminate the watch).
213 (
214 # Reconnect loop: scutil can exit noisily on certain state changes (rare,
215 # but observed). Without the wrapper, the network listener's pipe EOFs and
216 # every subsequent roam goes unseen until launchd restarts us.
217 while ;; do
218 (
219 # Watch both IPv4 and IPv6 umbrella state IPv6-only or dual-stack
220 # networks never fire on the IPv4 key alone, and the user's first roam
221 # in a hotel / airport / on cellular hotspot is often IPv6-first under
222 # NAT64.
223 echo "n.add State:/Network/Global/IPv4"
224 echo "n.add State:/Network/Global/IPv6"
225 echo "n.watch"
226 while true; do sleep 86400; done
227 ) | script -q /dev/null scutil 2>/dev/null \
228 | while IFS= read -r line; do
229 case "$line" in
230 *changedKey*|*State:/Network/Global/IPv*|*n.state*|*SCEventUpdate*)
231 detect_lan_p2p_mode
232 run_recover "NET: $(echo "$line" | head -c 100)"
233 ;;
234 *) ;;
235 esac
236 done
237 echo "[$(date -u +%FT%TZ)] scutil pipe exited; reconnecting in 5s"
238 sleep 5
239 done
240 ) &
241 # Listener 3: TUNNEL_FAILED_CLOSED.marker Watcher
242 # Polls for the fail-closed marker dropped by the routing-fix container.
243 (
244 last_notified=0
245 while ;; do
246 if [ -f "$SCRIPT_DIR/../TUNNEL_FAILED_CLOSED.marker" ]; then
247 now=$(date +%s)
248 # Notify once every 5 minutes (300s) if it stays failed
249 if [ "$(now - last_notified)" -ge 300 ]; then
250 osascript -e 'display notification "VPN Tunnel Health Check Failed. Internet traffic is blackholed to
prevent IP leak." with title "NullExit Alert"' || true
251 last_notified=$now
252 fi
253 else
254 last_notified=0
255 fi
256 sleep 3
257 done
258 ) &
259 # Lifetime
260 # `wait` blocks until both background pipelines exit (which they normally
261 # never do). launchctl `bootout`/SIGTERM triggers the trap, which kills the
262 # whole process group so the sleep-forever in the scutil pipe also dies.
263 # We catch TERM/INT/HUP for explicit signals and EXIT for any normal-exit path
264 # (so listeners always get cleaned up even if `exit 0` runs somewhere).
265 # `rm -f $LOCK_FILE` first so a stale lock can never block the next launch.
266 # `pkill -TERM -P $$` then targets direct listener-children, then `kill 0`
267 # takes care of any backgrounded group-mates not reachable via the parent.
268 trap 'echo "[$(date -u +%FT%TZ)] watcher.sh terminating (signal/exit=$?)"; rm -f "$LOCK_FILE" 2>/dev/null ||
true; pkill -TERM -P $$ 2>/dev/null || true; kill 0 2>/dev/null || true; exit 0' TERM INT HUP EXIT
270 wait

```

15 scripts/crypto.sh

```

1 #!/usr/bin/env bash
2 # crypto.sh - Cryptographic Integrity Check
3
4 set -euo pipefail
5 cd "$(dirname "$0")/.."
6
7 if [ "$#" -ne 1 ] || { [ "$1" != "--sign" ] && [ "$1" != "--verify" ]; }; then
8 echo "Usage: $0 --sign | --verify"
9 exit 1
10 fi
11
12 if [ ! -f .env ]; then

```

```

13  echo "Error: .env not found."
14  exit 1
15  fi
16
17  SEED=$(grep "^NULLEXIT_SEED=" .env | cut -d '=' -f2)
18  if [ -z "$SEED" ]; then
19      echo "Error: NULLEXIT_SEED not found in .env."
20      exit 1
21  fi
22
23  if [ "$1" == "--sign" ]; then
24      echo "Generating cryptographic signatures using seed..."
25      rm -f .signatures
26      for file in toggle.sh recover.sh scripts/common.sh scripts/routing-fix.sh scripts/toggle-linux.sh scripts/
27          recover-linux.sh scripts/diagnose-host-leak.sh scripts/watcher.sh scripts/pf.conf post-rules.txt docker-
28          compose.yml; do
29          if [ -f "$file" ]; then
30              hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
31              echo "$file:$hash" >> .signatures
32              echo " [Signed] $file"
33          else
34              echo " [Skipped] $file not found"
35          fi
36      done
37      echo "Signatures saved to .signatures"
38      exit 0
39  fi
40
41  if [ "$1" == "--verify" ]; then
42      if [ ! -f .signatures ]; then
43          echo "Error: .signatures file missing. Run $0 --sign first."
44          exit 1
45      fi
46      FAIL=0
47      while IFS=: read -r file expected_hash; do
48          if [ -z "$file" ]; then continue; fi
49          if [ ! -f "$file" ]; then
50              echo "[FAIL] $file is missing!"
51              FAIL=1
52              continue
53          fi
54          actual_hash=$(openssl dgst -sha256 -hmac "$SEED" "$file" | awk '{print $NF}')
55          if [ "$actual_hash" != "$expected_hash" ]; then
56              echo "[FAIL] $file has been modified! Hash mismatch."
57              FAIL=1
58          fi
59      done < .signatures
60
61      if [ "$FAIL" -eq 1 ]; then
62          echo ""
63          echo "CRITICAL: Script integrity verification failed!"
64          echo "One or more core files have been modified without authorization."
65          echo "To authorize these changes, run: ./scripts/crypto.sh --sign"
66          echo ""
67          exit 1
68      fi
69      exit 0
70  fi

```

16 .aiignore

```

1  # Environment variables and secrets
2  .env
3  tor-bridges.txt

```

17 .gitignore

```

1  # Environment variables and secrets
2  .env
3
4  # wgcf generated files containing keys and account info
5  wgcf-account.toml

```



```

6 wgcf-profile.conf
7
8 # Tailscale persistent state containing node identity
9 tailscale-state/
10
11 # Binaries
12 wgcf
13
14 # macOS App bundles
15 *.app/
16 *.app
17
18 # AdGuard Home persistent state
19 adguard/
20 .gateway_ip
21 logs.txt
22
23 #
24 stability_incident_report.md
25 output.log
26 .DS_Store
27 *.desktop
28 blocked.log
29 wtf.md
30 host-leak-diagnostic-*.txt
31 nullexit_review.md
32 nullexit_unified.tex
33 nullexit_unified.txt
34
35 nullexit_unified.synctex.gz
36 tor-bridges.txt
37
38 # Runtime state files
39 .host_ips
40 .lan_p2p_detected
41 .signatures
42 TUNNEL_FAILED_CLOSED.marker
43
44 # Python cache
45 __pycache__/
46 *.py[cod]

```

18 .host_ips

```

1 172.17.52.223
2 100.109.94.19
3 192.168.64.1

```

19 Recover-Gateway.applescript

```

1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
4 on error
5     return
6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & "; bash recover.sh"
10 end tell

```

20 Toggle-Gateway.applescript

```

1 tell application "Finder" to set scriptDir to POSIX path of (container of (path to me) as alias)
2 try
3     do shell script "true" with administrator privileges
4 on error
5     return

```

```

6 end try
7 tell application "Terminal"
8     activate
9     do script "cd " & quoted form of scriptDir & " ; ./toggle.sh"
10 end tell

```

21 benchmarks/benchmark.sh

```

1  #!/usr/bin/env bash
2
3  # benchmark.sh
4  # Automates the full benchmarking suite (both Python download test and Lighthouse HTML test)
5  # Runs the suite with Gateway ON, toggles OFF, runs again, then toggles back ON.
6
7  set -e
8
9  # Change to project root so we can access toggle.sh reliably
10 cd "$(dirname "$0")/.."
11
12 # Ensure jq is installed
13 if ! command -v jq &> /dev/null; then
14     echo "Error: 'jq' is not installed. Please run 'brew install jq'"
15     exit 1
16 fi
17
18 run_lighthouse() {
19     local url=$1
20     local file_prefix=$2
21     echo " [Lighthouse] Testing $url ..."
22
23     # Run lighthouse headlessly, silencing standard output
24     npx -y lighthouse "$url" --chrome-flags="--headless" --only-categories=performance --output=json --output-
25     path="${file_prefix}.json" >/dev/null 2>&1
26
27     # Parse the results
28     local score=$(jq -r '.categories.performance.score' "${file_prefix}.json")
29     local tti=$(jq -r '.audits["interactive"].displayValue' "${file_prefix}.json")
30     local si=$(jq -r '.audits["speed-index"].displayValue' "${file_prefix}.json")
31
32     # Convert score to percentage
33     local score_pct=$(awk "BEGIN {print $score*100}")
34
35     echo "    -> Performance Score: ${score_pct}%"
36     echo "    -> Time to Interactive: $tti"
37     echo "    -> Speed Index: $si"
38
39     # Cleanup
40     rm -f "${file_prefix}.json"
41 }
42
43 echo "=====
44 echo " PHASE 1: Testing with Gateway ON"
45 echo "=====
46 # 1. Run raw Python download test
47 python3 benchmarks/test_load.py
48
49 # 2. Run Lighthouse HTML render test
50 echo ""
51 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
52 run_lighthouse "https://www.cnet.com/" "on_cnet"
53 run_lighthouse "https://www.independent.co.uk/" "on_ind"
54
55 echo ""
56 echo "=====
57 echo " PHASE 2: Toggling Gateway OFF"
58 echo "=====
59 ./toggle.sh
60 echo "Waiting 10 seconds for macOS Wi-Fi to stabilize after DNS reset..."
61 sleep 10
62
63
64 echo ""
65 echo "=====
66 echo " PHASE 3: Testing with Gateway OFF"
67 echo "=====

```

```

68 # 1. Run raw Python download test
69 python3 benchmarks/test_load.py
70
71 # 2. Run Lighthouse HTML render test
72 echo ""
73 echo "Starting Lighthouse HTML Tests (this takes ~30-60s per site)..."
74 run_lighthouse "https://www.cnet.com/" "off_cnet"
75 run_lighthouse "https://www.independent.co.uk/" "off_ind"
76
77
78 echo ""
79 echo "=====
80 echo "    PHASE 4: Restoring Gateway ON"
81 echo "=====
82 ./toggle.sh
83
84 echo ""
85 echo "=====
86 echo " BENCHMARKS COMPLETE!"
87 echo "You can scroll up to compare Phase 1 (AdGuard ON) vs Phase 3 (AdGuard OFF)."
88 echo "=====

```

22 benchmarks/test_load.py

```

1 import urllib.request
2 import urllib.error
3 import time
4 from html.parser import HTMLParser
5 import concurrent.futures
6
7 class ResourceParser(HTMLParser):
8     def __init__(self):
9         super().__init__()
10        self.urls = set()
11
12    def handle_starttag(self, tag, attrs):
13        attrs = dict(attrs)
14        url = None
15        if tag in ['img', 'script']:
16            url = attrs.get('src')
17        elif tag == 'link' and attrs.get('rel') == 'stylesheet':
18            url = attrs.get('href')
19
20        if url and url.startswith('http'):
21            self.urls.add(url)
22
23 def fetch_url(url):
24     try:
25         req = urllib.request.Request(url, headers={'User-Agent': 'Mozilla/5.0'})
26         urllib.request.urlopen(req, timeout=3)
27         return True
28     except Exception:
29         return False
30
31 def measure_url(target_url):
32     print(f"\nTesting {target_url} Load...")
33     start_total = time.time()
34
35     # Fetch main HTML
36     try:
37         req = urllib.request.Request(target_url, headers={'User-Agent': 'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36'})
38         response = urllib.request.urlopen(req, timeout=5)
39         html = response.read().decode('utf-8', errors='ignore')
40     except Exception as e:
41         print(f"Failed to fetch {target_url}: {e}")
42         return
43
44     parser = ResourceParser()
45     parser.feed(html)
46     urls = list(parser.urls)
47     print(f"Found {len(urls)} subresources to fetch (scripts, images, css)...")
48
49     # Concurrently fetch subresources
50     success_count = 0
51     fail_count = 0

```

```

52     max_threads = min(64, max(1, len(urls)))
53     with concurrent.futures.ThreadPoolExecutor(max_workers=max_threads) as executor:
54         results = list(executor.map(fetch_url, urls))
55
56     success_count = sum(1 for r in results if r)
57     fail_count = len(results) - success_count
58
59     total_time = time.time() - start_total
60     print(f"Time taken: {total_time:.2f} seconds")
61     print(f"Successfully fetched: {success_count}, Failed/Blocked: {fail_count}")
62
63 urls_to_test = [
64     "https://www.dailymail.co.uk/home/index.html",
65     "https://www.cnet.com/",
66     "https://www.independent.co.uk/"
67 ]
68
69 for u in urls_to_test:
70     measure_url(u)

```

23 black_list.txt

```

1 ad.doubleclick.net
2 adclick.g.doubleclick.net
3 adeventtracker.spotify.com
4 adfstat.yandex.ru
5 ads-api.tiktok.com
6 ads-api.twitter.com
7 ads-fa.spotify.com
8 ads-sg.tiktok.com
9 ads.tiktok.com
10 adserver.unityads.unity3d.com
11 adsfs.oppomobile.com
12 an.facebook.com$important
13 connect.facebook.net$important
14 pixel.facebook.com$important
15 analytics.query.yahoo.com
16 analytics.s3.amazonaws.com
17 analytics.spotify.com
18 analyticsengine.s3.amazonaws.com
19 appmetrica.yandex.ru
20 audio2.spotify.com
21 bdapi-ads.realmemobile.com
22 bdapi-in-ads.realmemobile.com
23 bounceexchange.com
24 bs.serving-sys.com
25 business-api.tiktok.com
26 ck.ads.oppomobile.com
27 crashdump.spotify.com
28 data.ads.oppomobile.com
29 data.mistat.india.xiaomi.com
30 data.mistat.rus.xiaomi.com
31 data.mistat.xiaomi.com
32 desktop.spotify.com
33 doubleclick.net
34 ds.metric.spotify.com
35 gemini.yahoo.com
36 googleads.g.doubleclick.net
37 googleadservices.com
38 googlesyndication.com
39 grs.hicloud.com
40 gtm.spotify.com
41 iot-eu-logser.realme.com
42 iot-logser.realme.com
43 log.byteoversea.com
44 log.fc.yahoo.com
45 log.spotify.com
46 m.doubleclick.net
47 mediavisor.doubleclick.net
48 metrics.spotify.com
49 metrika.yandex.ru
50 omaze.com
51 pagead2.googlesyndication.com
52 securepubads.g.doubleclick.net
53 static.doubleclick.net
54 stats.g.doubleclick.net

```

```

55 tracking.rus.miui.com
56 udcml.yahoo.com
57 v.jwpcdn.com
58 weblb-wg.gslb.spotify.com
59 webview.unityads.unity3d.com
60 diagassets.apple.com
61 iphonesubmissions.apple.com
62 radarsubmissions.apple.com
63 metrics.apple.com
64 xp.apple.com
65 heads-fa.spotify.com
66 heads4.spotify.com
67 pixel.spotify.com
68 segs.spotify.com
69 audio-fa.spotify.com
70 events.data.microsoft.com
71 datarouter-prod.ak.epicgames.com
72 nel.cloudflare.com
73 eu-mobile.events.data.microsoft.com
74 stocks-analytics-events.apple.com

```

24 cloud-init.yaml

```

1 #cloud-config
2 # Generic Cloud-Init Bootstrap Script for Remote Exit Nodes (Ubuntu/Debian)
3 # Paste this into the "User Data" or "Initialization Script" section when creating a VPS.
4
5 package_update: true
6 package_upgrade: true
7
8 packages:
9   - apt-transport-https
10   - ca-certificates
11   - curl
12   - gnupg
13   - lsb-release
14   - git
15   - python3
16   - python3-pip
17
18 runcmd:
19   # 1. Install Docker natively
20   - curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /usr/share/keyrings/docker-
    archive-keyring.gpg
21   - echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/docker-archive-keyring.gpg]
    https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /etc/apt/sources.list.d/
    docker.list > /dev/null
22   - sudo apt-get update
23   - sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-compose-plugin
24
25   # 2. Add default user (ubuntu) to docker group
26   - usermod -aG docker ubuntu || true
27   - usermod -aG docker debian || true
28
29   # 3. Create the installation directory
30   - mkdir -p /opt/nullexit
31   - chown -R ubuntu:ubuntu /opt/nullexit || chown -R debian:debian /opt/nullexit
32
33   # 4. Enable IPv4 and IPv6 forwarding for VPN routing
34   - echo "net.ipv4.ip_forward=1" >> /etc/sysctl.d/99-tailscale.conf
35   - echo "net.ipv6.conf.all.forwarding=1" >> /etc/sysctl.d/99-tailscale.conf
36   - sysctl -p /etc/sysctl.d/99-tailscale.conf
37
38 final_message: "The remote exit node has been fully bootstrapped. SSH in, clone the repo to /opt/nullexit, add
    your .env, and run ./setup-linux.sh!"

```

25 docker/tor/Dockerfile

```

1 FROM debian:bookworm-slim
2
3 RUN apt-get update && \
4     apt-get install -y tor obfs4proxy gosu bash && \

```

```

5   rm -rf /var/lib/apt/lists/*
6
7   COPY entrypoint.sh /entrypoint.sh
8   RUN chmod +x /entrypoint.sh
9
10  ENTRYPOINT ["/entrypoint.sh"]

```

26 docker/tor/entrypoint.sh

```

1  #!/bin/bash
2  set -e
3
4  TORRC="/etc/tor/torrc"
5  mkdir -p /etc/tor /var/lib/tor
6
7  # Base configuration (SOCKS proxy on 9050, Control port on 9051, strictly localhost/container boundaries)
8  cat <<EOF > $TORRC
9  SocksPort 0.0.0.0:9050
10 ControlPort 0.0.0.0:9051
11 CookieAuthentication 0
12 Log notice stdout
13 DataDirectory /var/lib/tor
14
15 # Transparent proxying & DNS resolution for .onion mapping
16 TransPort 0.0.0.0:9040
17 DNSPort 0.0.0.0:5353
18 AutomapHostsOnResolve 1
19 VirtualAddrNetworkIPv4 198.18.0.0/15
20 EOF
21
22 if [ -n "$TOR_PASSWORD" ]; then
23     HASH=$(tor --hash-password "$TOR_PASSWORD" | tail -n 1)
24     echo "HashedControlPassword $HASH" >> $TORRC
25 fi
26
27 if [ -n "$TOR_EXCLUDE_EXIT_NODES" ]; then
28     echo "ExcludeExitNodes $TOR_EXCLUDE_EXIT_NODES" >> $TORRC
29     echo "StrictNodes 1" >> $TORRC
30 fi
31
32 if [ "${TOR_USE_BRIDGES:-false}" = "true" ]; then
33     BRIDGE_FILE="${TOR_BRIDGE_LINES_FILE:-/tor-bridges.txt}"
34
35     if ! grep -vE "^[[:space:]]*#" "$BRIDGE_FILE" | grep -q '^[[:space:]]'; then
36         MSG="[$(date '+%Y-%m-%d %H:%M:%S')] [CRITICAL] [Tor Proxy] TOR_USE_BRIDGES is enabled, but no bridge
37         lines found at $BRIDGE_FILE. Proxy startup ABORTED. Get bridges from https://bridges.torproject.org."
38         echo "$MSG"
39         if [ -w "/output.log" ]; then
40             echo "$MSG" >> /output.log
41         fi
42         # Sleep indefinitely to prevent docker-compose 'unless-stopped' from triggering a crash loop
43         exec sleep infinity
44     fi
45
46     echo "UseBridges 1" >> $TORRC
47     echo "ClientTransportPlugin obfs4 exec /usr/bin/obfs4proxy" >> $TORRC
48
49     while IFS= read -r line; do
50         # Ignore comments and empty lines
51         [[ -z "$line" || "$line" == \#* ]] && continue
52         echo "Bridge $line" >> $TORRC
53     done < "$BRIDGE_FILE"
54 fi
55
56 # Debian tor package uses 'debian-tor' user
57 chown -R debian-tor:debian-tor /var/lib/tor /etc/tor
58
59 exec gosu debian-tor /usr/bin/tor -f $TORRC

```

27 launchd/com.nullexit.wake-recovery.plist

```

1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
3 <plist version="1.0">
4 <dict>
5   <key>Label</key>
6   <string>com.nullexit.wake-recovery</string>
7
8   <!-- WorkingDirectory + relative program arg keeps this plist install-agnostic.
9        The source-of-truth for the install path is __NULLEXIT_HOME__, which
10       setup.sh or the user must substitute with their absolute nullexit
11       install root at install time. Program arg is `scripts/watcher.sh`
12       RELATIVE to WorkingDirectory, so launchd's evaluated cwd IS the
13       install root and BASH_SOURCE inside scripts/watcher.sh still
14       resolves to the real <install>/scripts/watcher.sh, preserving the
15       SCRIPT_DIR pattern. After install-time `sed -i ' ' 's|__NULLEXIT_HOME__|<abs_path>|'`,
16       the plist is portable to any clone location. -->
17   <key>WorkingDirectory</key>
18   <string>__NULLEXIT_HOME__</string>
19
20   <key>ProgramArguments</key>
21   <array>
22     <string>/bin/bash</string>
23     <string>scripts/watcher.sh</string>
24   </array>
25
26   <!-- Start the daemon immediately on launchctl load (instead of waiting
27        for the next user login). Combined with keepAlive below this means
28        once installed: load once, runs forever across logouts/reboots. -->
29   <key>RunAtLoad</key>
30   <true/>
31
32   <!-- Restart policy:
33        SuccessfulExit=false    SIGTERM / clean exit does NOT relaunch (so
34                                `launchctl bootout` cleanly stops, not loops).
35        Crashed=false          HARD crashes ALSO do NOT relaunch. We've seen
36                                that repeated sleep/wake cycles on macOS
37                                Sonoma+ accumulate orphan watcher.sh PIDs
38                                because SIGCONT/resume doesn't reliably kill
39                                the prior instance. The watcher.sh script
40                                now enforces single-instance via a manual
41                                PID-file and process-identity check at the
42                                top, so a re-launch on crash is
43                                actively dangerous (it would skip the lock
44                                and exit; better to surface the crash in
45                                logs than to silently 0-process). -->
46   <key>KeepAlive</key>
47   <dict>
48     <key>SuccessfulExit</key>
49     <false/>
50     <key>Crashed</key>
51     <false/>
52   </dict>
53
54   <!-- Background hint to App Nap so it doesn't get suspended when
55        the user is inactive. We're the entire reason this LaunchAgent
56        needs to NOT nap. -->
57   <key>ProcessType</key>
58   <string>Background</string>
59
60   <!-- launchd-level stdout/stderr (the bash launcher shell). The script
61        itself writes to /tmp/nullexit-watcher.log via exec >>. These
62        separate channels help us tell "launchd crashed" from "the bash
63        wrapper piped to recover.sh errored". -->
64   <key>StandardOutPath</key>
65   <string>/tmp/nullexit-watcher.out.log</string>
66   <key>StandardErrorPath</key>
67   <string>/tmp/nullexit-watcher.err.log</string>
68 </dict>
69 </plist>

```

28 post-rules.txt

```

1 iptables -A FORWARD -i tailscale0 -o tun0 -j ACCEPT
2 iptables -A FORWARD -i tun0 -o tailscale0 -j ACCEPT
3 iptables -A INPUT -i tailscale0 -j ACCEPT
4 iptables -A INPUT -i eth0 -p udp --dport 41642 -j ACCEPT

```



```

5 iptables -t nat -A POSTROUTING -o tun0 -j MASQUERADE
6 # Clamp MSS to 1120. With Tailscale overhead on top of WARP (each WireGuard encapsulation adds ~60 bytes),
7 # the safe MSS for double-tunneled traffic is 1120 to prevent strict-MSS endpoints from stalling.
8 iptables -t mangle -A POSTROUTING -p tcp --tcp-flags SYN,RST SYN -j TCPMSS --set-mss 1120
9
10 iptables -t nat -A PREROUTING -i tailscale0 -p udp --dport 53 -j REDIRECT --to-port 5335
11 iptables -t nat -A PREROUTING -i tailscale0 -p tcp --dport 53 -j REDIRECT --to-port 5335
12
13 # Also redirect DNS from Docker bridge interface so the host can reach AdGuard via localhost:53
14 iptables -t nat -A PREROUTING -i eth0 -p udp --dport 53 -j REDIRECT --to-port 5335
15 iptables -t nat -A PREROUTING -i eth0 -p tcp --dport 53 -j REDIRECT --to-port 5335
16
17 # Block IPv6 exit-node traffic to prevent leakage (Gluetun/WARP is IPv4-only)
18 ip6tables -A FORWARD -i tailscale0 -j DROP
19
20 # Block DNS-over-TLS (Port 853) to force Android Private DNS to fallback to Port 53
21 iptables -A FORWARD -i tailscale0 -p tcp --dport 853 -j REJECT --reject-with tcp-reset
22 iptables -A FORWARD -i tailscale0 -p udp --dport 853 -j REJECT
23
24 # Block QUIC (UDP 443) to force Facebook/Google to fallback to TCP.
25 # UDP bypasses TCP MSS clamping, causing fragmentation. Over weak Wi-Fi (far from gateway),
26 # dropping a single fragment destroys the entire packet, stalling image downloads.
27 iptables -A FORWARD -i tailscale0 -p udp --dport 443 -j REJECT

```

29 scripts/Dockerfile.routing-fix

```

1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY logger/ ./logger/
4 RUN cd logger && go build -o logger main.go
5
6 FROM alpine:3.20
7 RUN apk add --no-cache iptables iproute2 curl ipset
8 COPY --from=builder /app/logger/logger /usr/local/bin/nullexit-logger
9 COPY routing-fix.sh /routing-fix.sh
10 CMD ["sh", "/routing-fix.sh"]

```

30 scripts/Dockerfile.rule-compiler

```

1 FROM golang:1.22-alpine AS builder
2 WORKDIR /app
3 COPY rule-compiler/ ./rule-compiler/
4 RUN cd rule-compiler && go build -o sync-rules main.go
5
6 FROM alpine:3.20
7 COPY --from=builder /app/rule-compiler/sync-rules /usr/local/bin/sync-rules
8 WORKDIR /app
9 CMD ["sync-rules"]

```

31 scripts/Toggle-Gateway.bat

```

1 <# :
2 @echo off
3 powershell -NoProfile -ExecutionPolicy Bypass -Command "Invoke-Expression ([System.IO.File]::ReadAllText('%-f0
4   ') -replace '^(?s).*?#\s*>\s*\r?\n','')
5 exit /b
6 #>
7 # Toggle-Gateway.bat (Hybrid Batch-PowerShell Toggler) nullexit Gateway Toggler for Windows
8 # Mirrors the sophisticated error handling, checks, and automation of toggle.sh
9 # 1. Administrator Check & Elevation
10 $isAdmin = ([Security.Principal.WindowsPrincipal][Security.Principal.WindowsIdentity]::GetCurrent()).IsInRole([
11   Security.Principal.WindowsBuiltInRole]::Administrator)
12 if (-not $isAdmin) {
13   Write-Host "Requesting Administrator privileges..." -ForegroundColor Yellow
14   Start-Process PowerShell -ArgumentList "-NoProfile -ExecutionPolicy Bypass -Command `\"Invoke-Expression ([
15     System.IO.File]::ReadAllText('$PSCommandPath') -replace '^(?s).*?#\s*>\s*\r?\n','')`\" -Verb RunAs
16   Exit

```

```

15 }
16
17 # Set console output encoding to UTF8 for clean symbols
18 [Console]::OutputEncoding = [System.Text.Encoding]::UTF8
19 Clear-Host
20
21 Write-Host "" -ForegroundColor Green
22 Write-Host "          nullexit Windows Toggler          " -ForegroundColor Green
23 Write-Host "" -ForegroundColor Green
24 Write-Host ""
25
26 # 2. Helpers
27 function Reset-HostDNS {
28     Write-Host "Restoring default DNS (DHCP/DHCP-assigned DNS)..." -ForegroundColor Cyan
29     $ActiveAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
30     foreach ($adapter in $ActiveAdapters) {
31         Write-Host "    -> Resetting DNS on adapter: $($adapter.Name)" -ForegroundColor Gray
32         Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ResetServerAddresses -ErrorAction
33         SilentlyContinue
34     }
35 }
36
37 function Hijack-HostDNS ($ip) {
38     Write-Host "Hijacking DNS to route through AdGuard ($ip)..." -ForegroundColor Yellow
39     $ActiveAdapters = Get-NetAdapter | Where-Object { $_.Status -eq 'Up' }
40     foreach ($adapter in $ActiveAdapters) {
41         Write-Host "    -> Pointing DNS to $ip on adapter: $($adapter.Name)" -ForegroundColor Gray
42         Set-DnsClientServerAddress -InterfaceIndex $adapter.InterfaceIndex -ServerAddresses $ip -ErrorAction
43         SilentlyContinue
44     }
45 }
46
47 function Get-HostTailscalePath {
48     $paths = @(
49         "$env:ProgramFiles\Tailscale\tailscale.exe",
50         "$env:ProgramFiles (x86)\Tailscale\tailscale.exe",
51         "tailscale.exe"
52     )
53     foreach ($path in $paths) {
54         if (Test-Path $path) { return $path }
55     }
56     return $null
57 }
58
59 # 3. Detect State
60 # Prevent deadlocks by resetting DNS to default temporarily during checks
61 Reset-HostDNS
62
63 Write-Host "Checking Docker daemon status..." -ForegroundColor Cyan
64 & docker info >$null 2>&1
65 if ($LASTEXITCODE -ne 0) {
66     Write-Host "Docker is not running. Starting Docker Desktop..." -ForegroundColor Yellow
67     $dockerPath = "C:\Program Files\ Docker\ Docker\ Docker Desktop.exe"
68     if (Test-Path $dockerPath) {
69         Start-Process $dockerPath
70         Write-Host "Waiting for Docker daemon to become responsive..." -ForegroundColor Gray
71         $timeout = 60
72         while ($timeout -gt 0) {
73             Start-Sleep -Seconds 2
74             & docker info >$null 2>&1
75             if ($LASTEXITCODE -eq 0) {
76                 Write-Host "Docker is ready!" -ForegroundColor Green
77                 break
78             }
79             $timeout -= 2
80         }
81         if ($timeout -le 0) {
82             Write-Error "Docker failed to start in time. Aborting."
83             Exit 1
84         }
85     } else {
86         Write-Error "Docker Desktop executable not found at standard path. Please start Docker manually."
87         Exit 1
88     }
89 }
90
91 # Detect if gateway is already running
92 $runningWarp = docker compose ps --status running --format json | ConvertFrom-Json -ErrorAction
93     SilentlyContinue | Where-Object { $_.Service -eq "warp" }
94 $isGatewayActive = ($null -ne $runningWarp)

```

```

93 # 4. Execution
94 if ($isGatewayActive) {
95     # STOP PATH
96     Write-Host "Gateway is currently RUNNING. Disabling now..." -ForegroundColor Yellow
97     Write-Host ""
98
99     # Disconnect host Tailscale from exit node if installed
100     $tsCli = Get-HostTailscalePath
101     if ($null -ne $tsCli) {
102         Write-Host "Disconnecting host Tailscale from exit node..." -ForegroundColor Cyan
103         & $tsCli up --exit-node= --accept-dns=true >$null 2>&1
104     }
105
106     Write-Host "Stopping Docker containers..." -ForegroundColor Cyan
107     docker compose down -t 5
108
109     Reset-HostDNS
110     Write-Host "Gateway has been successfully STOPPED." -ForegroundColor Green
111 } else {
112     # START PATH
113     Write-Host "Gateway is currently STOPPED. Enabling now..." -ForegroundColor Yellow
114     Write-Host ""
115
116     # Clean up corrupted AdGuardHome configurations
117     $confFile = "adguard\conf\AdGuardHome.yaml"
118     if (Test-Path $confFile) {
119         $fileSize = (Get-Item $confFile).Length
120         if ($fileSize -eq 0) {
121             Remove-Item $confFile -Force
122             Write-Host "Removed corrupted empty AdGuardHome.yaml to prevent container crash loop." -
123                 ForegroundColor Yellow
124         }
125     }
126
127     Write-Host "Starting Docker containers..." -ForegroundColor Cyan
128     docker compose up -d
129
130     Write-Host "Waiting for container's Tailscale connection to offer Exit Node..." -ForegroundColor Cyan
131     Write-Host " (This can take up to 60 seconds..." -ForegroundColor Gray
132
133     $elapsed = 0
134     $maxWait = 60
135     $resolvedIP = $null
136
137     while ($elapsed -lt $maxWait) {
138         $statusJson = docker compose exec -T tailscale tailscale status --json 2>$null
139         if ($null -ne $statusJson) {
140             $status = $statusJson | ConvertFrom-Json -ErrorAction SilentlyContinue
141             if ($null -ne $status -and $null -ne $status.Self) {
142                 # Check if it has a valid Tailscale IP and is running
143                 if ($status.Self.Online -and $status.Self.TailscaleIPs.Count -gt 0) {
144                     $resolvedIP = $status.Self.TailscaleIPs[0]
145                     break
146                 }
147             }
148             Start-Sleep -Seconds 2
149             $elapsed += 2
150         }
151     }
152
153     if ($null -ne $resolvedIP) {
154         Write-Host "Tailscale IP resolved: $resolvedIP" -ForegroundColor Green
155         Write-Host ""
156
157         # Configure host Tailscale client to route through this exit node
158         $tsCli = Get-HostTailscalePath
159         if ($null -ne $tsCli) {
160             Write-Host "Configuring host Tailscale to use Exit Node ($resolvedIP)..." -ForegroundColor Cyan
161             & $tsCli up --exit-node=$resolvedIP --accept-dns=false >$null 2>&1
162         }
163
164         # Hijack DNS
165         Hijack-HostDNS $resolvedIP
166         Write-Host ""
167         Write-Host "Gateway has been successfully STARTED." -ForegroundColor Green
168     } else {
169         Write-Host "Warning: Could not resolve gateway Tailscale IP. DNS hijacking skipped." -ForegroundColor
170         Red
171
172         # Fallback to .gateway_ip if it exists
173         if (Test-Path ".gateway_ip") {
174             $fallbackIP = Get-Content ".gateway_ip" -TotalCount 1
175         }
176     }
177 }

```

```

172         if (-not [string]::IsNullOrEmpty($fallbackIP)) {
173             Write-Host "[Fallback] Using static IP from .gateway_ip: $fallbackIP" -ForegroundColor Yellow
174             Hijack-HostDNS $fallbackIP
175             Write-Host "Gateway STARTED (with static fallback IP)." -ForegroundColor Green
176         }
177     } else {
178         Write-Host "Gateway started in offline/unauthenticated state. Run setup.sh if this is a fresh setup"
179         ." -ForegroundColor Yellow
180     }
181 }
182
183 Write-Host ""
184 Write-Host "Press any key to exit..."
185 [void]$Host.UI.RawUI.ReadKey("NoEcho,IncludeKeyDown")

```

32 scripts/check_circuits.py

```

1  #!/usr/bin/env python3
2  import socket
3  import os
4  import re
5
6  def query_tor(port, command, password=""):
7      try:
8          s = socket.socket()
9          s.settimeout(2)
10         s.connect(('127.0.0.1', port))
11         s.sendall(f"AUTHENTICATE \'{password}\' \'{command}\' \nQUIT\n".encode())
12         response = b""
13         while True:
14             chunk = s.recv(4096)
15             if not chunk:
16                 break
17             response += chunk
18         return response.decode()
19     except Exception as e:
20         return f"Error connecting to Tor Control Port: {e}"
21
22 def get_ports():
23     ports = {}
24     ports_file = "/tmp/nullexit-ports.env"
25     if os.path.exists(ports_file):
26         with open(ports_file, 'r') as f:
27             for line in f:
28                 if line.startswith("export "):
29                     # Split at first '=' to handle potential multiple '=' in value
30                     parts = line.replace("export ", "").strip().split("=", 1)
31                     if len(parts) == 2:
32                         key, val = parts
33                         try:
34                             ports[key] = int(val)
35                         except ValueError:
36                             ports[key] = val
37     return ports
38
39 def get_password(ports):
40     # 1. Try ports.env
41     password = ports.get("TOR_PASSWORD")
42     if password:
43         return password
44
45     # 2. Try .env
46     if os.path.exists(".env"):
47         try:
48             with open(".env", "r") as f:
49                 for line in f:
50                     if line.startswith("ADGUARD_PASSWORD="):
51                         return line.split("=", 1)[1].strip().strip('\'')
52                     if line.startswith("TOR_PASSWORD="):
53                         return line.split("=", 1)[1].strip().strip('\'')
54         except Exception:
55             pass
56
57     # 3. Fallback
58     return "nullexit"

```

```

59
60 def main():
61     ports = get_ports()
62     control_port = ports.get("TOR_CONTROL_PORT", 9051)
63     password = get_password(ports)
64
65     print(f"Connecting to Tor Control Port on 127.0.0.1:{control_port}...")
66     status = query_tor(control_port, "GETINFO circuit-status", password)
67
68     if "Error" in status:
69         print(status)
70         return
71
72     print("\nActive Tor Circuits:")
73     circuits = re.findall(r"(\d+)\s+BUILT\s+(.+) \s+PURPOSE", status)
74     if not circuits:
75         print("No active built circuits found yet. Tor might still be bootstrapping.")
76         return
77
78     for circ_id, path_str in circuits:
79         print(f"\nCircuit #{circ_id}:")
80         nodes = path_str.split(",")
81         for idx, node in enumerate(nodes):
82             fingerprint = node.split("~")[0].replace("$", "")
83             nickname = node.split("~")[1] if "~" in node else "unknown"
84
85             # Fetch node IP
86             ns_info = query_tor(control_port, f"GETINFO ns/id/{fingerprint}", password)
87             ip = "unknown"
88             for line in ns_info.split("\n"):
89                 if line.startswith("r "):
90                     parts = line.split(" ")
91                     if len(parts) >= 7:
92                         ip = parts[6]
93                         break
94
95             # Fetch country
96             country = "unknown"
97             if ip != "unknown":
98                 country_info = query_tor(control_port, f"GETINFO ip-to-country/{ip}", password)
99                 for line in country_info.split("\n"):
100                     if line.startswith("250-ip-to-country/"):
101                         country = line.split("=")[1].strip().upper()
102                         break
103
104             role = "Guard" if idx == 0 else ("Exit" if idx == len(nodes) - 1 else "Middle")
105             print(f" [{role}] {nickname} ({ip}) - Country: {country}")
106
107 if __name__ == "__main__":
108     main()

```

33 scripts/diagnose-host-leak.sh

```

1 #!/bin/bash
2 # scripts/diagnose-host-leak.sh nullexit host-routing diagnostic
3 #
4 # Symptom this addresses: tests like https://www.whatismyip.com on the
5 # *HOST MAC* return the underlying physical ISP/ASN
6 # local ISP / campus network") even though the nullexit gateway is active and remote
7 # tailnet clients correctly egress via Cloudflare WARP. The container and
8 # the gateway itself are healthy only the host's own routing is leaking.
9 #
10 # This script runs 7 checks in sequence, classifies the result into ONE
11 # of the three known leak scenarios, writes the full report to a
12 # timestamped file in the project root, and prints a verdict + a
13 # ready-to-run fix command on stdout. Pass --fix to apply the matching
14 # remediation in the same invocation and re-verify egress afterwards.
15 # See devref.md 10.30 for the deep-dive rationale.
16 #
17 # Usage:
18 #   bash scripts/diagnose-host-leak.sh           # diagnose + write report
19 #   bash scripts/diagnose-host-leak.sh --fix      # diagnose + fix + re-verify
20 #   bash scripts/diagnose-host-leak.sh --watch    # full baseline then loop checks every 60s
21 #   bash scripts/diagnose-host-leak.sh --watch 30 # same, with custom interval (seconds)
22 #   bash scripts/diagnose-host-leak.sh --help    # this message
23 #

```

```

24 # Multiple runs produce timestamped files (one per run) so historical
25 # reports can be diffed. The newest file is the most recent.
26
27 set -uo pipefail
28 # NOTE: errexit (`set -e`) is intentionally NOT enabled. Several checks
29 # below are EXPECTED to fail and that's diagnostic data (e.g. `tailscale
30 # status` returns non-zero when the daemon is wedged). Every command
31 # pipes to a helper that records the exit code without killing the script.
32
33 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
34 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
35 TIMESTAMP="$(date -u +%Y%m%dT%H%M%SZ)"
36 source "$SCRIPT_DIR/common.sh"
37 LOG_FILE="$PROJECT_ROOT/output.log"
38 REPORT_FILE="$PROJECT_ROOT/host-leak-diagnostic-$TIMESTAMP.txt"
39
40 # Argument parsing
41 DO_FIX=false
42 DO_WATCH=false
43 WATCH_INTERVAL=60
44 WATCH_LOG="$PROJECT_ROOT/host-leak-watch.log"
45 case "${1:-}" in
46     --fix) DO_FIX=true ;;
47     --watch)
48         DO_WATCH=true
49         # Optional second argument: custom interval in seconds
50         if [ -n "${2:-}" ] && [[ "$2" =~ ^[0-9]+$ ]]; then
51             WATCH_INTERVAL="$2"
52         fi
53         ;;
54     --help|-h)
55         sed -n '2,31p' "$0"
56         exit 0
57         ;;
58     *)
59         : # default: diagnose only
60         ;;
61     *)
62         echo "Unknown argument: $1" >&2
63         echo "Run with --help for usage." >&2
64         exit 2
65         ;;
66 esac
67
68 # Colours (auto-disabled when stdout is not a tty, e.g. piped)
69 if [ -t 1 ]; then
70     RED='\033[0;31m'; GREEN='\033[0;32m'; YELLOW='\033[1;33m'
71     BOLD='\033[1m'; NC='\033[0m'
72 else
73     RED=''; GREEN=''; YELLOW=''; BOLD=''; NC=''
74 fi
75
76 # Output helpers write to BOTH stdout and the report file
77 exec 3>> "$REPORT_FILE" || {
78     printf '\n[FATAL] cannot open %s for write\n' "$REPORT_FILE" >&2
79     exit 1
80 }
81 section() {
82     local title="$1"
83     printf '\n\b %s \b\n' "$BOLD" "$title" "$NC" >&3
84     printf '\n\b %s \b\n' "$BOLD" "$title" "$NC"
85 }
86 line() { printf ' %b\n' "$1" >&3; printf ' %b\n' "$1"; }
87 ok() { line "${GREEN}${NC} $1"; }
88 warn() { line "${YELLOW}${NC} $1"; }
89 fail() { line "${RED}${NC} $1"; }
90 note() { line "(no ${1})"; }
91
92 # Watch-mode header (only shown when entering watch loop)
93 if [ "$DO_WATCH" = "true" ]; then
94     echo ""
95     echo ""
96     echo " n u l l e x i t   w a t c h   m o d e "
97     echo ""
98     echo " Interval:          ${WATCH_INTERVAL}s"
99     echo " Watch log:         $WATCH_LOG"
100     echo ""
101 else
102     echo ""
103     echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c "
104     echo ""

```

```

105 echo " Timestamp (UTC): $TIMESTAMP"
106 echo " Project root:      $PROJECT_ROOT"
107 echo " Report file:       $REPORT_FILE"
108 echo " Mode:              $([ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only (pass --
    fix to remediate)")"
109 echo ""
110 echo "" >&3
111 echo "" >&3
112 echo " n u l l e x i t   h o s t - r o u t i n g   d i a g n o s t i c" >&3
113 echo "" >&3
114 echo " Timestamp (UTC): $TIMESTAMP" >&3
115 echo " Mode:              $([ "$DO_FIX" = true ] && echo "diagnose + apply fix" || echo "diagnose only")" >&3
116 fi
117
118 # Always add Homebrew to PATH (same as toggle.sh / recover.sh)
119 export PATH="/opt/homebrew/bin:/usr/local/bin:$PATH"
120
121 # Pure-bash timeout (macOS lacks GNU `timeout`)
122 run_with_timeout() {
123     local timeout_sec="$1"
124     shift
125     if [ "$#" -eq 0 ]; then return 1; fi
126     "$@" &
127     local cmd_pid=$!
128     (
129         sleep "$timeout_sec"
130         if kill -0 "$cmd_pid" 2>> "$LOG_FILE"; then
131             kill -9 "$cmd_pid" 2>> "$LOG_FILE" || true
132         fi
133     ) >> "$LOG_FILE" 2>&1 &
134     local watcher_pid=$!
135     set +e
136     wait "$cmd_pid"
137     local exit_status=$?
138     set -uo pipefail
139     kill "$watcher_pid" 2>> "$LOG_FILE" || true
140     return "$exit_status"
141 }
142
143 # Resolve the active network service (same heuristic as toggle.sh)
144 resolve_active_service() {
145     get_active_service
146 }
147
148 #
149 # Quick-check functions (used by --watch loop; return simple status strings)
150 # Each echoes its result so the caller can capture and compare.
151 #
152
153 quick_warp() {
154     # Returns: "on", "off", "unknown"
155     local out
156     out=$(run_with_timeout 8 curl -s --max-time 6 \
157         https://www.cloudflare.com/cdn-cgi/trace 2>/dev/null || echo "")
158     local warp
159     warp=$(printf '%s' "$out" | awk -F=' ' '/^warp=/ {print $2; exit}')
160     printf '%s' "${warp:-unknown}"
161 }
162
163 quick_ipv6() {
164     # Returns: the IPv6 address if leaking, or empty string if clean
165     local out
166     out=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>/dev/null || echo "")
167     if [ -n "$out" ] && printf '%s' "$out" | grep -qE '[0-9a-fA-F:]+$'; then
168         printf '%s' "$out"
169     fi
170 }
171
172 quick_default_route() {
173     # Returns: "utun" if default route is via Tailscale utun*,
174     #           "wifi" if via physical Wi-Fi (192.168.x.x),
175     #           "other" otherwise
176     local route
177     route=$(netstat -rn 2>/dev/null | awk '/^default/ {print $0; exit}')
178     if [ -z "$route" ]; then
179         printf '%s' "none"
180     elif printf '%s' "$route" | grep -qE 'utun[0-9]+'; then
181         printf '%s' "utun"
182     elif printf '%s' "$route" | grep -qE '^(default|0\.\0\.\0\.\0).*\b192\.\168\.'; then
183         printf '%s' "wifi"
184     else

```



```

185     printf '%s' "other"
186 fi
187 }
188
189 # Watch loop (runs after baseline diagnostic when --watch is passed)
190 run_watch_loop() {
191     local baseline_warp="$1"
192     local baseline_default="$2"
193     local baseline_ipv6_leak="$3"
194     local interval="$4"
195
196     echo ""
197     echo ""
198     echo " Baseline established. Monitoring every ${interval}s..."
199     echo " warp:      $baseline_warp"
200     echo " default:    $baseline_default"
201     echo ""
202
203     # Safety: warn if interval is very short (avoids hammering APIs)
204     if [ "$interval" -lt 30 ]; then
205         printf '%b' Short interval(%ss) rate-limiting risk on external APIs%b\n' "$YELLOW" "$interval" "$NC"
206         printf ' Consider 30s or longer for production use.\n'
207     fi
208     printf '[%s] WATCH START warp=%s default=%s interval=%ss\n' \
209         "$(date -u +%Y-%m-%dT%H:%M:%SZ)" "$baseline_warp" "$baseline_default" "$interval" \
210         >> "$WATCH_LOG"
211
212     local iteration=0
213     local last_warp="$baseline_warp"
214     local last_default="$baseline_default"
215     local last_ipv6_ok=$(( [ "$baseline_ipv6_leak" = "false" ] && echo true || echo false ))
216
217     trap 'printf "\n%bWatch stopped by user. Log at: %s%b\n" "$YELLOW" "$WATCH_LOG" "$NC"; exit 0' INT TERM
218
219     while true; do
220         iteration=$((iteration + 1))
221
222         local warp_now ipv6_now default_now
223         warp_now=$(quick_warp)
224         ipv6_now=$(quick_ipv6)
225         default_now=$(quick_default_route)
226
227         local ts
228         ts=$(date -u +%H:%M:%S)
229
230         # WARP check
231         if [ "$warp_now" != "$last_warp" ]; then
232             if [ "$warp_now" = "off" ] || [ "$warp_now" = "unknown" ]; then
233                 printf '\n%b ALERT %b\n' "$RED$BOLD" "$NC"
234                 printf '%b[%s] %bWARP FLIPPED: %s %s%b YOUR IP IS LEAKING!\n' \
235                     "$RED" "$ts" "$BOLD" "$last_warp" "$warp_now" "$NC"
236                 printf '[%s] ALERT warp flipped %s%b\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
237             else
238                 printf '\n%b[%s] warp recovered: %s %s%b\n' \
239                     "$GREEN" "$ts" "$last_warp" "$warp_now" "$NC"
240                 printf '[%s] OK warp recovered %s%b\n' "$ts" "$last_warp" "$warp_now" >> "$WATCH_LOG"
241             fi
242             last_warp="$warp_now"
243         fi
244
245         # IPv6 check
246         if [ -n "$ipv6_now" ]; then
247             if [ "$last_ipv6_ok" = true ]; then
248                 printf '\n%b ALERT %b\n' "$RED$BOLD" "$NC"
249                 printf '%b[%s] %bIPv6 LEAK DETECTED %s%b\n' \
250                     "$RED" "$ts" "$BOLD" "$ipv6_now" "$NC"
251                 printf '[%s] ALERT IPv6 leak %s\n' "$ts" "$ipv6_now" >> "$WATCH_LOG"
252                 last_ipv6_ok=false
253             fi
254         else
255             if [ "$last_ipv6_ok" = false ]; then
256                 printf '%b[%s] IPv6 leak resolved%b\n' "$GREEN" "$ts" "$NC"
257                 printf '[%s] OK IPv6 resolved\n' "$ts" >> "$WATCH_LOG"
258             fi
259             last_ipv6_ok=true
260         fi
261
262         # Default route check
263         if [ "$default_now" != "$last_default" ]; then
264             if [ "$default_now" != "utun" ]; then
265                 printf '\n%b ALERT %b\n' "$RED$BOLD" "$NC"

```

```

266     printf '%b[%s] %bDEFAULT ROUTE CHANGED: %s %s\b traffic may bypass WARP!\n' \
267         "$RED" "$ts" "$BOLD" "$last_default" "$default_now" "$NC"
268     printf '[%s] ALERT route changed %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
269 else
270     printf '%b[%s] default route recovered: %s %s\b\n' \
271         "$GREEN" "$ts" "$last_default" "$default_now" "$NC"
272     printf '[%s] OK route recovered %s%s\n' "$ts" "$last_default" "$default_now" >> "$WATCH_LOG"
273 fi
274 last_default="$default_now"
275 fi
276
277 # Heartbeat (every 10th iteration or if all OK)
278 if [ $((iteration % 10)) -eq 0 ]; then
279     printf '[%s] #d warp=%s ipv6=%s route=%s\n' \
280         "$ts" "$iteration" "$warp_now" "${ipv6_now:-none}" "$default_now" >> "$WATCH_LOG"
281     printf '\r%b[%s] #d warp=%s route=%s (Ctrl+C to stop)%b\n' \
282         "$GREEN" "$ts" "$iteration" "$warp_now" "$default_now" "$NC"
283 fi
284
285 sleep "$interval"
286 done
287 }
288
289 #
290 # Begin main diagnostic (skipped in non-watch modes if we're just watching)
291 #
292
293 ACTIVE_SERVICE=$(resolve_active_service)
294 ENO_SERVICE=$(networksetup -listnetworkserviceorder 2>> "$LOG_FILE" \
295     | grep -B 1 "Device: en0" | head -1 \
296     | sed -E 's/^\([0-9\*]+\)/ /' || true)
297 [ -z "$ENO_SERVICE" ] && ENO_SERVICE="Wi-Fi"
298
299 # Resolve the gateway's Tailscale IP.
300 GATEWAY_TS_IP=""
301 if [ -f "$PROJECT_ROOT/.gateway_ip" ]; then
302     GATEWAY_TS_IP=$(cat "$PROJECT_ROOT/.gateway_ip" 2>> "$LOG_FILE" \
303         | tr -d '\r' | awk 'NR==1{print $1; exit}')
304 fi
305 if [ -z "$GATEWAY_TS_IP" ] && command -v docker >> "$LOG_FILE" 2>&1 \
306     && docker compose ps --status running 2>/dev/null | grep -q 'tailscale'; then
307     GATEWAY_TS_IP=$(run_with_timeout 10 docker compose exec -T tailscale \
308         tailscale ip -4 2>> "$LOG_FILE" \
309         | tr -d '\r' | awk 'NR==1{print $1; exit}')
310 fi
311
312 # Scenario classification state
313 SCENARIO=""
314 WARP_STATUS=""
315 SOCKS5_ENABLED=false
316 TS_ON_MESH=false
317 DEFAULT_VIA_UTUN=false
318 IPV6_LEAK=false
319
320 #
321 section "1/8 Host Tailscale mesh status"
322 #
323 if ! command -v tailscale >> "$LOG_FILE" 2>&1; then
324     fail "tailscale CLI not on PATH install via 'brew install tailscale' then re-run"
325     echo " (a Tailscale daemon is required for the exit-node path to work)"
326 else
327     set +e
328     TS_OUT=$(run_with_timeout 5 tailscale status 2>&1)
329     TS_EXIT=$?
330     set -uo pipefail
331     TS_HEAD=$(printf '%s\n' "$TS_OUT" | head -5 | tr -d '\r')
332     if [ "$TS_EXIT" -eq 137 ]; then
333         fail "tailscaled daemon is wedged/unresponsive (5s timeout)"
334         echo " Fix path: 'sudo brew services restart tailscale'"
335     elif printf '%s' "$TS_OUT" | grep -qE 'Logged out|not logged in|expired'; then
336         fail "tailscale is logged out / auth expired on this host"
337         echo " Fix path: 'sudo tailscale up' (browser auth once)"
338     elif printf '%s' "$TS_OUT" | grep -q '100\.'; then
339         ok "Host is on tailnet (100.x.x.x visible in status)"
340         TS_ON_MESH=true
341         if printf '%s' "$TS_OUT" | grep -q "$GATEWAY_TS_IP"; then
342             ok "Gateway $GATEWAY_TS_IP visible in host's peer list"
343         elif [ -n "$GATEWAY_TS_IP" ]; then
344             warn "Gateway $GATEWAY_TS_IP NOT in host's peer list possible auth/key mismatch"
345         fi
346         line " first 5 lines of 'tailscale status' "

```

```

347     line "${printf '%s' "$TS_HEAD" | awk '{print "    "$0}}'"
348 else
349     fail "tailscale output unrecognized:"
350     line "${printf '%s' "$TS_OUT" | head -3 | awk '{print "    "$0}}'"
351 fi
352 fi
353
354 #
355 section "2/8 Active SOCKS5 proxy on Wi-Fi"
356 #
357 SOCKS_OUT=$(networksetup -getsocksfirewallproxy "$ACTIVE_SERVICE" 2>> "$LOG_FILE" \
358     || networksetup -getsocksfirewallproxy Wi-Fi 2>> "$LOG_FILE" \
359     || echo "Could not query SOCKS5 proxy state")
360 if printf '%s' "$SOCKS_OUT" | grep -q "Enabled: Yes"; then
361     fail "SOCKS5 proxy IS enabled on $ACTIVE_SERVICE toggle.sh fell back to SOCKS5 mode last run"
362     SOCKS5_ENABLED=true
363     line " Full state:"
364     line "${printf '%s' "$SOCKS_OUT" | awk '{print "    "$0}}'"
365     line " Browse on macOS ignores system SOCKS5 by default traffic bypasses WARP."
366 else
367     ok "SOCKS5 proxy is OFF on $ACTIVE_SERVICE exit-node path should be active"
368 fi
369
370 #
371 section "3/8 Egress IP + WARP tunnel verification (definitive test)"
372 #
373 CDN_OUT=$(run_with_timeout 8 curl -s --max-time 6 \
374     https://www.cloudflare.com/cdn-cgi/trace 2>&1 || echo "(curl failed)")
375 CDN_WARP=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^warp={print $2; exit}')
376 CDN_IP=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^ip={print $2; exit}')
377 CDN_COLO=$(printf '%s' "$CDN_OUT" | awk -F=' ' /^colo={print $2; exit}')
378 WARP_STATUS="$CDN_WARP"
379 if [ "$CDN_WARP" = "on" ]; then
380     ok "warp=on tunnel is hot, host traffic IS going through Cloudflare WARP"
381     line " Public IP: $CDN_IP"
382     line " Cloudflare: ${CDN_COLO:-unknown colo}"
383     line " whatismyip.com's 'local ISP' result is its ISP-database error, not yours."
384 elif [ "$CDN_WARP" = "off" ]; then
385     fail "warp=off host traffic is NOT going through WARP"
386     line " Public IP: $CDN_IP (NOT Cloudflare)"
387     warn "This is the actual leak."
388 else
389     warn "could not determine warp status (curl failed or returned unexpected shape)"
390     line " Raw output: $(printf '%s' "$CDN_OUT" | head -3 | tr '\n' '|')
391 fi
392
393 #
394 section "4/8 IPv6 leak probe (bypasses IPv4 exit-node on campus networks)"
395 #
396 IPV6_OUT=$(run_with_timeout 7 curl -s --max-time 5 ifconfig.co 2>&1 || echo "")
397 if [ -n "$IPV6_OUT" ] && printf '%s' "$IPV6_OUT" | grep -qE '[0-9a-fA-F:]+\$'; then
398     fail "host has live IPv6 egress $IPV6_OUT (A6 record bypasses WARP)"
399     IPV6_LEAK=true
400     line " Tailscale exit-node advertisement is IPv4-only, so IPv6 traffic skips it."
401     line " Quick-fix: 'sudo networksetup -setv6off $ACTIVE_SERVICE'"
402 else
403     ok "no IPv6 egress detected (good IPv6 won't bypass the tunnel)"
404 fi
405
406 #
407 section "5/8 Host egress route for real traffic (route -n get 1.1.1.1)"
408 #
409 # WHY NOT 'netstat -rn | grep default':
410 # On macOS, Tailscale does NOT replace the physical default route installed by
411 # DHCP. Instead it adds a second interface-scoped route bound to the utun
412 # interface that the kernel prefers for real traffic forwarding. The literal
413 # "default" entry (192.168.x.x via en0) is left untouched in the table as a
414 # BSD routing quirk it's not lying, it's just answering a different question.
415 # The correct question is: "where does a real destination actually resolve?"
416 # `route -n get 1.1.1.1` asks the kernel's forwarding logic directly.
417 ROUTE_GET=$(route -n get 1.1.1.1 2>/dev/null)
418 ROUTE_IFACE=$(echo "$ROUTE_GET" | awk '/interface/{print $2}')
419 ROUTE_GATEWAY=$(echo "$ROUTE_GET" | awk '/gateway/{print $2}')
420 if [ -z "$ROUTE_IFACE" ]; then
421     warn "route -n get 1.1.1.1 returned no interface routing table may be empty"
422 else
423     line " interface: $ROUTE_IFACE gateway: ${ROUTE_GATEWAY:-n/a}"
424     if echo "$ROUTE_IFACE" | grep -qE '^utun[0-9]+'; then
425         ok "1.1.1.1 resolves via $ROUTE_IFACE Tailscale interface-scoped route is winning"
426         DEFAULT_VIA_UTUN=true
427     elif echo "$ROUTE_IFACE" | grep -qE '^en[0-9]+'; then

```

```

428     fail "1.1.1.1 resolves via $ROUTE_IFACE (physical Wi-Fi)  exit-node interface route not active"
429 else
430     warn "1.1.1.1 resolves via unexpected interface: $ROUTE_IFACE"
431 fi
432 fi
433
434 #
435 section "6/8  Last toggle.sh START-relevant lines in output.log (if any)"
436 #
437 if [ ! -f "$LOG_FILE" ]; then
438     note "output.log"
439     line "  toggle.sh was never run from this directory. Run './toggle.sh' first,"
440     line "  then re-run this diagnostic."
441 else
442     HITS=$(grep -E 'Exit node enabled|SKIP_EXIT_NODE|Pre-flight|Falling back to SOCKS|Starting|Successfully' \
443             "$LOG_FILE" 2>/dev/null | tail -8 || true)
444     if [ -z "$HITS" ]; then
445         note "matching lines in output.log"
446     else
447         line "$(printf '%s\n' "$HITS" | awk '{print "    "$0}')"
448     fi
449 fi
450
451 #
452 section "7/8  Gateway IP we should be pointing at"
453 #
454 if [ -n "$GATEWAY_TS_IP" ]; then
455     ok "gateway Tailscale IP: $GATEWAY_TS_IP"
456 else
457     warn "could not resolve gateway Tailscale IP"
458     line "  (looked in .gateway_ip and tried docker compose exec tailscale ip -4)"
459 fi
460
461 #
462 section "8/8  Tailscale System Extension (App Store conflict check)"
463 #
464 SE_PENDING_UNINSTALL=false
465 SE_ACTIVE=false
466 if command -v systemextensionsctl >/dev/null 2>&1; then
467     SE_OUT=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale')
468     if [ -n "$SE_OUT" ]; then
469         line "  Tailscale System Extension(s) detected:"
470         line "$(printf '%s' "$SE_OUT" | awk '{print "    "$0}')"
471         if printf '%s' "$SE_OUT" | grep -q 'waiting to uninstall'; then
472             SE_PENDING_UNINSTALL=true
473             warn "System Extension is pending uninstall on next reboot."
474         else
475             SE_ACTIVE=true
476             warn "Active Tailscale System Extension detected. This will conflict with Homebrew Tailscale."
477         fi
478     else
479         ok "no Tailscale System Extension detected"
480     fi
481 else
482     note "systemextensionsctl not available (not on macOS or permission restricted)"
483 fi
484
485 #
486 section "CLASSIFICATION  applies ONE of the four known scenarios"
487 #
488
489 if [ "$WARP_STATUS" = "on" ] && [ "$IPV6_LEAK" = "false" ]; then
490     SCENARIO="OK"
491 elif [ "$SE_PENDING_UNINSTALL" = "true" ]; then
492     SCENARIO="SE_PENDING"
493 elif [ "$SOCKS5_ENABLED" = "true" ]; then
494     SCENARIO="A"
495 elif [ "$IPV6_LEAK" = "true" ]; then
496     SCENARIO="B"
497 elif [ "$DEFAULT_VIA_UTUN" = "false" ] && [ "$TS_ON_MESH" = "true" ]; then
498     SCENARIO="C"
499 else
500     SCENARIO="UNKNOWN"
501 fi
502
503 case "$SCENARIO" in
504     SE_PENDING)
505         echo ""
506         echo -e "  ${RED}${BOLD}VERDICT:  Scenario SE_PENDING  Tailscale System Extension Pending Uninstall${NC}"
507         echo "  A prior App Store Tailscale System Extension is pending uninstall."
508         echo "  Since System Integrity Protection (SIP) is enabled, macOS blocks manual"

```

```

509 echo "  uninstallation of terminated system extensions until the next reboot."
510 echo "  Brew-managed tailscaled cannot engage the Network Extension slot until this is resolved."
511 echo ""
512 echo "  ${BOLD}Recommended fix:${NC}"
513 echo "    sudo shutdown -r now"
514 echo "    # After reboot, run: ./toggle.sh"
515 ;;
516 A)
517 echo ""
518 echo -e "  ${RED}${BOLD}VERDICT:  Scenario A  SOCKS5 fallback lane${NC}"
519 echo "  toggle.sh's pre-flight checks failed last run. The script activated"
520 echo "  SOCKS5 + local DNS proxy as a fallback. DNS gets hijacked but"
521 echo "  browsers on macOS ignore the system SOCKS5  traffic egresses"
522 echo "  direct through the local network."
523 echo ""
524 echo "  ${BOLD}Recommended fix:${NC}"
525 echo "    echo 'GATEWAY_BYPASS_PING=true' >> $PROJECT_ROOT/.env"
526 echo "    sudo tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \\"
527 if [ -n "$GATEWAY_TS_IP" ]; then
528 echo "      --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true"
529 else
530 echo "      --exit-node=$GATEWAY_TS_IP --exit-node-allow-lan-access=true # fill in $GATEWAY_TS_IP"
531 fi
532 echo "    sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
533 echo "    ./toggle.sh"
534 ;;
535 B)
536 echo ""
537 echo -e "  ${RED}${BOLD}VERDICT:  Scenario B  IPv6 leak over dual-stack campus/local IPv6${NC}"
538 echo "  Tailscale exit-node advertises only IPv4. macOS sends AAAA queries"
539 echo "  straight out en0, which is dual-stack on most campus APs  IPv6"
540 echo "  traffic bypasses the WARP tunnel entirely."
541 echo ""
542 echo "  ${BOLD}Recommended fix:${NC}"
543 echo "    sudo networksetup -setv6off $ACTIVE_SERVICE"
544 echo "    sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder"
545 echo "    # Re-enable IPv6 later with:  sudo networksetup -setv6automatic $ACTIVE_SERVICE"
546 ;;
547 C)
548 echo ""
549 echo -e "  ${RED}${BOLD}VERDICT:  Scenario C  Tailscale route-freeze${NC}"
550 echo "  Host's default route points at the Wi-Fi gateway instead of the"
551 echo "  Tailscale utun* interface. The exit-node preference is set but the"
552 echo "  routing-table assertion did not take effect (devref 10.26)."

```

```

589 #
590 #
591 # Write the verdict block to the report file
592 #
593 SECOND_BLOCK=$(cat <<EOF
594
595
596 C L A S S I F I C A T I O N   R E S U L T
597
598 Scenario: $SCENARIO
599 WARP egress: $WARP_STATUS (cdn-cgi/trace)
600 SOCKS5 lane: ${[ "$SOCKS5_ENABLED" = true ] && echo "ENABLED (browsers bypass)" || echo "off"}
601 IPv6 leak: ${[ "$IPv6_LEAK" = true ] && echo "YES (campus IPv6 egress)" || echo "no"}
602 Default via: ${[ "$DEFAULT_VIA_UTUN" = true ] && echo "utun* (Tailscale)" || echo "Wi-Fi gateway"}
603 Host on mesh: ${[ "$TS_ON_MESH" = true ] && echo "yes" || echo "no"}
604 SE pending: ${[ "$SE_PENDING_UNINSTALL" = true ] && echo "yes" || echo "no"}
605 Gateway IP: ${GATEWAY_TS_IP:-unresolved}
606 EOF
607 )
608 printf '%s\n' "$SECOND_BLOCK" >&3
609
610 #
611 # --fix mode
612 #
613 if [ "$DO_FIX" = "true" ] && [ "$SCENARIO" != "OK" ] && [ "$SCENARIO" != "UNKNOWN" ]; then
614     echo ""
615     echo ""
616     echo " A P P L Y I N G   F I X (scenario $SCENARIO)"
617     echo ""
618     echo ""
619
620     case "$SCENARIO" in
621         SE_PENDING)
622             warn "System Extension uninstall is pending on next reboot."
623             line "Please run 'sudo shutdown -r now' to reboot the system."
624             ;;
625         A)
626             if [ -n "$GATEWAY_TS_IP" ]; then
627                 line " writing GATEWAY_BYPASS_PING=true to .env"
628                 ENV="$PROJECT_ROOT/.env"
629                 [ -f "$ENV" ] || touch "$ENV"
630                 if grep -q '^GATEWAY_BYPASS_PING=' "$ENV" 2>/dev/null; then
631                     sed -i 's/^GATEWAY_BYPASS_PING=.*GATEWAY_BYPASS_PING=true/' "$ENV"
632                 else
633                     printf '\nGATEWAY_BYPASS_PING=true\n' >> "$ENV"
634                 fi
635                 line " re-asserting tailscale exit-node (with --reset)"
636                 sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \
637                     --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
638                     >> "$LOG_FILE" 2>&1 \
639                     && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
640                     || warn "tailscale up did not return success (check $LOG_FILE)"
641             else
642                 warn "no gateway Tailscale IP resolved skipping tailscale up"
643                 warn "fix manually: 'sudo tailscale up --reset ... --exit-node=<TS_IP>'"
644             fi
645             ;;
646         B)
647             line " disabling IPv6 on $ACTIVE_SERVICE while gateway is up"
648             sudo -n networksetup -setv6off "$ACTIVE_SERVICE" >> "$LOG_FILE" 2>&1 \
649                 && ok "IPv6 disabled on $ACTIVE_SERVICE (re-enable with 'setv6automatic')\" \
650                 || warn "networksetup -setv6off failed (check $LOG_FILE)"
651             ;;
652         C)
653             line " restarting tailscaled (route-table fix)"
654             run_with_timeout 15 brew services restart tailscale >> "$LOG_FILE" 2>&1 \
655                 || run_with_timeout 15 sudo -n brew services restart tailscale >> "$LOG_FILE" 2>&1
656             sleep 3
657             line " flushing stale host routes + DNS cache"
658             sudo -n route delete -net 0.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
659             sudo -n route delete -net 128.0.0.0/1 >> "$LOG_FILE" 2>&1 || true
660             sudo -n dscacheutil -flushcache >> "$LOG_FILE" 2>&1 || true
661             sudo -n killall -HUP mDNSResponder >> "$LOG_FILE" 2>&1 || true
662             if [ -n "$GATEWAY_TS_IP" ]; then
663                 line " re-asserting exit-node after restart"
664                 sudo -n tailscale up --reset --ssh=true --accept-dns=false --accept-routes=true \
665                     --exit-node="$GATEWAY_TS_IP" --exit-node-allow-lan-access=true \
666                     >> "$LOG_FILE" 2>&1 \
667                     && ok "tailscale exit-node re-asserted for $GATEWAY_TS_IP" \
668                     || warn "tailscale up did not return success (check $LOG_FILE)"
669             fi

```

```

670     ;;
671     esac
672
673     echo ""
674     echo "Re-verifying after fix..."
675     sleep 3
676     CDN_WARP_AFTER=$(run_with_timeout 8 curl -s --max-time 6 \
677         https://www.cloudflare.com/cdn-cgi/trace 2>&1 \
678         | awk -F'=' '/^warp={print $2; exit}')
679     IPV6_AFTER=$(run_with_timeout 7 curl -6 -s --max-time 5 ifconfig.co 2>&1 || echo "")
680     printf '\n[AFTER FIX]\n' >&3
681     printf ' warp status:    %s\n' "$CDN_WARP_AFTER" >&3
682     printf ' IPv6 egress:      %s\n' "${IPV6_AFTER:-none}" >&3
683     line " warp status:    ${CDN_WARP_AFTER:-unknown}"
684     line " IPv6 egress:    ${IPV6_AFTER:-none}"
685     if [ "$CDN_WARP_AFTER" = "on" ] && [ -z "$IPV6_AFTER" ]; then
686         echo ""
687         ok "Fix verified host is now routing through WARP. Remote clients unaffected."
688     else
689         echo ""
690         warn "Fix did not fully take effect on first try. Likely causes:"
691         line " tailscaled needed a longer settling window (re-run diagnostic in 30s)"
692         line " The classified scenario was wrong; paste this report for triage."
693         line " Tailscale needs a full auth refresh: 'sudo tailscale up'"
694     fi
695 fi
696
697 echo ""
698 echo ""
699 echo " Full report written to:"
700 echo " $REPORT_FILE"
701 echo ""
702 echo ""
703
704 #
705 # --watch mode: enter the continuous monitoring loop after baseline diagnostic
706 #
707 if [ "$DO_WATCH" = "true" ]; then
708     # Build baseline from the full diagnostic just completed
709     BASELINE_WARP="$WARP_STATUS"
710     BASELINE_DEFAULT=$( [ "$DEFAULT_VIA_UTUN" = "true" ] && echo "utun" || echo "wifi/other")
711
712     if [ "$SCENARIO" != "OK" ]; then
713         warn "Baseline diagnostic found issues (scenario: $SCENARIO)."
714         line " Watch loop will still run alerts fire on any STATE CHANGE from current baseline."
715         line " Fix the issue first? Re-run with: bash scripts/diagnose-host-leak.sh --fix"
716         echo ""
717     fi
718
719     run_watch_loop "$BASELINE_WARP" "$BASELINE_DEFAULT" "$IPV6_LEAK" "$WATCH_INTERVAL"
720 fi
721
722 exit 0

```

34 scripts/dns-proxy.py

```

1  #!/usr/bin/env python3
2  """
3  dns-proxy.py Local DNS proxy for nullexit gateway.
4
5  Listens on UDP port 53, receives DNS queries, forwards them over TCP
6  to AdGuard at 127.0.0.1:5354 (the Docker port mapping), and returns
7  the response via UDP. Handles the DNS-over-TCP wire format (2-byte
8  length prefix) correctly unlike socat which just forwards raw bytes.
9  """
10
11 import socket
12 import struct
13 import sys
14 import os
15 from concurrent.futures import ThreadPoolExecutor
16
17 LISTEN_ADDR = os.environ.get("LISTEN_ADDR", "127.0.0.1")
18 LISTEN_PORT = int(os.environ.get("LISTEN_PORT", 53))
19 TARGET_HOST = os.environ.get("TARGET_HOST", "127.0.0.1")
20 TARGET_PORT = int(os.environ.get("TARGET_PORT", 5354))

```



```

21
22
23 def handle_query(data: bytes, client_addr: tuple, sock: socket.socket) -> None:
24     """Forward a single DNS query over TCP and send the response back via UDP."""
25     try:
26         tcp = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
27         tcp.settimeout(5)
28         tcp.connect((TARGET_HOST, TARGET_PORT))
29
30         # DNS over TCP: prepend 2-byte length (big-endian)
31         tcp.sendall(struct.pack("!H", len(data)) + data)
32
33         # Read the 2-byte response length
34         raw_len = tcp.recv(2)
35         if len(raw_len) < 2:
36             tcp.close()
37             return
38         resp_len = struct.unpack("!H", raw_len)[0]
39
40         # Read the full response
41         response = b""
42         while len(response) < resp_len:
43             chunk = tcp.recv(resp_len - len(response))
44             if not chunk:
45                 break
46             response += chunk
47
48         tcp.close()
49
50         # Send the response back over UDP (strip TCP length prefix)
51         if response:
52             sock.sendto(response, client_addr)
53     except Exception:
54         # Silently drop failed queries (malformed, timeout, etc.)
55         pass
56
57
58 def main() -> None:
59     # Create UDP socket
60     sock = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
61     sock.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
62
63     try:
64         sock.bind((LISTEN_ADDR, LISTEN_PORT))
65     except PermissionError:
66         print(f"dns-proxy: ERROR need root to bind port {LISTEN_PORT}", flush=True)
67         sys.exit(1)
68     except OSError as e:
69         print(f"dns-proxy: ERROR {e}", flush=True)
70         sys.exit(1)
71
72     print(f"dns-proxy: listening on UDP {LISTEN_ADDR}:{LISTEN_PORT} TCP {TARGET_HOST}:{TARGET_PORT}", flush=True)
73
74     # Use a thread pool to handle concurrent DNS queries efficiently without thread exhaustion
75     executor = ThreadPoolExecutor(max_workers=64)
76
77     while True:
78         try:
79             data, client_addr = sock.recvfrom(4096) # Support EDNS0 (up to 4096 bytes) to prevent truncation
80             executor.submit(handle_query, data, client_addr, sock)
81         except KeyboardInterrupt:
82             break
83         except Exception:
84             pass
85
86     executor.shutdown(wait=False)
87     sock.close()
88
89
90 if __name__ == "__main__":
91     main()

```

35 scripts/fix-docker-bridge-collision.sh

```
1 #!/bin/bash
```

```

2 # scripts/fix-docker-bridge-collision.sh nullexit fix for devref 9.13
3 #
4 # Symptom: host default route points at Docker's bridge gateway (172.17.0.1)
5 # instead of the real Wi-Fi gateway. ALL host internet egress fails. Remote
6 # tailnet clients route through the gateway correctly (their tunnel is
7 # independent of the host's broken route table), so they look healthy while
8 # the host itself cannot reach anything.
9 #
10 # Root cause: Docker's default bridge claims 172.17.0.0/16 by default. When the
11 # host's physical Wi-Fi also lands in the same /16 extremely common on
12 # university networks which hand out 172.16.0.0/12 the kernel
13 # installs Docker's bridge as the default route, and every packet bound for
14 # the internet is silently absorbed by docker0 instead of egressing.
15 #
16 # This script:
17 # 1. Detects the actual collision (Docker's 172.17.0.0/16 vs the host's
18 # Wi-Fi subnet, queried live via the ACTIVE interface not hardcoded en0)
19 # 2. Picks a non-colliding Docker bip using /8-family routing (works for
20 # 172.16.0.0/12 campus networks; naive prefix-match would pick 172.26/24
21 # which still lives INSIDE a /12 of 172.16.0.0/12)
22 # 3. Backs up ~/.colima/default/colima.yaml with an ISO-8601 timestamp
23 # 4. Edits the file in place, idempotent (replace existing docker.bip /
24 # append new docker.bip without disturbing other keys; skip YAML comments)
25 # 5. Restarts Colima so the VM rebuilds with the new bridge subnet
26 # 6. Rebinds Wi-Fi DHCP so the host re-learns its real Wi-Fi gateway
27 # 7. Verifies the new default route is no longer the Docker bridge
28 # 8. Re-runs toggle.sh to bring the gateway back up cleanly
29 # 9. Re-runs scripts/diagnose-host-leak.sh and prints the new verdict
30 #
31 # Usage:
32 # bash scripts/fix-docker-bridge-collision.sh # apply the fix
33 # bash scripts/fix-docker-bridge-collision.sh --dry # show what would change, then exit
34 # bash scripts/fix-docker-bridge-collision.sh --skip-toggle # fix but don't restart gateway
35 # bash scripts/fix-docker-bridge-collision.sh --help # usage
36
37 set -uo pipefail
38 # NOTE: errexit (`set -e`) is intentionally NOT enabled so intermediate
39 # `command || warn` patterns can continue past non-fatal failures.
40 # Every critical failure (colima restart, sed, cp backup, toggle.sh)
41 # uses an explicit `|| die` so a half-applied state never escapes.
42
43 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
44 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
45 LOG_FILE="$PROJECT_ROOT/output.log"
46 source "$SCRIPT_DIR/common.sh"
47 TIMESTAMP="$(date -u +%Y%m%dT%H%M%SZ)"
48 COLIMA_YAML="$HOME/.colima/default/colima.yaml"
49 PLATFORM="$(uname -s)"
50
51 # Argument parsing
52 DRY_RUN=false
53 SKIP_TOGGLE=false
54 case "${1:-}" in
55   --dry|-n) DRY_RUN=true ;;
56   --skip-toggle) SKIP_TOGGLE=true ;;
57   --help|-h)
58     sed -n '28,37p' "$0"
59     exit 0
60   ;;
61   *)
62     : # default: apply
63     ;;
64   *)
65     printf 'Unknown argument: %s\n' "$1" >&2
66     exit 2
67   ;;
68 esac
69
70
71 # Helper: print "[dry-run] $cmd" or actually run $cmd. Wraps the redirect
72 # INSIDE the function so outer `>> $FILE` statements never accidentally
73 # append dry-run noise to the real file. (Earlier draft didn't and the
74 # dry-run polluted $COLIMA_YAML.)
75 run() {
76   if [ "$DRY_RUN" = "true" ]; then
77     printf '[dry-run] %s\n' "$*"
78     return 0
79   fi
80   printf '%s\n' "$*"
81 }
82

```

```

83 }
84
85 # Platform-aware sed in-place. macOS BSD sed needs an empty argument;
86 # GNU sed (Linux) does not. Earlier draft used '-i' everywhere and
87 # would have failed on Linux silently.
88 sed_inplace() {
89     local expr="$1"
90     local file="$2"
91     case "$PLATFORM" in
92         Darwin) run sed -i '' "$expr" "$file" ;;
93         *)      run sed -i "$expr" "$file" ;;
94     esac
95 }
96
97 # 0. Resolve the ACTIVE physical interface once (used everywhere)
98 # Replaces the earlier `en0` hardcoding. Same algorithm recover.sh uses:
99 # `route get default` if utun/tun, walk en0..en5 first with `status: active`
100 # + `inet` fall back to en0. This means the script also works on hosts
101 # whose primary NIC is en1/en2 (Thunderbolt Ethernet, USB-LAN, etc.).
102 resolve_active_iface() {
103     local iface
104     iface=$(route get default 2>> "$LOG_FILE" | awk '/interface:/{print $2; exit}')
105     if [ -z "$iface" ] || [ "${iface#utun}" != "$iface" ] || [ "${iface#tun}" != "$iface" ]; then
106         for i in en0 en1 en2 en3 en4 en5; do
107             if ifconfig "$i" 2>> "$LOG_FILE" | grep -q "status: active" \
108                 && ifconfig "$i" 2>> "$LOG_FILE" | grep -q "inet" ; then
109                 iface="$i"; break
110             fi
111         done
112     fi
113     [ -z "$iface" ] && iface="en0"
114     printf '%s\n' "$iface"
115 }
116 ACTIVE_IFACE=$(resolve_active_iface)
117
118 # 1. Detect collision
119 step "1/9 Detecting Docker-bridge vs Wi-Fi subnet collision"
120
121 # Host subnet on the ACTIVE interface (not hardcoded en0).
122 HOST_SUBNET=$(ifconfig "$ACTIVE_IFACE" 2>> "$LOG_FILE" \
123     | awk '/inet /{print $2; exit}')
124 HOST_GATEWAY=$(route get default 2>> "$LOG_FILE" \
125     | awk '/gateway:/{print $2; exit}')
126 DEFAULT_DEV=$(route get default 2>> "$LOG_FILE" \
127     | awk '/interface:/{print $2; exit}')
128
129 printf ' active iface: %s\n' "$ACTIVE_IFACE"
130 printf ' iface inet: %s\n' "${HOST_SUBNET:-none}"
131 printf ' default via: %s\n' "${HOST_GATEWAY:-none}"
132 printf ' default dev: %s\n' "${DEFAULT_DEV:-none}"
133
134 # Collision fires when the host's default-route gateway is 172.17.0.1 (Docker's
135 # bridge) OR when the host subnet is in 172.17.0.0/16. The full-bridge-collision
136 # case (gateway is literally the docker bridge IP) is the smoking gun.
137 COLLISION=false
138 if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#172.17.}" != "$HOST_GATEWAY" ]; then
139     COLLISION=true
140     fail "default gateway $HOST_GATEWAY collides with Docker's default bridge (172.17.0.0/16)"
141 elif [ -n "$HOST_SUBNET" ] && [ "${HOST_SUBNET#172.17.}" != "$HOST_SUBNET" ]; then
142     COLLISION=true
143     fail "host subnet $HOST_SUBNET overlaps Docker's 172.17.0.0/16 bridge"
144 else
145     ok "no collision detected between host Wi-Fi and Docker's default bridge"
146 fi
147
148 if [ "$COLLISION" = "false" ]; then
149     echo ""
150     echo "Nothing to fix. If you still see egress issues, run:"
151     echo "    bash scripts/diagnose-host-leak.sh"
152     exit 0
153 fi
154
155 # 2. Pick non-colliding bip via /8-family routing
156 step "2/9 Picking a non-colliding Docker bip for ~/.colima/default/colima.yaml"
157
158 # Earlier draft tried naive prefix-match: "if candidate's 3-octet prefix is
159 # not a literal prefix of host gateway, pick it". That breaks on /12 campus
160 # networks (172.16.0.0/12 covers 172.16-172.31): a candidate like
161 # 172.26.0.1/24 LIVES INSIDE the host's /12 even though the prefix test
162 # passes. Fix: classify the host's address family FIRST, then choose ALL
163 # candidates from a DIFFERENT RFC1918 family. /8 boundaries are coarse

```

```

164 # enough that no /24 picked this way can ever collide.
165 case "$HOST_GATEWAY" in
166     172.16.*|172.17.*|172.18.*|172.19.*|172.20.*|172.21.*|172.22.*|172.23.*|\
167     172.24.*|172.25.*|172.26.*|172.27.*|172.28.*|172.29.*|172.30.*|172.31.*)
168     HOST_FAMILY="172"
169     # Host is in 172.x jump past 172.31 entirely. Pick from 10.x / 192.168.x.
170     CANDIDATES="10.200.0.1/24 10.201.0.1/24 192.168.99.1/24"
171     ;;
172 10.*)
173     HOST_FAMILY="10"
174     # Host is in 10.x pick from 172.x (well outside any commonly-deployed
175     # 10.0.0.0/8 slice) / 192.168.x.
176     CANDIDATES="172.26.0.1/24 172.28.0.1/24 192.168.99.1/24"
177     ;;
178 192.168.*)
179     HOST_FAMILY="192"
180     CANDIDATES="172.26.0.1/24 10.200.0.1/24 10.201.0.1/24"
181     ;;
182 *)
183     HOST_FAMILY="unknown"
184     warn "host gateway $HOST_GATEWAY is not in any RFC1918 range (/8 family)"
185     warn "falling back to abstract candidate set; verify collision-free manually"
186     CANDIDATES="172.26.0.1/24 10.200.0.1/24 192.168.99.1/24"
187     ;;
188 esac
189
190 printf ' host address family: %s\n' "$HOST_FAMILY"
191 printf ' candidate pool: %s\n' "$CANDIDATES"
192
193 # Sanity-check the chosen pool has at least one entry that doesn't literally
194 # share a /24 with the host gateway. Naive check (just /24 prefix) is fine
195 # here because we already picked across families.
196 CHOSEN=""
197 for cand in $CANDIDATES; do
198     PREFIX3=$(printf '%s' "$cand" | cut -d. -f1-3)
199     if [ -n "$HOST_GATEWAY" ] && [ "${HOST_GATEWAY#$PREFIX3.}" != "$HOST_GATEWAY" ]; then
200         warn "skip $cand (literal /24 collision with host gateway $HOST_GATEWAY)"
201         continue
202     fi
203     CHOSEN="$cand"
204     ok "chose bip=$CHOSEN (cross-family, no /24 literal collision)"
205     break
206 done
207
208 if [ -z "$CHOSEN" ]; then
209     CHOSEN="172.26.0.1/24"
210     warn "could not find a clean candidate; defaulting to $CHOSEN"
211     warn "verify with 'route get default' after restart"
212 fi
213
214 # 3. Backup colima.yaml
215 step "3/9 Backing up $COLIMA_YAML"
216 mkdir -p "$(dirname "$COLIMA_YAML")"
217 BACKUP="$HOME/.colima/default/colima.yaml.bak.$TIMESTAMP"
218 if [ -f "$COLIMA_YAML" ]; then
219     # `cp` is safe in dry-run because run() suppresses the inner command
220     # entirely no chance of accidentally writing a real backup.
221     run cp -p "$COLIMA_YAML" "$BACKUP" || die "backup failed"
222     if [ "$DRY_RUN" = "false" ]; then ok "backup: $BACKUP"; fi
223 else
224     warn "no existing $COLIMA_YAML (will create fresh)"
225 fi
226
227 # 4. Edit colima.yaml in place (idempotent)
228 step "4/9 Setting docker.bip=$CHOSEN in colima.yaml (in place)"
229
230 # Resolve the existing docker.bip (if any) without picking up commented-out
231 # YAML keys. awk-based extraction; the negated-comment check (~! /^[[:space:]]*#`)
232 # ensures `# bip: 1.2.3.4/24` is treated as a comment, not the active value.
233 EXISTING_BIP=""
234 if [ -f "$COLIMA_YAML" ]; then
235     EXISTING_BIP=$(awk '
236         /^[[:space:]]*#/{next}
237         /^[[:space:]]*bip:[[:space:]]*/{print $2; exit}
238         ' "$COLIMA_YAML" 2>> "$LOG_FILE" || true)
239 fi
240
241 if [ -n "$EXISTING_BIP" ] && [ "$EXISTING_BIP" = "$CHOSEN" ]; then
242     ok "colima.yaml already has docker.bip=$CHOSEN no edit needed"
243 elif [ -n "$EXISTING_BIP" ]; then
244     printf ' (replacing existing docker.bip=%s with %s)\n' "$EXISTING_BIP" "$CHOSEN"

```

```

245 # Use sed_inplace so Linux / macOS get the right syntax. Skip YAML comment
246 # lines so existing `# bip:` comments aren't rewired.
247 sed_inplace "s|~\{[:space:]]*bip:[[:space:]]*\}|.*|\1$CHOSEN|" "$COLIMA_YAML" \
248 || die "sed replacement failed (colima.yaml malformed?)"
249 ok "colima.yaml updated"
250 else
251 # No existing `bip:` key. Branch:
252 # - file has `docker:` block append ` bip:` under it
253 # - file has no `docker:` block append new docker: section
254 # - file does not exist create fresh
255 # Each branch is dry-run-aware so no branch can accidentally leak noise
256 # into $COLIMA_YAML during a preview.
257 HAS_DOCKER_BLOCK=false
258 if [ -f "$COLIMA_YAML" ] && grep -q '^docker:' "$COLIMA_YAML"; then
259     HAS_DOCKER_BLOCK=true
260 fi
261
262 if [ "$HAS_DOCKER_BLOCK" = "true" ]; then
263     if [ "$DRY_RUN" = "true" ]; then
264         printf ' [dry-run] would append to %s:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
265     else
266         printf '\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
267         || die "append failed"
268         ok "appended bip under existing docker: block"
269     fi
270 elif [ -f "$COLIMA_YAML" ]; then
271     if [ "$DRY_RUN" = "true" ]; then
272         printf ' [dry-run] would append to %s:\ndocker:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
273     else
274         printf '\ndocker:\n bip: %s\n' "$CHOSEN" >> "$COLIMA_YAML" \
275         || die "append failed"
276         ok "appended new docker: block at end of file"
277     fi
278 else
279     if [ "$DRY_RUN" = "true" ]; then
280         printf ' [dry-run] would create %s with:\ndocker:\n bip: %s\n' "$COLIMA_YAML" "$CHOSEN"
281     else
282         printf 'docker:\n bip: %s\n' "$CHOSEN" > "$COLIMA_YAML" \
283         || die "create failed"
284         ok "created fresh $COLIMA_YAML with docker.bip=$CHOSEN"
285     fi
286 fi
287 fi
288
289 # Always show the final colima.yaml docker section for visibility
290 printf '\n%b colima.yaml `docker:` section now reads:%b\n' "$BOLD" "$NC"
291 awk '
292 /~docker:/{p=1}
293 p && /^[^ ]/ && !/^docker:/{exit}
294 p
295 ' "$COLIMA_YAML" 2>/dev/null | sed 's/~ / /' || true
296
297 if [ "$DRY_RUN" = "true" ]; then
298     printf '\n %b(dry-run: exiting before colima restart / dhcp rebind / toggle.sh)%b\n' "$YELLOW" "$NC"
299     exit 0
300 fi
301
302 # 5. Restart Colima
303 step "5/9 Restarting Colima VM with new bip"
304 if ! command -v colima >> "$LOG_FILE" 2>&1; then
305     die "colima not on PATH install with 'brew install colima' first"
306 fi
307 # No timeout wrapper (would mask actual VM startup time). If colima hangs,
308 # the user kills with Ctrl+C and can manually `colima restart` from a fresh
309 # terminal the rest of this script is idempotent so re-running works.
310 colima restart >> "$LOG_FILE" 2>&1 || die "colima restart failed; check $LOG_FILE"
311 ok "colima restarted"
312
313 # 6. Rebind DHCP on the ACTIVE interface (not hardcoded en0)
314 step "6/9 Rebinding DHCP on $ACTIVE_IFACE so host re-learns its real gateway"
315
316 # Map ifname macOS network service name (Wi-Fi, USB 10/100 LAN, etc.).
317 # Same logic as the diagnostic + toggle.sh.
318 ACTIVE_SVC=""
319 if [ "$PLATFORM" = "Darwin" ]; then
320     ACTIVE_SVC=$(get_active_service)
321 fi
322
323 case "$PLATFORM" in
324     Darwin)
325         sudo -n networksetup -setdhcp "$ACTIVE_SVC" renew >> "$LOG_FILE" 2>&1 \

```

```

326     && ok "DHCP rebind issued on $ACTIVE_SVC" \
327     || warn "setdhcp renew failed; try 'sudo ipconfig set $ACTIVE_IFACE DHCP' manually"
328     ;;
329 Linux)
330     sudo -n dhclient -r "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 || true
331     sudo -n dhclient "$ACTIVE_IFACE" >> "$LOG_FILE" 2>&1 \
332     && ok "dhclient rebind on $ACTIVE_IFACE" \
333     || warn "dhclient rebind failed"
334     ;;
335 *)
336     warn "unsupported platform $PLATFORM; skip DHCP rebind"
337     ;;
338 esac
339
340 # 7. Verify new default route
341 step "7/9 Verifying the new default route is NOT the Docker bridge"
342 sleep 2 # give DHCP + kernel a beat
343 NEW_DEFAULT=$(netstat -rn 2>> "$LOG_FILE" | awk '/^default/ {print $0; exit}')
344 printf '%s\n' "${NEW_DEFAULT:-no default route detected}"
345 NEW_GATEWAY=$(printf '%s' "$NEW_DEFAULT" | awk '{print $2; exit}')
346
347 if [ -n "$NEW_GATEWAY" ] && [ "${NEW_GATEWAY#172.17.}" != "$NEW_GATEWAY" ]; then
348     fail "default route STILL points at 172.17.x.x DHCP rebind may need manual touch"
349     warn "try: sudo ipconfig set $ACTIVE_IFACE DHCP (or reconnect Wi-Fi in the menu bar)"
350 elif [ -n "$NEW_GATEWAY" ]; then
351     ok "default route is now via $NEW_GATEWAY (no longer Docker's bridge)"
352 else
353     warn "could not determine new gateway; verify with 'route get default'"
354 fi
355
356 # 8. Re-run toggle.sh to bring the gateway back up
357 if [ "$SKIP_TOGGLE" = "true" ]; then
358     step "8/9 Skipping toggle.sh (--skip-toggle)"
359     warn "gateway is currently down run ./toggle.sh manually when ready"
360 else
361     step "8/9 Re-running toggle.sh to bring the gateway back up"
362     if bash "$PROJECT_ROOT/toggle.sh" 2>&1 | tee -a "$LOG_FILE"; then
363         ok "toggle.sh completed"
364     else
365         warn "toggle.sh returned non-zero see $LOG_FILE for details"
366     fi
367 fi
368
369 # 9. Re-run the diagnostic + show verdict
370 step "9/9 Re-running host-leak diagnostic to confirm fix"
371 if bash "$SCRIPT_DIR/diagnose-host-leak.sh" 2>&1 | tee -a "$LOG_FILE"; then
372     LATEST_REPORT=$(ls -t "$PROJECT_ROOT"/host-leak-diagnostic-*.txt 2>/dev/null | head -1)
373     if [ -n "$LATEST_REPORT" ]; then
374         VERDICT=$(awk '/^ Scenario:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
375         WARP=$(awk -F': *' '/^ WARP egress:/ {print $2; exit}' "$LATEST_REPORT" 2>/dev/null)
376         printf '\n Most recent verdict: Scenario=%s WARP=%s\n' \
377             "${VERDICT:-?}" "${WARP:-?}"
378         if [ "$VERDICT" = "OK" ]; then
379             ok "fix confirmed by diagnostic host traffic is going through WARP"
380         fi
381     fi
382 else
383     warn "diagnostic exited non-zero; verify manually"
384 fi
385
386 printf '\n%b\n' "$BOLD" "$NC"
387 printf '%bDone.%b Full transcript at %s\n' "$BOLD" "$NC" "$LOG_FILE"
388 printf '%b\n' "$BOLD" "$NC"
389 exit 0

```

36 scripts/fix-ssh-delay.sh

```

1 #!/bin/bash
2 # scripts/fix-ssh-delay.sh Fix ~20s SSH connection delay caused by UseDNS
3 #
4 # macOS OpenSSH defaults UseDNS to "yes". When Tailscale SSH connects from
5 # a peer (phone/laptop), sshd performs a reverse-DNS (PTR) lookup on the
6 # client's 100.x.x.x Tailscale IP. Since DNS is hijacked to the nullexit
7 # gateway AdGuard Cloudflare WARP, and 100.64.0.0/10 is CGNAT private
8 # space, the PTR query times out (~15-20s) before SSH proceeds.
9 #

```

```

10 # This script uncomments "UseDNS no" in /etc/ssh/sshd_config and reloads
11 # sshd. Existing SSH sessions are NOT dropped.
12 #
13 # Usage:
14 #   sudo bash scripts/fix-ssh-delay.sh
15
16 set -euo pipefail
17
18 SSHD_CONFIG="/etc/ssh/sshd_config"
19
20 # Check we're root
21 if [ "$(id -u)" -ne 0 ]; then
22     echo "ERROR: This script must be run with sudo (it edits $SSHD_CONFIG)"
23     echo "Usage: sudo bash scripts/fix-ssh-delay.sh"
24     exit 1
25 fi
26
27 # Apply config (idempotent)
28 if grep -qE '^UseDNS no' "$SSHD_CONFIG" 2>/dev/null; then
29     echo "UseDNS is already set to 'no' in $SSHD_CONFIG"
30 elif grep -qE '^#UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
31     echo "Uncommenting 'UseDNS no' in $SSHD_CONFIG"
32     sed -i 's/^#UseDNS no/UseDNS no/' "$SSHD_CONFIG"
33 elif grep -qEi '^UseDNS' "$SSHD_CONFIG" 2>/dev/null; then
34     echo "Changing existing UseDNS line to 'UseDNS no'"
35     sed -i 's/^UseDNS.*/UseDNS no/' "$SSHD_CONFIG"
36 else
37     echo "Appending 'UseDNS no' to $SSHD_CONFIG"
38     echo "" >> "$SSHD_CONFIG"
39     echo "UseDNS no" >> "$SSHD_CONFIG"
40 fi
41
42 # Verify
43 if ! grep -qE '^UseDNS no' "$SSHD_CONFIG"; then
44     echo "Failed to apply UseDNS no check $SSHD_CONFIG manually"
45     exit 1
46 fi
47
48 # Reload sshd (does NOT drop existing connections)
49 echo "Reloading sshd..."
50
51 SSHD_PID=$(pgrep -f /usr/sbin/sshd 2>/dev/null | head -1 || true)
52 if [ -n "$SSHD_PID" ]; then
53     kill -HUP "$SSHD_PID"
54     echo "Sent SIGHUP to sshd (PID $SSHD_PID) config reloaded"
55 else
56     echo "No running sshd found. macOS starts sshd on-demand per connection."
57     echo "The new 'UseDNS no' config will take effect on the next SSH connection."
58 fi
59
60 echo ""
61 echo "Done. SSH connections should now connect instantly."
62 echo "Existing sessions were not interrupted."

```

37 scripts/generate_tex.py

```

1 #!/usr/bin/env python3
2 import os
3
4 PROJECT_ROOT = os.path.abspath(os.path.join(os.path.dirname(__file__), '..'))
5
6 IGNORE_DIRS = {'.git', '.github', 'adguard', '__pycache__', 'Recover Gateway.app', 'Toggle Gateway.app'}
7 IGNORE_FILES = {'.env', 'nullexit_unified.tex', 'nullexit_unified.pdf', 'output.log', 'blocked.log', '.DS_Store',
8                 '.lan_p2p_detected', '.signatures', '.gateway_ip', 'TUNNEL_FAILED_CLOSED.marker'}
9 IGNORE_EXTS = {'.pdf', '.tex', '.png', '.jpg', '.jpeg', '.zip', '.tar', '.gz', '.log', '.aux', '.fls', '.fdb_latexmk',
10                '.out', '.toc', '.xdv', '.synctex(busy)'}
11
12 # Exclude from code block inclusion, but keep in tree
13 SKIP_CONTENT_FILES = {'LICENSE'}
14
15 PRIORITY_FILES = [
16     'README.md',
17     'agent.md',
18     'devref.md',
19     'diagrams.md',
20     'toggle.sh',

```



```

19     'recover.sh',
20     'setup.sh',
21     'scripts/toggle-linux.sh',
22     'scripts/recover-linux.sh',
23     'scripts/setup-linux.sh',
24     'docker-compose.yml',
25     'scripts/common.sh',
26     'scripts/watcher.sh',
27     'scripts/crypto.sh'
28 ]
29
30 def get_language(filename):
31     ext = os.path.splitext(filename)[1].lower()
32     if ext == '.sh' or ext == '.applescript':
33         return 'bash'
34     elif ext == '.py':
35         return 'Python'
36     elif ext == '.go':
37         return 'Go'
38     elif ext in {' .yml', '.yaml'}:
39         return 'bash'
40     elif ext == '.plist':
41         return 'XML'
42     elif ext == '.md':
43         return ''
44     return ''
45
46 def escape_tex(text):
47     return text.replace('_', '\\_')
48
49 def generate_tree(dir_path, prefix=""):
50     tree_str = ""
51     entries = sorted(os.listdir(dir_path))
52     entries = [e for e in entries if e not in IGNORE_DIRS and e not in IGNORE_FILES and os.path.splitext(e)[1].lower() not in IGNORE_EXTS]
53
54     for i, entry in enumerate(entries):
55         path = os.path.join(dir_path, entry)
56         is_last = (i == len(entries) - 1)
57         connector = "-- " if is_last else "|-- "
58         tree_str += f"{prefix}{connector}{entry}\n"
59
60         if os.path.isdir(path):
61             extension = " " if is_last else "|  "
62             tree_str += generate_tree(path, prefix + extension)
63
64     return tree_str
65
66 def main():
67     tex_path = os.path.join(PROJECT_ROOT, 'nullexit_unified.tex')
68
69     preamble = r"""
70 \documentclass{article}
71 \usepackage[utf8]{inputenc}
72 \usepackage[margin=1in]{geometry}
73 \usepackage{listings}
74 \usepackage{xcolor}
75 \usepackage{hyperref}
76 \usepackage{tocloft}
77
78 \definecolor{codegreen}{rgb}{0,0.6,0}
79 \definecolor{codegray}{rgb}{0.5,0.5,0.5}
80 \definecolor{codepurple}{rgb}{0.58,0,0.82}
81 \definecolor{backcolour}{rgb}{0.97,0.97,0.96}
82
83 \lstdefinestyle{mystyle}{
84     backgroundcolor=\color{backcolour},
85     commentstyle=\color{codegreen},
86     keywordstyle=\color{blue}\bfseries,
87     numberstyle=\tiny\color{codegray},
88     stringstyle=\color{codepurple},
89     basicstyle=\ttfamily\scriptsize,
90     breakatwhitespace=false,
91     breaklines=true,
92     captionpos=b,
93     keepspaces=true,
94     numbers=left,
95     numbersep=5pt,
96     showspaces=false,
97     showstringspaces=false,
98     showtabs=false,
99     tabsize=2,

```


38 scripts/logger/go.mod

```
1 module nullexit/logger
2
3 go 1.22
```

39 scripts/logger/main.go

```
1 package main
2
3 import (
4     "bufio"
5     "encoding/json"
6     "fmt"
7     "io"
8     "log"
9     "os"
10    "regexp"
11    "strings"
12    "time"
13)
14
15 const (
16     queryLogPath = "/adguard_work/data/querylog.json"
17     procKmsgPath = "/proc/kmsg"
18     outputLogPath = "/app/blocked.log"
19 )
20
21 func logMessage(msg string) {
22     timestamp := time.Now().Format("2006-01-02 15:04:05")
23     formattedMsg := fmt.Sprintf("%s %s\n", timestamp, msg)
24     fmt.Print(formattedMsg)
25
26     f, err := os.OpenFile(outputLogPath, os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0644)
27     if err != nil {
28         log.Printf("Error writing to log file: %v\n", err)
29         return
30     }
31     defer f.Close()
32     f.WriteString(formattedMsg)
33 }
34
35 func tailFile(filepath string, out chan<- string) {
36     for {
37         if _, err := os.Stat(filepath); os.IsNotExist(err) {
38             time.Sleep(1 * time.Second)
39             continue
40         }
41         break
42     }
43
44     f, err := os.Open(filepath)
45     if err != nil {
46         log.Printf("Error opening file %s: %v\n", filepath, err)
47         return
48     }
49     defer func() {
50         if f != nil {
51             f.Close()
52         }
53     }()
54
55     // Seek to end
56     f.Seek(0, io.SeekEnd)
57     reader := bufio.NewReader(f)
58
59     for {
60         line, err := reader.ReadString('\n')
61         if err != nil {
62             if err == io.EOF {
63                 stat, err := os.Stat(filepath)
64                 if err == nil {
65                     currentOffset, _ := f.Seek(0, io.SeekCurrent)
66                     if stat.Size() < currentOffset {
67                         // File truncated or rotated

```

```

68         f.Close()
69         for {
70             if _, err := os.Stat(filepath); os.IsNotExist(err) {
71                 time.Sleep(1 * time.Second)
72                 continue
73             }
74             break
75         }
76         f, _ = os.Open(filepath)
77         reader = bufio.NewReader(f)
78         continue
79     }
80 }
81 time.Sleep(500 * time.Millisecond)
82 continue
83 }
84 log.Printf("Error reading file %s: %v\n", filepath, err)
85 time.Sleep(1 * time.Second)
86 continue
87 }
88 out <- line
89 }
90 }
91
92 type adguardLogEntry struct {
93     IP      string `json:"IP"`
94     QH      string `json:"QH"`
95     QT      string `json:"QT"`
96     Result struct {
97         IsFiltered bool `json:"IsFiltered"`
98         Reason      int  `json:"Reason"`
99         Rules      []struct {
100             Text string `json:"Text"`
101         } `json:"Rules"`
102     } `json:"Result"`
103 }
104
105 func monitorDNS() {
106     logMessage("[System] Starting DNS block logger...")
107     lines := make(chan string)
108     go tailFile(queryLogPath, lines)
109
110     reasonMap := map[int]string{
111         2: "CustomRule",
112         3: "BlockList",
113         4: "SafeBrowsing",
114         5: "ParentalControl",
115         6: "SafeSearch",
116         7: "BlockedService",
117     }
118
119     for line := range lines {
120         var data adguardLogEntry
121         if err := json.Unmarshal([]byte(line), &data); err != nil {
122             continue
123         }
124
125         res := data.Result
126         if res.IsFiltered || (res.Reason != 0 && res.Reason != 1) {
127             qh := data.QH
128             if qh == "" {
129                 qh = "unknown"
130             }
131             qt := data.QT
132             if qt == "" {
133                 qt = "unknown"
134             }
135             clientIP := data.IP
136             if clientIP == "" {
137                 clientIP = "unknown"
138             }
139
140             ruleText := "unknown rule"
141             if len(res.Rules) > 0 && res.Rules[0].Text != "" {
142                 ruleText = res.Rules[0].Text
143             }
144
145             reasonStr, ok := reasonMap[res.Reason]
146             if !ok {
147                 reasonStr = fmt.Sprintf("Reason-%d", res.Reason)
148             }

```

```

149     logMessage(fmt.Sprintf("[DNS] Blocked %s (Type: %s) for client %s | Reason: %s | Rule: %s", qh, qt,
150         clientIP, reasonStr, ruleText))
151 }
152 }
153 }
154
155 func monitorIPs() {
156     logMessage("[System] Starting IP block logger...")
157
158     prefixRe := regexp.MustCompile(`(IP_BLOCK_[A-Z_]+):`)
159     srcRe := regexp.MustCompile(`SRC=([0-9a-fA-F\.:]+)`)
160     dstRe := regexp.MustCompile(`DST=([0-9a-fA-F\.:]+)`)
161     protoRe := regexp.MustCompile(`PROTO=([A-Z0-9]+)`)
162     sptRe := regexp.MustCompile(`SPT=(\d+)`)
163     dptRe := regexp.MustCompile(`DPT=(\d+)`)
164     inRe := regexp.MustCompile(`IN=([a-zA-Z0-9\.\_]+)`)
165     outRe := regexp.MustCompile(`OUT=([a-zA-Z0-9\.\_]+)`)
166
167     f, err := os.Open(procKmsgPath)
168     if err != nil {
169         if os.IsPermission(err) {
170             logMessage("[System] ERROR: Insufficient permissions to read /proc/kmsg. Enable CAP_SYSLOG.")
171         } else {
172             logMessage(fmt.Sprintf("[System] ERROR in IP logger: %v", err))
173         }
174         return
175     }
176     defer f.Close()
177
178     reader := bufio.NewReader(f)
179     for {
180         line, err := reader.ReadString('\n')
181         if err != nil {
182             if err == io.EOF {
183                 time.Sleep(100 * time.Millisecond)
184                 continue
185             }
186             logMessage(fmt.Sprintf("[System] ERROR reading kmsg: %v", err))
187             time.Sleep(1 * time.Second)
188             continue
189         }
190
191         if strings.Contains(line, "IP_BLOCK") {
192             match := prefixRe.FindStringSubmatch(line)
193             if len(match) > 1 {
194                 prefix := match[1]
195
196                 src := "unknown"
197                 if m := srcRe.FindStringSubmatch(line); len(m) > 1 {
198                     src = m[1]
199                 }
200
201                 dst := "unknown"
202                 if m := dstRe.FindStringSubmatch(line); len(m) > 1 {
203                     dst = m[1]
204                 }
205
206                 proto := "unknown"
207                 if m := protoRe.FindStringSubmatch(line); len(m) > 1 {
208                     proto = m[1]
209                 }
210
211                 spt := ""
212                 if m := sptRe.FindStringSubmatch(line); len(m) > 1 {
213                     spt = m[1]
214                 }
215
216                 dpt := ""
217                 if m := dptRe.FindStringSubmatch(line); len(m) > 1 {
218                     dpt = m[1]
219                 }
220
221                 inIf := ""
222                 if m := inRe.FindStringSubmatch(line); len(m) > 1 {
223                     inIf = m[1]
224                 }
225
226                 outIf := ""
227                 if m := outRe.FindStringSubmatch(line); len(m) > 1 {
228                     outIf = m[1]

```

```

229     }
230
231     direction := "inbound"
232     if strings.HasSuffix(prefix, "DST") {
233         direction = "outbound"
234     }
235
236     listType := "MALICIOUS"
237     if !strings.Contains(prefix, "MALICIOUS") {
238         parts := strings.Split(prefix, "_")
239         if len(parts) >= 3 {
240             listType = "GEO_" + parts[2]
241         }
242     }
243
244     portStr := ""
245     if dpt != "" {
246         portStr = ":" + dpt
247     }
248
249     srcPortStr := ""
250     if spt != "" {
251         srcPortStr = ":" + spt
252     }
253
254     ifStr := ""
255     if inIf != "" && outIf != "" {
256         ifStr = fmt.Sprintf("IF: %s->%s", inIf, outIf)
257     } else if inIf != "" || outIf != "" {
258         ifStr = fmt.Sprintf("IF: %s%s", inIf, outIf)
259     }
260
261     logMessage(fmt.Sprintf("[IP] Blocked %s to %s%s (Proto: %s) from %s%s (%s) | List: %s", direction, dst,
262         portStr, proto, src, srcPortStr, ifStr, listType))
263 }
264 }
265 }
266
267 func main() {
268     logMessage("[System] Nullexit Block Logger starting...")
269
270     go monitorDNS()
271     monitorIPs()
272 }

```

40 scripts/pf.conf

```

1 # scripts/pf.conf - Packet Filter configuration for nullexit killswitch
2 # Dynamically parameterized by pfctl macro override
3
4 # ext_if must be passed via command line, e.g. -D ext_if=en0
5 # If not passed, default to en0
6 ext_if = "en0"
7 ts_if = "utun+"
8
9 # Tables populated dynamically via pfctl
10 table <vpn_endpoints> persist
11 table <derp_relays> persist
12
13 # Core Rules
14 set skip on lo0 # Ensure local loopback is never blocked
15 scrub out all max-mss 1160 # TCP MSS Clamping: Prevents fragmentation on host TCP flows (like Tailscale
16     control plane)
17 pass quick on $ts_if all # Allow all Tailscale traffic (essential to prevent SSH lockout)
18
19 # Default deny outbound WAN traffic
20 block out on $ext_if all
21
22 # Block inbound Screen Sharing (VNC) and SSH from local Wi-Fi, forcing them to use Tailscale
23 block in on $ext_if proto tcp from any to any port { 22, 5900 }
24
25 # Exceptions for tunnel handshakes and coordination
26 pass out quick on $ext_if proto udp to <vpn_endpoints> port 2408
27 pass out quick on $ext_if proto udp to <derp_relays>
28 pass out quick on $ext_if proto tcp to 192.200.0.0/24 port 443

```

```

28
29 # Allow local LAN traffic (AirDrop, local Tailscale P2P, etc)
30 pass out quick on $ext_if proto tcp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
31 pass out quick on $ext_if proto udp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
32 pass out quick on $ext_if proto icmp to { 192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12 }
33
34 # Allow DHCP outbound (client port 68 to server port 67) so we can obtain/renew IP address
35 pass out quick on $ext_if proto udp from any port 68 to any port 67

```

41 scripts/reinstall-host-tailscale.sh

```

1 #!/usr/bin/env bash
2 #
3 # scripts/reinstall-host-tailscale.sh
4 #
5 # Complete uninstall + reinstall of the host-side Homebrew Tailscale.
6 #
7 # Use this when the brew LaunchAgent is wedged, Login Items entries are
8 # duplicated, "Tailscale is stopped" persists despite
9 # `brew services restart tailscale`, or you simply want a clean baseline
10 # before re-engaging the gateway as exit node.
11 #
12 # Idempotent. Safe to re-run. Use --dry to preview every step with zero
13 # changes. Use --yes to skip the per-step confirmation prompts.
14 #
15 # This script ONLY touches the HOST-side Tailscale (the one installed by
16 # Homebrew `brew install tailscale` and managed by launchd as
17 # homebrew.mxcl.tailscale). The nullexit gateway container's tailscaled
18 # (which lives inside the Colima VM at 100.100.21.8 and offers the exit
19 # node to this host) is unaffected.
20
21 set -u
22
23 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
24
25 DRY=0
26 YES=0
27
28 for arg in "$@"; do
29     case "$arg" in
30         --dry) DRY=1 ;;
31         --yes) YES=1 ;;
32         --help|-h) # NOTE: heredoc is UNQUOTED so $SCRIPT_DIR expands in the body. If you
33                 # add new $-style placeholders here, they'll expand the same way.
34                 cat <<USAGE
35 Usage: bash "$SCRIPT_DIR/reinstall-host-tailscale.sh" [--dry] [--yes]
36
37 --dry  Print every step, make zero changes. Use to audit before
38        committing.
39 --yes  Auto-confirm the prompts at every destructive step. Useful in
40        CI / fully-scripted recovery; do not use until you've read
41        the script.
42
43 After this script completes cleanly, you still must (MANUALLY):
44
45 1. Open System Settings Privacy & Security Local Network and
46    enable Tailscale. macOS will prompt automatically the first time
47    the fresh install of `tailscale` is invoked.
48
49 2. Re-engage the tailnet + exit node:
50    sudo tailscale up --ssh=true --accept-dns=false \
51      --exit-node="$(cat .gateway_ip | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
52      --exit-node-allow-lan-access=true
53
54 3. Re-run ./toggle.sh from the project root.
55
56 4. Final verification:
57    curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)='
58    expected: warp=on, ip=<Cloudflare>, colo=YYZ/YUL
59 USAGE
60     exit 0
61     ;;
62     *)
63     echo "Unknown flag: $arg" >&2
64     exit 2
65     ;;

```



```

66     esac
67 done
68
69 # ----- helpers -----
70
71 confirm() {
72     if [ "$YES" = 1 ] || [ "$DRY" = 1 ]; then
73         return 0
74     fi
75     printf "        %s [y/N] " "$1"
76     read -r ans
77     case "$ans" in
78         y|Y|yes|YES) return 0 ;;
79         *)
80             echo "        aborted by user"
81             exit 1
82             ;;
83     esac
84 }
85
86 header() {
87     printf '\n %s\n' "$1"
88 }
89
90 # with_timeout <seconds> <cmd...>
91 # Run a command in the background and kill it if it exceeds <seconds>.
92 # Returns the command's exit code if it finished in time, 124 (matching
93 # GNU `timeout` convention) if it timed out. macOS doesn't ship GNU
94 # coreutils `timeout`, so this is the bash-3.2-friendly replacement.
95 # stdout + stderr are discarded (we only care about exit code); redirect
96 # or pipe externally if you need the output.
97 with_timeout() {
98     local timeout_s="${1:-30}"
99     shift
100     "$@" >/dev/null 2>&1 &
101     local cmdpid=$!
102     local elapsed=0
103     while kill -0 "$cmdpid" 2>/dev/null; do
104         if [ "$elapsed" -ge "$timeout_s" ]; then
105             kill -9 "$cmdpid" 2>/dev/null
106             wait "$cmdpid" 2>/dev/null
107             return 124
108         fi
109         sleep 1
110         elapsed=$((elapsed+1))
111     done
112     wait "$cmdpid" 2>/dev/null
113     return $?
114 }
115
116 # ----- 0. Pre-flight -----
117 header "Pre-flight current host tailscale state"
118 echo "    brew: $(command -v brew >/dev/null 2>&1 && brew --version | head -1 || echo 'not on
119     PATH')"
119 echo "    tailscaled running: $(pgrep -lf tailscaled 2>/dev/null | wc -l | tr -d ' ') process(es)"
120 echo "    brew service tailscale: $(brew services list 2>/dev/null | awk ' $1=="tailscale" {print $2, $3}' | tr
121     '\n' ' ' | sed 's/ $//')
121 echo "    LaunchAgent plists (under ~/Library/LaunchAgents/):"
122 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's/~/ /' || echo "    (none)"
123
124 # Detect Tailscale's macOS Network Extension (residual from prior App Store
125 # Tailscale.app installs). Lives under /Library/Apple/SystemExtensions/ and
126 # is managed by `systemextensionsctl` NOT launchd. If loaded, brew
127 # tailscaled will NOT engage the Network Extension framework slot because
128 # the SE still has it claimed. Symptom: 'sudo tailscale status' returns
129 # "Tailscale is stopped" even when the daemon is alive and listening.
130 se_found=0
131 if command -v systemextensionsctl >/dev/null 2>&1; then
132     # Grab any line matching tailscale and network-extension
133     se_line=$(systemextensionsctl list 2>/dev/null | grep -iE '(io|com)\.tailscale\..*network-extension' | head -
134         n 1)
134     if [ -n "$se_line" ]; then
135         se_found=1
136         # Extract the teamID (10-char uppercase/numeric word)
137         se_teamID=$(echo "$se_line" | tr -s ' \t' '\n' | grep -E '^[A-Z0-9]{10}$' | head -n 1)
138         # Extract the bundleID (starts with io.tailscale or com.tailscale)
139         se_bundleID=$(echo "$se_line" | tr -s ' \t' '\n' | grep -E '^(io|com)\.tailscale\.' | head -n 1 | sed 's
140             /(.*//')
141         # If teamID is not found, fallback to '-'
142         if [ -z "$se_teamID" ]; then

```

```

143     se_teamID="-"
144     fi
145 fi
146 fi
147
148 if [ "$se_found" = 1 ] && [ -n "$se_bundleID" ]; then
149     echo "    Tailscale Network Extension System Extension is loaded (residual from App Store Tailscale.app
install)"
150     echo "    Team ID: $se_teamID, Bundle ID: $se_bundleID"
151     echo "    brew tailscaled cannot engage the NE slot while this SE has it claimed."
152     if [ "$SDRY" = 1 ]; then
153         echo "    [dry] (would prompt) sudo systemextensionsctl uninstall $se_teamID $se_bundleID"
154     else
155         confirm "Run 'sudo systemextensionsctl uninstall $se_teamID $se_bundleID'? (frees the NE slot for brew
tailscale; non-fatal if it errors)"
156         sudo systemextensionsctl uninstall "$se_teamID" "$se_bundleID" 2>&1 \
157         || echo "    ! uninstall returned non-zero (may require reboot, non-fatal)"
158     fi
159 else
160     echo "    no Tailscale System Extension loaded"
161 fi
162
163 # Pattern used by the Step-1 bootout loops, in both dry + real branches.
164 # Widened from a bare /tailscale/ so the same loops also pick up any
165 # nullexit-labeled LaunchAgents (e.g., com.nullexit.wake-recovery, the
166 # caffeinate + post-wake recovery hook). The wake-recovery LaunchAgent
167 # is re-installed by ./toggle.sh / setup.sh on re-engage, so wiping it
168 # here is non-fatal and is part of restoring a clean baseline.
169 TAILSCALE_LABEL_RE='tailscale'
170 NULLEXIT_LABEL_RE='nullexit'
171
172 # ----- 1. brew services stop -----
173 header "Step 1 Stop the brew-managed service + unload LaunchAgents"
174 # Detect whether the brew-managed service is registered as $USER (gui/$UID
175 # domain) or as root (system domain). NOPASSWD sudoers can leave a stale
176 # root-owned registration after a prior 'sudo brew services ...' round;
177 # 'brew services stop' without sudo refuses to touch a root-owned plist
178 # (Error: Service 'tailscale' is started as 'root').
179 brew_status_line="$(brew services list 2>/dev/null | awk '$1=="tailscale"')"
180 brew_status_user="$(echo "$brew_status_line" | awk '{print $3}')"
181 if [ -n "$brew_status_line" ]; then
182     if [ "$SDRY" = 1 ]; then
183         echo "    [dry] brew services stop tailscale (user=$brew_status_user)"
184     else
185         if [ "$brew_status_user" = "root" ]; then
186             confirm "Run 'sudo brew services stop tailscale' (service is registered as root)?"
187             sudo brew services stop tailscale 2>&1 || echo "    (sudo brew services stop returned non-zero non-fatal
)"
188         else
189             echo "    brew services stop tailscale (user=$brew_status_user, no sudo needed)"
190             brew services stop tailscale 2>&1 || echo "    (brew reported it wasn't actually running)"
191         fi
192     fi
193     echo "    unload any tailscale-related LaunchAgents (prevents launchd from respawning after kill)"
194     if [ "$SDRY" = 1 ]; then
195         echo "    [dry] for each 'tailscale|nullexit'-matched label in gui/$UID domain:"
196         while IFS= read -r label; do
197             [ -z "$label" ] && continue
198             echo "    [dry] launchctl bootout gui/$UID/$label"
199             done < <(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 &&
($3 ~ ts || $3 ~ nx) {print $3}')
200             echo "    [dry] for each 'tailscale|nullexit'-matched label in system domain (root-owned brew plist):"
201             while IFS= read -r label; do
202                 [ -z "$label" ] && continue
203                 echo "    [dry] (would prompt) sudo launchctl bootout system/$label"
204                 done < <(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" '
NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
205                 if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
206                     echo "    [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macos"
207                 fi
208                 if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
209                     echo "    [dry] (would prompt) sudo launchctl bootout system/com.tailscale.ipn.macsys"
210                 fi
211             else
212                 # Bootout every gui-domain LaunchAgent whose label matches 'tailscale'
213                 # OR 'nullexit'. The nullexit arm catches com.nullexit.wake-recovery,
214                 # which holds caffeinate/watcher.sh alive across a daemon reset not
215                 # desired while the script is establishing a fresh baseline. Awake-
216                 # recovery is re-installed by ./toggle.sh / setup.sh on re-engage, so
217                 # wiping here is non-fatal. The pattern stays narrower than a bare
218                 # /tail/ or /null/ so unrelated Apple services are not unloaded.

```

```

219 while IFS= read -r label; do
220     [ -z "$label" ] && continue
221     echo "      launchctl bootout gui/$UID/$label"
222     launchctl bootout "gui/$UID/$label" 2>/dev/null || true
223 done < <(launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" 'NR > 1 &&
($3 ~ ts || $3 ~ nx) {print $3}')
224 # Bootout every system-domain LaunchDaemon/Agent whose label matches
225 # 'tailscale' OR 'nullexit'. This catches the root-owned brew plist left
226 # behind by prior 'sudo brew services ...' invocations that was the
227 # case that left tailscaled PID 47447 57734 respawning between sudo
228 # pkill runs. Defensive nullexit catch in case the user previously
229 # `sudo ln -s`'d the wake-recovery plist into /Library/LaunchDaemons.
230 while IFS= read -r label; do
231     [ -z "$label" ] && continue
232     confirm "Run 'sudo launchctl bootout system/$label'?"
233     sudo launchctl bootout "system/$label" 2>/dev/null || true
234 done < <(sudo -n launchctl list 2>/dev/null | awk -v ts="$TAILSCALE_LABEL_RE" -v nx="$NULLEXIT_LABEL_RE" '
NR > 1 && ($3 ~ ts || $3 ~ nx) {print $3}')
235 # Legacy Tailscale.app system-domain entries. Each one is checked and
236 # confirmed separately so we don't sudo-prompt or sudo-bootout for nothing.
237 if launchctl print system/com.tailscale.ipn.macos >/dev/null 2>&1; then
238     confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macos'?"
239     sudo launchctl bootout system/com.tailscale.ipn.macos 2>/dev/null || true
240 fi
241 if launchctl print system/com.tailscale.ipn.macsys >/dev/null 2>&1; then
242     confirm "Run 'sudo launchctl bootout system/com.tailscale.ipn.macsys'?"
243     sudo launchctl bootout system/com.tailscale.ipn.macsys 2>/dev/null || true
244 fi
245 fi
246 echo "      waiting up to 10s for tailscaled to exit"
247 for i in 1 2 3 4 5 6 7 8 9 10; do
248     if [ "$DRY" = 1 ]; then
249         echo "      [dry] sleep 1; loop check would terminate here"
250         break
251     fi
252     sleep 1
253     if ! pgrep -lf tailscaled >/dev/null 2>&1; then
254         echo "      tailscaled exited in ${i}s"
255         break
256     fi
257     [ "$i" = 10 ] && echo "      tailscaled still alive after 10s Step 2 will deal with it"
258 done
259 else
260     echo "      brew reports no tailscale service loaded"
261 fi
262
263 # ----- 2. Kill any orphan tailscaled -----
264 header "Step 2 Kill any orphan tailscaled process(es) (may escalate to sudo)"
265 if ! pgrep -lf tailscaled >/dev/null 2>&1; then
266     echo "      no orphans"
267 else
268     echo "      Stray process(es):"
269     pgrep -lf tailscaled | sed 's/^/      /'
270     echo "      no-sudo: pkill tailscaled"
271     if [ "$DRY" = 1 ]; then
272         echo "      [dry] pkill tailscaled on failure: sudo pkill -9 tailscaled"
273     else
274         if pkill tailscaled 2>/dev/null; then
275             echo "      no-sudo pkill succeeded"
276         else
277             echo "      ! no-sudo pkill insufficient; cannot escalate yet"
278         fi
279         sleep 1
280     fi
281 fi
282 # Recheck; if still alive, offer sudo escalation
283 if { [ "$DRY" = 0 ] && pgrep -lf tailscaled >/dev/null 2>&1; } || [ "$DRY" = 1 ]; then
284     if [ "$DRY" = 1 ]; then
285         echo "      [dry] sudo pkill -9 tailscaled (escalation would fire here)"
286     else
287         confirm "Run 'sudo pkill -9 tailscaled' to escalate?"
288         sudo pkill -9 tailscaled 2>&1
289         sleep 1
290     fi
291 fi
292
293 if [ "$DRY" = 0 ]; then
294     if pgrep -lf tailscaled >/dev/null 2>&1; then
295         echo "      Failed to kill stragglers bailing out"
296         pgrep -lf tailscaled | sed 's/^/      /'
297         exit 3

```

```

298     fi
299     echo "        all tailscaled processes gone"
300 fi
301 fi
302
303 # ----- 3. brew uninstall -----
304 header "Step 3  brew uninstall tailscale"
305 if ! brew list tailscale >/dev/null 2>&1; then
306     echo "        not currently installed via brew  skip"
307 else
308     # Detect whether the keg directory is owner=$USER or owner=root. If the
309     # keg is root-owned (typically a residual of some prior 'sudo brew ...'
310     # cycle that did perms fixups), 'brew uninstall' without sudo refuses
311     # to remove the files and prints: 'Error: Could not remove tailscale
312     # keg! Do so manually: sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>'.
313     # Detect that and escalate to a sudoed uninstall, with a manual rm -rf
314     # fallback if `sudo brew uninstall` itself errors out (e.g., partial
315     # state where the keg dir is gone but brew still thinks it's installed).
316     # Detect keg state. Treat `(dir absent)` AND `(dir present + empty)` as the
317     # same `<unknown>` bucket so both escalate to sudo. The empty-parent case
318     # is what you get after a manual `sudo rm -rf /opt/homebrew/Cellar/tailscale/<v>`
319     # from inside: brew's parent dir is left present-but-empty and brew still
320     # thinks the package is installed.
321     keg_dir="/opt/homebrew/Cellar/tailscale"
322     if [ -d "$keg_dir" ] && [ -n "$(ls -A "$keg_dir" 2>/dev/null)" ]; then
323         keg_owner="$(stat -f '%Su' "$keg_dir" 2>/dev/null || echo '<unknown>')"
324     else
325         keg_owner="<unknown>"
326     fi
327     if [ "$DRY" = 1 ]; then
328         echo "        [dry] brew uninstall tailscale (keg_owner=$keg_owner)"
329     else
330         # 'root' AND '<unknown>' both require sudo escalation. The '<unknown>'
331         # case is the partial-state path: keg dir was already removed (manually
332         # or by a prior failed uninstall) but brew's internal state still says
333         # installed. Without this branch the script would fall through to a
334         # non-sudo brew uninstall that fails again with the same 'Could not
335         # remove tailscale keg!' error.
336         if [ "$keg_owner" = "root" ] || [ "$keg_owner" = "<unknown>" ]; then
337             confirm "Run 'sudo brew uninstall tailscale' (keg=$keg_owner)?"
338             sudo brew uninstall tailscale 2>&1 || {
339                 echo "        ! sudo brew uninstall failed; offering manual rm -rf fallback"
340                 # Guard the glob so an empty/absent dir doesn't literal-expand the
341                 # '*' and confuse the user with a sudo error about a non-existent path.
342                 if [ -d /opt/homebrew/Cellar/tailscale ] && [ -n "$(ls -A /opt/homebrew/Cellar/tailscale 2>/dev/null)" ]; then
343                     confirm "Run 'sudo rm -rf /opt/homebrew/Cellar/tailscale/*'?"
344                     sudo rm -rf /opt/homebrew/Cellar/tailscale/* 2>&1
345                 else
346                     echo "        keg dir already empty or absent  nothing more to rm"
347                 fi
348             }
349         else
350             confirm "Run 'brew uninstall tailscale' (removes LaunchAgent plist + linked symlinks)?"
351             brew uninstall tailscale 2>&1
352         fi
353     fi
354 fi
355
356 # ----- 4. Wipe state directories -----
357 header "Step 4  Wipe stale state"
358 # User-owned
359 declare -a USER_PATHS
360 USER_PATHS=(
361     "$HOME/.local/share/tailscale"
362     "$HOME/.local/run/tailscaled.socket"
363     "$HOME/.local/run/tailscaled.state"
364     "$HOME/.local/state/tailscale"
365     "$HOME/Library/Application Support/Tailscale"
366     "$HOME/Library/Caches/com.tailscale.ipn.macos"
367     "$HOME/Library/Caches/Tailscale"
368     "$HOME/Library/Logs/Tailscale"
369 )
370 echo "        User-owned paths (no sudo):"
371 for p in "${USER_PATHS[@]}; do
372     if [ -e "$p" ]; then
373         if [ "$DRY" = 1 ]; then
374             echo "        [dry] rm -rf '$p'"
375         else
376             rm -rf "$p"
377             echo "        rm -rf '$p'"

```

```

378     fi
379 fi
380 done
381
382 # System
383 declare -a SUDO_PATHS
384 SUDO_PATHS=(
385     "/var/lib/tailscale"
386     "/var/cache/tailscale"
387 )
388 echo "    System paths under /var (will prompt for sudo if any are present):"
389 eligible_sudo=0
390 for p in "${SUDO_PATHS[@]"; do
391     if [ -e "$p" ]; then
392         eligible_sudo=1
393         if [ "$DRY" = 1 ]; then
394             echo "        [dry] sudo rm -rf '$p'"
395         else
396             if confirm "Run 'sudo rm -rf $p'?" ; then
397                 sudo rm -rf "$p" && echo "            rm -rf '$p'"
398             fi
399         fi
400     fi
401 done
402 [ "$eligible_sudo" = 0 ] && echo "        (none of /var/lib/tailscale, /var/cache/tailscale are present)"
403
404 # ----- 5. Wipe stale LaunchAgent plists -----
405 header "Step 5 Wipe stale LaunchAgent plist files"
406 declare -a PLIST_FILES
407 PLIST_FILES=(
408     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
409     "$HOME/Library/LaunchAgents/com.tailscale.ipn.macos.plist"
410     "$HOME/Library/LaunchAgents/homebrew.mxcl.tailscale.plist.bak"
411     "$HOME/Library/LaunchAgents/com.nullexit.wake-recovery.plist"
412 )
413 for f in "${PLIST_FILES[@]"; do
414     if [ -e "$f" ]; then
415         if [ "$DRY" = 1 ]; then
416             echo "        [dry] rm -f '$f'"
417         else
418             rm -f "$f"
419             echo "            rm -f '$f'"
420         fi
421     fi
422 done
423 echo "    Remaining tailscale|nullexit plists (should be empty):"
424 ls -l ~/Library/LaunchAgents/ 2>/dev/null | grep -i 'tail' | sed 's/~/ /' || echo "        (none)"
425
426 # ----- 5.5. Drain Background-Items (Login Items) -----
427 header "Step 5.5 Drain Background-Items (Login Items)"
428 # macOS 13+ keeps a parallel "Background Items" database alongside
429 # classic LaunchAgents. Even when we wipe ~/Library/LaunchAgents/homebrew.mxcl.tailscale.plist
430 # in Step 5, the .btm file at
431 # ~/Library/Application Support/com.apple.backgroundtaskmanagementagent.backgrounditems.btm
432 # can still hold a Background-Items entry pointing at the (now-dead) plist
433 # path. Next login: macOS sees the entry, tries to bootstrap, finds the
434 # plist missing, and either silently regenerates it via brew's post-install
435 # hook (causing tailscaled respawn at next login) or refuses outright.
436 # `sfltool reset-login-items` is Apple's canonical API for wiping the
437 # entire .btm file. It DOES wipe all Login Items the user re-adds what
438 # they want via System Settings General Login Items. Acceptable here
439 # because the goal of this script is "fresh baseline for Tailscale".
440 BTM="$HOME/Library/Application Support/com.apple.backgroundtaskmanagementagent/backgrounditems.btm"
441 if [ -f "$BTM" ]; then
442     if [ "$DRY" = 1 ]; then
443         echo "        [dry] found $BTM; would prompt sfltool reset-login-items"
444     else
445         confirm "Run 'sfltool reset-login-items' (wipes ALL Login Items; rebuild via System Settings General Login Items)?"
446         if sfltool reset-login-items 2>&1; then
447             echo "            Login Items drained"
448         else
449             echo "        ! sfltool reset-login-items failed (non-fatal; do manually: System Settings General Login Items)"
450         fi
451     fi
452 else
453     echo "        no backgrounditems.btm present"
454 fi
455
456 # ----- 6. brew install -----

```

```

457 header "Step 6 brew install tailscale (fresh)"
458 if brew list tailscale >/dev/null 2>&1; then
459     echo "    already installed skip"
460 else
461     confirm "Run 'brew install tailscale'?"
462     if [ "$DRY" = 1 ]; then
463         echo "    [dry] brew install tailscale"
464     else
465         brew install tailscale 2>&1
466     fi
467 fi
468
469 # ----- 6.5. Strip quarantine xattr from keg -----
470 header "Step 6.5 Strip quarantine xattr from tailscale install"
471 # brew's freshly poured bottles don't carry com.apple.quarantine (the
472 # bottles are content-addressed and Homebrew-verified), but the upstream
473 # installer (Tailscale.app from App Store) and any manual download do.
474 # Strip the xattr from the entire keg as a defensive safety net so the
475 # freshly-installed tailscaled binary and any helper binaries are not
476 # Gatekeeper-blocked on first launch. Idempotent no-op if no quarantine
477 # xattr is present.
478 quarantine_count="$(xattr -lr /opt/homebrew/Cellar/tailscale 2>/dev/null | grep -c '^[^]* com.apple.quarantine'
479 ' || echo 0)"
480 if [ "${quarantine_count:-0}" -gt 0 ]; then
481     if [ "$DRY" = 1 ]; then
482         echo "    [dry] found $quarantine_count file(s) with com.apple.quarantine; would prompt xattr -dr"
483     else
484         confirm "Run 'xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale'? (strips quarantine xattr from
485         $quarantine_count file(s); non-fatal if it errors)"
486         if xattr -dr com.apple.quarantine /opt/homebrew/Cellar/tailscale 2>&1; then
487             echo "    quarantine xattr stripped"
488         else
489             echo "    ! xattr -dr failed (non-fatal; open /opt/homebrew/Cellar/tailscale in Finder, right-click
490             tailscaled Open Anyway)"
491         fi
492     fi
493 else
494     echo "    no com.apple.quarantine xattr on tailscale install"
495 fi
496
497 # ----- 7. brew services start (NO sudo!) -----
498 header "Step 7 Start the service as $USER (NO sudo, with multi-tier self-healing)"
499 if ! command -v tailscale >/dev/null 2>&1; then
500     echo "    tailscale not on PATH after install brew install errored; aborting"
501     exit 4
502 fi
503
504 # Aliveness fan-out: ANY one of these is taken as proof of life. None of
505 # them alone is reliable pgrep misses tailscaled if it does setproctitle
506 # or fork-detach; launchctl's reported PID can be a shim that exited
507 # within 1s; tailscale-cli hangs while macOS Local-Network permission is
508 # unresolved. Combined, they cover each other's blind spots.
509
510 tailscaled_api_up() {
511     # Strategy A: tailscale CLI can talk to the local API. Canonical proof.
512     # Bound at 8s tailscale status itself returns quickly on success OR
513     # failure, but if the post-install state is wedged it can hang.
514     with_timeout 8 /opt/homebrew/bin/tailscale status >/dev/null 2>&1
515 }
516
517 tailscaled_proc_up() {
518     # Strategy B: any process has 'tailscaled' substring in argv.
519     pgrep -lf tailscaled >/dev/null 2>&1
520 }
521
522 tailscaled_lc_up() {
523     # Strategy C: launchctl-managed service has a non-dash, non-zero PID.
524     local lc_pid
525     lc_pid="$(launchctl list 2>/dev/null | awk '$3 == "homebrew.mxcl.tailscale" {print $1; exit}')"
526     if [ -n "$lc_pid" ] && [ "$lc_pid" != "-" ] && [ "$lc_pid" -gt 0 ] 2>/dev/null; then
527         return 0
528     fi
529     return 1
530 }
531
532 tailscaled_is_alive() {
533     tailscaled_api_up && return 0
534     tailscaled_proc_up && return 0
535     tailscaled_lc_up && return 0
536     return 1
537 }

```

```

535 confirm "Run 'brew services start tailscale'?"
536 if [ "$DRY" = 1 ]; then
537     echo " [dry] brew services start tailscale"
538     echo " [dry] (then poll up to 30s for tailscaled_is_alive)"
539     echo " [dry] (recovery tier 1: launchctl kickstart)"
540     echo " [dry] (recovery tier 2: launchctl bootout + bootstrap of explicit plist)"
541     echo " [dry] (recovery tier 3: brew services restart (stop+start cycle))"
542     echo " [dry] (recovery tier 4: direct /opt/homebrew/bin/tailscaled spawn (bypasses launchd))"
543 else
544     brew services start tailscale 2>&1
545     # Primary poll tailscaled_is_alive covers all 3 detection strategies.
546     echo " polling for tailscaled to come up (up to ~30s)"
547     tailscaled_up=0
548     for i in 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15; do
549         if tailscaled_is_alive; then
550             tailscaled_up=1
551             echo " tailscaled alive after ${i}*2s"
552             break
553         fi
554         sleep 2
555     done
556
557     # Recovery tier 1: kickstart the launchd-managed service. This is the
558     # cheapest and most-aligned-with-launchd fix when brew has loaded the
559     # LaunchAgent but tailscaled didn't fork. Kickstart tells launchd
560     # "re-spawn if not running" without forcing a teardown.
561     if [ "$tailscaled_up" = 0 ]; then
562         echo " ! Recovery tier 1: launchctl kickstart"
563         if launchctl kickstart -kp "gui/$UID/homebrew.mxcl.tailscale" 2>&1; then
564             for i in 1 2 3 4 5 6 7 8 9 10; do
565                 if tailscaled_is_alive; then
566                     tailscaled_up=1
567                     echo " tailscaled alive after kickstart (${i}*2s)"
568                     break
569                 fi
570                 sleep 2
571             done
572         else
573             echo " ! kickstart returned non-zero (LaunchAgent may not be loaded)"
574         fi
575     fi
576
577     # Recovery tier 2: explicit plist bootout + bootstrap. Forces a fresh
578     # launchd registration cycle for the brew-managed service, which fixes
579     # the wider class of "brew says started but daemon never spawned"
580     # wedge states that kickstart alone can't dig out of.
581     if [ "$tailscaled_up" = 0 ]; then
582         echo " ! Recovery tier 2: launchctl bootout + bootstrap of explicit plist"
583         launchctl bootout "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null || true
584         sleep 1
585         # Brew installs the canonical plist template at $HOMEBREW_PREFIX/Library/.
586         # On Apple Silicon that's /opt/homebrew/Library/LaunchAgents/. On Intel
587         # it's /usr/local/Library/LaunchAgents/. Detect which and pick the
588         # right one so this script works across both architectures.
589         if [ -f /opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
590             plist_path="/opt/homebrew/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
591         elif [ -f /usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist ]; then
592             plist_path="/usr/local/Library/LaunchAgents/homebrew.mxcl.tailscale.plist"
593         else
594             plist_path=""
595         fi
596         if [ -n "$plist_path" ]; then
597             if launchctl bootstrap "gui/$UID" "$plist_path" 2>&1; then
598                 for i in 1 2 3 4 5 6 7 8 9 10; do
599                     if tailscaled_is_alive; then
600                         tailscaled_up=1
601                         echo " tailscaled alive after bootstrap (${i}*2s)"
602                         break
603                     fi
604                     sleep 2
605                 done
606             else
607                 echo " ! explicit bootstrap returned non-zero"
608             fi
609         else
610             echo " ! could not locate brew plist template; skipping bootstrap tier"
611         fi
612     fi
613
614     # Recovery tier 3: brew services restart (forces a stop + start cycle,
615

```



```

616 # which unsticks LaunchAgents with stale 'started' state from prior
617 # half-cycles). This is the canonical brew-doc fix for 'services don't
618 # actually start even though brew says started'.
619 if [ "$tailscaled_up" = 0 ]; then
620     echo "    ! Recovery tier 3: brew services restart (forces stop+start cycle)"
621     if brew services restart tailscale 2>&1; then
622         for i in 1 2 3 4 5 6 7 8 9 10; do
623             if tailscaled_is_alive; then
624                 tailscaled_up=1
625                 echo "        tailscaled alive after brew restart (${i}*2s)"
626                 break
627             fi
628             sleep 2
629         done
630     else
631         echo "        ! brew services restart returned non-zero"
632     fi
633 fi
634
635 # Recovery tier 4: last-resort direct spawn of the daemon binary,
636 # bypassing launchd entirely. If tailscaled crashes here too, we'll
637 # capture its actual stderr in /tmp for the user to read (NOPASSWD
638 # sudoers often have local-network permission denied the captured
639 # stderr makes it obvious what to fix in System Settings).
640 if [ "$tailscaled_up" = 0 ]; then
641     echo "    ! Recovery tier 4: direct spawn (bypasses launchd; tail stderr to /tmp)"
642     /opt/homebrew/bin/tailscaled >/tmp/nullexit-tailscaled-spawn.log 2>&1 &
643     spawned_pid=$!
644     disown "$spawned_pid" 2>/dev/null || true
645     sleep 3
646     if tailscaled_is_alive; then
647         tailscaled_up=1
648         echo "        tailscaled alive via direct spawn (pid=$spawned_pid)"
649         echo "        (NOTE: not under launchd supervision a reboot, re-run of"
650         echo "        this script, or ./toggle.sh is required to recreate."
651         echo "        Last 5 lines of stderr captured to /tmp/nullexit-tailscaled-spawn.log:)"
652         tail -n 5 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/~/ /' || true
653     fi
654 fi
655
656 if [ "$tailscaled_up" = 0 ]; then
657     echo "    HARD FAIL: tailscaled refuses to stay alive across every recovery tier"
658     echo "    launchctl-managed state (if available):"
659     launchctl print "gui/$UID/homebrew.mxcl.tailscale" 2>/dev/null \
660     | grep -E 'state =|last exit code =|runs =|path =' \
661     | sed 's/~/ /' \
662     || echo "        (no launchctl state available for homebrew.mxcl.tailscale)"
663     echo "    Last 10 lines of /tmp/nullexit-tailscaled-spawn.log (if any):"
664     tail -n 10 /tmp/nullexit-tailscaled-spawn.log 2>/dev/null | sed 's/~/ /' || echo "        (no log)"
665     echo "    Most likely remaining cause: macOS Local Network permission denied."
666     echo "    System Settings Privacy & Security Local Network toggle Tailscale ON."
667     exit 7
668 fi
669 fi
670 sleep 3
671
672 # ----- 8. Verify -----
673 header "Step 8 Verify fresh install"
674 echo "    brew service tailscale: $(brew services list 2>/dev/null | awk '$1=="tailscale"{{print $2, $3}}' | tr
675 '\n' ' ' | sed 's/ $//')"
```

```

694 if { [ "$DRY" = 0 ] && /opt/homebrew/bin/tailscale status 2>/dev/null | grep -q '^Tailscale is stopped'; };
    then
695     echo
696     echo "      NOTICE: 'tailscale status' still reports stopped after a fresh install."
697     echo "      Two equally likely causes:"
698     echo "      (a) macOS hasn't yet shown the first-run 'Local Network' prompt for"
699     echo "          Tailscale open System Settings Privacy & Security Local
700     echo "          Network. Toggle Tailscale ON."
701     echo "      (b) macOS denied Local Network permission at some prior point."
702     echo "          Either way: System Settings Privacy & Security Local Network."
703     echo "          If the entry isn't present, invoking any tailscale command will"
704     echo "          re-trigger the prompt automatically."
705 fi
706
707 # ----- 9. Done -----
708 header "Done"
709 if [ "$DRY" = 1 ]; then
710     echo "      (dry-mode finished; no follow-up commands were issued.)"
711     exit 0
712 fi
713
714 cat <<'DONE'
715
716 Manual follow-ups (all of these; the script stops short of running sudo
717 tailscale up so you can verify the brew install is actually live first):
718
719 1. System Settings Privacy & Security Local Network Tailscale ON
720    macOS will normally pop the prompt the first time you run `tailscale`.
721    If it didn't, trigger it via the menu bar check.
722
723 2. Re-engage the tailnet + exit node:
724    sudo tailscale up --ssh=true --accept-dns=false \
725    --exit-node="$(cat .gateway_ip | tr -d '\r' | awk 'NR==1{print $1; exit}')" \
726    --exit-node-allow-lan-access=true
727
728 3. Re-run: ./toggle.sh
729    (re-installs the com.nullexit.wake-recovery LaunchAgent wiped
730    in Step 5 to clear the slate and re-engages the gateway)
731
732 4. Final verification:
733    curl -s https://www.cloudflare.com/cdn-cgi/trace | grep -E '^(warp|ip|colo)= '
734    expect: warp=on, ip=<Cloudflare IP>, colo=YYZ/YUL
735
736 Container-side (gateway / 100.100.21.8) Tailscale is unaffected by this
737 script. Other tailnet devices (S24, iPhone, Windows) are unaffected.
738
739 DONE

```

42 scripts/routing-fix.sh

```

1  #!/bin/sh
2  # routing-fix: Maintains routing for SOCKS5 proxy traffic through WARP (tun0)
3  # and FORWARD rules for Tailscale exit-node return traffic.
4  #
5  # Runs inside the warp container's network namespace. This script ensures:
6  # 1. Table 200 has default via tun0 (for SOCKS5 proxy traffic from 172.18.0.2)
7  # with the WARP endpoint exception via eth0 (prevents tunnel loop)
8  # 2. The CGNAT ip rule (100.64.0.0/10 table 52 tailscale routes) is maintained for Tailscale
9  # 3. FORWARD RELATED,ESTABLISHED rule allows return traffic for exit-node
10 # forwarded connections (S24 tailscale0 eth0 host internet)
11 #
12 # CRITICAL: Does NOT touch the main table's default route. Gluetun handles that
13 # internally via its own routing rules and WireGuard fwmark (0xca6c table 51820).
14 # Overriding the main table default would create a loop: encrypted WireGuard
15 # packets exiting tun0 would match default dev tun0 and re-enter the tunnel.
16
17 set -e
18
19 trap 'echo "routing-fix: Caught signal, exiting..."; exit 0' TERM INT QUIT
20
21 sleep 15 &
22 wait $! || true
23
24 echo "routing-fix: Setting up routing for SOCKS5 proxy through WARP (tun0)..."
25
26 # Dynamically detect Docker subnet from eth0

```

```

27 DOCKER_NET=$(ip route show dev eth0 | grep -v default | head -1 | awk '{print $1}')
28 DOCKER_GW=$(ip route show default | head -1 | awk '{print $3}')
29 WARP_ENDPOINT="${WARP_ENDPOINT}"
30 TAILSCALE_ALLOW_LAN_P2P="${TAILSCALE_ALLOW_LAN_P2P:-false}"
31 IP_BLOCKLIST_FILE="/userfilters/ip_blocklist.ipset"
32 LAST_IP_MTIME=""
33 echo "routing-fix: Docker network: ${DOCKER_NET}, gateway: ${DOCKER_GW}"
34
35 # Table 200: Used by ip rule 100 (from 172.18.0.2) for SOCKS5 proxy traffic
36 # WARP endpoint goes via eth0 to avoid tunnel loop, everything else via tun0
37 ip route add "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || \
38 ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
39 ip route replace default dev tun0 table 200 2>/dev/null || true
40
41 # Main table: Just ensure the Docker subnet route exists via eth0
42 # (gluetun owns the default route here do NOT touch it)
43 ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
44
45 # FORWARD RELATED, ESTABLISHED: Allow return traffic for exit-node connections.
46 # The container has BOTH iptables (nftables backend) and iptables-legacy
47 # loaded. Gluetun's post-rules.txt uses the nftables backend, while Tailscale
48 # uses the legacy backend. Packets go through BOTH stacks, so we add the rule
49 # to both to ensure it survives regardless of which backend has policy DROP.
50 #
51 # Packet flow: S24 outgoing (tailscale0->eth0) ACCEPTed by first rule.
52 # Return traffic (eth0->eth0 via table 52 to host) needs RELATED, ESTABLISHED
53 # to match the conntrack entry created by the outgoing flow.
54 add_fwd_related_established() {
55     # nftables backend (used by gluetun's post-rules.txt)
56     if ! iptables -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
57         iptables -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
58     fi
59     # legacy backend (used by Tailscale's ts-forward)
60     if command -v iptables-legacy >/dev/null 2>&1; then
61         if ! iptables-legacy -C FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null; then
62             iptables-legacy -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT 2>/dev/null || true
63         fi
64     fi
65 }
66
67 add_fwd_related_established
68
69 # Policy Routing for Tailscale P2P:
70 # Tailscale sends its encrypted mesh UDP packets from port 41642.
71 # If these go out tun0 (WARP), Cloudflare's strict NAT drops the returning
72 # hole-punch packets, causing Tailscale to fail P2P and fall back to DERP relays (high latency).
73 # We mark packets originating from port 41642 and force them out the main table (eth0).
74 #
75 # LAN P2P is controlled by two sources (in priority order):
76 # 1. /app/.lan_p2p_detected written by watcher.sh on macOS on every network change
77 #    using airport -I (802.1x detection) + AP isolation probe. Overrides .env.
78 # 2. TAILSCALE_ALLOW_LAN_P2P env var (from .env) static fallback.
79 # When disabled, we explicitly DROP local RFC1918-destined Tailscale UDP so the
80 # phone cleanly falls back to DERP rather than getting poisoned by macOS gvpnproxy SNAT.
81 add_tailscale_p2p_bypass() {
82     # Dynamic override from watcher.sh (network auto-detection) takes priority over .env
83     local allow_p2p="${TAILSCALE_ALLOW_LAN_P2P:-false}"
84     if [ -f "/app/.lan_p2p_detected" ]; then
85         allow_p2p=$(cat "/app/.lan_p2p_detected" 2>/dev/null | tr -d '[:space:]')
86     fi
87
88     # Always allow P2P directly to the Docker host gateway (the Mac running this container).
89     # The gateway container is always co-located with the host Mac on the same machine,
90     # so direct Tailscale P2P via the Docker bridge is always safe and avoids DERP overhead.
91     # This RETURN rule must be inserted BEFORE the general DROP rules so it takes priority.
92     local default_gw
93     default_gw=$(ip route show default | awk '{print $3}')
94     if [ -n "$default_gw" ]; then
95         if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null; then
96             iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${default_gw}/32" -j RETURN 2>/dev/null || true
97         fi
98     fi
99
100     # Also explicitly whitelist all physical IPs of the host Mac (e.g. 172.17.52.223).
101     # The Tailscale control plane registers the Mac using its physical IP, so the
102     # container will attempt P2P handshakes using that physical IP instead of the
103     # Docker bridge IP. If this physical IP falls within 172.16.0.0/12, it would be dropped.
104     if [ -f "/app/.host_ips" ]; then
105         while IFS= read -r host_ip; do
106             [ -z "$host_ip" ] && continue
107             if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null; then

```

```

108     iptables -t mangle -I OUTPUT 1 -p udp --sport 41642 -d "${host_ip}/32" -j RETURN 2>/dev/null || true
109     fi
110     done < "/app/.host_ips"
111 fi
112
113 if [ "${allow_p2p}" != "true" ]; then
114     # Drop Tailscale UDP packets destined for local subnets to prevent macOS gvproxy SNAT
115     # endpoint poisoning. When disabled (campus/enterprise WPA2-Enterprise with AP isolation),
116     # dropping these forces the S24 to use DERP relays reliably.
117     for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
118         if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; then
119             iptables -t mangle -A OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null || true
120         fi
121     done
122 else
123     # Remove DROP rules if they exist (switching from restricted to allowed)
124     for subnet in "192.168.0.0/16" "10.0.0.0/8" "172.16.0.0/12"; do
125         while iptables -t mangle -D OUTPUT -p udp --sport 41642 -d "$subnet" -j DROP 2>/dev/null; do ;; done
126     done
127 fi
128
129 # Bypass STUN and P2P packets to public IPs (DERP relays and remote peers) out eth0
130 if ! iptables -t mangle -C OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null; then
131     iptables -t mangle -A OUTPUT -p udp --sport 41642 -j MARK --set-mark 0x8888 2>/dev/null || true
132 fi
133 if ! ip rule show | grep -q 'fwmark 0x8888'; then
134     ip rule add fwmark 0x8888 lookup 254 pref 98 2>/dev/null || true
135 fi
136 }
137
138 add_tailscale_p2p_bypass
139
140 add_onion_transparent_proxy() {
141     for ipt in iptables iptables-legacy; do
142         command -v "$ipt" >/dev/null 2>&1 || continue
143         if ! $ipt -t nat -C PREROUTING -d 198.18.0.0/15 -p tcp -j REDIRECT --to-ports 9040 2>/dev/null; then
144             $ipt -t nat -I PREROUTING -d 198.18.0.0/15 -p tcp -j REDIRECT --to-ports 9040 2>/dev/null || true
145         fi
146         if ! $ipt -C FORWARD -d 198.18.0.0/15 -j ACCEPT 2>/dev/null; then
147             $ipt -I FORWARD -d 198.18.0.0/15 -j ACCEPT 2>/dev/null || true
148         fi
149     done
150 }
151
152 add_tailscale_p2p_bypass
153 add_onion_transparent_proxy
154
155 create_log_drop_chain() {
156     local ipt="$1"
157     local chain="$2"
158     local prefix="$3"
159
160     if ! $ipt -N "$chain" 2>/dev/null; then
161         $ipt -F "$chain" 2>/dev/null || true
162     fi
163     $ipt -A "$chain" -j LOG --log-prefix "$prefix"
164     $ipt -A "$chain" -j DROP 2>/dev/null || true
165 }
166
167 add_country_block() {
168     # To add a country to the blocklist, simply define the 'BLOCKED_COUNTRIES' variable in .env with 2-letter ISO
169     # country codes.
170     # You can find the country code by looking at the filename on http://www.ipdeny.com/ipblocks/data/countries/
171     # For example, to block China (cn) and Russia (ru), set: BLOCKED_COUNTRIES="cn ru" in your .env file
172     for zone in $BLOCKED_COUNTRIES; do
173         set_name="block_${zone}"
174         zone_upper=$(echo "$zone" | tr 'a-z' 'A-Z')
175         chain_name="LOG_DROP_${zone_upper}"
176         log_prefix="IP_BLOCK_${zone_upper}: "
177
178         # Create the ipset if it doesn't exist
179         if ! ipset list "$set_name" >/dev/null 2>&1; then
180             ipset create "$set_name" hash:net 2>/dev/null || true
181             echo "routing-fix: Downloading ${zone_upper} IPs for blocklist..."
182             curl -sS "http://www.ipdeny.com/ipblocks/data/countries/${zone}.zone" | grep -v '^#' | while read -r
183                 subnet; do
184                 [ -n "$subnet" ] && ipset add "$set_name" "$subnet" 2>/dev/null || true
185             done
186         fi
187
188         # Apply LOG_DROP rule to both backends

```

```

187     for ipt in iptables iptables-legacy; do
188         command -v "$ipt" >/dev/null 2>&1 || continue
189
190         create_log_drop_chain "$ipt" "$chain_name" "$log_prefix"
191
192         # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
193         while $ipt -D FORWARD -m set --match-set "$set_name" dst -j DROP 2>/dev/null; do ;; done
194
195         if ! $ipt -C FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null; then
196             $ipt -I FORWARD -m set --match-set "$set_name" dst -j "$chain_name" 2>/dev/null || true
197         fi
198     done
199 done
200 }
201
202 add_ip_blocklist() {
203     # Only reload when the compiled file has actually changed (mtime check).
204     # This prevents a pointless ipset restore on every 30-second loop tick.
205     if [ ! -f "$IP_BLOCKLIST_FILE" ]; then
206         return 0
207     fi
208
209     CURRENT_MTIME=$(stat -c %Y "$IP_BLOCKLIST_FILE" 2>/dev/null || echo "0")
210     if [ "$CURRENT_MTIME" != "$LAST_IP_MTIME" ]; then
211         echo "routing-fix: IP blocklist changed, reloading..."
212         # ipset restore handles the atomic swap internally:
213         # create_new populate swap with live destroy_new
214         if ipset restore < "$IP_BLOCKLIST_FILE" 2>/dev/null; then
215             LAST_IP_MTIME="$CURRENT_MTIME"
216             echo "routing-fix: IP blocklist loaded ($(ipset list block_malicious | grep -c '[0-9]') entries)."
217         else
218             echo "routing-fix: Warning: ipset restore failed. Retrying next cycle."
219             return 1
220         fi
221     fi
222
223     # Apply FORWARD rules in BOTH iptables backends.
224     for ipt in iptables iptables-legacy; do
225         command -v "$ipt" >/dev/null 2>&1 || continue
226
227         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_DST "IP_BLOCK_MALICIOUS_DST: "
228         create_log_drop_chain "$ipt" LOG_DROP_MALICIOUS_SRC "IP_BLOCK_MALICIOUS_SRC: "
229
230         # Clean up old plain DROP rules to ensure we hit the log-and-drop chains
231         while $ipt -D FORWARD -m set --match-set block_malicious dst -j DROP 2>/dev/null; do ;; done
232         while $ipt -D FORWARD -m set --match-set block_malicious src -j DROP 2>/dev/null; do ;; done
233
234         if ! $ipt -C FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null; then
235             $ipt -I FORWARD -m set --match-set block_malicious dst -j LOG_DROP_MALICIOUS_DST 2>/dev/null || true
236         fi
237
238         if ! $ipt -C FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null; then
239             $ipt -I FORWARD -m set --match-set block_malicious src -j LOG_DROP_MALICIOUS_SRC 2>/dev/null || true
240         fi
241     done
242 }
243
244 # Start background block logger (DNS + IP)
245 if ! pgrep -f "nullexit-logger" >/dev/null; then
246     echo "routing-fix: Starting background block logger..."
247     nullexit-logger &
248 fi
249
250 add_country_block
251 add_ip_blocklist
252
253 echo 'routing-fix: Routes applied.'
254
255 UNHEALTHY_COUNT=0
256
257 # Re-assert loop (every 30 seconds)
258 while true; do
259     # Table 200 routes
260     ip route replace "${WARP_ENDPOINT}" via "${DOCKER_GW}" dev eth0 table 200 2>/dev/null || true
261
262     # Tunnel health check
263     if ! curl -m 3 -sS --interface tun0 https://cloudflare.com/cdn-cgi/trace >/dev/null 2>&1; then
264         UNHEALTHY_COUNT=$((UNHEALTHY_COUNT + 1))
265     else
266         UNHEALTHY_COUNT=0
267         if [ -f "/app/TUNNEL_FAILED_CLOSED.marker" ]; then

```

```

268     rm -f "/app/TUNNEL_FAILED_CLOSED.marker"
269 fi
270 fi
271
272 if [ "$SUNHEALTHY_COUNT" -ge 3 ]; then
273     echo "routing-fix: Tunnel health check failed $SUNHEALTHY_COUNT times. Failing closed."
274     ip route del default dev tun0 table 200 2>/dev/null || true
275     ip route replace blackhole default table 200 2>/dev/null || true
276     touch "/app/TUNNEL_FAILED_CLOSED.marker"
277 else
278     ip route replace default dev tun0 table 200 2>/dev/null || true
279 fi
280
281 # Main table: keep Docker subnet route
282 ip route replace "${DOCKER_NET}" dev eth0 2>/dev/null || true
283
284 # CGNAT ip rule for Tailscale
285 if ! ip rule show | grep -q '100.64.0.0/10'; then
286     ip rule add to 100.64.0.0/10 lookup 52 pref 99 2>/dev/null || true
287     echo 'routing-fix: CGNAT rule missing, re-injected'
288 fi
289
290 # FORWARD RELATED, ESTABLISHED: Allow return traffic in both iptables
291 # backends. Gluetun may reset the nftables ruleset on VPN reconnect,
292 # and Tailscale may reset the legacy ruleset on auth refresh.
293 add_fwd_related_established
294
295 # Ensure P2P packets bypass WARP to maintain low latency
296 add_tailscale_p2p_bypass
297
298 # Ensure transparent proxying for .onion is active
299 add_onion_transparent_proxy
300
301 # Enforce country blocklist
302 add_country_block
303
304 # Enforce IP blocklist (reloads automatically when file changes)
305 add_ip_blocklist
306
307 sleep 30 &
308 wait $! || true
309 done

```

43 scripts/rule-compiler/go.mod

```

1 module nullexit/rule-compiler
2
3 go 1.22

```

44 scripts/rule-compiler/main.go

```

1 package main
2
3 import (
4     "bufio"
5     "bytes"
6     "crypto/md5"
7     "encoding/base64"
8     "encoding/hex"
9     "fmt"
10    "io"
11    "net/http"
12    "net/netip"
13    "os"
14    "os/exec"
15    "path/filepath"
16    "regexp"
17    "sort"
18    "strings"
19    "sync"
20    "time"
21 )

```

```

22
23 const (
24     BlackListPath = "black_list.txt"
25     WhiteListPath = "white_list.txt"
26     OutputPath    = "adguard/work/userfilters/compiled_rules.txt"
27     IpOutputPath  = "adguard/work/userfilters/ip_blocklist.ipset"
28     IpCacheDir    = "adguard/work/userfilters/cache/ip"
29     CacheDir      = "adguard/work/userfilters/cache"
30 )
31
32 var CoreLists = []string{
33     "https://raw.githubusercontent.com/anudeepND/blacklist/master/adservers.txt",
34     "https://raw.githubusercontent.com/anudeepND/blacklist/master/facebook.txt",
35     "https://raw.githubusercontent.com/crazy-max/WindowsSpyBlocker/master/data/hosts/spy.txt",
36     "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.samsung.txt",
37     "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/adblock/native.apple.txt",
38     "https://abp.oisd.nl/basic/",
39     "https://adaway.org/hosts.txt",
40     "https://pgl.yoyo.org/adserver/serverlist.php?hostformat=adblockplus&showintro=1&mimetype=plaintext",
41     "https://raw.githubusercontent.com/nextdns/cname-cloaking-blocklist/master/domains",
42 }
43
44 var MediumAdditions = []string{
45     "https://raw.githubusercontent.com/lightswitch05/hosts/master/docs/lists/facebook-extended.txt",
46     "https://someonewhocares.org/hosts/zero/hosts",
47     "https://urlhaus.abuse.ch/downloads/hostfile/",
48 }
49
50 var HeavyAdditions = []string{
51     "https://raw.githubusercontent.com/StevenBlack/hosts/master/hosts",
52     "https://raw.githubusercontent.com/Perflyst/PiHoleBlocklist/master/SmartTV-AGH.txt",
53     "https://raw.githubusercontent.com/DandelionSprout/adfilt/master/GameConsoleAdblockList.txt",
54 }
55
56 var IpBlockLists = []string{
57     "https://feodotracker.abuse.ch/downloads/ipblocklist.txt",
58     "https://www.spamhaus.org/drop/drop.txt",
59     "https://rules.emergingthreats.net/fwrules/emerging-Block-IPs.txt",
60     "https://cinsscore.com/list/ci-badguys.txt",
61 }
62
63 func getProfiles() map[string][]string {
64     profiles := make(map[string][]string)
65
66     light := append([]string(nil), CoreLists...)
67     light = append(light, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/light.txt")
68     profiles["light"] = light
69
70     medium := append([]string(nil), CoreLists...)
71     medium = append(medium, MediumAdditions...)
72     medium = append(medium, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/multi.txt")
73     profiles["medium"] = medium
74
75     heavy := append([]string(nil), CoreLists...)
76     heavy = append(heavy, MediumAdditions...)
77     heavy = append(heavy, HeavyAdditions...)
78     heavy = append(heavy, "https://raw.githubusercontent.com/hagezi/dns-blocklists/main/domains/pro.txt")
79     profiles["heavy"] = heavy
80
81     return profiles
82 }
83
84 func loadEnvProfile() string {
85     profile := "medium"
86
87     if bytesData, err := os.ReadFile(".env"); err == nil {
88         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
89         for scanner.Scan() {
90             line := strings.TrimSpace(scanner.Text())
91             if line != "" && !strings.HasPrefix(line, "#") {
92                 parts := strings.SplitN(line, "=", 2)
93                 if len(parts) == 2 {
94                     key := strings.TrimSpace(parts[0])
95                     val := strings.TrimSpace(parts[1])
96                     if key == "GATEWAY_RULE_PROFILE" {
97                         val = strings.Trim(val, "\"'")
98                         profile = strings.ToLower(val)
99                     }
100                 }
101             }
102         }
103     }

```



```

103 } else if !os.IsNotExist(err) {
104     fmt.Printf("Warning: Failed to read .env file for profile configuration (%v). Using default.\n", err)
105 }
106
107 if envVal := os.Getenv("GATEWAY_RULE_PROFILE"); envVal != "" {
108     profile = strings.ToLower(envVal)
109 }
110
111 profiles := getProfiles()
112 if _, exists := profiles[profile]; !exists {
113     fmt.Printf("Warning: Profile '%s' is invalid. Falling back to 'medium'.\n", profile)
114     profile = "medium"
115 }
116 return profile
117 }
118
119 func loadDomains(filepath string) map[string]bool {
120     domains := make(map[string]bool)
121     if bytesData, err := os.ReadFile(filepath); err == nil {
122         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
123         for scanner.Scan() {
124             line := strings.TrimSpace(scanner.Text())
125             if line != "" && !strings.HasPrefix(line, "#") {
126                 domains[strings.ToLower(line)] = true
127             }
128         }
129     }
130     return domains
131 }
132
133 func parseDomainsFromContent(content string) map[string]bool {
134     domains := make(map[string]bool)
135     scanner := bufio.NewScanner(strings.NewReader(content))
136     ipRegex := regexp.MustCompile(`^\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3}$`)
137     agRegex := regexp.MustCompile(`^\|([a-z0-9.-]+)\|` )
138     domainRegex := regexp.MustCompile(`^[a-z0-9.-]+$`)
139
140     for scanner.Scan() {
141         line := strings.TrimSpace(scanner.Text())
142         if line == "" || strings.HasPrefix(line, "!") || strings.HasPrefix(line, "[") {
143             continue
144         }
145         parts := strings.SplitN(line, "#", 2)
146         line = strings.TrimSpace(parts[0])
147         if line == "" {
148             continue
149         }
150
151         fields := strings.Fields(line)
152         var rule string
153         if len(fields) >= 2 {
154             if ipRegex.MatchString(fields[0]) {
155                 rule = fields[1]
156             } else {
157                 rule = fields[0]
158             }
159         } else {
160             rule = fields[0]
161         }
162
163         rule = strings.TrimSpace(strings.ToLower(rule))
164
165         var domain string
166         if m := agRegex.FindStringSubmatch(rule); m != nil {
167             domain = m[1]
168         } else if domainRegex.MatchString(rule) {
169             domain = rule
170         } else {
171             domain = rule
172         }
173
174         if domain != "" && domain != "localhost" && domain != "0.0.0.0" && domain != "127.0.0.1" && domain != "
175             broadcasthost" {
176             domains[domain] = true
177         }
178     }
179     return domains
180 }
181
182 // WARNING: Parsing AdGuardHome.yaml using regex instead of a proper YAML parser
183 // is brittle and assumes the exact format currently emitted by AdGuard.

```

```

183 // If an AdGuard version bump updates the configuration schema format,
184 // this parser will fail silently or loudly and should be refactored to use gopkg.in/yaml.v3.
185 func getAdguardNativeLists() []string {
186     yamlPath := "adguard/conf/AdGuardHome.yaml"
187     var enabledUrls []string
188     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
189         return enabledUrls
190     }
191
192     contentBytes, err := os.ReadFile(yamlPath)
193     if err != nil {
194         fmt.Printf("Warning: Failed to parse AdGuardHome.yaml for native lists (%v)\n", err)
195         return enabledUrls
196     }
197     content := string(contentBytes)
198
199     filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*?)(?:^[a-zA-Z_]+:|\z)`
200     match := filtersRegex.FindStringSubmatch(content)
201     if len(match) > 1 {
202         filtersText := match[1]
203         blocks := strings.Split(filtersText, "\n - ")
204         urlRegex := regexp.MustCompile(`url:\s*(\S+)`)
205
206         for _, block := range blocks {
207             if strings.TrimSpace(block) == "" {
208                 continue
209             }
210             if strings.Contains(block, "enabled: true") {
211                 urlMatch := urlRegex.FindStringSubmatch(block)
212                 if len(urlMatch) > 1 {
213                     url := urlMatch[1]
214                     if !strings.Contains(url, "compiled_rules.txt") {
215                         enabledUrls = append(enabledUrls, url)
216                     }
217                 }
218             }
219         }
220     }
221     return enabledUrls
222 }
223
224 func getAdguardCompiledRulesCachePath() string {
225     yamlPath := "adguard/conf/AdGuardHome.yaml"
226     if _, err := os.Stat(yamlPath); os.IsNotExist(err) {
227         return ""
228     }
229
230     contentBytes, err := os.ReadFile(yamlPath)
231     if err != nil {
232         fmt.Printf("Warning: Failed to resolve compiled_rules.txt cache path (%v)\n", err)
233         return ""
234     }
235     content := string(contentBytes)
236
237     filtersRegex := regexp.MustCompile(`(?ms)^filters:(.*?)(?:^[a-zA-Z_]+:|\z)`
238     match := filtersRegex.FindStringSubmatch(content)
239     if len(match) > 1 {
240         blocks := strings.Split(match[1], "\n - ")
241         idRegex := regexp.MustCompile(`id:\s*(\d+)`)
242         for _, block := range blocks {
243             if !strings.Contains(block, "compiled_rules.txt") {
244                 continue
245             }
246             idMatch := idRegex.FindStringSubmatch(block)
247             if len(idMatch) > 1 {
248                 return fmt.Sprintf("adguard/work/data/filters/%s.txt", idMatch[1])
249             }
250         }
251     }
252     return ""
253 }
254
255 func handleFetchError(urlStr string, cacheFile string, err error) map[string]bool {
256     if _, statErr := os.Stat(cacheFile); statErr == nil {
257         fmt.Printf("-> Warning: Failed to fetch %s (%v). Falling back to expired local cache.\n", urlStr, err)
258         content, readErr := os.ReadFile(cacheFile)
259         if readErr == nil {
260             domains := parseDomainsFromContent(string(content))
261             fmt.Printf("-> Loaded from expired cache (%d domains).\n", len(domains))
262             return domains
263         }
264     }

```

```

264     fmt.Printf(" -> Warning: Failed to read expired cache (%v). Using local lists only.\n", readErr)
265 } else {
266     fmt.Printf(" -> Warning: Failed to fetch %s (%v). Using local lists only for this source.\n", urlStr, err)
267 }
268 return make(map[string]bool)
269 }
270
271 func fetchRemoteDomains(urlStr string) map[string]bool {
272     os.MkdirAll(CacheDir, 0755)
273
274     hash := md5.Sum([]byte(urlStr))
275     urlHash := hex.EncodeToString(hash[:])
276     cacheFile := filepath.Join(CacheDir, fmt.Sprintf("%s.txt", urlHash))
277
278     if stat, err := os.Stat(cacheFile); err == nil {
279         fileAge := time.Since(stat.ModTime()).Seconds()
280         if fileAge < 86400 {
281             fmt.Printf("Loading remote blacklist from cache: %s\n", urlStr)
282             content, err := os.ReadFile(cacheFile)
283             if err == nil {
284                 domains := parseDomainsFromContent(string(content))
285                 fmt.Printf(" -> Loaded from cache (copy is %.1f hours old, %d domains).\n", fileAge/3600.0, len(domains))
286             }
287             return domains
288         }
289         fmt.Printf(" -> Warning: Failed to read cache file %s (%v). Re-fetching...\n", cacheFile, err)
290     }
291
292     fmt.Printf("Fetching remote blacklist from: %s ...\n", urlStr)
293
294     req, err := http.NewRequest("GET", urlStr, nil)
295     if err != nil {
296         return handleFetchError(urlStr, cacheFile, err)
297     }
298     req.Header.Set("User-Agent", "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7)")
299
300     client := &http.Client{Timeout: 15 * time.Second}
301     resp, err := client.Do(req)
302     if err != nil {
303         return handleFetchError(urlStr, cacheFile, err)
304     }
305     defer resp.Body.Close()
306
307     if resp.StatusCode != 200 {
308         return handleFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
309     }
310
311     bodyBytes, err := io.ReadAll(resp.Body)
312     if err != nil {
313         return handleFetchError(urlStr, cacheFile, err)
314     }
315
316     content := string(bodyBytes)
317     domains := parseDomainsFromContent(content)
318
319     if len(domains) < 10 {
320         return handleFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only found %d domains. Possible 404 or bad URL", len(domains)))
321     }
322
323     err = os.WriteFile(cacheFile, bodyBytes, 0644)
324     if err != nil {
325         fmt.Printf(" -> Warning: Failed to save cache file (%v)\n", err)
326     }
327
328     fmt.Printf(" -> Successfully fetched and cached %d domains.\n", len(domains))
329     return domains
330 }
331
332 func optimizeSubdomains(domains map[string]bool, listName string) map[string]bool {
333     fmt.Printf("Optimizing %s by removing redundant subdomains...\n", listName)
334     rawCount := len(domains)
335     if rawCount == 0 {
336         return make(map[string]bool)
337     }
338
339     optimized := make(map[string]bool)
340     for domain := range domains {
341         if strings.ContainsAny(domain, "|^@$/") {
342             optimized[domain] = true

```

```

343     continue
344 }
345
346 parts := strings.Split(domain, ".")
347 hasParent := false
348
349 for i := 1; i < len(parts)-1; i++ {
350     parent := strings.Join(parts[i:], ".")
351     if domains[parent] {
352         hasParent = true
353         break
354     }
355 }
356
357 if !hasParent {
358     optimized[domain] = true
359 }
360 }
361
362 saved := rawCount - len(optimized)
363 reduction := float64(0)
364 if rawCount > 0 {
365     reduction = (float64(saved) / float64(rawCount)) * 100
366 }
367 fmt.Printf(" -> Reduced %s from %d to %d domains (-%d / %.1f%% reduction).\n", listName, rawCount, len(
    optimized), saved, reduction)
368 return optimized
369 }
370
371 func parseIpsFromContent(content string) map[string]bool {
372     ips := make(map[string]bool)
373     scanner := bufio.NewScanner(strings.NewReader(content))
374
375     for scanner.Scan() {
376         line := strings.TrimSpace(scanner.Text())
377         if line == "" || strings.HasPrefix(line, "#") || strings.HasPrefix(line, ";") {
378             continue
379         }
380
381         fields := strings.Fields(line)
382         if len(fields) > 0 {
383             token := strings.TrimRight(fields[0], ",")
384             if _, err := netip.ParsePrefix(token); err == nil {
385                 ips[token] = true
386             } else if _, err := netip.ParseAddr(token); err == nil {
387                 ips[token] = true
388             }
389         }
390     }
391     return ips
392 }
393
394 func handleIpFetchError(urlStr string, cacheFile string, err error) map[string]bool {
395     if _, statErr := os.Stat(cacheFile); statErr == nil {
396         fmt.Printf(" -> Warning: fetch failed (%v). Falling back to stale cache.\n", err)
397         content, readErr := os.ReadFile(cacheFile)
398         if readErr == nil {
399             ips := parseIpsFromContent(string(content))
400             fmt.Printf(" -> %d IPs loaded from stale cache.\n", len(ips))
401             return ips
402         }
403         fmt.Printf(" -> Warning: stale cache also failed (%v).\n", readErr)
404     } else {
405         fmt.Printf(" -> Warning: %s failed (%v). No cache available. Skipping.\n", urlStr, err)
406     }
407     return make(map[string]bool)
408 }
409
410 func fetchRemoteIps(urlStr string) map[string]bool {
411     os.MkdirAll(IpCacheDir, 0755)
412
413     hash := md5.Sum([]byte(urlStr))
414     urlHash := hex.EncodeToString(hash[:])
415     cacheFile := filepath.Join(IpCacheDir, fmt.Sprintf("%s.txt", urlHash))
416
417     if stat, err := os.Stat(cacheFile); err == nil {
418         fileAge := time.Since(stat.ModTime()).Seconds()
419         if fileAge < 86400 {
420             fmt.Printf("Loading IP feed from cache: %s\n", urlStr)
421             content, err := os.ReadFile(cacheFile)
422             if err == nil {

```

```

423     ips := parseIpsFromContent(string(content))
424     fmt.Printf(" -> %d IPs (cache is %.1fh old).\n", len(ips), fileAge/3600.0)
425     return ips
426 }
427     fmt.Printf(" -> Warning: cache read failed (%v). Re-fetching...\n", err)
428 }
429 }
430
431 fmt.Printf("Fetching IP feed: %s ...\n", urlStr)
432 req, err := http.NewRequest("GET", urlStr, nil)
433 if err != nil {
434     return handleIpFetchError(urlStr, cacheFile, err)
435 }
436 req.Header.Set("User-Agent", "Mozilla/5.0")
437
438 client := &http.Client{Timeout: 15 * time.Second}
439 resp, err := client.Do(req)
440 if err != nil {
441     return handleIpFetchError(urlStr, cacheFile, err)
442 }
443 defer resp.Body.Close()
444
445 if resp.StatusCode != 200 {
446     return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("HTTP %d", resp.StatusCode))
447 }
448
449 bodyBytes, err := io.ReadAll(resp.Body)
450 if err != nil {
451     return handleIpFetchError(urlStr, cacheFile, err)
452 }
453
454 content := string(bodyBytes)
455 ips := parseIpsFromContent(content)
456 if len(ips) < 1 {
457     return handleIpFetchError(urlStr, cacheFile, fmt.Errorf("Sanity check failed: only %d IPs found. Possible 404", len(ips)))
458 }
459
460 err = os.WriteFile(cacheFile, bodyBytes, 0644)
461 if err != nil {
462     fmt.Printf(" -> Warning: cache write failed (%v)\n", err)
463 }
464
465 fmt.Printf(" -> %d IPs fetched and cached.\n", len(ips))
466 return ips
467 }
468
469 func compileIpBlocklist() int {
470     fmt.Println("\n IP Blocklist Compilation ")
471
472     allIps := make(map[string]bool)
473     var mu sync.Mutex
474     var wg sync.WaitGroup
475
476     maxThreads := 8
477     if len(IpBlockLists) < 8 {
478         maxThreads = len(IpBlockLists)
479     }
480     semaphore := make(chan struct{}, maxThreads)
481
482     for _, urlStr := range IpBlockLists {
483         wg.Add(1)
484         go func(u string) {
485             defer wg.Done()
486             semaphore <- struct{}{}
487             defer func() { <-semaphore }()
488
489             ips := fetchRemoteIps(u)
490             mu.Lock()
491             for ip := range ips {
492                 allIps[ip] = true
493             }
494             mu.Unlock()
495         }(urlStr)
496     }
497     wg.Wait()
498
499     fmt.Printf("Total unique entries before filtering: %d\n", len(allIps))
500
501     reservedStr := []string{
502         "10.0.0.0/8",

```

```

503     "172.16.0.0/12",
504     "192.168.0.0/16",
505     "127.0.0.0/8",
506     "169.254.0.0/16",
507     "100.64.0.0/10",
508     "0.0.0.0/8",
509 }
510 var reserved []netip.Prefix
511 for _, s := range reservedStr {
512     p, _ := netip.ParsePrefix(s)
513     reserved = append(reserved, p)
514 }
515
516 cleanIps := make(map[string]bool)
517 for entry := range allIps {
518     var prefix netip.Prefix
519     p, err := netip.ParsePrefix(entry)
520     if err != nil {
521         addr, err2 := netip.ParseAddr(entry)
522         if err2 != nil {
523             continue
524         }
525         prefix = netip.PrefixFrom(addr, addr.BitLen())
526     } else {
527         prefix = p
528     }
529
530     overlaps := false
531     for _, r := range reserved {
532         if r.Overlaps(prefix) {
533             overlaps = true
534             break
535         }
536     }
537     if !overlaps {
538         cleanIps[entry] = true
539     }
540 }
541
542 removed := len(allIps) - len(cleanIps)
543 if removed > 0 {
544     fmt.Printf(" -> Removed %d private/reserved entries.\n", removed)
545 }
546 fmt.Printf(" -> Final IP blocklist: %d entries.\n", len(cleanIps))
547
548 os.MkdirAll(filepath.Dir(IpOutputPath), 0755)
549
550 f, err := os.Create(IpOutputPath)
551 if err != nil {
552     fmt.Printf("Error writing IP output file: %v\n", err)
553     return len(cleanIps)
554 }
555 defer f.Close()
556
557 f.WriteString("# nullexit Compiled IP Blocklist\n")
558 f.WriteString("# Sources: Feodo Tracker, Spamhaus DROP/EDROP, Emerging Threats, CINS\n")
559 f.WriteString(fmt.Sprintf("# Entries: %d\n\n", len(cleanIps)))
560
561 f.WriteString("create block_malicious_new hash:net maxelem 200000 -exist\n")
562
563 var sortedIps []string
564 for ip := range cleanIps {
565     sortedIps = append(sortedIps, ip)
566 }
567 sort.Strings(sortedIps)
568
569 for _, ip := range sortedIps {
570     f.WriteString(fmt.Sprintf("add block_malicious_new %s -exist\n", ip))
571 }
572
573 f.WriteString("create block_malicious hash:net maxelem 200000 -exist\n")
574 f.WriteString("swap block_malicious block_malicious_new\n")
575 f.WriteString("destroy block_malicious_new\n")
576
577 fmt.Printf("IP blocklist written to %s\n", IpOutputPath)
578 return len(cleanIps)
579 }
580
581 func main() {
582     profile := loadEnvProfile()
583     fmt.Printf("Active memory profile: '%s'\n", strings.ToUpper(profile))

```

```

584 localBlackList := loadDomains(BlackListPath)
585 whiteList := loadDomains(WhiteListPath)
586
587
588 blackList := make(map[string]bool)
589 for k := range localBlackList {
590     blackList[k] = true
591 }
592
593 profiles := getProfiles()
594 urlsToFetch := profiles[profile]
595
596 fmt.Printf("\nStarting concurrent downloads for %d lists...\n", len(urlsToFetch))
597 startTime := time.Now()
598
599 var mu sync.Mutex
600 var wg sync.WaitGroup
601
602 maxThreads := 16
603 if len(urlsToFetch) < 16 {
604     maxThreads = len(urlsToFetch)
605 }
606 semaphore := make(chan struct{}, maxThreads)
607
608 for _, urlStr := range urlsToFetch {
609     wg.Add(1)
610     go func(u string) {
611         defer wg.Done()
612         semaphore <- struct{}{}
613         defer func() { <-semaphore }()
614
615         domains := fetchRemoteDomains(u)
616         mu.Lock()
617         for d := range domains {
618             blackList[d] = true
619         }
620         mu.Unlock()
621     }(urlStr)
622 }
623 wg.Wait()
624
625 fmt.Printf("Finished concurrent downloads in %.2f seconds using %d threads.\n", time.Since(startTime).Seconds(), maxThreads)
626
627 adguardNativeUrls := getAdguardNativeLists()
628 var adguardNativeDomains map[string]bool
629 if len(adguardNativeUrls) > 0 {
630     fmt.Printf("\nFetching %d AdGuard native list(s) for deduplication...\n", len(adguardNativeUrls))
631     adguardNativeDomains = make(map[string]bool)
632
633     maxNativeThreads := 4
634     if len(adguardNativeUrls) < 4 {
635         maxNativeThreads = len(adguardNativeUrls)
636     }
637     nativeSemaphore := make(chan struct{}, maxNativeThreads)
638
639     for _, urlStr := range adguardNativeUrls {
640         wg.Add(1)
641         go func(u string) {
642             defer wg.Done()
643             nativeSemaphore <- struct{}{}
644             defer func() { <-nativeSemaphore }()
645
646             domains := fetchRemoteDomains(u)
647             mu.Lock()
648             for d := range domains {
649                 adguardNativeDomains[d] = true
650             }
651             mu.Unlock()
652         }(urlStr)
653     }
654     wg.Wait()
655
656     if len(adguardNativeDomains) > 0 {
657         fmt.Println("Deduplicating compiled rules against AdGuard native lists...")
658         originalSize := len(blackList)
659         for d := range adguardNativeDomains {
660             delete(blackList, d)
661         }
662         fmt.Printf(" -> Removed %d redundant rules already covered by AdGuard.\n", originalSize-len(blackList))
663     }

```



```

664 }
665
666 var contradictions []string
667 for d := range whiteList {
668     if blackList[d] {
669         contradictions = append(contradictions, d)
670     }
671 }
672
673 if len(contradictions) > 0 {
674     fmt.Printf("Detected %d contradictions. Whitelist taking priority.\n", len(contradictions))
675     sort.Strings(contradictions)
676     for _, domain := range contradictions {
677         if localBlackList[domain] {
678             fmt.Printf(" -> Removed local blacklist domain: %s\n", domain)
679         }
680         delete(blackList, domain)
681     }
682 }
683
684 blackList = optimizeSubdomains(blackList, "blacklist")
685 whiteList = optimizeSubdomains(whiteList, "whitelist")
686
687 os.MkdirAll(filepath.Dir(OutputPath), 0755)
688 outFile, err := os.Create(OutputPath)
689 if err != nil {
690     fmt.Printf("Error creating output file: %v\n", err)
691     return
692 }
693 defer outFile.Close()
694
695 outFile.WriteString("! Custom Compiled Rules (Auto-Generated)\n")
696 outFile.WriteString(fmt.Sprintf("! Memory Profile: %s\n", strings.ToUpper(profile)))
697 outFile.WriteString(fmt.Sprintf("! Total Block Rules: %d\n", len(blackList)))
698 outFile.WriteString(fmt.Sprintf("! Native AdGuard Rules: %d\n", len(adguardNativeDomains)))
699 outFile.WriteString(fmt.Sprintf("! Total Whitelist Rules: %d\n\n", len(whiteList)))
700
701 outFile.WriteString("! --- Blacklist Rules ---\n")
702
703 var sortedBlacklist []string
704 for d := range blackList {
705     sortedBlacklist = append(sortedBlacklist, d)
706 }
707 sort.Strings(sortedBlacklist)
708
709 for _, domain := range sortedBlacklist {
710     if strings.Contains(domain, "$") {
711         parts := strings.SplitN(domain, "$", 2)
712         dom, mod := parts[0], parts[1]
713         if strings.HasPrefix(dom, "|") {
714             if strings.HasSuffix(dom, "^") {
715                 outFile.WriteString(fmt.Sprintf("%s%s\n", dom, mod))
716             } else {
717                 outFile.WriteString(fmt.Sprintf("%s^%s\n", dom, mod))
718             }
719         } else {
720             outFile.WriteString(fmt.Sprintf("||%s^%s\n", dom, mod))
721         }
722     } else if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|") || strings.HasSuffix(domain, "^") {
723         outFile.WriteString(fmt.Sprintf("%s\n", domain))
724     } else {
725         outFile.WriteString(fmt.Sprintf("||%s^", domain))
726     }
727 }
728
729 outFile.WriteString("\n! --- Whitelist Rules ---\n")
730 var sortedWhitelist []string
731 for d := range whiteList {
732     sortedWhitelist = append(sortedWhitelist, d)
733 }
734 sort.Strings(sortedWhitelist)
735
736 for _, domain := range sortedWhitelist {
737     if strings.HasPrefix(domain, "/") || strings.HasPrefix(domain, "|") || strings.HasSuffix(domain, "|") || strings.HasSuffix(domain, "^") || strings.HasPrefix(domain, "@@") {
738         rule := domain
739         if !strings.HasPrefix(domain, "@@") {
740             rule = "@@" + domain
741         }
742         outFile.WriteString(fmt.Sprintf("%s\n", rule))

```

```

743     } else {
744         outFile.WriteString(fmt.Sprintf("@@||%s^\n", domain))
745     }
746 }
747
748 fmt.Printf("\nSuccessfully compiled %d block rules and %d allow rules to %s\n", len(blackList), len(whiteList)
749 ), OutputPath)
750
751 cachedFilterPath := getAdguardCompiledRulesCachePath()
752 if cachedFilterPath != "" {
753     input, err := os.ReadFile(OutputPath)
754     if err == nil {
755         err = os.WriteFile(cachedFilterPath, input, 0644)
756         if err == nil {
757             fmt.Printf("Updated AdGuard filter cache: %s\n", cachedFilterPath)
758         } else {
759             fmt.Printf("Warning: Could not update AdGuard filter cache %s (%v)\n", cachedFilterPath, err)
760         }
761     } else {
762         fmt.Printf("Warning: Could not read OutputPath to copy to cache (%v)\n", err)
763     }
764 } else {
765     fmt.Println("Warning: Could not determine compiled_rules.txt cache path from AdGuardHome.yaml; skipping
766     cache update.")
767 }
768
769 adguardReachable := false
770
771 dockerComposeYamlExists := false
772 if _, err := os.Stat("docker-compose.yml"); err == nil {
773     dockerComposeYamlExists = true
774 }
775 dockerPath, err := exec.LookPath("docker")
776 if err == nil && dockerComposeYamlExists {
777     cmd := exec.Command(dockerPath, "compose", "ps", "--status", "running", "-q", "adguardhome")
778     out, err := cmd.Output()
779     if err == nil && strings.TrimSpace(string(out)) != "" {
780         fmt.Println("Force-recreating AdGuard Home container to drop persisted filter state...")
781         upCmd := exec.Command(dockerPath, "compose", "up", "-d", "--force-recreate", "--no-deps", "adguardhome")
782         err := upCmd.Run()
783         if err == nil {
784             fmt.Println("AdGuard Home force-recreated successfully.")
785             time.Sleep(8 * time.Second)
786             adguardReachable = true
787         } else {
788             fmt.Println("Failed to restart AdGuard Home. Is it running?")
789         }
790     }
791 }
792
793 if adguardReachable {
794     agPass := "nullexit"
795     if bytesData, err := os.ReadFile(".env"); err == nil {
796         scanner := bufio.NewScanner(strings.NewReader(string(bytesData)))
797         for scanner.Scan() {
798             line := scanner.Text()
799             if strings.HasPrefix(line, "ADGUARD_PASSWORD=") {
800                 parts := strings.SplitN(line, "=", 2)
801                 if len(parts) == 2 {
802                     agPass = strings.Trim(strings.TrimSpace(parts[1]), "\"'")
803                 }
804                 break
805             }
806         }
807     }
808
809     creds := base64.StdEncoding.EncodeToString([]byte(fmt.Sprintf("admin:%s", agPass)))
810     req, err := http.NewRequest("POST", "http://127.0.0.1:3000/control/filtering/refresh", bytes.NewBuffer([]
811     byte("{}")))
812     if err == nil {
813         req.Header.Set("Authorization", fmt.Sprintf("Basic %s", creds))
814         req.Header.Set("Content-Type", "application/json")
815         client := &http.Client{Timeout: 15 * time.Second}
816         resp, err := client.Do(req)
817         if err == nil {
818             fmt.Printf("Triggered AdGuard filter refresh via REST API (HTTP=%d).\n", resp.StatusCode)
819             resp.Body.Close()
820         } else {
821             fmt.Printf("Warning: AdGuard filter refresh API call failed (%v). New whitelist will not load until
822             next refresh tick (~24h) or manual refresh.\n", err)
823         }
824     }
825 }

```

```

820 }
821 }
822
823 compileIpBlocklist()
824 }

```

45 scripts/setup-common.sh

```

1  #!/bin/bash
2  # 1. Docker
3  step "Checking Docker"
4
5  if ! command -v docker >> output.log 2>&1; then
6      echo ""
7      if [[ "$OSTYPE" == "darwin"* ]]; then
8          echo " Docker is not installed."
9          # Note: Homebrew aggressively rolls forward and discourages version pinning.
10         # This setup is confirmed stable on Colima v0.10.3 and Docker v29.6.1.
11         read -rp " Would you like to automatically install Colima & Docker via Homebrew? [y/N]: "
12         INSTALL_DOCKER
13         if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
14             step "Installing Colima and Docker..."
15             brew install colima docker docker-compose
16             step "Starting Colima VM..."
17             colima start --memory 0.6 --vm-type vz --network-address --network-mode bridged
18         else
19             echo ""
20             echo " Please install Docker manually. Two options:"
21             echo " A) Colima (Recommended): brew install colima docker docker-compose && colima start --memory
22             0.6 --vm-type vz --network-address --network-mode bridged"
23             echo " B) Docker Desktop: https://www.docker.com/products/docker-desktop/"
24             die "Install Docker, then re-run this script."
25         fi
26     else
27         echo " Docker is not installed."
28         # Note: The Docker convenience script always fetches the latest stable release.
29         # This setup is confirmed stable on Docker Engine v29.6.1.
30         read -rp " Would you like to automatically install Docker? (Requires sudo) [y/N]: " INSTALL_DOCKER
31         if [[ "$INSTALL_DOCKER" == "y" || "$INSTALL_DOCKER" == "Y" ]]; then
32             step "Installing Docker..."
33             curl -fsSL https://get.docker.com | sh
34             sudo usermod -aG docker "$USER"
35             echo -e "\n\033[0;32m[OK] Docker installed successfully!\033[0m"
36             echo -e "\033[0;33m[!] CRITICAL: You must log out of your SSH session and log back in for Docker
37             permissions to apply.\033[0m"
38             die "Please log out, log back in, and re-run setup.sh."
39         else
40             echo " Please install Docker manually:"
41             echo " curl -fsSL https://get.docker.com | sh"
42             echo " sudo usermod -aG docker \"$USER && newgrp docker"
43             die "Install Docker, then re-run this script."
44         fi
45     fi
46 fi
47
48 if ! docker info >> output.log 2>&1; then
49     echo ""
50     if [[ "$OSTYPE" == "darwin"* ]]; then
51         echo " Docker is installed but not running."
52         echo " If using Colima: colima start --memory 0.6 --vm-type vz --network-address --network-mode
53         bridged"
54         echo " If using Docker Desktop: open the app."
55     else
56         echo " Docker is installed but not running."
57         echo " Run: sudo systemctl start docker"
58     fi
59     die "Start Docker, then re-run this script."
60 fi
61
62 if ! docker compose version >> output.log 2>&1; then
63     die "Docker Compose v2 not found. Update Docker or install the compose plugin:\n https://docs.docker.com/
64     compose/install/"
65 fi
66
67 ok "Docker $(docker --version | grep -oE '[0-9.]+' | head -1) is running."

```

```

64 # 1.5 Python 3
65 step "Checking Python 3"
66
67 if ! command -v python3 >> output.log 2>&1; then
68     echo ""
69     if [[ "$OSTYPE" == "darwin"* ]]; then
70         echo " Python 3 is not installed."
71         echo " macOS includes it via Xcode Command Line Tools, or you can install it via Homebrew:"
72         echo " brew install python3"
73         die "Install Python 3, then re-run this script."
74     else
75         echo " Python 3 is not installed."
76         echo " Please install it using your system's package manager. For example:"
77         echo " Debian/Ubuntu: sudo apt install python3"
78         echo " Fedora: sudo dnf install python3"
79         echo " Arch: sudo pacman -S python"
80         die "Install Python 3, then re-run this script."
81     fi
82 fi
83
84 # Note: This setup is confirmed stable on Python 3.9.6 (macOS default).
85 PYTHON_VER=$(python3 --version 2>&1 | head -n 1)
86 ok "$PYTHON_VER is installed."
87
88 # 2. Tailscale (host)
89 step "Checking Tailscale (host)"
90 # The host needs its own Tailscale installation separate from the Docker
91 # container so Toggle-Gateway can run 'tailscale down' before stopping
92 # containers and 'tailscale up' after starting them. Without this, the
93 # exit-node deadlock that setup.sh is designed to prevent can still occur.
94
95 # Resolve the tailscale binary: CLI (brew/package) or bundled macOS app
96 TAILSCALE_BIN=""
97 if command -v tailscale >> output.log 2>&1; then
98     TAILSCALE_BIN="tailscale"
99 elif [[ -x "/Applications/Tailscale.app/Contents/MacOS/Tailscale" ]]; then
100     TAILSCALE_BIN="/Applications/Tailscale.app/Contents/MacOS/Tailscale"
101 fi
102
103 RECOMMENDED_TS_VERSION="1.98.5"
104
105 if [[ -z "$TAILSCALE_BIN" ]]; then
106     warn "Tailscale not found on host installing..."
107
108     if [[ "$OSTYPE" == "darwin"* ]]; then
109         if command -v brew >> output.log 2>&1; then
110             step "Installing Tailscale CLI formula via Homebrew (standalone, no .app GUI)..."
111             # Note: We install Tailscale via Homebrew, targeting the stable version (recommended: 1.98.5)
112             brew install tailscale -q
113             TAILSCALE_BIN="tailscale"
114
115             step "Starting tailscaled as a per-user LaunchAgent (auto-starts at login)..."
116             if brew services start tailscale 2>&1 | grep -qE 'Successfully|started'; then
117                 ok "tailscaled LaunchAgent registered."
118             else
119                 warn "Could not auto-start tailscaled. Run 'brew services start tailscale' manually."
120                 warn "For a system-wide LaunchDaemon that runs before login, run instead:"
121                 warn " sudo brew services start tailscale"
122             fi
123             echo ""
124         else
125             die "Homebrew not found. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION})\n https://tailscale.com/download/mac\n Then re-run this script."
126         fi
127     elif [[ "$OSTYPE" == "linux-gnu"* ]]; then
128         curl -fsSL https://tailscale.com/install.sh | sh
129         sudo systemctl enable --now tailscaled 2>> output.log || true
130         TAILSCALE_BIN="tailscale"
131     else
132         die "Unsupported OS. Install Tailscale manually (recommended version: ${RECOMMENDED_TS_VERSION})\n https://tailscale.com/download\n Then re-run this script."
133     fi
134
135     TAILSCALE_VER=$(("${TAILSCALE_BIN}" version 2>> output.log | head -n 1))
136     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then
137         warn "Tailscale installed, but version (${TAILSCALE_VER}) differs from recommended version ( ${RECOMMENDED_TS_VERSION})."
138     else
139         ok "Tailscale installed (version ${TAILSCALE_VER})."
140     fi
141

```

```

142 fi
143 else
144     TAILSCALE_VER=$( ${TAILSCALE_BIN} version 2>> output.log | head -n 1)
145     if [[ "$TAILSCALE_VER" != "$RECOMMENDED_TS_VERSION" ]]; then
146         warn "Tailscale is already installed ($TAILSCALE_VER), but recommended version is
147             $RECOMMENDED_TS_VERSION for host/container consistency."
148     else
149         ok "Tailscale is already installed at recommended version ($TAILSCALE_VER)."
150     fi
151 fi
152 # Authenticate the host machine if it isn't already connected.
153 # This is an interactive step it opens a browser login URL.
154 # We do it now, before containers start, so Toggle-Gateway can
155 # safely run 'tailscale down/up' from day one.
156 if ! ${TAILSCALE_BIN} status >> output.log 2>&1; then
157     echo ""
158     echo " Your host machine needs to join your Tailscale network."
159     echo " A browser window or login URL will appear authenticate, then"
160     echo " return here. The script will continue automatically."
161     echo ""
162     if [[ "$OSTYPE" == "linux-gnu*" ]]; then
163         sudo ${TAILSCALE_BIN} up
164     else
165         ${TAILSCALE_BIN} up
166     fi
167     ok "Host machine connected to Tailscale."
168 else
169     ok "Host machine is already connected to Tailscale."
170 fi
171
172 # 3. wgcf
173 step "Checking wgcf (Cloudflare WARP key tool)"
174
175 if ! command -v wgcf >> output.log 2>&1; then
176     warn "wgcf not found installing..."
177
178     if [[ "$OSTYPE" == "darwin*" ]]; then
179         command -v brew >> output.log 2>&1 || die "Homebrew not found.\nInstall wgcf manually: https://github.
180             com/ViRb3/wgcf/releases"
181         brew install wgcf -q
182     elif [[ "$OSTYPE" == "linux-gnu*" ]]; then
183         ARCH=$(uname -m)
184         case $ARCH in
185             x86_64) WGCF_ARCH="amd64" ;;
186             aarch64) WGCF_ARCH="arm64" ;;
187             armv7l) WGCF_ARCH="armv7" ;;
188             *) die "Unsupported architecture: $ARCH.\nInstall wgcf manually: https://github.com/ViRb3/wgcf/
189                 releases" ;;
190         esac
191
192         echo " Fetching latest release..."
193         WGCF_VERSION=$(curl -sf https://api.github.com/repos/ViRb3/wgcf/releases/latest \
194             | grep 'tag_name' | cut -d'"' -f4)
195         [[ -z "$WGCF_VERSION" ]] && die "Could not fetch wgcf version. Check your internet connection."
196
197         sudo curl -sL \
198             "https://github.com/ViRb3/wgcf/releases/download/${WGCF_VERSION}/wgcf_${WGCF_VERSION#v}_linux_${
199             WGCF_ARCH}" \
200             -o /usr/local/bin/wgcf
201         sudo chmod +x /usr/local/bin/wgcf
202     else
203         die "Unsupported OS. Install wgcf manually: https://github.com/ViRb3/wgcf/releases"
204     fi
205 fi
206
207 ok "wgcf is ready."
208
209 # 4. WARP profile
210 step "Generating Cloudflare WARP profile"
211
212 if [[ -f "wgcf-profile.conf" ]]; then
213     warn "wgcf-profile.conf already exists skipping key generation."
214 else
215     wgcf register --accept-tos 2>> output.log || die "wgcf register failed. Check your internet connection."
216     wgcf generate 2>> output.log || die "wgcf generate failed."
217     ok "WARP profile generated."
218 fi
219
220 # Parse keys (IPv4 address only gluetun doesn't need the IPv6 one)

```

```

219 PRIVATE_KEY=$(awk '/PrivateKey/{print $3}' wgcf-profile.conf)
220 PUBLIC_KEY=$(awk '/PublicKey/{print $3}' wgcf-profile.conf)
221 ADDRESSES=$(awk '/Address/{print $3}' wgcf-profile.conf | grep -v ':' | head -1)
222
223 [[ -z "$PRIVATE_KEY" ]] && die "Could not parse PrivateKey from wgcf-profile.conf"
224 [[ -z "$PUBLIC_KEY" ]] && die "Could not parse PublicKey from wgcf-profile.conf"
225 [[ -z "$ADDRESSES" ]] && die "Could not parse IPv4 Address from wgcf-profile.conf"
226
227 ok "WARP keys extracted."
228
229 # 5. Tailscale auth key
230 step "Tailscale authentication"
231
232 TS_AUTHKEY=""
233 if [[ -f ".env" ]]; then
234     EXISTING_TS_KEY=$(read_env_var TS_AUTHKEY)
235     if [[ -n "$EXISTING_TS_KEY" ]]; then
236         TS_AUTHKEY="$EXISTING_TS_KEY"
237         warn "Found existing TS_AUTHKEY in .env. Skipping prompt."
238     fi
239 fi
240
241 if [[ -z "$TS_AUTHKEY" ]]; then
242     echo ""
243     echo "Generate a key at: https://login.tailscale.com/admin/settings/keys"
244     echo "'Generate auth key' enable Reusable if you plan to re-run setup"
245     echo ""
246     read -rp "Paste your Tailscale auth key: " TS_AUTHKEY
247     echo ""
248 fi
249
250 [[ -z "$TS_AUTHKEY" ]] && die "Tailscale auth key is required."
251 [[ "$TS_AUTHKEY" != tskey-auth-* ]] && warn "Key doesn't look like a Tailscale auth key proceeding anyway."
252 ok "Auth key accepted."
253
254 # 6. AdGuard Home admin password
255 step "AdGuard Home admin account"
256
257 # Hardcoded default credentials to prevent lockout
258 # Username: admin
259 # Password: nullexit
260 if [[ -f ".env" ]]; then
261     EXISTING_AG_PASS=$(read_env_var ADGUARD_PASSWORD)
262     if [[ -n "$EXISTING_AG_PASS" ]]; then
263         ADGUARD_PASSWORD="$EXISTING_AG_PASS"
264         ADGUARD_PWD_ESC="$EXISTING_AG_PASS"
265         ok "Found existing ADGUARD_PASSWORD in .env."
266     else
267         ADGUARD_PASSWORD="nullexit"
268         ADGUARD_PWD_ESC="nullexit"
269         ok "Default password set to 'nullexit'."
270     fi
271 else
272     ADGUARD_PASSWORD="nullexit"
273     ADGUARD_PWD_ESC="nullexit"
274     ok "Default password set to 'nullexit'."
275 fi
276
277 # 7. Write .env
278 step "Writing .env"
279
280 if [[ -f ".env" ]]; then
281     read -rp ".env already exists. Overwrite? [y/N]: " OVERWRITE
282     if [[ "$OVERWRITE" != "y" && "$OVERWRITE" != "Y" ]]; then
283         warn "Keeping existing .env."
284     else
285         WRITE_ENV=1
286     fi
287 else
288     WRITE_ENV=1
289 fi
290
291 if [[ "${WRITE_ENV:-0}" == "1" ]]; then
292     # Preserve existing configuration flags if .env already exists
293     EXISTING_PROFILE="medium"
294     EXISTING_MSS="1120"
295     EXISTING_HIJACK="true"
296     EXISTING_EXIT="true"
297     EXISTING_THRESH="6"
298     EXISTING_KILL="false"

```

```

300 EXISTING_BLOCKED="kp il"
301 EXISTING_HOST_MTU="1200"
302
303 if [[ -f ".env" ]]; then
304     [[ -n "$(read_env_var GATEWAY_RULE_PROFILE)" ]] && EXISTING_PROFILE=$(read_env_var GATEWAY_RULE_PROFILE
305 )
306     [[ -n "$(read_env_var GATEWAY_MSS)" ]] && EXISTING_MSS=$(read_env_var GATEWAY_MSS)
307     [[ -n "$(read_env_var GATEWAY_HIJACK_HOST)" ]] && EXISTING_HIJACK=$(read_env_var GATEWAY_HIJACK_HOST)
308     [[ -n "$(read_env_var GATEWAY_USE_EXIT_NODE)" ]] && EXISTING_EXIT=$(read_env_var GATEWAY_USE_EXIT_NODE)
309     [[ -n "$(read_env_var WARP_FAIL_THRESHOLD)" ]] && EXISTING_THRESH=$(read_env_var WARP_FAIL_THRESHOLD)
310     [[ -n "$(read_env_var KILL_SWITCH)" ]] && EXISTING_KILL=$(read_env_var KILL_SWITCH)
311     [[ -n "$(read_env_var BLOCKED_COUNTRIES)" ]] && EXISTING_BLOCKED=$(read_env_var BLOCKED_COUNTRIES)
312     [[ -n "$(read_env_var HOST_MTU)" ]] && EXISTING_HOST_MTU=$(read_env_var HOST_MTU)
313 fi
314
315 cat > .env <<EOF
316 WIREGUARD_PRIVATE_KEY=${PRIVATE_KEY}
317 WIREGUARD_PUBLIC_KEY=${PUBLIC_KEY}
318 WIREGUARD_ADDRESSES=${ADDRESSES}
319 TS_AUTHKEY=${TS_AUTHKEY}
320 GATEWAY_RULE_PROFILE=${EXISTING_PROFILE}
321 # Safe MSS for double-tunneled traffic (Tailscale + WARP). Change to 1180 if you experience slow speeds and
322 # have a healthy path.
323 GATEWAY_MSS=${EXISTING_MSS}
324 # Lower the host Tailscale MTU to 1200 to prevent fragmentation when passing through the 1280-byte WARP tunnel.
325 HOST_MTU=${EXISTING_HOST_MTU}
326 # Set to false to run as a 'Headless Server' (Docker acts as an exit node, but the host Mac/Linux's own
327 # internet is NOT hijacked).
328 GATEWAY_HIJACK_HOST=${EXISTING_HIJACK}
329 GATEWAY_USE_EXIT_NODE=${EXISTING_EXIT}
330 WARP_FAIL_THRESHOLD=${EXISTING_THRESH}
331 KILL_SWITCH=${EXISTING_KILL}
332 ADGUARD_PASSWORD=${ADGUARD_PASSWORD}
333 BLOCKED_COUNTRIES="${EXISTING_BLOCKED}"
334 EOF
335 ok ".env written."
336 fi
337
338 # 8. Prepare directories
339 step "Preparing AdGuard directories"
340
341 mkdir -p adguard/conf adguard/work/userfilters
342
343 # Remove empty AdGuardHome.yaml that would cause a crash loop (same logic as Toggle-Gateway scripts)
344 if [[ -f "adguard/conf/AdGuardHome.yaml" ]] && ! -s "adguard/conf/AdGuardHome.yaml" ]; then
345     rm "adguard/conf/AdGuardHome.yaml"
346     warn "Removed empty AdGuardHome.yaml to prevent crash loop."
347 fi
348
349 ok "Directories ready."
350
351 # 9. Compile DNS filter rules
352 step "Compiling DNS filter rules (this may take a minute)"
353
354 # 10. Start containers
355 step "Starting containers"
356 # Note: tailscale depends on adguardhome being healthy (port 5335 up),
357 # so Docker Compose will hold it back automatically until the wizard is done.
358 docker compose up -d
359 ok "Containers starting."
360
361 # 11. Wait for AdGuard Home setup wizard
362 step "Waiting for AdGuard Home setup wizard"
363 echo " (up to 60 seconds...)"
364
365 MAX=60; ELAPSED=0
366 until docker exec routing-fix curl -sf http://127.0.0.1:3000 >> output.log 2>&1; do
367     sleep 2; ELAPSED=$((ELAPSED + 2))
368     [[ $ELAPSED -ge $MAX ]] && die "AdGuard Home didn't come up.\nDebug with: docker compose logs adguardhome"
369 done
370
371 ok "AdGuard Home is up."
372
373 # 12. Configure AdGuard Home via API
374 step "Configuring AdGuard Home"
375
376 # Run all API calls in a single docker exec sh -c so the session cookie file
377 # is shared across calls without leaving the container.
378 docker exec routing-fix sh -c "
379     set -e

```



```

378
379 # Step 1: Complete the setup wizard (sets admin creds + DNS port to 5335)
380 curl -sf -X POST http://127.0.0.1:3000/control/install/configure \
381   -H 'Content-Type: application/json' \
382   -d '{"web":{"ip":"0.0.0.0","port":3000},"dns":{"ip":"0.0.0.0","port":5335},"username":"admin","password":"'${ADGUARD_PWD_ESC}'"}' \
383   >/dev/null || true
384
385 # Step 2: Wait for AdGuard to restart after wizard
386 sleep 4
387 WAITED=0
388 until curl -sf http://127.0.0.1:3000 >/dev/null 2>&1; do
389   sleep 2; WAITED=$((WAITED + 2))
390   [ \${WAITED} -ge 30 ] && { echo 'AdGuard did not restart in time'; exit 1; }
391 done
392
393 # Step 3: Login to get session cookie
394 curl -sf -X POST http://127.0.0.1:3000/control/login \
395   -H 'Content-Type: application/json' \
396   -d '{"name":"admin","password":"'${ADGUARD_PWD_ESC}'"}' \
397   -c /tmp/agh.cookie >/dev/null
398
399 # Step 4: Point upstream DNS back through the WARP tunnel
400 curl -sf -X POST http://127.0.0.1:3000/control/dns_config \
401   -H 'Content-Type: application/json' \
402   -b /tmp/agh.cookie \
403   -d '{"upstream_dns":["127.0.0.1:53","[/onion/]127.0.0.1:5353"],"bootstrap_dns":["1.1.1.1","8.8.8.8"],"upstream_mode":"load_balance"}' \
404   >/dev/null
405
406 # Step 5: Register the compiled rules file as a filter list
407 curl -sf -X POST http://127.0.0.1:3000/control/filtering/add_url \
408   -H 'Content-Type: application/json' \
409   -b /tmp/agh.cookie \
410   -d '{"name":"Custom Compiled Rules","url":"/opt/adguardhome/work/userfilters/compiled_rules.txt"}' \
411   >/dev/null || true
412
413 # Step 6: Force a filter refresh to load the rules immediately
414 curl -sf -X POST http://127.0.0.1:3000/control/filtering/refresh \
415   -H 'Content-Type: application/json' \
416   -b /tmp/agh.cookie \
417   -d '{"whitelist":false}' \
418   >/dev/null || true
419
420 rm -f /tmp/agh.cookie
421
422
423 ok "AdGuard Home configured."
424
425 # 13. Wait for Tailscale
426 step "Waiting for Tailscale to authenticate"
427 echo " (Tailscale only starts once AdGuard's healthcheck passes up to 90s)"
428
429 MAX=90; ELAPSED=0; TS_IP=""
430 until [[ -n "$TS_IP" ]]; do
431   TS_IP=$(docker exec tailscale tailscale ip -4 2>> output.log || true)
432   if [[ -z "$TS_IP" ]]; then
433     sleep 3; ELAPSED=$((ELAPSED + 3))
434     if [[ $ELAPSED -ge $MAX ]]; then
435       warn "Tailscale hasn't authenticated yet."
436       warn "Once it does, re-run the toggle script it will resolve the IP dynamically."
437       break
438     fi
439   fi
440 done
441
442 if [[ -n "$TS_IP" ]]; then
443   ok "Tailscale IP: ${TS_IP} (resolved dynamically by the toggle script on each startup)"
444 fi

```

46 scripts/unlock-files.sh

```

1 #!/bin/bash
2 # Resolve project root relative to this script's location, so the script
3 # works regardless of the caller's CWD.

```

```

4 SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
5 PROJECT_ROOT="$(cd "$SCRIPT_DIR/.." && pwd)"
6
7 echo "Unlocking .env..."
8 if [ -f "$PROJECT_ROOT/.env" ]; then
9     sudo cat "$PROJECT_ROOT/.env" > "$PROJECT_ROOT/.env.tmp" && mv "$PROJECT_ROOT/.env.tmp" "$PROJECT_ROOT/.env"
10    echo ".env unlocked"
11 fi
12
13 echo "Unlocking AdGuard cache files..."
14 for f in "$PROJECT_ROOT"/adguard/work/userfilters/compiled_rules.txt \
15         "$PROJECT_ROOT"/adguard/work/userfilters/ip_blocklist.ipset \
16         "$PROJECT_ROOT"/adguard/work/userfilters/cache/*.txt \
17         "$PROJECT_ROOT"/adguard/work/userfilters/cache/ip/*.txt \
18         "$PROJECT_ROOT"/adguard/work/data/filters/*.txt; do
19     if [ -f "$f" ]; then
20         cat "$f" > "$f.tmp" && mv "$f.tmp" "$f"
21         echo "Unlocked $f"
22     fi
23 done
24
25 echo ""
26 echo "All locked files have been successfully bypassed via atomic rename!"

```

47 tor-bridges.txt

48 white_list.txt

```

1 0.client-channel.google.com
2 idrv.com
3 2.android.pool.ntp.org
4 akamaihd.net
5 akamaitechnologies.com
6 akamaized.net
7 amazonaws.com
8 android.clients.google.com
9 api-adservices.apple.com
10 api.ipify.org
11 api.rlje.net
12 app-api.ted.com
13 appleid.apple.com
14 apps.skype.com
15 appsbackup-pa.clients6.google.com
16 appsbackup-pa.googleapis.com
17 appspot-preview.l.google.com
18 apt.sonarr.tv
19 aspnetscdn.com
20 attestation.xboxlive.com
21 ax.phobos.apple.com.edgesuite.net
22 books-analytics-events.apple.com
23 brightcove.net
24 c.s-microsoft.com
25 cdn.cloudflare.net
26 cdn.embedly.com
27 cdn.optimizely.com
28 cdn.vidible.tv
29 cdn2.optimizely.com
30 cdn3.optimizely.com
31 cdnjs.cloudflare.com
32 cert.mgt.xboxlive.com
33 clientconfig.passport.net
34 clients1.google.com
35 clients2.google.com
36 clients3.google.com
37 clients4.google.com
38 clients5.google.com
39 clients6.google.com
40 connectivitycheck.android.com
41 connectivitycheck.gstatic.com
42 continuum.dds.microsoft.com
43 cpms.spop10.ams.plex.bz

```

```
44 cpms35.spop10.ams.plex.bz
45 cse.google.com
46 ctldl.windowsupdate.com
47 cws.conviva.com
48 d2c8v52ll5s99u.cloudfront.net
49 d2gatte9o95jao.cloudfront.net
50 dashboard.plex.tv
51 dataplicity.com
52 def-vef.xboxlive.com
53 delivery.vidible.tv
54 dev.virtualearth.net
55 device.auth.xboxlive.com
56 display.ugc.bazaarvoice.com
57 displaycatalog.mp.microsoft.com
58 dl.delivery.mp.microsoft.com
59 dl.dropbox.com
60 dl.dropboxusercontent.com
61 dns.msftncsi.com
62 download.sonarr.tv
63 drift.com
64 driftt.com
65 dynupdate.no-ip.com
66 ecn.dev.virtualearth.net
67 edge.api.brightcove.com
68 eds.xboxlive.com
69 fonts.gstatic.com
70 forums.sonarr.tv
71 g.live.com
72 geo-prod.do.dsp.mp.microsoft.com
73 geo3.ggpht.com
74 gfws1.geforce.com
75 giphy.com
76 github.com
77 github.io
78 googleapis.com
79 gravatar.com
80.gstatic.com
81 help.ui.xboxlive.com
82 hls.ted.com
83 i.ytimg.com
84 il.ytimg.com
85 iadsdk.apple.com
86 imagesak.secureserver.net
87 img.vidible.tv
88 imgix.net
89 imgs.xkcd.com
90 instantmessaging-pa.googleapis.com
91 intercom.io
92 jquery.com
93 jsdelivr.net
94 keystone.mwbsys.com
95 lastfm-img2.akamaized.net
96 licensing.xboxlive.com
97 live.com
98 livepassdl.conviva.com
99 login.live.com
100 login.microsoftonline.com
101 manifest.googlevideo.com
102 meta-db-worker02.pop.ric.plex.bz
103 meta.plex.bz
104 meta.plex.tv
105 microsoftonline.com
106 msftncsi.com
107 my.plexapp.com
108 nexusrules.officeapps.live.com
109 nine.plugins.plexapp.com
110 no-ip.com
111 node.plexapp.com
112 notes-analytics-events.apple.com
113 notify.xboxlive.com
114 npr-news.streaming.adswizz.com
115 ns1.dropbox.com
116 ns2.dropbox.com
117 o1.email.plex.tv
118 o2.sg0.plex.tv
119 ocsp.apple.com
120 office.com
121 office.net
122 office365.com
123 officeclient.microsoft.com
124 om.cbsi.com
```

```

125 onedrive.live.com
126 outlook.live.com
127 outlook.office365.com
128 pings.conviva.com
129 placeholder.it
130 placeholder.imgix.net
131 players.brightcove.net
132 pricelist.skype.com
133 products.office.com
134 proxy.plex.bz
135 proxy.plex.tv
136 proxy02.pop.ord.plex.bz
137 pubsub.plex.bz
138 pubsub.plex.tv
139 raw.githubusercontent.com
140 redirector.googlevideo.com
141 res.cloudinary.com
142 s.gateway.messenger.live.com
143 s.marketwatch.com
144 s.youtube.com
145 s.ytimg.com
146 s1.wp.com
147 s2.youtube.com
148 s3.amazonaws.com
149 sa.symcb.com
150 secure.avangate.com
151 secure.brightcove.com
152 secure.surveymonkey.com
153 services.sonarr.tv
154 skyhook.sonarr.tv
155 spclient.wg.spotify.com
156 ssl.p.jwpcdn.com
157 staging.plex.tv
158 status.plex.tv
159 t.co
160 t0.ssl.ak.dynamic.tiles.virtualearth.net
161 t0.ssl.ak.tiles.virtualearth.net
162 tawk.to
163 tedcdn.com
164 themoviedb.com
165 thetvdb.com
166 tinyurl.com
167 title.auth.xboxlive.com
168 title.mgt.xboxlive.com
169 traffic.libsyn.com
170 tvdb2.plex.tv
171 tvthemes.plexapp.com
172 twimg.com
173 ui.skype.com
174 video-stats.l.google.com
175 videos.vidible.tv
176 vidtech.cbsinteractive.com
177 weather-analytics-events.apple.com
178 widget-cdn.rpxnow.com
179 win10.ipv6.microsoft.com
180 wp.com
181 ws.audioscrobbler.com
182 www.dataplicity.com
183 www.googleapis.com
184 www.msftconnecttest.com
185 www.msftncsi.com
186 www.no-ip.com
187 www.youtube-nocookie.com
188 xbox.ipv6.microsoft.com
189 xboxexperiencesprod.experimentation.xboxlive.com
190 xflight.xboxlive.com
191 xkms.xboxlive.com
192 xsts.auth.xboxlive.com
193youtu.be
194 youtube-nocookie.com
195 yt3.ggpht.com
196 zee.cws.conviva.com
197
198 # YouTube family base domains so every subdomain (video stream servers on
199 # *.googlevideo.com, image hosts on *.ytimg.com, profile thumbnails on
200 # *.ggpht.com, etc.) is automatically allowlisted via the AdGuard '|||domain^'
201 # wildcard. The specific s.youtube.com / manifest.googlevideo.com entries
202 # above are kept for clarity but will be optimized away by rule-compiler
203 # because their parents are now in this list.
204 youtube.com
205 googlevideo.com

```

```
206 yting.com
207 ggphht.com
208 gvt1.com
209 gvt2.com
210 gvt3.com
211 gvt1.net
212 gvt2.net
213 gvt3.net
214
215 edge-mqtt.facebook.com
216 gateway.instagram.com
217 i.instagram.com
218
219 facebook.com
220 fb.com
221 fbcdn.net
222 fbsbx.com
223 messenger.com
224 whatsapp.com
225 web.whatsapp.com
226 instagram.com
227 cdninstagram.com
```